
INVENTORY SHEET

AS350 B3 2B1 FLIGHT MANUAL

EASA CERTIFICATION

RFM dated on 10/01/2025

The following chapters are subjected to export control regulations.
Classified sections or appendices are provided within this Flight Manual only if relevant to the aircraft (equipment installed/not installed) and if authorized by the proper export licence.

The presence of this Inventory Sheet means the documentation has been checked and meets Export Control requirements.

US extraterritorial jurisdiction (ITAR)
US_EC_NoUScontent

US extraterritorial jurisdiction (Dual Use)
US_EC_NotAssessed

French Jurisdiction (ML)
FR_EC_NotListed

AS 350 B3 2B1
SITUATION DES REVISIONS DU MANUEL DE VOL
FLIGHT MANUAL REVISIONS STATUS
CERTIFICATION EASA
EASA CERTIFICATION

Ce manuel doit contenir la révision normale (RN) et les révisions rapides (RR) référencées dans l'édition (EDIT) considérée.

This manual must contain the normal revision (RN) and rush revisions (RR) listed under the relevant issue (EDIT).

PARTIE REGLEMENTAIRE PRESCRIBED SECTION Volume 1		
SECT. / SUP.	.	.
	EDIT	.
	.	DATE
0 => 5.1	RN7	24-30
SUP.0	RN5	23-40
SUP.4	RN0	15-45
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SUP.7	RN1	19-40
SUP.12	RN1	16-24
SUP.13	RN1	16-24
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SUP.13.2	RN1	16-24
SUP.14	RN0	15-45
SUP.16	RN0	15-45
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SUP.18	RN1	19-47
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SUP.19.1	RN1 RR2a	19-47 20-28
SUP.20	RN0	15-45
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PARTIE REGLEMENTAIRE PRESCRIBED SECTION Volume 1		
SECT. / SUP.	.	.
	EDIT	.
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SUP.28	RN0	15-45
SUP.29	RN0	15-45
SUP.40	RN0	16-33
SUP.50	RN0	15-45
SUP.52	RN0	15-45
SUP.55.2	RN0	15-45
SUP.56	RN0	15-45

PARTIE COMPLEMENTAIRE COMPLEMENTARY SECTION Volume 2		
SECT.	EDIT	DATE
0, 5.2, 6, 7, 8, 9	RN8	24-30

R

Merci de vous référer à la Fiche d'Inventaire de ce Manuel de Vol pour la classification relative au contrôle des exportations.
 Please refer to the Inventory Sheet of this flight manual for export control classification information.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

EASA TYPE CERTIFICATE No. EASA.R.008

REGISTRATION No.

SERIAL No.

APPROVED BY:
European Aviation Safety Agency

BY:
EASA No. 2004 - 7544

DATE:
on JULY 16, 2004

THE EFFECTIVITY OF THIS MANUAL AT THE LATEST REVISION IS SPECIFIED ON THE LIST OF EFFECTIVE PAGES.

IT IS THE OPERATOR'S RESPONSIBILITY TO MAINTAIN THIS MANUAL IN A CURRENT STATUS IN ACCORDANCE WITH THE LIST OF EFFECTIVE PAGES.

THIS HANDBOOK INCLUDES THE MATERIAL TO BE FURNISHED TO THE PILOT AS REQUIRED BY FAR-27 AND ADDITIONAL INFORMATION PROVIDED BY THE MANUFACTURER AND HAS BEEN APPROVED IN ACCORDANCE WITH THE EASA CERTIFICATION PROCEDURES.

THE EASA FLIGHT MANUAL CONSISTS OF ALL UNCODED AND CODED A PAGES MARKED "APPROVED".



Airbus Helicopters Direction Technique Support
Aéroport international Marseille-Provence 13725 Marignane Cedex - France

APPROVED

350 B3 2B1

0.0.P1

A

15-45

Page 1

REVISION TO AIRCRAFT PUBLICATION: AS350 B3 2B1
PUBLICATION CONCERNED: FLIGHT MANUAL
CUSTOMIZATION AIRCRAFT:

PMVR	REVISION No. : 7	DATE CODE: 24-30	CERTIFICATION CODE:	A
PMVN	REVISION No. : 8	DATE CODE: 24-30	CERTIFICATION CODE:	/

- The outline of the revision is given below :
 - . Sections or supplements affected (added or modified),
 - . Major points of the revision.
- Check that pages in each section are those specified in the list of effective pages.
- Withdraw old and insert new pages affected by this revision.
- Return the acknowledgement card.
- This list of amended pages may be filed (apart from the manual).

**THE CONTENT OF THE FLIGHT MANUAL REVISION
MUST BE BROUGHT TO THE ATTENTION OF FLIGHT CREWS.**

	DELETED PAGES			INSERTED PAGES		
	Section, SUP or APP	Pages	DATE CODE	Section, SUP or APP	Pages	DATE CODE
Inventory sheet EASA	-		23/07/2024	-		10/01/2025
SRD EASA	-		23/07/2024	-		10/01/2025
Inventory sheet TC	-		23/07/2024	-		10/01/2025
SRD TC	-		23/07/2024	-		10/01/2025
Inventory sheet FATA	-		23/07/2024	-		10/01/2025
SRD FATA	-		23/07/2024	-		10/01/2025
NORMAL REVISION	0.0.P5 (PMVR)	1 to 4	23-40	0.0.P5	1 to 5	24-30
	2.0.P6	1 to 2	21-08	2.0.P6	1 to 2	24-30
	2.1	1 to 2	23-40	2.1	1 to 2	24-30
	0.0.P5 (PMVN)	1 to 4	23-04	0.0.P5	1 to 4	24-30
	6.0.P6	1 to 2	15-45	6.0.P6	1 to 2	24-30
	6.2	1 to 7	15-45	6.2	1 to 8	24-30
	7.10	1 to 8	15-45	7.10	1 to 8	24-30

DESCRIPTION OF THE REVISION	Section	§
PMVR		
Update of the list of effective pages and the log of normal revisions (PMVR).	0.0.P5	-
Addition of new paragraph: 4 CABIN LOADING INSTRUCTIONS	2.0.P6	2.1
New paragraph: CABIN LOADING INSTRUCTIONS	2.1	4
PMVN		
Update of the list of effective pages and the log of normal revisions (PMVN).	0.0.P5	-
Update of § 6.2 (title)	6.0.P6	1
Update of figure 2 and Addition of a Note	6.2	2
Removal of low hydraulic pressure value	7.10	2.1
Removal of low hydraulic pressure value ("between 20 and 30 bar (290 psi and 435 psi)" replaced by "above minimum threshold")	7.10	3
Removal of low hydraulic pressure value	7.10	4.2

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COMPOSITION

OF APPROVED CONDITIONAL REVISIONS (RC)

This Manual assigned to the helicopter mentioned on the title page contains the following pink pages except those cancelled when the conditions are complied with.

CAUTION

The reader will have to insert the pink pages incorporating the paragraph(s) affected by the conditional revision so as the paragraph(s) cover(s) the paragraph(s) of the standard version or of the variant of standard definition.

(1) Paragraph Revision Code:

- **R** Revised, to be replaced
- **N** New, to be inserted

RC No.	SECTION or SUP.	PARAGRAPH	DATE CODE	Number of pages	(1)	Applicable before condition is met:
a	5.1	3.2.3 *RC*	15-45	1	R	MOD 07-3354
b	2.3	2 *RC*	16-28	1		MOD AL-4236
c	2.4	1 *RC*	21-42	1		MOD 07-8036
	2.4	4 *RC*	21-42	1		
	3.6	4 *RC*	21-42	1		

COMPOSITION

OF NON APPROVED CONDITIONAL REVISIONS (RC)

This Manual assigned to the helicopter mentioned on the title page contains the following pink pages except those cancelled when the conditions are complied with.

CAUTION

The reader will have to insert the pink pages incorporating the paragraph(s) affected by the conditional revision so as the paragraph(s) cover(s) the paragraph(s) of the standard version or of the variant of standard definition.

(1) Paragraph Revision Code:

- **R** Revised, to be replaced
- **N** New, to be inserted

RC No.	SECTION	PARAGRAPH	DATE CODE	Number of pages	(1)	Applicable before condition is met:
	0.0.P3	*RC*	15-45	1		
a	7.3	2 *RC*	15-45	1		MOD 07.4622

COMPOSITION
OF (APPROVED OR NON APPROVED)
RUSH REVISIONS (RR)

The Manual contains the following additional yellow page(s):

CAUTION

The reader will have to insert the yellow pages incorporating the paragraph(s) affected by the rush revision opposite the existing paragraph(s) of the standard version or of the variant of standard definition.

(1) Paragraph Revision Code:

- **R** Revised, to be replaced.
- **N** New, to be inserted.

RR No.	SECTION or SUP.	PARAGRAPH	DATE CODE	Number of pages	(1)

LIST OF APPROVED EFFECTIVE PAGES - EASA CERTIFICATION

(1) AIRWORTHINESS EFFECTIVITY:

- Without indication..... Applicable to all aircraft
- **A**..... Specific to EASA.

(2) VARIANT OF STANDARD DEFINITION EFFECTIVITY:

- Without indication..... Applicable to all aircraft
- XXX..... Specific to aircraft equipped with XXX

SECTION	PAGES	DATE CODE	(1)	(2)
0.0.P1	1 to 1	15-45	A	
0.0.P2	1 to 1	15-45		
0.0.P3	1 to 1	23-40		
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1.0.P6	1 to 1	15-45		
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1.2	1 to 2	15-45		
1.3	1 to 4	15-45		
1.4	1 to 1	15-45		
2.0.P6	1 to 2	24-30		
2.1	1 to 2	24-30		
2.2	1 to 1	15-45		
2.3	1 to 5	23-40		
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3.6	1 to 12	23-40		
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4.0.P6	1 to 2	23-40		
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4.3	1 to 5	23-40		
4.4	1 to 1	19-40		

LIST OF APPROVED EFFECTIVE PAGES - EASA CERTIFICATION

SECTION	PAGES	DATE CODE	(1)	(2)
4.5	1 to 2	23-40		
4.6	1 to 2	23-40		
4.7	1 to 1	19-23		
4.8	1 to 1	15-45		
5.1.P6	1 to 1	15-45		
5.1	1 to 14	23-40		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) - EASA

ISSUE 1: NR 0 to NR 8:

NORMAL REVISION 8 - JUNE 2014	Approved on September 03, 2015 under the authority of EASA DOA No 21J056
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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016 under the authority of EASA DOA No 21J056
Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 16-28		EASA approved No.10064924 on March 09, 2018
Title	Insertion of limitations "icing and snow", modification 07-3368 (altimeter), limitation T4 in flight Max Transient; process for MGB and TGB oil temperature checking when preheating system is used for oil "O-155" and "O-156"	
Revised information	Pages: 0.0.P5 pages 1 to 3, 2.1 page 1, 2.4 page 3 and 2.5 page 5	
Deleted information	None	
NORMAL REVISION 2 date code 19-23		Approved on November 07, 2019 under the authority of EASA DOA No. 21J700
Title	Installation of crash resistant fuel system	
Revised information	Pages: 0.0.P5 pages 1 to 3, 4.7 page 1	
Deleted information	None	

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) - EASA

NORMAL REVISION 3 date code 19-40		Approved on October 23, 2020 under the authority of EASA DOA No. 21J700
Title	Data transfer from 0.0.P3*RC* pink page to 0.0.P3 white page. 0.0.P3*RC* is cancelled (no longer necessary) - Added an "Engine Starter/Generator" paragraph in the limitation section - Harmonization of NOTE section 2.3 - Improvement of the hydraulic emergency procedure - Harmonization of P2 air bleed procedures at take-off and landing.	
Revised information	0.0.P3 page 1; 0.0.P5 pages 1 to 4; 2.0.P6 page 2; 2.3 pages 1 and 4; 2.4 pages 2 and 3; 2.5 pages 6 and 7; 3.0.P6 page 2; 3.6 pages 1 to 7 and 10 to 11; 4.3 pages 1 to 3; 4.4 page 1; 4.5 page 1.	
Deleted information	None	
NORMAL REVISION 4 date code 21-08		EASA Approval No. 10078088 on January 11, 2022
Main points	Modification of Chinese fuel, deletion of Fuel placard	
Revised information	Sections: 0.0.P5; 2.0.P6; 2.5 and 2.6.	
Deleted information	Deletion of Fuel placard	
NORMAL REVISION 5 date code 21-42		EASA Approval No. 10082223 on June 23, 2023
Title	New NR/Nf indicator (MOD 07.8036).	
Revised information	0.0.P3 page 1; 0.0.P5 pages 1 to 4; 2.4 pages 1 and 2; 3.6 pages 1 and 9.	
Deleted information	None	
NORMAL REVISION 6 date code 23-40		Approved on June 21, 2024 under the authority of EASA DOA No. 21J700
Title	- Modification allowing the use of NATO 0.155 lubricant below -20°C, "Engine Starter/Generator", "Tail rotor failure", "Engine health check" and "cranking" procedures, - Addition of "After cockpit fire extinguisher use" procedure and VEMD caution message, - Minor corrections and wording improvement.	
Revised information	0.0.P3; 0.0.P5; 2.1 page 1; 2.3 page 4; 2.5 pages 1, 4 to 6; 3.0.P6 pages 1 to 2; 3.1 page 2; 3.2 pages 1 to 3; 3.3 page 2; 3.4 page 2; 3.5 pages 1 to 2; 3.6 pages 4, 6 to 7; 3.7 pages 1 and 3; 4.0.P6 page 1; 4.3 pages 1 to 3 and 5; 4.5 page 1; 4.6 page 1; 5.1 page 2.	
Deleted information	None	

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) - EASA

NORMAL REVISION 7 date code 24-30		Approved on November 13, 2024 under the authority of EASA DOA No. 21J700
Title	Addition of freight transportation information.	
Revised information	0.0.P5; 2.0.P6 page 1; 2.1 page 2.	
Deleted information	None	

LIST OF EFFECTIVE PAGES

(1) AIRWORTHINESS EFFECTIVITY:

- Without indication..... Applicable to all aircraft
- Indicated Specific to indicated civilian airworthiness.

(2) VARIANT OF STANDARD DEFINITION EFFECTIVITY:

- Without indication..... Applicable to all aircraft
- XXX..... Specific to aircraft equipped with XXX

SECTION	PAGES	DATE CODE	(1)	(2)
0.0.P3	1 to 1	22-06		
0.0.P5	1 to 4	24-30		
0.0	1 to 2	22-06		
5.2.P6	1 to 1	15-45		
5.2	1 to 18	15-45		
6.0.P6	1 to 1	24-30		
6.1	1 to 2	15-45		
6.2	1 to 8	24-30		
6.3	1 to 3	15-45		
6.4	1 to 3	15-45		
6.5	1 to 4	15-45		
7.0.P6	1 to 3	20-36		
7.1	1 to 3	19-23		
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8.0.P6	1 to 1	22-24		
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8.2	1 to 4	19-23		
8.3	1 to 19	22-24		

LIST OF EFFECTIVE PAGES

8.4	1 to 1	22-24		
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LOG OF NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2)

ISSUE 1: NR 0 to NR 10:

NORMAL REVISION 10 - SEPTEMBER 2015	
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ISSUE 2:

NORMAL REVISION 0 - date code 15-45		
Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 - date code 16-33		
Title	Revision of section 7.6	
Revised information	0.0.P.5: pages 1, 2 and 3; Section 7 page 7.0.P6, Section 7.6 pages 1, 2, 3, 4, 5 and 6	
Deleted information	None	
NORMAL REVISION 2 - date code 19-23		
Title	Installation of crash resistant fuel system, minor corrections	
Revised information	0.0.P.5: pages 1 to 3, Section 7.1 page 3, Section 7.11 pages 5, 6, 8 and 9, Section 8.2 page 1	
Deleted information	None	
NORMAL REVISION 3 - date code 20-36		
Title	Integration of camera APPAREO VISION 1000	
Revised information	0.0.P.5 pages 1 to 3, 0.0 page 1, Section 7.0.P6 page 3, Section 7.14 pages 1 and 2	
Deleted information	None	

LOG OF NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2)

NORMAL REVISION 4 - date code 22-06	
Title	-Addition of MOD 07-20060: "Blade-shedding improvement with Arriel 2B1 engine and EMC101" and MOD 07-20061: "Blade-shedding improvement with Arriel 2B1 engine and FADEC B". -Data transfer from 0.0.P3*RC* pink page to 0.0.P3 white page. 0.0.P3*RC* is canceled (no longer necessary).
Revised information	0.0.P3 page 1, 0.0.P.5 pages 1 to 4, 0.0 pages 1 and 2, Section 7.8 pages 3 and 4.
Deleted information	None
NORMAL REVISION 5 - date code 22-22	
Title	Relocation of "FLOAT ARM" and "FLOAT FIRE" pushbuttons (MOD 07.20175)
Revised information	0.0.P.5 pages 1 to 4, Section 7.2 pages 1 to 3.
Deleted information	None
NORMAL REVISION 6 - date code 22-24	
Title	Modification of sheets 2A, 2C, 2D, 3A, 3B, 4, 5, 6B and 7 in Section 8.3 Addition of Section 8.4
Revised information	0.0.P5 pages 1 to 4, Section 8.0.P6 page 1, Section 8.3 pages 6, 8, 9, 12 to 15 and 17 to 18, Section 8.4 page 1.
Deleted information	None
NORMAL REVISION 7 - date code 23-04	
Title	Introduction of a warning and a caution in the lighting section
Revised information	0.0.P5 pages 1 to 4, Section 7.6 page 5.
Deleted information	None
NORMAL REVISION 8 - date code 24-30	
Title	Identification of eligible area for cabin freight and baggage and deletion of hydraulic pressure values (flight manual harmonization).
Revised information	0.0.P5 pages 1 to 4, 6.0.P6 page 1, 6.2 page 3, 7.10 pages 2, 5 and 6.
Deleted information	None

LIST OF MODIFICATIONS OR SERVICE BULLETINS MENTIONED IN THE FLIGHT MANUAL

This list includes all modifications or service bulletins that are or have been referenced in the flight manual.

MODIFICATION / SB	DESCRIPTION	Embodiment of MOD / SB	
		Yes	No
MOD 07-3195	VEMD 2003		
MOD 07-3273	New SCU panel		
MOD 07-3274	DC distribution modifications (direct BATT busbar and breakers panels instead of fuses panels)		
MOD 07-3317	HYDR warning light flashes when the hydraulic test switch is depressed		
MOD 07-3354	VEMD 2006		
MOD 07-3363	AFCS installation without collective pitch to yaw coupling		
MOD 07-3368	Instrument panel and console standardization		
MOD 07-3409	AFCS with new monitoring panel		
MOD 07-4614	New load indicator.		
MOD 07-4622	Yaw servo test lighting modification on dual hydraulic H/C		
MOD 07-4719	Modification of "ACCU_TST" button in unstable button for SMS bloc and Multibloc (applicable only for aircraft equipped with dual hydraulic)		

This list includes all modifications or service bulletins that are or have been referenced in the flight manual.

MOD MC 8024	Camera APPAREO VISION 1000		
MOD 07-20060 or MOD 07-20061	Blade-shedding improvement with Arriel 2B1 engine and EMC101 or Blade-shedding improvement with Arriel 2B1 engine and FADEC B		

SECTION 1
GENERAL
CONTENTS

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1.1 PRELIMINARY NOTES	
1 GENERAL.....	1
2 DESCRIPTION OF THE MANUAL	1
1.2 UPDATING	
1 GENERAL.....	1
2 REVISIONS	1
1.3 SYMBOLS AND CONVERSION FACTORS	
1 SYMBOLS AND ABBREVIATIONS	1
2 CONVERSION FACTORS.....	4
1.4 TERMINOLOGY	
1 GENERAL.....	1
2 USE OF PROCEDURAL WORDS	1

SECTION 1.1

PRELIMINARY NOTES

1 GENERAL

To achieve the required degree of safety, this manual must be used in conjunction with the relevant regulations covering aircraft operation, such as aerial navigation laws in the operator's country.

It is essential for the crew to become familiar with the contents of this manual, particularly with the information specific to customized configurations, and to check all revisions and related requirement.

2 DESCRIPTION OF THE MANUAL

This manual contains legally approved information, together with additional manufacturer's information not subject to approval.

- The approved information is contained in PART 1 "FLIGHT MANUAL", in SECTIONS 1, 2, 3, 4, 5.1, in the SUPPLEMENTS and the Appendix.
- The information not subject to Approval is contained in PART 2 "COMPLEMENTARY FLIGHT MANUAL", as a complement to PART 1. This information is covered by SECTIONS 5.2, 6, 7, 8, and 9.

Each PART, each Supplement and each Appendix of the Manual make up a whole and, for this reason, incorporate its own list of effective pages and are revised separately.

The list of effective pages (P5) identifies all the pages which compose the manual.

The total number of P5 pages is shown on the List of Effective Pages, identified 1/xy where xy is a number between 01 and 99 corresponding to the number of P5 pages.

2.1 BASIC AIRCRAFT

The basic helicopter specifications are covered by SECTIONS 1 through 9.

2.2 SPECIAL SYSTEMS AND PROCEDURES

Information concerning optional equipment systems and operational procedures is covered by SUPPLEMENTS. These are mini Flight Manuals covering any differences from the basic aircraft information, SECTION by SECTION. The SUPPLEMENTS are approved on an individual basis.

2.3 ADAPTATION OF MANUAL TO CERTIFICATION REQUIREMENTS

Specific certification requirements may necessitate modifications to the text or layout of certain pages.

Therefore, a specific Flight Manual (PART 1) is drawn up for each certification.

Each Flight Manual includes its own particular title page; the alphabetical code, corresponding to the relevant certification, appears in the lower left-hand corner of each page of the approved PART 1.

SECTION 1.2

UPDATING

1 GENERAL

This Manual is updated periodically through Rush Revisions (RR) or Normal Revisions (RN).

2 REVISIONS

The Manufacturer makes every effort to keep this Manual updated by revisions to complete the user's information and capabilities. Each revision is accompanied by instructions summarizing the major points affected by the change and advising the person responsible for incorporating the revised pages in the Manual. (The instruction sheet can be filed separately from the Manual).

The user is responsible for ensuring proper updating of the Manual complying with the List of Pages given at the beginning of PART 1, PART 2 and of each supplement, since each of these PARTS or Supplements is revised separately.

The composition must be checked by number of page and by the date code. The date code is composed of the last two digits of the year, followed by the number of the week in that year.

2.1 NORMAL REVISIONS (RN) PRINTED ON WHITE PAPER

Normal revisions fully or partially update the Manual. The pages may be new pages or may supersede the existing pages.

They are printed on white paper.

The Manual effectivity is specified on the new list of approved effective pages (0.0.P5, SUP.0.P5 and/or APP.X.X.P5).

Normal revisions are identified in numerical order.

2.2 RUSH REVISIONS (RR) PRINTED ON YELLOW PAPER

Rush revisions partially update a few major points in the Manual.

The new information is given on a page which must face the former text to be modified or completed.

The Rush Revision is printed on yellow paper.

No white page is deleted.

The revised pages are specified on a separate list (0.0.P4 or SUP.0.P4).

Rush revisions are identified by the number of the next normal revision and a letter suffix in normal alphabetical order. Several rush revisions may be issued between two normal revisions. All rush revisions are cancelled when the normal revision bearing the same number is issued. If certain rush revision provisions remain after the subsequent normal revision, they are confirmed by a new rush revision with another identification code.

2.3 CONDITIONAL REVISIONS (RC) PRINTED ON PINK PAPER

The revised manual issued on white pages, corresponds to the recommended standard.

For helicopters authorized to fly at an earlier standard, the conditional revision (RC) retains the previous standard.

The user is responsible for embodiment of the aircraft modification(s) required for compliance with the recommended standard, after which the pink pages may be deleted under the user's responsibility.

The pink pages are specified on a separate list (0.0.P3 or SUP.0.P3).

NOTE

These pages are unaffected by normal and rush revisions or by customization.

2.4 THE "ERRATUM" PROCEDURE

In the case of minor errors (typing errors, bad printing) likely to affect the understanding of the text, the "ERRATUM" procedures are used to make quick corrections between revisions. In this case, the pages affected by the procedures are re-issued completely and the date code is underlined for identification. These pages are summarized on an accompanying sheet which is not identified.

SECTION 1.3

SYMBOLS AND CONVERSION FACTORS

1 SYMBOLS AND ABBREVIATIONS

DESIGNATION	SYMBOL OR ABBREVIATION
<u>SPEEDS</u>	
Calibrated airspeed	CAS
Indicated airspeed	IAS
True airspeed	TAS
Never exceed speed	VNE
Best rate of climb speed	V _y
Rate of climb	R/C
Rate of descent	R/D
<u>METEOROLOGY</u>	
International Standard Atmosphere	ISA
Outside Air Temperature	OAT
Outside air pressure	p
Relative air density	σ
Wind velocity	V _w
<u>ALTITUDE / HEIGHT</u>	
Geometric altitude	H
Pressure altitude	H _p
Density altitude	H _{σ}
Radio altimeter height	HRA
Height	h
<u>POWER / ENGINE PARAMETERS</u>	
Maximum Continuous Power	MCP
Maximum Takeoff Power (5 min.)	MTOP
Power	PWR
Engine Power Check	EPC
Rotor speed	NR
Engine generator speed	Ng
Engine generator deviation indication	ΔNg
Free turbine speed	Nf
Torque	Tq
Power turbine inlet temperature	t ₄
First Limitation Indicator	FLI

DESIGNATION	SYMBOL OR ABBREVIATION
<u>HOVER / TAKEOFF / LANDING</u>	
Hover In Ground Effect	HIGE
Hover Out Of Ground Effect	HOGF
<u>WEIGHT AND BALANCE</u>	
Center of Gravity	CG
Empty Weight	EW
Equipped Empty Weight	EEW
Operating Empty Weight	OEW
Useful load	UL
Payload	P/L
All-Up Weight	AUW
Maximum Take-Off Weight	MTOW
<u>MISCELLANEOUS</u>	
Automatic Direction Finder	ADF
Automatic Flight Control System	AFCS
Ancillary System Unit	ASU
Battery Contactor	BATC
Caution and Warning Panel	CWP
Direct Current	DC
Emergency Locator Transmitter	ELT
Electrical Master Box	EMB
Engine	ENG
Engine Back-up Control Ancillary Unit	EBCAU
Electronic Engine Control Unit	EECU
Equivalent	≅
Essential contactor	ESSC
External Power Line Contactor	EPLC
External Power Unit	EPU
Full Authority Digital Engine Control	FADEC
Generator Line Contactor	GLC
Global Positioning System	GPS

DESIGNATION	SYMBOL OR ABBREVIATION
MISCELLANEOUS (cont'd)	
Hall effect sensors	HECS
High Load Contactor	HLC
Horizontal Situation Indicator	HSI
Height-Velocity	HV
Intercommunication system	ICS
Left hand side	LH
Main gear box	MGB
Main bus voltage	U bus
Minute	min.
Right hand side	RH
Radio magnetic indicator	RMI
Second	sec.
Shed bus contactor	SBC
Starting contactor	SC
Systems Control Unit	SCU
To be confirmed	TBC
To be defined	PENDING
Tail gear box	TGB
Vehicle and engine management display	VEMD

- Symbol used for switches or pushbuttons
(example)

: [HORN]
- Symbol used for warning lights
(example)

:

GOV

GOV

Light ON

Light OFF
- Symbol used for VEMD indications
(example)

:

P2

P2

indication ON

indication OFF

2 CONVERSION FACTORS

2.1 METRIC UNITS TO OTHER UNITS

Multiply	By	To obtain
Centimeter (cm).....	0.3937	Inch (in)
Meter (m).....	3.2808	Foot (ft)
Meter per second (m/s)	196.85	Foot per minute (ft/min)
Kilometer (km).....	0.5400	Nautical mile (Nm)
Liter (l).....	0.2642	US gallon (US gal)
Liter (l).....	0.2200	UK gallon (UK gal)
Kilogram (kg).....	2.2046	Pound (lb)
Bars (bar)	14.504	Pound per Square Inch (psi)
Kilometer per hour (km/h).....	0.5400	Knot (kt)
Hecto pascal (hPa).....	0.02953	Inch of Mercury (inHg)

Conversion of degree centigrade (°C) into degree Fahrenheit (°F): °F = (°C x 9/5) + 32

2.2 OTHER UNITS TO METRIC UNITS

Multiply	By	To obtain
Inch (in)	2.5400	Centimeter (cm)
Foot (ft).....	0.3048	Meter (m)
Foot per minute (ft/min)	0.00508	Meter per second (m/s)
Nautical mile (Nm).....	1.8520	Kilometer (km)
US gallon (US gal).....	3.7850	Liter (l)
UK gallon (UK gal).....	4.5460	Liter (l)
Pound (lb).....	0.4536	Kilogram (kg)
Pound per Square Inch (psi).....	0.0689	Bar (bar)
Knot (kt).....	1.8520	Kilometer per hour (km/h)
Inch of Mercury (inHg).....	33.864	Hecto pascal (hPa)

Conversion of degree Fahrenheit (°F) into degree centigrade (°C): °C = 5/9 x (°F – 32)

SECTION 1.4

TERMINOLOGY

1 GENERAL

Unless otherwise specified in the text, altitudes are pressure-altitudes (Hp), speeds are indicated airspeeds (IAS).

Warnings, Cautions and Notes are used throughout this manual to emphasize important and critical instructions and are used as follows:

WARNING

AN OPERATING PROCEDURE, PRACTICE, ETC., WHICH, IF NOT CORRECTLY FOLLOWED, COULD RESULT IN PERSONAL INJURY OR LOSS OF LIFE.

CAUTION

An operating procedure, practice, etc., which, if not strictly observed, could result in damage to, or destruction of helicopter parts or equipment.

NOTA

An operating procedure, condition, etc., which is essential to highlight.

2 USE OF PROCEDURAL WORDS

The concept of procedural word usage and intended meaning which has been adhered to in preparing this manual is as follows:

- "Shall" or "must" has been used only when application of a procedure is mandatory.
- "Should" has been used only when application of procedure is recommended.
- "May" and "Need not" have been used only when application of a procedure is optional.
- "Will" has been used only to indicate future event or action, never to indicate a mandatory procedure.

SECTION 2

LIMITATIONS

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2 OCCUPANTS	1
3 INSTRUMENT MARKINGS	2
4 CABIN LOADING INSTRUCTIONS	2
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SECTION 2.1

GENERAL LIMITATIONS

The helicopter is approved on the basis of the FAR part 27 "NORMAL" rotorcraft category. The helicopter shall be operated in compliance with the limitations of this section.

1 TYPE OF OPERATIONS

The helicopter is approved to operate:

- By day and night in VFR.

NOTE

Additional equipment may be required by operational regulations.

The following are forbidden:

- Aerobatic maneuvers.
- Engine starting when snow or ice accumulations are in or around the engine air intake.
- Flight in freezing rain or icing conditions.
(visible moisture and temperature conducive to producing ice).
- In-flight engine power reduction using twist grip control except for engine failure training, emergency procedures referring to it.
- In-flight intentional complete VEMD cut-off (lane 1 + 2).

2 OCCUPANTS

- Minimum flight crew One pilot in right seat
- Maximum number of occupants
(including flight crew) Six

3 INSTRUMENT MARKINGS

Limitations are marked on instruments with the following color code:

FM 50.0078.00

Red		: Safety limit or takeoff limitation
Red with white hatching		: VNE, power off
Yellow or amber	 or 	: Caution range or takeoff power
Green		: Normal operating range
White mark		: Equipment operating limit
Red triangle		: Transient limit

On the VEMD, related numerical values of parameters are underlined:

- In yellow, when the parameter is in caution or takeoff power range,
- In red, when at or above a safety limit or maximum takeoff power. Moreover, to attract attention, red underlining flashes.

4 CABIN LOADING INSTRUCTIONS

The cabin floor rings are certified as provisions only - Refer to Section 6.2.
There is no restraint system, cabin freight and baggage configuration certified by Airbus Helicopters.

SECTION 2.2
WEIGHT AND BALANCE LIMITS

1 WEIGHT LIMITS

- Maximum permissible internal weight in flight : 2250 kg (4961 lb).
- Maximum permissible internal weight for IGE, take-off and landing : 2250 kg (4961 lb).

2 LONGITUDINAL CG

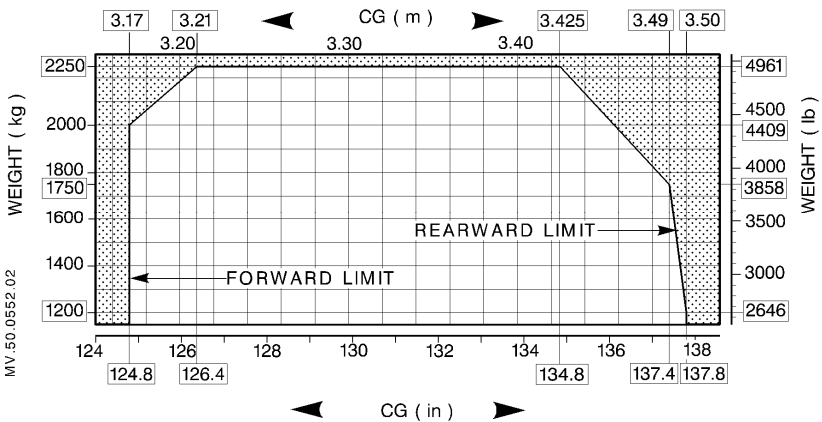


Figure 1: Longitudinal CG chart

NOTE

The datum is located 3.40 m (133.8 in) forward of the main rotor head center line.

3 LATERAL CG

- Maximum left CG 0.18 m (7.08 in).
- Maximum right CG 0.14 m (5.51 in).

NOTE

The datum is located in the plane of symmetry of the helicopter.

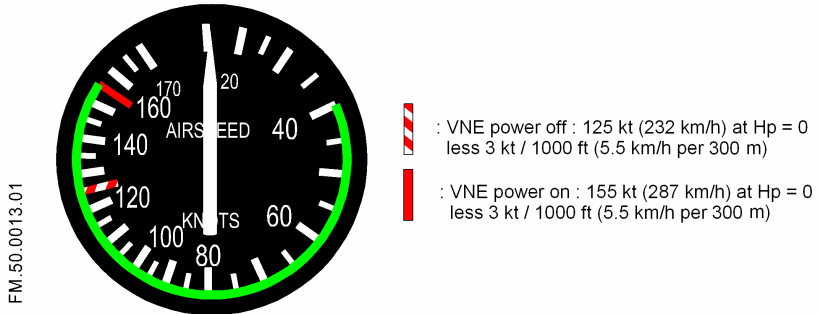
SECTION 2.3

FLIGHT ENVELOPE LIMITS

1 AIRSPEED LIMITS

All airspeed limitations are Indicated Airspeeds.

1.1 WITH DOORS CLOSED



- When OAT < -30°C, calculated VNE power on must be reduced by 10 kt (18.5 km/h),
- When OAT < -20°C, calculated VNE power off must be reduced by 20 kt (37 km/h), with a minimum of 65 kt (120 km/h).

1.2 WITH DOORS OPEN OR REMOVED

NOTE

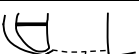
Flight with any configuration not shown is prohibited.

In configurations with at least one door opened or removed, loose objects shall not be in the cabin, cushions of unoccupied seats shall be properly secured or removed and seat belts shall be stowed.

Caption code used for open or removed doors limitations:

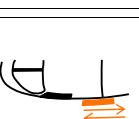
 : door closed
  : door removed
  : sliding door closed
 : sliding door maneuvering in flight
  : sliding door open-locked or removed

• **Aircraft fitted with four standard doors (LH and RH hand doors)**

RH doors \ LH doors			
	VNE	VNE	110 kt (204 km/h) or VNE*
	VNE	110 kt (204 km/h) or VNE*	110 kt (204 km/h) or VNE*
	100 kt (185 km/h) or VNE*	110 kt (204 km/h) or VNE*	110 kt (204 km/h) or VNE*

(*) lowest value

• **Aircraft fitted with LH sliding door (optional) and RH standard doors**

RH doors \ LH doors				
	VNE	VNE	70 kt (130 km/h) or VNE*	110 kt (204 km/h) or VNE*
	135 kt (250 km/h) or VNE*	110 kt (204 km/h) or VNE*	100 kt (185 km/h) or VNE*	110 kt (204 km/h) or VNE*
	100 kt (185 km/h) or VNE*	110 kt (204 km/h) or VNE*	100 kt (185 km/h) or VNE*	110 kt (204 km/h) or VNE*
	 FLIGHT PROHIBITED			
	opening : 110 kt (204 km/h) or VNE* closing : 80 kt (148 km/h) or VNE*			

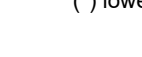
(*) lowest value

Caption code used for open or removed doors limitations:

 : door closed  : door removed  : sliding door closed

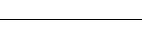
 : sliding door maneuvering in flight  : sliding door open-locked or removed

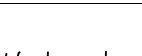
• Aircraft fitted with LH standard doors and RH sliding door (optional)

RH doors \ LH doors					
					
	VNE	110 kt (204 km/h) or VNE*	100 kt (185 km/h) or VNE*		opening : 60 kt (111 km/h) or VNE*
	VNE	110 kt (204 km/h) or VNE*	110 kt (204 km/h) or VNE*		closing : 60 kt (111 km/h) or VNE*
	100 kt (185 km/h) or VNE*	110 kt (204 km/h) or VNE*	110 kt (204 km/h) or VNE*		 FLIGHT PROHIBITED

(*) lowest value

• Aircraft fitted with RH and LH sliding doors (optional)

RH doors \ LH doors					
					
	VNE	110 kt (204 km/h) or VNE*	100 kt (185 km/h) or VNE*		opening : 60 kt (111 km/h) or VNE*
	135 kt (250 km/h) or VNE*	110 kt (204 km/h) or VNE*	110 kt (204 km/h) or VNE*		closing : 60 kt (111 km/h) or VNE*
	100 kt (185 km/h) or VNE*	110 kt (204 km/h) or VNE*	110 kt (204 km/h) or VNE*		



opening : 60 kt (111 km/h) or VNE*
Closing : 60 kt (111 km/h) or VNE*


FLIGHT PROHIBITED


FLIGHT PROHIBITED

(*) lowest value

2 ALTITUDE LIMITS

Maximum operating altitude in flight Hp = 23000 ft (7010 m)

3 TEMPERATURE LIMITS

- Minimum temperature - 40°C
- Maximum temperature ISA+35°C
limited to +50°C

For cold weather operations, refer to SUP.4.

4 LANDING AND ROTOR STOPPING LIMITATIONS ON SLOPE

- Nose up..... 10°
- Nose down 6°
- Sideways..... 8°

5 RUNNING LANDING LIMITATIONS

Excluding emergencies and failures, maximum speed for performing running landings 40 kt (74 km/h).

6 MANEUVERING LIMITATIONS

Continued operation in servo transparency (where load feedback is felt in the controls) is prohibited.

WARNING

SERVO-TRANSPARENCY PHENOMENON COULD LEAD TO A HAZARDOUS SITUATION AT LOW CLEARANCE FROM GROUND, AND PARTICULARLY DURING RIGHT TURN MANEUVER.

RC b

The paragraph 2 - **ALTITUDE LIMITS**, is superseded by:

2 ALTITUDE LIMITS

Post MOD 07-3368

Maximum operating altitude in flight..... Hp = 20000 ft (6096 m)

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION AL-4236.

7 FLIGHT IN FALLING SNOW

- Flight when visibility is greater than 1500 m (0.81 NM) :
 - Flight in falling snow is authorized.
- Flight when visibility is within 800 to 1500 m (0.43 to 0.81 NM) :

- Total flying time in falling snow is limited to 10 min.

This time limit includes the time required to leave all snowy conditions, irrespective of the visibility.

- Flight when visibility is lower than 800 m (0.43 NM) :
 - Flight in falling snow is prohibited.

NOTE

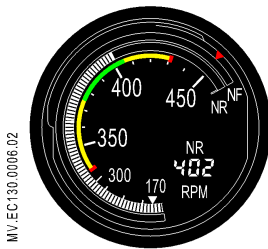
Refer to SUP.4 for check before flight.

RC c

The paragraph 1 - **MAIN ROTOR LIMITATIONS**, is superseded by the following paragraph:

1 MAIN ROTOR LIMITATIONS

It is prohibited to use the rotor brake prior to engine shutdown.
Minimum time between two consecutive brake applications: 5 min.



- ▽ : 170 rpm Rotor brake operation max. speed
- : 320 rpm min. power-off
- : 320 to 375 rpm Caution range
- : 375 to 405 Normal operating range
- : 405 to 430 Caution range
- : 430 rpm max. power-off

NOTE

Low NR aural warning ≤ 360 rpm
High NR aural warning ≥ 410 rpm

CAUTION

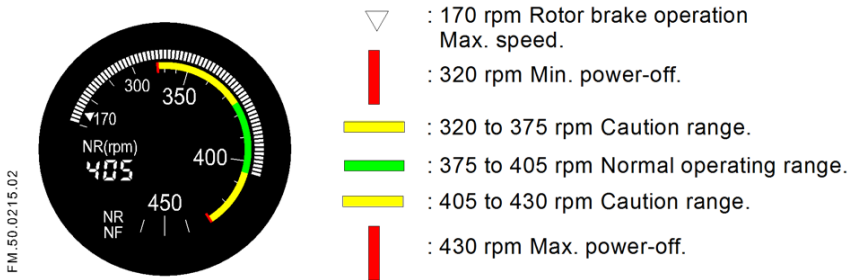
THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.8036.

SECTION 2.4

VEHICLE LIMITATIONS

1 MAIN ROTOR LIMITATIONS

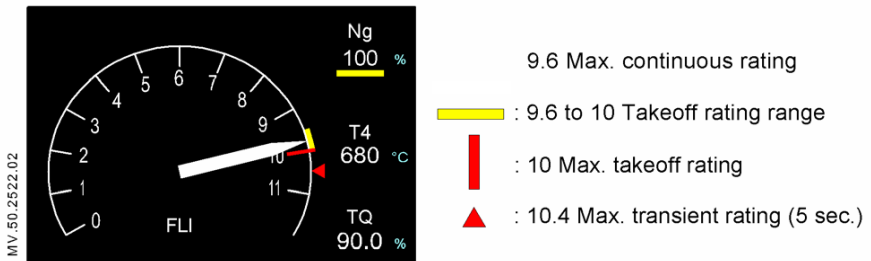
It is prohibited to use the rotor brake prior to engine shutdown.
Minimum time between two consecutive brake applications: 5 min.



NOTE

Low NR aural warning ≤ 360 rpm
High NR aural warning ≥ 410 rpm

2 FIRST LIMITATION INSTRUMENT



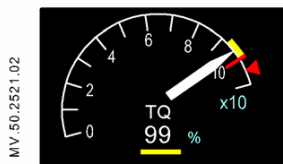
Use of heating and demisting is forbidden above the engine maximum continuous rating (Ng or t4).

NOTE

The values (Ng = 100 %, t4 = 680 °C, Tq = 90%) are given as examples.

3 MAIN TRANSMISSION LIMITATIONS

- TORQUE LIMITATIONS



92.7 % Max. continuous rating

- █ : 92.7 % to 100 % Takeoff rating torque range from 0 to 40 kt
- █ : 100 % Max. takeoff rating torque
- ▲ : 104 % Max. transient torque (5 sec.)

NOTE

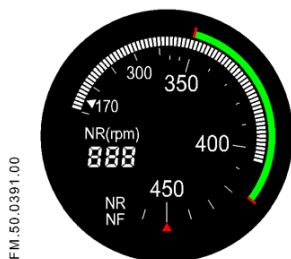
Use of takeoff rating torque range has no time limit.
100 % torque corresponds to 535 kW at 386 NR rpm.

4 ENGINE LIMITATIONS

Use of heating and demisting is forbidden above the engine maximum continuous rating (Ng or t4).

Use of engine takeoff rating range, is limited to 5 continuous minutes.

- Nf LIMITATIONS



- █ : 349 rpm Mini.
- █ : 349 to 418 rpm Normal operating range
- █ : 418 rpm Max. continuous
- ▲ : 450 rpm Max. transient limit (5 sec.)

NOTE

A NR rotor speed of 386 rpm corresponds to a Nf speed of 39158 rpm.

RC c

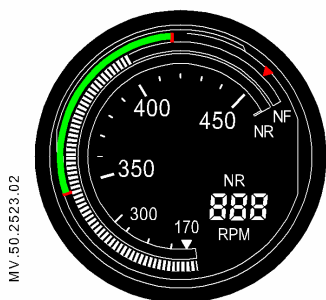
The paragraph 4 - **ENGINE LIMITATIONS**, is modified as follows:

4 ENGINE LIMITATIONS

Use of heating and demisting is forbidden above the engine maximum continuous rating (Ng or t4).

Use of engine takeoff rating range, is limited to 5 continuous minutes.

- Nf LIMITATIONS



- : 349 rpm Min.
- : 349 to 418 rpm Normal operating range
- : 418 rpm Max. continuous
- : 450 rpm Max. transient limit (5 sec.)

NOTE

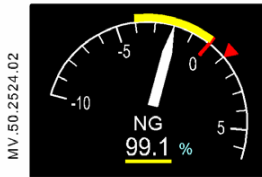
A NR rotor speed of 386 rpm corresponds to a Nf speed of 39158 rpm.

The rest of the paragraph is unchanged.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.8036.

- Ng LIMITATIONS



Ng = 67 % : Mini stabilized rating
 $\Delta Ng = -4 \%$ Max. continuous rating
 (Ng = 97.1 %, Hp = 0, ISA)

▬ : $\Delta Ng = -4 \%$ to $\Delta Ng = 0 \%$ Takeoff rating range

▬ : $\Delta Ng = 0 \%$ Max. takeoff rating
 (Ng = 101.1 %, Hp = 0, ISA)

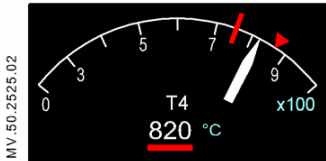
▲ : $\Delta Ng = +1 \%$ Max. transient rating (5 sec.)
 (Ng = 102.3 %, Hp = 0, ISA).

NOTE

100 % Ng corresponds to 52110 rpm.

- t4 LIMITATIONS

Starting limitations:



▬ : 750 °C Max. continuous

▲ : 865 °C Max. transient (10 sec.)

Flight limitations:



849 °C Max. continuous rating

▬ : 849 °C to 915 °C Takeoff rating range

▬ : 915 °C Max. temperature takeoff rating

▲ : 941 °C Max. transient (20 sec.)

- OIL TEMPERATURE LIMITATIONS



Minimum oil temperature before power application: 0°C (Oil 5 cSt).

- OIL PRESSURE LIMITATIONS



5 ELECTRICAL CIRCUIT LIMITATIONS

- Maximum voltage31.5 V
(Rated voltage 26-29 V)
- Maximum current150 A Max. continuous
For 200A Starter Generator (if fitted)
Refer to SUP.29

SECTION 2.5

MISCELLANEOUS LIMITATIONS

1 APPROVED FUELS

NOTE 1

Commercial designations of authorized fuels and additives are specified in the engine manufacturer documentation.

- NORMAL FUELS

(Fuels approved to operate throughout the flight envelope with no restrictions).

Type of fuel	NATO code	Specifications				Anti-ice additive included
		FRANCE	USA	UK	OTHER	
Kerosene - 50 (AVTUR-FSII) (JP8)	F 34	DCSEA 134 AIR 3405	MIL-T-83133 (JP8)	DEF STAN 91-87 D.ENG.RD 2453	-	Yes
Kerosene - 50 (AVTUR) (JP1)	F 35	DCSEA 134 AIR 3405	ASTM-D-1655 JET A1	DEF STAN 91-91 D.ENG.RD 2494	-	No
Kerosene	-	-	ASTM-D-1655 JET A	-	-	No
High flash point (JP5) (AVCAT)	F 43	DCSEA 144 AIR 3404	MIL-T-5624 (JP5)	D.ENG.RD 2498	-	No
High flash point (JP5) (AVCAT FSII)	F 44	DCSEA 144 AIR 3404	MIL-T-5624 (JP5)	DEF STAN 91-86 D.ENG.RD 2452	-	Yes
Chinese fuel Jet fuel No.3	-	-	-	-	CHINA GB 6537-2018	No

NOTE 2

All specifications are effective at latest issue or amendment.

All fuels not mentionned are to be excluded from use.

- REPLACEMENT FUELS

(Fuels allowing operations in a restricted flight envelope.)

- Fuels with restricted flight envelope of:
OAT $\leq 25^{\circ}\text{C}$ and Hp $\leq 5000\text{ ft}$ (1524 m)

Type of fuel	NATO code	Specifications			Anti-ice additive included
		FRANCE	USA	UK	
WIDE CUT (JP4) (AVTAG FSII)	F 40	AIR 3407	MIL-T-5624 (JP4)	D.ENG.RD 2454	Yes
WIDE CUT (JET B) (AVTAG)	-	-	ASTM-D-1655 (JET B)	-	No

- Fuels with restricted flight envelope of:
OAT $\leq 40^{\circ}\text{C}$ and Hp $\leq 15000\text{ ft}$ (4572 m)

Type of fuel	NATO code	Specifications			Anti-ice additive included
		-	RUSSIA	-	
KEROSENE TS 1 (TC1)	-	-	GOST 10227	-	No
KEROSENE RT (PT)	-	-	GOST 10227	-	No

- REFUELING

Fuel temperature must be inside the above OAT limitations specified for the fuel actually used for refuelling.

- ADDITIVES

CAUTION

ONLY THE FUEL ADDITIVES LISTED IN THIS SECTION ARE APPROVED FOR USE. ANY OTHER ADDITIVE NOT LISTED IN THIS SECTION IS NOT APPROVED AND SHOULD NOT BE USED.

Anti-ice additives:

If the fuel does not contain a freezing inhibitor and if the OAT is below -20 °C, the use of an anti-icing additive is mandatory.

The additive shall comply with French specification AIR 3652B (equivalent to : MIL-I 27686, D-ENG-RD 2451, MB-NATO S 748, MIL-I 85470A).

Concentration shall be between 0.10 % and 0.15 % by volume.

Fungicide additive:

BIOBOR JF.

Jet fuel thermal stability improver additive:

Type +100: NATO symbol 1749, MIL-DTL 83133E, SPEC AID 8Q462, APA 101.

Concentration by volume: 256 mg/l.

NOTE 1

Additive for F34 and F35

NOTE 2

The Chinese additives T1502 and T1602 for Jet fuel N°3 are not approved for use.

2 APPROVED LUBRICANTS AND FLUIDS

- ENGINE LUBRICANTS

NORMAL USE (forbidden for OAT < -30°C)						
Oil type	NATO Code	Specification			Class	Approved oil trademarks
		FRANCE	USA	UK		
Recommended use Average synthetic 5 cSt at 98.9° C	0.156	-	MIL-L-23699	DERD 2499	HTS (High thermal stability)	Aero SHELL Turbine Oil 560 EXXON Turbo Oil 2197 MOBIL Jet Oil 254 MOBIL Jet Oil 291
Normal use Average synthetic 5 cSt at 98.9° C	0.156	-	MIL-L-23699	DERD 2499	(Corrosion inhibiting)	CASTROL Aerojet 5
					Standard	CASTROL 5000 Aero SHELL 500 EXXON Turbo Oil 2380 MOBIL Jet Oil 2 TURBO NYCOIL 600 TOTAL Aero turbine 535 ELF Turbo Jet 2
OTHER OILS (forbidden for OAT > 30°C)						
Oil type	NATO Code	Specification			Class	Approved oil trademarks
		FRANCE	USA	UK		
Synthetic fluid 3 to 3.5 cSt at 98.9° C	0.148	-	MIL-L-7808	-	-	EXXON Turbo Oil 2389 MOBIL AVREX 256 CASTROL 325 CASTROL 3C TURBO NYCOIL 160
	0.150	AIR 3514	-	-	-	ELF Jet Synthetic Oil 15 TURBO NYCOIL 13B
Synthetic fluid 3.9 cSt at 98.9°C	-	-	-	DEF STAN 91-94	-	Aero SHELL Turbine Oil 390

NOTE 1

When the oil specification/grade/trademark differs from the approved one, engine manufacturer approval shall be obtained before using this oil.

NOTE 2

In case of oil change with trademark/NATO code/category/grade or specification change, apply instructions as prescribed in the engine manufacturer maintenance manual.

NOTE 3

All specifications are effective at latest issue or amendment.

- MAIN AND TAIL GEAR BOX LUBRICANTS

NORMAL USE						
Oil type	NATO Code	Specification			MGB oil temperature use	TGB oil temperature use
		FRANCE	USA	UK		
Mineral oil	0.155	AIR 3525	MIL-L-6086	DTD 581	-15°C * ≤ OAT ≤ +50°C	-20°C ** ≤ OAT ≤ +50°C
NOTE : The "SHELL" trademark is prohibited						

(*) Operation at OAT < -15°C is approved provided that MGB oil temperature at engine start is ≥ -15°C.

(**) Operation at OAT < -20°C is approved provided that TGB oil temperature at engine start is ≥ -20°C.

A procedure to measure the oil temperatures is provided in the Aircraft Maintenance Manual (AMM 60-00-00, 6-3).

USE IN COLD WEATHER					
Oil type	NATO Code	Specification			Approved operating temperatures
		FRANCE	USA	UK	
Synthetic oil	0.150	AIR 3514	-	-	-40°C ≤ OAT ≤ +0°C
Synthetic oil	0.148	-	MIL-L-7808	-	-40°C ≤ OAT ≤ +0°C

- TAIL GEAR BOX LUBRICANTS

NORMAL USE					
Oil type	NATO Code	Specification			Approved operating temperatures
		FRANCE	USA	UK	
Synthetic oil 5 cSt	0.156	-	MIL-L-23699	DERD 2499	-20°C * ≤ OAT ≤ +50°C

(*) Operation at OAT < -20°C is approved provided that TGB oil temperature at engine start is ≥ -20°C.

A procedure to measure the oil temperatures is provided in the Aircraft Maintenance Manual (AMM 60-00-00, 6-3).

- SERVO CONTROL FLUID

NORMAL USE					
Fluid type	NATO Code	Specification			Notes
		FRANCE	USA	UK	
Synthetic fluid	H 537	-	MIL-PRF-83282	-	Recommended
Mineral fluid	H 515	AIR 3520	MIL-H-5606	DTD 585	

NOTE 1

All specifications are effective at latest issue or amendment.

NOTE 2

In case of fluid change with trademark / NATO code/category/grade or specification change, apply instructions as prescribed in the maintenance manual.

3 BAGGAGE COMPARTMENT LOAD LIMITATIONS

With a max. distributed load of 300 kg/m² (62.5 lb/ft²)

- RH cargo compartment 100 kg (220 lb)
- LH cargo compartment..... 120 kg (264 lb)

With a max. distributed load of 145 kg/m² (30 lb/ft²)

- Rear cargo compartment 80 kg (176 lb)

4 CABIN COMPARTMENT LOAD LIMITATIONS

With a max. distributed load of 300 kg/m² (62.5 lb/ft²)

- Rear cabin floor 310 kg (682 lb)
- Forward left cabin floor..... 150 kg (330 lb)

5 ENGINE STARTER/GENERATOR

To prevent starter overheat damage, limit starter time to the following:

1st Start	Cooling time	Crank	Cooling time	2nd Start	Cooling time	Crank	Cooling time
Failed	1 min	30 s	1 min	OK			
Failed	1 min	30 s	1 min	Failed	30 min	30 s	1 min

When performing an engine crank, the starter time is the duration that the [CRANK] is pressed.

6 MANDATORY EQUIPMENT

A minimum of two adequate radio/audio headsets shall be on-board the helicopter, one worn by the pilot at the controls to monitor the audio warnings delivered through the ICS system, and a spare one.

7 OPTIONAL EQUIPMENT

When optional equipment items are installed, refer to the supplements for additional limitations, procedures and performance data.

SECTION 2.6

PLACARDS

All placards shown hereafter are usually presented in bilingual form French/English. However, the State of Registry may approve markings and placards in local language intended for:

- Emergency passenger information and instruction,
- Instruction for operation of passenger doors.

The following illustrations of placards and decals are typical presentations. Slight formal differences from the real placards and decals do not affect information presented therein.

1 VNE PLACARDS

MV.50.0553.03

VNE EN PUISSANCE	
Zp (ft) Vi (kts)	
0	155
2000	149
4000	143
6000	137
8000	131
10 000	125
12 000	119
14 000	113
16 000	107
18 000	101
20 000	95
22 000	89
SANS PUISSANCE: -30 kts	

VNE POWER ON	
Hp (ft) IAS (kts)	
0	155
2000	149
4000	143
6000	137
8000	131
10 000	125
12 000	119
14 000	113
16 000	107
18 000	101
20 000	95
22 000	89
POWER OFF: -30 kts	

Location: Inside cabin, on center post.

2 OPERATING LIMITATION PLACARD

MV.50.0054.00

LES REPERES ET PLAQUETTES INDICATRICES INSTALLEES SUR CET HELICOPTERE CONTIENNENT LES LIMITATIONS D'UTILISATION QUI DOIVENT ETRE RESPECTEES LORS DE L'UTILISATION DE CE GIRAVION. LES AUTRES LIMITATIONS D'UTILISATION QUI DOIVENT ETRE RESPECTEES LORS DE L'UTILISATION DE CE GIRAVION SONT CONTENUES DANS LE MANUEL DE VOL DU GIRAVION. LA SECTION LIMITATIONS DE NAVIGABILITE DU MANUEL D'ENTRETIEN DU GIRAVION DOIT ETRE RESPECTEE

THE MARKINGS AND PLACARDS INSTALLED ON THIS HELICOPTER CONTAIN OPERATING LIMITATIONS WHICH MUST BE COMPLIED WITH WHEN OPERATING THIS ROTORCRAFT. OTHER OPERATING LIMITATIONS WHICH MUST BE COMPLIED WITH WHEN OPERATING THIS ROTORCRAFT ARE CONTAINED IN THE ROTORCRAFT FLIGHT MANUAL THE AIRWORTHINESS LIMITATIONS SECTION OF THE ROTORCRAFT MAINTENANCE MANUAL MUST BE COMPLIED WITH.

Location: Inside cabin, on overhead canopy bow.

3 OTHER PLACARDS DISPLAYED IN THE COCKPIT

MV.50.0728.02



LH side



RH side

Location: Inside cabin, at bottom of doors.

MV.50.0729.00

N° APPAREIL A/C SERIAL N°:		WERK Nr.
MASSE WEIGHT :		GEWICHT
C. OF G. :		SCHWERPKT
DATE :		DATUM

Location: Console RH side.

MV.50.2530.00

COMPASS AIRCRAFT DATE	
HEADING	
MAGNETIC	CORRECTED
000	
045	
090	
135	
180	
225	
270	
315	

Location: Inside cabin,
on center post.

4 FLOOR LOADING PLACARDS

MV.50.0051.00

CHARGES REPARTIES MAXI

DISTRIBUTED LOADS MAXI

SUR PLANCHER CABINE ARRIERE

ON REAR CABIN FLOOR

SUR PLANCHER AVANT GAUCHE

ON L.H. FORWARD CABIN FLOOR

310 kg

682 lb

150 kg

330 lb

Location: On console.

MV.50.0052.00

CHARGE MAXI. 120 kg

MAX. LOAD 264 lb

MV.50.0053.00

CHARGE MAXI. 100 kg

MAX. LOAD 220 lb

Location: Cargo hold LH side.

Location: Cargo hold RH side.

MV.50.0050.00

CHARGE REPARTIE MAXI ——— 80 kg

DISTRIBUTED LOAD MAXI ——— 176lb

Location: Rear cargo hold.

5 ELECTRICAL PLACARD

MV.50.0730.00

F

28

V.d.c

Location: RH side, on ground power receptacle cover.

SECTION 3

EMERGENCY PROCEDURES

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SECTION 3.1

EMERGENCY PROCEDURES

1 GENERAL

Emergency procedures describe the actions that the pilot must take relative to the various possible failures that can occur.

Meanwhile, depending on the many variable external environments, such as the type of terrain overflown, the pilot may have to adapt to the situation according to his experience.

To help the pilot in his decision process, four recommendations are used:

LAND IMMEDIATELY

Self explanatory.

LAND AS SOON AS POSSIBLE

Emergency conditions are urgent and require landing at the nearest landing site at which a safe landing can be made.

LAND AS SOON AS PRACTICABLE

Emergency conditions are less urgent and in the pilot's judgment, he may proceed to the nearest airfield where he can expect appropriate assistance.

CONTINUE FLIGHT

Continue flight as planned. Repair at the destination according to the maintenance manual.

NOTE

Immediate actions that the pilot shall take are written in bold characters.

2 AUDIO WARNINGS

On the SCU, a [**HORN**] pushbutton is used to activate the audio warning.

When [**HORN**] pushbutton is pressed: **HORN**.

NOTE

The pilot at the controls shall wear an adequate radio/ICS audio headset to monitor the audio warnings through the ICS system.

- **GONG**

A gong is generated each time a red warning appears on the warning panel.

- **CONTINUOUS TONE**

Two continuous tones can be heard:

- A 310 Hz tone when NR is below 360 rpm.
- A 285 Hz tone when maximum take-off limitations are exceeded:

* After 1.5 sec. delay if power remains within transient range.

* Immediately when transient power limitations are exceeded.

In addition, for VEMD after embodiment of MOD. 07-3195 only:

* Immediately when max. transient rating is or will be exceeded during fast power increase.

1. Collective REDUCE to maintain NR in normal operating range or power within limitations

2. Engine parameters CHECK

- **INTERMITTENT TONE**

An intermittent tone (310 Hz) is heard when the NR is above 410 rpm.

Collective INCREASE
to maintain NR in normal operating range

Apply procedure according to the situation.

SECTION 3.2

ENGINE FLAME-OUT

1 CRUISE FLIGHT

AUTOROTATION PROCEDURE OVER LAND

1. Collective **REDUCE**
to maintain NR in normal operating range
 2. IAS **V_y**
- After loss of tail rotor thrust
 3. Twist grip IDLE position
 4. Maneuver the aircraft into the wind on final approach
 - At height $\cong$ 70 ft (21 m)
 5. Cyclic **FLARE**
 - At 20/25 ft (6/8 m) and at constant attitude
 6. Collective **GRADUALLY INCREASE**
to reduce the rate of descent and forward speed
 7. Cyclic **FORWARD** to adopt a slightly nose-up landing attitude ($< 10^\circ$)
 8. Pedals **ADJUST**
to cancel any sideslip tendency
 9. Collective **INCREASE**
to cushion touch-down
 - After touch-down
 10. Cyclic, collective, pedals **ADJUST**
to control ground run
 - Once the aircraft has stopped
 11. Collective **FULL LOW PITCH**
 12. Rotor brake **APPLY** below 170 rotor rpm

AUTOROTATION PROCEDURE OVER WATER

Apply same procedure as over land, except items 10, 11 and 12, but maneuver to head the aircraft equally between the wind and wave direction on final approach. Ditch with minimum forward speed (IAS < 30 kt (56 km/h)) and rate of descent. Then apply following check list for items 10, 11, 12.

- After touch-down
 - 10. Collective MAINTAIN
 - 11 Doors jettison handles PULL-UP
 - 12. Rotor brake APPLY

Abandon aircraft once the rotor has stopped.

2 HOVER IGE

- 1. Collective MAINTAIN
- 2. Pedals CONTROL YAW
- 3. Collective INCREASE as needed to cushion touch-down

3 HOVER OGE

WARNING

SAFE AUTOROTATIVE LANDING CANNOT BE ENSURED IN CASE OF A FAILURE IN HOGE BELOW THE TOP POINT OF THE HV DIAGRAM (REFER TO SECTION 5.1) OR IN CONFINED AREA.

- 1. Collective FULL LOW PITCH
- When NR stops decreasing
 - 2. Cyclic FORWARD
to gain airspeed according to available height
 - 3. Autorotation procedure APPLY

4 IN FLIGHT RELIGHTING

According to available height and cause of flame-out:

1. Engine starting selector **OFF position** |
2. [FUEL P] or [FUEL PUMP] **ON**
3. Engine starting selector **ON position** |

The relighting sequence will then run automatically as soon as $N_g < 17\%$.

- After relighting

4. [FUEL P] or [FUEL PUMP] **OFF**

At least 1000 ft (300 m) are necessary to complete relighting procedure after flame-out.

SECTION 3.3

TAIL ROTOR FAILURES

1 COMPLETE LOSS OF TAIL ROTOR THRUST

Symptom: the helicopter will yaw to the left with a rotational speed depending on the amount of power and the forward speed set at the time of the failure.

WARNING

SAFE AUTOROTATIVE LANDING CANNOT BE ENSURED IN CASE OF A FAILURE IN HOGE BELOW THE TOP POINT OF THE HV DIAGRAM (REFER TO SECTION 5.1) OR IN CONFINED AREA.

1.1 HOVER IGE (OR OGE WITHIN HV DIAGRAM)

LAND IMMEDIATELY

1. Twist grip IDLE position
2. Collective INCREASE to cushion touch-down

1.2 HOVER OGE (CLEAR AREA, OUTSIDE HV DIAGRAM)

Simultaneously,

1. Collective REDUCE depending on available height
2. Cyclic FORWARD to gain speed
3. Airspeed MAINTAIN Vy or higher
4. Collective ADJUST to obtain minimum sideslip angle

LAND AS SOON AS POSSIBLE

If a go-around was performed, carry out an autorotative landing on an area suitable for the autorotation procedure.

1.3 IN CRUISE FLIGHT

1. Airspeed MAINTAIN Vy or higher
2. Collective ADJUST to obtain minimum sideslip angle

LAND AS SOON AS POSSIBLE

APPROACH AND LANDING

On a suitable area for autorotative landing:

1. Carry out an autorotative landing according to the autorotation procedure
(Refer to SECTION 3.2 § 1).
2. During descent, twist grip IDLE position

2 LOSS OF TAIL ROTOR CONTROL

Symptom: Jamming of pedals or loss of pedal effectiveness. These conditions make it impossible to change tail rotor thrust with the pedals.

WARNING

WHEN AIRSPEED IS LESS THAN 20 KT (37 KM/H) AND PARTICULARLY NEAR TO THE GROUND, GO-AROUND MAY BE IMPOSSIBLE DUE TO LOSS OF THE VERTICAL FIN EFFICIENCY.

1. Cyclic and collective ADJUST to set IAS to 70 kt
(130 km/h) in level flight
2. [HYD TEST] or [ACCU TST] ON, load compensator depressurizes
After 5 sec.:
3. [HYD TEST] or [ACCU TST] RESET to OFF position

On a suitable area for a running landing procedure:

Make a shallow approach with a slight left sideslip. Perform a running landing, the sideslip will be reduced progressively as airspeed is reduced and collective is applied to cushion the landing.

NOTE

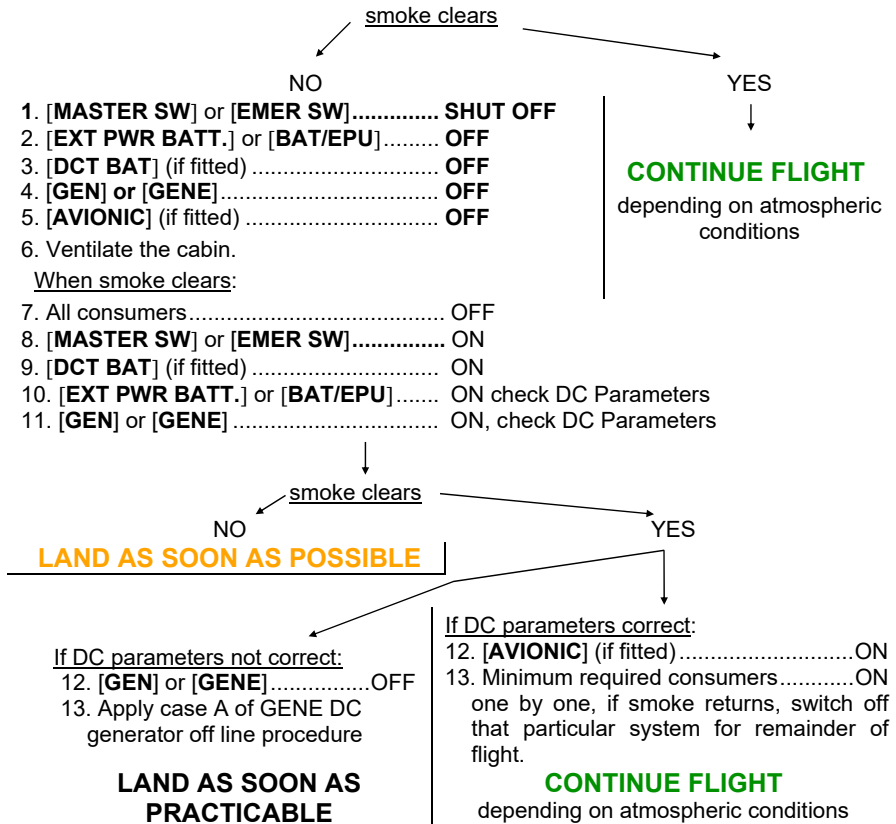
Landing is made easier with a RH wind component.

SECTION 3.4

SMOKE IN THE CABIN

1 SOURCE NOT IDENTIFIED

Heating, Demisting.....OFF



CAUTION

When [MASTER SW] or [EMER SW] is actuated or battery and generator are off line, the VEMD goes off. Apply the procedure for failure of both screens (Section 3.5 § 1 VEMD screen failures).

2 SOURCE IDENTIFIED

- 1. Corresponding system OFF
- 2. Ventilate the cabin.

CONTINUE FLIGHT

depending on system failed.

NOTE

After DC has been switched-off and on in flight, **GOV** light will remain on until the next normal full engine shut down and battery switch off on the ground.

3 AFTER COCKPIT FIRE EXTINGUISHER USE

- 1. Avoid as much as possible extinguisher agent inhalation.
- 2. Ventilate the cabin.

SECTION 3.5

VEMD FAILURES, CAUTION MESSAGES DISPLAYED ON VEMD AND ABNORMAL NR/Nf INDICATIONS

1 VEMD SCREEN FAILURES

- **Failure of one screen**

Failed Screen OFF

Read all available information on the other screen.

Informations are available using the [**SCROLL**] pushbutton either on the VEMD or on the collective pitch lever.

If top screen fails, the 3-parameter engine page will be automatically displayed on lower screen but with torque and ΔN_g off line. Only T4 and digital N_g will be available, refer to paragraph 4 for compliance with limitations.

- **Failure of both screens**

To avoid any power overlimit, the maximum authorized power will be the power needed to establish level flight with the following law:

IAS kt = 100 kt at 0 Hp - (2 kt / 1000 ft).

IAS km/h = 185 km/h at 0 Hp - (4 km/h per 300 m).

LAND AS SOON AS PRACTICABLE

Landing procedure: Carry out a no hover landing.

2 CAUTION MESSAGES ON VEMD

When a parameter is off line, the parameter value is not displayed on the corresponding VEMD screen and the parameter scale symbology is displayed in yellow.

Caution messages are self explanatory and the pilot shall comply with the action requested. If no light is lit on the caution and warning panel, no other action is required from the pilot.

LANE 1 (or 2) FAILED
-----> **PRESS OFF 1 (or 2)** : Self explanatory

VEH PARAM OUT RANGE : Abnormal vehicle parameter

ENG PARAM OUT RANGE : Abnormal engine parameter

These messages appear when a parameter usually displayed on these pages reaches a limitation, as the relevant (vehicle or engine) pages are not displayed.

- **[SCROLL]** : PRESS to reach the relevant page and check the parameter.

CROSSTALK FAILED
-----> **PRESS OFF 1 (or 2)** : Self explanatory.

BRT CNTRL FAILED : Brightness control failed.

FLI FAILED
-----> **CHECK PARAM** : One power parameter (Ng, t4, Tq) not consistent.

- Parameter consistency : CHECK
- Relevant procedures in paragraph ABNORMAL ENGINE PARAMETER INDICATION (§ 4) : APPLY

FLI VALID
-----> **PRESS RESET** : The "FLI" function becomes valid again after failure.

GEN PARAM OUT RANGE : Abnormal generator parameter.

BAT PARAM OUT RANGE : Abnormal battery parameter.

These messages appear when the relevant parameter is not displayed on the vehicle page and when an electrical limitation is reached.

GPS NOT AVAILABLE : GPS system data are not available.
(no absolute time reference)

- GPS navigation system : CHECK ON.

OVERLIMIT DETECTED : Engine or vehicle overlimit recorded.

This message appears as soon as a parameter overlimit is recorded in the VEMD. It will be displayed on the FLI or engine page until 40 % Ng during the next engine start. After the flight, check the recorded overlimit data and perform the associated maintenance actions.

For all of these messages, unless otherwise required by accompanying caution/warning lights or procedures:

CONTINUE FLIGHT

3 ABNORMAL NR/Nf INDICATIONS

- **NR indication Failure**

Collective : **MAINTAIN** Tq > 10 %

NR reading is given by Nf pointer.

LAND AS SOON AS PRACTICABLE

- **Nf indication Failure**

NR gauge : **CHECK** in normal operating range with Tq > 0

CONTINUE FLIGHT

NOTE 1

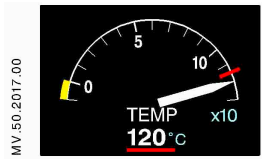
The Nf value can be read on the VEMD screen. Press [SELECT], then [+] as many times as required to display the parameter in the rectangular window at the bottom of the FLI or 3-data screen.

NOTE 2

In case of Nf indication failure, the EBCAU may not be available.

4 ABNORMAL ENGINE PARAMETER INDICATIONS

- Engine Oil Temperature over 115° C



AirspeedSET to 80 Kt (148 km/h)

Temperature decreases,

YES

NO

LAND AS SOON AS PRACTICABLE

LAND AS SOON AS POSSIBLE

Check oil cooler fan operation

- Low Engine Oil Pressure



CWPCHECK **ENG P**

NO

YES

CWP light test..... COMPLETED

ENG P

ENG P

LAND AS SOON AS PRACTICABLE

LAND IMMEDIATELY

- Autorotation procedure..... APPLY
- Shut down engine, time permitting

- **Loss of Ng, Torque or t4 parameters**

When a parameter is off line, the parameter value is not displayed on the VEMD upper screen and the parameter scale symbology is displayed in yellow.

The First Limitation Indicator (FLI) is replaced by the 3 data symbology (Ng/ Δ Ng, t4 and torque) and a failure message is displayed.

• **Ng/ Δ Ng Indicator Failure:**

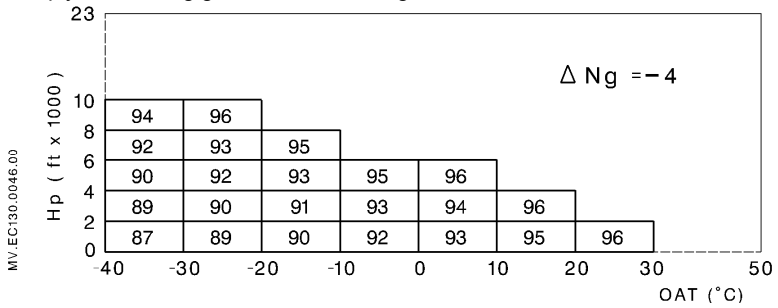
Comply with the maximum torque value and t4 limit of 810°C.

NOTE

In this case, the t4 limitations displayed are the starting limitations.

• **Torquemeter Failure:**

Comply with the Ng given in the following table:



• **Δ Ng and Torquemeter indications failure:**

GOV warning can also induce loss of Δ Ng and Torque indications.

The VEMD switches to 3-data symbology with only t4 and numeric Ng as valid parameter. Comply with Ng limitations in the above table, and respect a t4 limit of 810°C.

• **T4 Indicator Failure:**

Comply with Ng and torque limitations.

Switch off heating and demisting system.

On ground: do not try to start the engine.

For all these failures:

LAND AS SOON AS PRACTICABLE

5 ABNORMAL ELECTRICAL PARAMETER INDICATIONS

VEMD	CORRECTIVE ACTIONS
U/BUS 32.0 V	Over voltage on the generator (> 31.5 V) : 1. [GEN] or [GENE]OFF 2. CWPMONITOR for 3. GENE procedure case AApply BATT TEMP LAND AS SOON AS PRACTICABLE
U/BUS 30.0 V	Over voltage on the generator (29.0 V < U/BUS ≤ 31.5 V) : 1. U/BUS voltageMONITOR CONTINUE FLIGHT
U/BUS 24.0 V	Under voltage on DC bus (≤ 26.0 V) : 1. GENE procedureApply. 2. U/BUS voltageCHECK > 26.0 V YES NO CONTINUE FLIGHT 3. GENE procedure case AApply LAND AS SOON AS PRACTICABLE NOTE Probable cause of such a failure is a generator defect that does not make GENE come on.

VEMD	CORRECTIVE ACTIONS
 Red underline is flashing	Generator current over limit: 1. Unnecessary equipment..... OFF CONTINUE FLIGHT NOTE After engine starting on battery, a temporary high generator current is normal because the battery is being recharged. During this phase, avoid using high electrical load consumers.

SECTION 3.6

CAUTION AND WARNING PANEL

1 ENGINE ALARMS

WARNING PANEL	CORRECTIVE ACTIONS
ENG FIRE Fire in engine bay	- At Start-up: 1. Engine starting selector OFF position 2. Emergency fuel shut-off handle..... AFT 3. [FUEL P] or [FUEL PUMP] OFF 4. [CRANK]..... PRESS (10 sec.) 5. Rotor brake APPLY (≤ 170 rpm) 6. [MASTER SW] or [EMER SW]..... OFF 7. Evacuate aircraft and fight fire from outside - Hover, Takeoff, Final: LAND IMMEDIATELY Carry out a no hover powered landing. Once on ground, apply same procedure as above. - In Flight: LAND IMMEDIATELY 1. Collective LOWER 2. IAS..... V_y 3. Autorotation procedure APPLY 4. Emergency fuel shut-off handle AFT 5. [FUEL P] or [FUEL PUMP] CHECK OFF 6. Engine starting selector OFF position - After Landing: 7. Rotor brake APPLY (≤ 170 rpm) 8. [MASTER SW] or [EMER SW]..... OFF 9. Evacuate aircraft and fight fire from outside

WARNING PANEL	CORRECTIVE ACTIONS
GOV Major governor failure Emergency mode engaged	- IN FLIGHT: 1. Flight parameters..... CHECK Emergency mode automatically self-engages GOV illuminates. 2. CollectiveAVOID abrupt changes Maintain Ng > 80% Hp < 20000 ft (6096 m) Maintain Ng > 85% Hp ≥ 20000 ft (6096 m) LAND AS SOON AS PRACTICABLE Approach and Landing: . Make a powered approach . Avoid steep angle After touch down, shut-down: . Collective SLOWLY down to low pitch . Engine starting selector OFF NOTE On VEHICLE page: FF and END. This failure can also result in loss of ΔNg and torque parameters on the VEMD. - DURING ENGINE STARTING: Engine starting selector OFF position immediately

WARNING PANEL	CORRECTIVE ACTIONS
Minor FADEC failure	- Continuously on: Governing function degraded 1. CollectiveAVOID abrupt power changes 2. IASMAINTAIN below VNE power off LAND AS SOON AS PRACTICABLE On ground: Do not start engine. - Flashing at idle or during starting or shutdown: Governor redundancy failure, no impact on governing function. . Start-up procedure: Abort, refer to Maintenance Manual . Autorotation training: Cancel training, return to base
Engine oil pressure < 1.1 bar (16 psi)	Oil pressureCHECK gauge. LOW or NIL NORMAL LAND IMMEDIATELY LAND AS SOON AS PRACTICABLE . Autorotation procedure ... APPLY . Shutdown engine time permitting
Twist grip out of FLIGHT position	Twist gripTURN to FLIGHT position CONTINUE FLIGHT

WARNING PANEL	CORRECTIVE ACTIONS
ENG CHIP Metal particles in engine oil circuit	Collective..... REDUCE power LAND AS SOON AS POSSIBLE Low power approach and landing Be prepared in case of a loss of engine power. NOTE Takeoff is prohibited until checks specified in engine manufacturer maintenance manual have been completed.

2 TRANSMISSION ALARMS

WARNING PANEL	CORRECTIVE ACTIONS
MGB P Main Gear Box low oil pressure < 1 bar (14.5 psi)	Collective REDUCE power LAND AS SOON AS POSSIBLE If a safe landing is not possible, continue flight to the nearest appropriate landing site, reduce power to fly at minimum power speed (Vy). NOTE At low power (Vy) a maximum of 55 min. of simulated flight time has been demonstrated during bench tests
MGB TEMP Main Gear Box oil overheating (> 115°C)	1. IAS SET TO Vy 2. CWP MONITOR MGB TEMP MGB TEMP LAND AS SOON AS PRACTICABLE LAND AS SOON AS POSSIBLE
MGB CHIP Metal particles in MGB oil circuit	Collective REDUCE power MGB P and MGB TEMP MONITOR LAND AS SOON AS POSSIBLE
TGB CHIP Metal particles in TGB oil circuit	Avoid prolonged hovering CONTINUE FLIGHT

3 HYDRAULIC ALARMS

WARNING PANEL	CORRECTIVE ACTIONS
HYDR Loss of hydraulic pressure or Low hydraulic pressure	WARNING DO NOT USE [HYD TEST] OR [ACCU TST] PUSHBUTTON AS THIS WILL DEPRESSURIZE THE YAW LOAD COMPENSATOR RESULTING IN HEAVY PEDAL CONTROL LOADS. FAILURE TO COMPLY WITH THE FOLLOWING PROCEDURE MAY LEAD TO LOSS OF CONTROL. • <u>HIGE, Takeoff, Final:</u> (if landing within 30 sec. is possible) 1. Land normally 2. Collective LOCK 3. Twist grip..... IDLE position 4. Shutdown procedure..... Apply • <u>In flight:</u> 1. Set and maintain angle of bank below 30° 2. Avoid abrupt maneuvers Then, smoothly: 3. IAS SET between 40 and 60 KIAS (74 km/h and 111 km/h) NOTE Hydraulic failure safety speed: 40 to 60 KIAS (74 km/h and 111 km/h) <u>Once hydraulic failure safety speed is established:</u> 4. Hydraulic cut-off switch (collective grip) OFF WARNING AS CONTROL LOADS INCREASE, BE CAREFUL NOT TO INADVERTENTLY MOVE TWIST GRIP OUT OF FLIGHT POSITION (TWT GRIP LIGHT OFF). Pilot has to exert forces: - On collective increase or decrease around no force feedback point, - On forward and left cyclic.

LAND AS SOON AS POSSIBLE

NOTE

Accumulators will compensate hydraulic pressure loss for a limited operation of the main rotor controls allowing to secure flight conditions and to establish hydraulic failure safety speed.

Airspeed may be increased beyond safety speed as necessary but control loads will increase with speed.

Approach and landing:

WARNING

DO NOT ATTEMPT TO CARRY OUT HOVER FLIGHT OR ANY LOW SPEED MANEUVER BECAUSE THE INTENSITY AND DIRECTION OF THE CONTROL FEEDBACK FORCE WILL CHANGE RAPIDLY.

FAILURE TO COMPLY MAY RESULT IN POOR AIRCRAFT CONTROL AND POSSIBLE LOSS OF CONTROL.

Over a clear and flat area:

- Set IAS between 40 and 60 KIAS (74 km/h and 111 km/h)
- Perform a flat approach into the wind
- Make a slow no-hover running landing at around 10 kt ground speed (18.5 km/h)
- **Do not hover or taxi without hydraulic pressure**

After landing:

- Collective LOCK
- Twist grip IDLE position
- Shutdown procedure Apply

4 ELECTRICAL ALARMS

WARNING PANEL	CORRECTIVE ACTIONS
BATT TEMP Battery temperature above maximum	1. [EXT PWR BATT] or [BAT/EPU] OFF 2. U bus voltage CHECK NORMAL ABOVE U max 1. EXT PWR BATT] or [BAT/EPU] ON 2. [GEN] or [GENE] OFF 3. Unnecessary equipment OFF LAND AS SOON AS PRACTICABLE
BATT Battery off line	[EXT PWR BATT] or [BAT/EPU]CHECK ON YES CHECK voltage on VEMD LAND AS SOON AS PRACTICABLE NO [EXT PWR BATT] or [BAT/EPU] ON BATT CONTINUE FLIGHT NOTE [BATT] comes on if the DC system is supplied with external power.

RC c

The paragraph 4 - ELECTRICAL ALARMS, is modified as follows:

4 ELECTRICAL ALARMS

Supersede the table **GENE** by the following:

WARNING PANEL	CORRECTIVE ACTIONS
GENE DC generator off line	1. U bus on VEMDCHECK 2. [GEN] or [GENE].....CHECK ON YES NO "GENE RESET" circuit breakerCheck not popped out (30α panel) [GENE RESET] or [GENE RST].....PRESS Case A: GENE Case B: GENE Unnecessary equipment OFF U bus on VEMD MONITOR LAND AS SOON AS PRACTICABLE CONTINUE FLIGHT CAUTION If the battery fails, the VEMD will go out and only the analogue NR indication will remain. Apply the procedure for failure of both screens (Section 3.5 § 1 VEMD screen failures). NOTE GENE comes on if the DC system is supplied with external power.

The rest of the paragraph is unchanged.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.8036.

WARNING PANEL	CORRECTIVE ACTIONS
DC generator off line	1. U bus on VEMDCHECK 2. [GEN] or [GENE].....CHECK ON <pre> graph TD A[1. U bus on VEMDCHECK 2. [GEN] or [GENE].....CHECK ON] --> B{YES / NO} B -- YES --> C["'GENE RESET' circuit breakerCheck not popped out (30α panel) [GENE RESET] or [GENE RST]PRESS"] B -- NO --> D["[GEN] or [GENE] ON"] C --> E{Case A: GENE / Case B: GENE} E -- Case A: GENE --> F["Unnecessary equipmentOFF U bus on VEMDMONITOR"] F --> G[LAND AS SOON AS PRACTICABLE] E -- Case B: GENE --> H[CONTINUE FLIGHT] D --> H </pre> WARNING IF THE BATTERY FAILS, THE VEMD WILL GO OUT AND NR/NF INDICATION IS LOST. APPLY THE PROCEDURE FOR FAILURE OF BOTH SCREENS (SECTION 3.5 § 1 VEMD SCREEN FAILURE). AVOID AUTOROTATION, PERFORM A SHALLOW APPROACH WITH A CAUTIOUS LANDING. NOTE GENE comes on if the DC system is supplied with external power.
AC power supply failure	<u>Inverter* AC system:</u> <pre> graph TD A["[INV]CHECK ON"] --> B{YES / NO} B -- YES --> C["Loss of all AC consumers NOTE AFCS* disengages automatically ControlsHANDS ON"] B -- NO --> D["[INV]ON"] C --> E[CONTINUE FLIGHT] D --> F[INV] F --> G[CONTINUE FLIGHT] </pre>

(*) If installed

5 FUEL ALARMS

WARNING PANEL	CORRECTIVE ACTIONS
FUEL Fuel quantity < 48 kg (106 lb)	LAND AS SOON AS POSSIBLE NOTE 15 min. of flight time remains at MCP WARNING AVOID LARGE ATTITUDE CHANGES AS THIS COULD LEAD TO AN ENGINE FLAME-OUT.
FUEL P Low fuel pressure	- IN FLIGHT: 1. CollectiveREDUCE POWER 2. [FUEL P] or [FUEL PUMP]ON LAND AS SOON AS POSSIBLE Low power approach and landing WARNING BE PREPARED IN CASE OF AN ENGINE FLAME-OUT.

WARNING PANEL	CORRECTIVE ACTIONS
FUEL FILT Fuel filter clogged	WARNING FUEL FILTER BY-PASS OPENING DRIVES POLLUTION INTO THE FUEL LINES, AND THE GOVERNOR, WHICH MAY INDUCE OSCILLATIONS, LIMITED POWER OR EVENTUALLY FLAME-OUT. Collective Reduce POWER FUEL FILT FUEL FILT Establish flight at reduced power. LAND AS SOON AS PRACTICABLE LAND AS SOON AS POSSIBLE VEMD MONITOR Ng If Ng oscillations occur: LAND IMMEDIATELY Autorotation procedureAPPLY

6 MISCELLANEOUS ALARMS

WARNING PANEL	CORRECTIVE ACTIONS
PITOT Pitot heating not operative	[PITOT]CHECK ON YES ↓ Monitor airspeed indicator NO ↓ [PITOT] ON CONTINUE FLIGHT
HORN Aural warning not operative	[HORN]CHECK ON. YES ↓ Aural warning failure NO ↓ [HORN] ON CONTINUE FLIGHT
DOOR One or both cargo hold doors unlocked	AirspeedREDUCE to 70 kt (130 km/h) LAND AS SOON AS PRACTICABLE Descent and approach at low rate of descent.

SECTION 3.7

VARIOUS WARNINGS, FAILURES AND INCIDENTS NOT INDICATED ON THE CWP

1 ROTOR BRAKE INOPERATIVE

WARNING

WAIT UNTIL ROTOR COMES TO A STANDSTILL BEFORE LEAVING THE AIRCRAFT.

Rotor stopping with wind blowing:

1. Aircraft..... Head into the wind
2. Cyclic..... Slightly into the wind

2 FLIGHT CONTROL HARDOVER OR SERVOJAM

A hardover results in uncommanded movements of one or two flight controls (including yaw).

A servojam results in a higher than normal force to move the flight controls.

- **HIGE, Takeoff, Final**: (if immediate landing is possible)

LAND IMMEDIATELY

After landing:

1. Hydraulic cut-off switch (collective grip)..... OFF
2. Engine and rotor shut down procedure APPLY

- **In flight**

1. IAS SET to between 40 and 60 kt
(74 km/h and 111 km/h)
Enter sideslip if necessary
2. Hydraulic cut-off switch (collective grip)..... OFF, apply **HYDR** procedure

LAND AS SOON AS POSSIBLE

NOTE

With no hydraulic pressure, the aerodynamic loads to be counteracted may be heavy at high airspeed:

- Collective: Up to 20 daN (45 lbf) to pull,
- Cyclic: 7 to 12 daN (16 to 27 lbf) to push left,
- Cyclic: 2 to 4 daN (4.5 to 9 lbf) to push forward.

3 BLEED VALVE FAILURE

When the bleed valve opens, a  flag appears on the FLI or 3-data page.

The flag disappears when the bleed valve closes.

The bleed valve is normally open when the engine is shut down, during starting and at low power settings.

- If the flag does not disappear at high power settings (i.e. near MCP or above), the maximum available engine power is reduced, specifically in cold weather.
- If the flag does not reappear at low power settings, the engine may surge. Avoid abrupt changes in power settings.
- Bleed valve failure results in **GOV**.

LAND AS SOON AS PRACTICABLE

4 ICS INOPERATIVE (GMA 340H)*

- 1. ICS OFF
- 2. COM 1 Check ON, adjust volume

NOTE 1

VHF communications will remain available for the RH pilot only via COM 1 transceiver. Audio warnings will be transmitted via the COM 1 audio system.

NOTE 2

Abort or cancel hoisting operations in case of ICS failure.

(*) If installed

5 ABNORMAL NR GOVERNING

- Abnormal low NR and Nf (Parameters aligned)

FADEC governs Nf to a value that is significantly lower than usual.

This failure can be caused by an erroneous collective anticipator signal (XPC). NR/Nf values may decrease to 360 rpm before the failure is detected by the FADEC (**GOV** comes on).

If NR/Nf drop below 375 rpm in flight:

1. Engine parameters.....CHECK
2. NR/NfCHECK both parameters aligned

LAND AS SOON AS PRACTICABLE

Due to reduced tail rotor control margins avoid hover flights with left crosswind and carry out a no hover landing.

6 SERVO TRANSPARENCY

Self-correcting, the phenomenon will induce an un-commanded right cyclic load and an associated collective down reaction. However, even if the transparency feedback loads are controllable, immediate action is required to relieve the feedback loads:

- Reduce the severity of the maneuver,
- Follow the aircraft's natural reaction,
- Let the collective decrease naturally (avoid low pitch) and smoothly counteract the right cyclic motion.

Transparency will disappear as soon as excessive loads are relieved.

SECTION 4

NORMAL PROCEDURES

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4.8 EXTREME WEATHER OPERATIONS

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2 COLD WEATHER OPERATION	1

SECTION 4.1

GENERAL

This section contains instructions and procedures for operating the helicopter from the planning stage, through actual flight conditions, to secure the helicopter after landing.

Normal and standard conditions are assumed in these procedures. Pertinent data in other sections is referenced when applicable.

The instructions and procedures contained herein are written for the purpose of standardization and are not applicable to all situations.

1 OPERATING LIMITATIONS

For minimum and maximum limits, refer to SECTION 2.

Each time an operating limitation is exceeded, an appropriate entry shall be made in the logbook (helicopter, engine, etc.). The entry shall state which limit was exceeded, the duration, the extreme value attained, and any additional information essential in determining the maintenance action required.

2 FLIGHT PLANNING

Each flight should be planned adequately to ensure safe operations and to provide the pilot with the data to be used during flight. Flight planning must comply with helicopter limitations and performance (Refer to SECTIONS 2, 5, 6 and supplements).

3 TAKEOFF AND LANDING DATA

Refer to SECTION 2 - LIMITATIONS and
SECTION 5 - REGULATORY & ADDITIONAL PERFORMANCE DATA.

4 WEIGHT AND BALANCE DATA

Ascertain proper weight and balance of the helicopter as follows:

- Consult SECTION 6 - WEIGHT AND BALANCE.
- Ascertain weight of fuel, oil, payload, etc.
- Compute takeoff and anticipated landing gross weights.
- Check helicopter center of gravity (CG) locations.
- Check that the weight and CG limitations in SECTION 2 are not exceeded.

SECTION 4.2

PREFLIGHT CHECK

- Make sure that all flightworthiness-required corrective maintenance operations have been performed.
- These preflight checks can be done without opening any cowlings unless the helicopter had been parked for more than 2 days or in case of any visible leak or doubt.
- Check that the aircraft area is clean and unobstructed.
- Remove all picketing items if applicable
- Carry out the following checks:

1 EXTERIOR CHECK

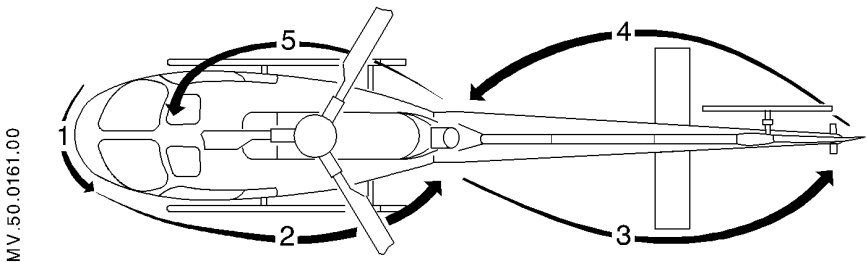


Figure 1: Sequence of checks

Station 1

- Transparent panelsCondition - Cleanliness.
- Windshield wiper (if installed).....Condition.
- MGB – Engine oil cooler air inlet.....Check no obstruction or foreign objects.
- Side slip indicatorCondition.
- Pitot tubeCover removed - Condition.
- Landing lightsCondition.

Station 2**WARNING**

ICE OR SNOW ACCUMULATIONS THAT REMAIN IN OR AROUND THE ENGINE AIR INTAKE MAY BE INGESTED AND CAN CAUSE A SUDDEN IN-FLIGHT ENGINE FAILURE.

- Front door Condition, jettison system check.
- Rear door Condition, closed or open locked (sliding door).
- Left cargo door Open.
- Loads and objects carried Stowed and secured.
- Left cargo door Closed, locked.
- Fuel tank filler cap Closed, locked.
- Fuel tank Bled (before the first flight, if OAT $\geq 0^{\circ}\text{C}$), no leakage at the bleed valve.
- MGB cowl MGB oil level - Cowl locked.
- All lower fairing panels Locked.
- Landing gear and foot step Attachment - Visual check.
- Static ports Clear, covers removed.
- OAT sensors, antennas Condition.
- Main rotor head and blades Visual inspection, no impact.
- Engine cowl Locked.
- Rear cargo door Open.
- Loads and objects carried Stowed and secured.
- ELT Check ARMED.
- Rear cargo door Closed, locked.
- Oil drain No oil under scupper.

Station 3

- Heat shield on tail rotor drive Condition, attachment.
- Tail boom, antennas Condition - Fairing fasteners locked.
- Stabilizer, fin, external lights General condition.
- Tail rotor guard (if fitted) Condition, attachment.
- TGB fairing Secured, fasteners locked.
- TGB oil level Checked.
- Tail skid Condition, attachment.

Station 4

- Tail rotor headCondition, laminated bearing.
Checked for separation, cracks, etc.
- Tail rotor bladesVisual inspection, no impact.
- Stabilizer, fin, external lights.....General condition.
- Tail boom, antennas.....Condition - Fairing fasteners locked.
- Heat shield on tail rotor drive.....Condition, attachment.

Station 5

- Oil drainNo oil under scupper.
- EPU doorClosed or EPU connected.
- Engine air intakeClean - No foreign objects or
accumulations of ice or snow in or
around the engine air intake and no
stagnant water at the drain hole.
- Engine cowl.....Locked.
- Exhaust coverRemoved.
- Right cargo door.....Open.
- Loads and objects carriedStowed and secured.
- Right cargo door.....Locked.
- Main rotor head and bladesVisual inspection, no impact.
- MGB cowl.....No foreign objects on transmission deck.
Cowl locked.
- Hydraulic oil levelCheck reservoir level.
- Engine oil levelCheck reservoir level.
- Landing gear and foot stepAttachment – Visual check.
- All lower fairing panelsLocked.
- DoorCondition, jettison system check.
- External mirror (if fitted).....Set to avoid dazzling (night flight).

2 INTERIOR CHECK

- Cabin.....Clean
- Fire extinguisher.....Secured - Checked
- Fuses or breakersAll set
- Loads and objects carried.....Stowed and secured
- Front door jettison systemsCheck - Plastic guard condition

3 TURN AROUND CHECK

- Overall aspectCondition, cleanliness.
- Engine / MGB / TGBOil level.
- Main and tail rotor bladesVisual inspection, no impact.
- LoadsSecured.
- All cowlingsLocked.
- DoorsClosed or sliding door open-locked.

NOTE

If the aircraft is to be parked for some time between flights, temporary picketing is recommended by fitting blanks, covers and blade socks (in winds above 40 kt (74 km/h)).
In this case, perform a complete pre-flight check.

SECTION 4.3

START UP

1 ENGINE PRESTART CHECK

- Seats and control pedals.....ADJUST and SECURE
- Seat beltsFASTEN

NOTE

Copilot seat belts shall be fastened in all cases.

1. Rotor brakeRELEASE, fully forward
2. Fuel shut-off leverFORWARD, plastic guard condition
3. Twist grip.....IDLE position
4. Hydraulic cut-off switch (collective grip).....ON
5. [MASTER SW] or [EMER SW].....CHECK UP position
6. Engine starting selector.....OFF
7. [EXT PWR BATT] or [BAT/EPU] [DCT/BAT] (if fitted), [GEN] or [GENE].....ON
8. Lighting circuits 1 and 2 test.....PERFORM (if night flight intended)
9. ICS and GPS nav. systemON
10. Electric mirror (if fitted).SET to avoid dazzling (night flight).
11. [W/LT TEST] or [W/LT TST]PERFORM
12. [FIRE TEST] or [FIRE TST].....PERFORM, CHECK

**ENG
FIRE**

Pre MOD 07.4719

13. [HYD TEST] or [ACCU TST]ON for 2 sec. then OFF

Post MOD 07.4719 (applicable only for aircraft equipped with dual hydraulic system):

13. [ACCU TST]PRESS for 2 sec.

14. CWP lightsCHECK:

- With battery power



- With EPU power: Same lights as above +



15. VEMD.....CHECK:

- . 3-data page: no message
- . Vehicle page: no message
- . Battery voltage > 22 V
- . Check fuel quantity
- .  (Bleed valve open)

16. Control pedals.....Free travel, then left pedal 2 cm (0.8 in) forward

17. CyclicCENTER, friction adjusted

18. CollectiveLOCK, friction adjusted

19. Heating, demisting,
air conditioning (if fitted).....OFF

2 ENGINE STARTING

1. CWP.....CHECK **GOV**
2. **[FUEL PUMP]** or **[FUEL P]**.....ON
3. **[A/COL LT]**.....ON
4. Cyclic control.....HAND ON
 - After 30 sec.
5. Engine starting selector.....ON position
6. Engine parameters.....CHECK:
 - . Ng increases
 - . T4 remains below its limits
 - . Rotor turns at $Ng \leq 25\%$
 - . Engine oil pressure increases
 - When $Ng \geq 67\%$
7. CWP.....CHECK:
 - ENG P** **MGB P** **HYDR**
8. **[PITOT]**.....ON, **PITOT**
9. **[FUEL PUMP]** or **[FUEL P]**.....OFF, CHECK **FUEL P**
10. Engine starting selector guardSET
11. All necessary systemsON - TESTED
(master avionics switch, lights..)

NOTE 1

In strong wind apply a little cyclic into wind.

NOTE 2

In case of failed engine start return the engine starting selector to OFF.
Observe the engine starter limitation given in SECTION 2.5 § 5.

NOTE 3

At $Ng > 60\%$ the VEMD upper screen automatically switches to FLI display.

12. EPU (if used).....DISCONNECT, make sure EPU door is closed and locked
13. CWP.....CHECK: **GENE** **BATT**

3 RUN-UP CHECK

1. Hydraulic checks:

CAUTION

If not locked, the collective pitch lever will move up when the accumulators are depleted or when the hydraulic cut-off switch on the collective is set to OFF.

- Accumulators checks:
 - **Collective**.....**CHECK correctly locked**
 - **[HYD TEST]** or **[ACCU TST]**.....ON
 - CWPCHECK **HYDR** flashes
(**HYDR** PRE MOD 07.3317)
 - Collective / cyclic controlsHANDS ON
 - Move the cyclic control 2 or 3 times on each axis (+/- 10% of travel) and check for accumulator hydraulic assistance on pitch and roll (no control loads).
 - **[HYD TEST]** or **[ACCU TST]**.....RESET to OFF position
 - CWPCHECK **HYDR**
- Hydraulic pressure isolation check:
 - Collective**CHECK correctly locked**
 - Hydraulic cut-off switch
(Collective).....OFF
 - CWPCHECK **HYDR**
 - Check that loads are felt immediately and that cyclic can be moved in pitch and roll with normal feedback loads.
 - Hydraulic cut-off switch
(Collective).....ON
 - CWPCHECK **HYDR** after 3 to 4 sec.

Maintenance action must be performed prior to flight if this time is reduced to 1 sec. (at least one of the accumulators is faulty).
- When minimum engine oil temperature is reached (Refer to SECTION 2.4 § 4):

2. Twist grip.....FLIGHT position

- When NR $\geq$ 340 rpm:

3. [HORN] ON, **HORN**
CHECK audio warning:
. ON for NR $\leq$ 360 rpm
. OFF for NR > 360 rpm
4. NR indication CHECK in lower normal operating range
5. [FIRE/TEST] or [FIRE/TST] PERFORM, CHECK:
ENG FIRE + Gong sounds
6. Parameter checks No warning light illuminated
Electrical system voltage and current
Engine oil pressure

4 CRANKING

The cranking procedure can be performed after a failed or aborted start and can be used for check or maintenance purposes.

Proceed as follows:

1. Engine starting selector OFF
2. Emergency fuel shut-off lever FORWARD
3. Ng CHECK $\leq$ 10 %
4. [FUEL PUMP] or [FUEL P] ON
5. [CRANK] PRESS for 30 sec. max
6. [FUEL PUMP] or [FUEL P] OFF

CAUTION

Do not crank the engine with the emergency fuel shutoff valve closed or with the fuel pump off as this could damage the engine high pressure fuel pump.

NOTE

Observe the engine starter limitation given in SECTION 2.5 § 5.

SECTION 4.4

TAKEOFF

1 BEFORE TAKEOFF CHECK

1. DoorsCLOSED or sliding doors
OPEN LOCKED
2. Collective, cyclic frictions locksAS REQUIRED
3. Landing light.....AS REQUIRED
4. Temperatures and pressuresNORMAL RANGE
5. CWP.....All lights OFF
6. CollectiveUNLOCKED
7. Heating, demisting,
air conditioning (if fitted).....AS REQUIRED

NOTE

Adjust collective and cyclic friction locks so that friction loads are felt by the pilot when moving the flight controls.

2 TAKEOFF CHECK AND PROCEDURE

CAUTION

Use of heating and demisting is forbidden above engine maximum continuous rating (Ng or t4).

- Gradually increase collective pitch to hover at 5 ft (1.5 m). Check engine and mechanical parameters, no warning light.
- Increase airspeed with HIGE power until IAS = 40 kt (74 km/h), then begin to climb so as to clear 40 ft (12 m) at IAS = 50 kt (93 km/h).

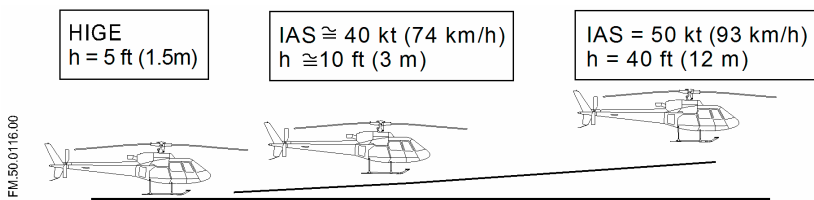


Figure 1: Takeoff procedure

CAUTION

For safe operation, takeoff path should avoid HV diagram (refer to Section 5.1).

SECTION 4.5

CLIMB - CRUISE FLIGHT, MANEUVERS - APPROACH - LANDING

1 CLIMB

Above 100 ft (30 m), for maximum climb performance, select up to Maximum Continuous Power and optimum climbing speed (V_y):

IAS kt = 65 kt at 0 Hp - (1 kt / 1000 ft).

IAS km/h = 120 km/h at 0 Hp - (2 km/h per 300 m).

2 CRUISE FLIGHT, MANEUVERS

2.1 CRUISING FLIGHT

Fast cruise is obtained by the first limitation reached corresponding to the beginning of the FLI amber area.

Corresponding mechanical or engine limits (T_q , N_g , t_4) are indicated by an underlined numerical value.

Economic cruise: refer to Section 5.2 "Additional performance data" (not approved).

2.2 LOAD FACTOR - SERVO TRANSPARENCY

Maximum load factor is determined by the servo-controls transparency limit.

Servo-transparency may be reached during maneuvers, turns, pull-up and/or when maneuvering near VNE.

Avoid a combination of TAS, H_σ , gross weight at high values associated with high collective.

2.3 OPERATION IN MAXIMUM POWER CONFIGURATION

Decrease collective slightly before initiating a turn, as for this maneuver the power requirement is increased.

2.4 HOVER TURNS

Avoid rotation faster than 6 sec. per full rotation.

2.5 OPERATION IN TURBULENCE

Reduce IAS.

3 APPROACH

CAUTION

Use of heating and demisting is forbidden above engine maximum continuous rating (Ng or t4).

- Begin approach at Vy.
- At approximately 100 ft (30 m), reduce airspeed down to HIGE at 5 ft (1.5 m).
 - Approach check:
 1. Landing lightAS REQUIRED
 2. All parameters.....CHECK
 3. Heating, demisting,
air conditioning (if fitted).....AS REQUIRED

4 LANDING

- In hover, gradually reduce collective until touchdown, then fully reduce collective.

SECTION 4.6

ENGINE AND ROTOR SHUTDOWN

1 ENGINE AND ROTOR SHUTDOWN

1. CyclicNEUTRAL
2. CollectiveLOCK
3. Twist grip.....IDLE position
4. Engine oil cooling.....WAIT for 30 sec.
5. [PITOT], [HORN], landing lightOFF
6. Non-required systems,
[AVIONIC] (if fitted)OFF
7. Engine starting selector.....OFF position
8. [GEN] or [GENE].....OFF

At $NR \leq 170$ rpm (for high wind conditions) or $NR \leq 140$ rpm (normal conditions).

9. Rotor brakeAPPLY
 - When rotor is stopped:
 10. GPS navigation system.....OFF
 11. [A/COL LT]OFF
 12. Yaw load compensator check:
 - Right pedal moves forward without
Pilot input, or right pedal can be
Moved forward with low forceCHECK
 - [HYD TEST] or Pre MOD 07.4719 [ACCU TST]ON for 2 sec. then OFF
- Post MOD 07.4719 (applicable only for aircraft equipped with dual hydraulic system):
- [ACCU TST].....PRESS for 2 sec
 - Pedals can be re-centered and
Remain centeredCHECK

NOTE

Yaw load compensator maintenance action is required if, before activation of [HYD TEST] or [ACCU TST] pushbutton, the right pedal cannot be moved forward with low force.

The [HYD TEST] or [ACCU TST] discharges the yaw load compensator. In order to repeat the check, it is necessary to re-pressurize the hydraulic system.

- **BEFORE LEAVING HELICOPTER**

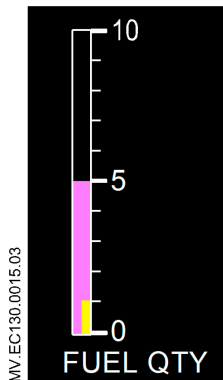
13. VEMD.....CHECK of Flight Report page data:
- Operating time (counted from Ng > 60 % after start, to Ng < 50 % at engine shutdown)
 - Ng and Nf cyclesCHECK (Indicated in white characters and above 0)
 - Advisory messages: **FAILURE DETECTED** or **OVERLIMIT DETECTED**.
14. **[DCT/BAT]** (if fitted),
[EXT PWR BATT] or **[BAT/EPU]**OFF
15. Pitot, static ports, air intake and exhaust covers, blade socks as required

SECTION 4.7

MISCELLANEOUS PROCEDURES AND DATA

1 TANK CAPACITY

- **Maximum capacity**
540 liters (142.7 US gal - 427 kg - 941 lb).
- **Fuel gauge**



10 = maximum usable fuel quantity, depending on type of fuel tank.
(Refer to section 7.1 § 2.4).

 : 15 min of flight time remains at MCP at the beginning of this range.

NOTE 1

The unusable fuel quantity is reached when zero is indicated on the fuel gauge.

NOTE 2

Fuel quantity indication in kg and fuel flow indication in kg/h is based on a fuel density of 0.79 kg/l.

SECTION 4.8

EXTREME WEATHER OPERATIONS

1 HIGH WIND OPERATION (WIND ABOVE 30 kt (56 km/h))

- **Parking**
 - Park the helicopter head into the wind. Maintain rotor brake applied with one blade at 12 o'clock. Keep blade socks until start up.
 - For wind above 40 kt (74 km/h) the helicopter must be tied down.
- **Start up**
 - When the rotor begins to turn, apply a small cyclic stick input into the wind.
 - As soon as Ng > 67 %:
Twist grip FLIGHT position
- **Run up check**
 - Perform the hydraulic checks with the twist grip in FLIGHT position and NR at nominal speed.
- **Engine and rotor shutdown**
 - Allow engine oil to cool with twist grip in FLIGHT position.

NOTE

Start up and shutdown have been demonstrated up to 40 kt (74 km/h) of wind from any directions and for 50 kt (93 km/h) headwind.

2 COLD WEATHER OPERATION

Refer to SUP.4 "Instructions for use in cold weather".

Start-up procedure:

Same start-up procedure as normal (refer to Section 4.3 § 2) except:

Engine oil temperature MONITOR.

When Engine oil temperature ≥ 0°C

CWP CHECK **ENG P**

Twist grip FLIGHT position

NOTE

Use of 3 cSt synthetic oil is recommended for low temperature operation.

SECTION 5.1
REGULATORY PERFORMANCE DATA
CONTENTS

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2 DEMONSTRATED WIND ENVELOPES.....	1
3 ENGINE POWER CHECK.....	1
4 AIR DATA SYSTEM CALIBRATION.....	7
5 HEIGHT - VELOCITY DIAGRAM.....	8
6 HOVER IN GROUND EFFECT.....	10
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9 RATE OF CLIMB	13
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11 NOISE LEVEL	14

SECTION 5.1

REGULATORY PERFORMANCE DATA

1 INTRODUCTION

The following performance curves apply to the basic version of the aircraft.
Refer to supplements when optional equipment is fitted.

NOTE

Values obtained on VEMD ENGINE POWER CHECK page can be checked with the ENGINE POWER CHECK and t4 CHECK curves.

2 DEMONSTRATED WIND ENVELOPES

2.1 STARTING AND STOPPING ROTOR WIND ENVELOPE

Starting and stopping the rotor has been demonstrated for winds of 40 kt (74 km/h) from any direction and 50 kt (93 km/h) with a headwind (forward center axis +/- 30°).

2.2 WIND ENVELOPE IN HOVER

Hovering with winds from any direction up to 17 kt (31 km/h) has been demonstrated over the entire flight envelope although it is not to be taken as a limit.

For example hover at sea level at maximum gross weight and for all CG locations has been substantiated at 30 kt (56 km/h).

3 ENGINE POWER CHECK

3.1 BEFORE TAKEOFF

In HIGE at 5ft (1.5 m) and before initiating forward flight, pull the collective pitch lever slightly to ensure that the Ng can increase by at least 1%, without exceeding the max. transient rating.

3.2 ENGINE POWER CHECK PROCEDURE

The engine power check consists in checking:

- The power margin (torque margin), and
- The t4 margin.

Checking can be performed using the VEMD data (Refer to paragraph 3.2.1) or manually recorded parameters (Refer to paragraph 3.2.2).

3.2.1 VEMD procedure

In stabilized level flight at a power rating close to MCP, perform an engine power check from the VEMD with $H_p \leq 12000$ ft (3657 m), heating and demisting system OFF.

Read the results:

- The engine power check is satisfactory if:
 - the "TRQ MARGIN" value is positive → "GOOD" displayed,
 - and
 - the "T4 MARGIN" value is negative → "GOOD" displayed.

NOTE

If the "T4 MARGIN" value is positive at the VEMD, calculate the T4 margin following the manual procedure below. If the value is negative (and the "TRQ MARGIN" is positive) then the engine health check is satisfactory.

3.2.2 Manual procedure

In stabilized level flight at a power rating close to MCP, record the following parameters: Tq, Ng, NR, Hp, OAT and t4. Refer to the ENGINE POWER CHECK chart (Figure 1) and the t4 CHECK chart (Figure 2). Use the chart in the direction shown by the arrows in the example.

The engine power check is satisfactory if:

- The point **P** is located within the "CORRECT" area on the ENGINE POWER CHECK chart → positive torque margin,
- and
- The point **T** is located within the "CORRECT" area on the t4 CHECK chart → negative t4 margin (*).

(*) If the point **T** is located within the "INCORRECT" area, calculate the torque margin and t4 margin values as indicated below, then calculate the corrected t4 margin as per paragraph 3.2.3.

If the corrected t4 margin is negative, result is acceptable.

Torque Margin Calculation:

- Mark the point **P** on the chart according to the recorded parameters.
- Mark the point **P'** on the separation line between the "CORRECT" and "INCORRECT" zones according to the recorded Ng and OAT values. Then carry over to the torque scale according to the recorded NR and Hp values.
- The torque margin is given by the torque value difference **P – P'** measured on the torque scale.

T4 Margin Calculation

- Mark the point **T** on the chart according to the recorded parameters.
- Mark the point **T'** on the separation line between the "CORRECT" and "INCORRECT" zones according to the recorded Ng and OAT values. Carry over to the t4 scale according to the recorded Hp value.
- The t4 margin is given by the t4 value difference **T – T'** measured on the t4 temperature scale on the LH side.

3.2.3 Corrected t4 Margin calculation

NOTE

This correction applies only if the "T4 MARGIN" (before correction), indicated on the VEMD or calculated manually, is less than +35°C. If the t4 margin is greater than +35°C, the engine state shall be considered to be unsatisfactory.

Find the relevant correction factor in the table (Hp, OAT) below.

OAT		- 40°C	-20°C	0°C	20°C	40°C
Hp (ft)	Hp (m)					
0	0	-2.5	-2.5	-2.5	-2.5	-2.5
2000	609	-3	-3	-3	-2.5	-2.5
4000	1219	-3.5	-3.5	-3.5	-3	-3
6000	1829	-4.5	-4.5	-4.5	-3.5	-3.5
8000	2438	-5.5	-5	-5	-4	-4
10000	3048	-6	-5.5	-5	-4	
12000	3658	-6.5	-6	-5	-4	
14000	4267	-6.5	-6	-5.5	-4.5	
16000	4877	-6.5	-6	-6	-5	

- t4 margin correction = torque margin x correction factor.
- Corrected t4 margin = t4 margin + t4 margin correction.

Example (Manual procedure):

Ng = 96.3%

t4 = 826°C

TRQ = 92.7%

Nf = 389 rpm

Hp = 420 ft (128 m)

OAT = +21.8°C

Calculated t4 margin: +4°C

Calculated TRQ margin: +5.7%

- . Enter table, find: Hp = 420 ft (128 m) and OAT = +21.8°C gives a correction factor of -2.5
- . t4 margin correction = Calculated TRQ margin x correction factor
= +5.7 x -2.5 = -14.25°C
- . Corrected t4 margin = Calculated t4 margin + t4 margin correction
= +4 - 14.25
= -10.25°C

RC a

The paragraph **3.2.3 - Corrected t4 Margin calculation**, is modified as follows:

- t4 margin correction = torque margin x correction factor.
- Corrected t4 margin = t4 margin + t4 margin correction.

Example (VEMD procedure):

MV 50 0542.02

ENGINE POWER CHECK RESULT			
NG	96.3%	NF	389 RPM
T4	826 °C	Zp	420 Ft
TRQ	92.7%	OAT	+21.8 °C
T4 MARGIN		TRQ MARGIN	
+4.0 °C		+5.7%	
BAD		GOOD	
EXIT -> PRESS RESET			

- Enter table, find : Hp = 420 ft and OAT = +21.8°C, gives a correction factor of -2.5
 - . t4 margin correction = TRQ MARGIN x correction factor
= +5.7 x -2.5 = -14.25°C
 - . Corrected t4 margin = T4 MARGIN + t4 margin correction
= +4 - 14.25
= -10.25°C

The rest of the paragraph is unchanged.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3354.

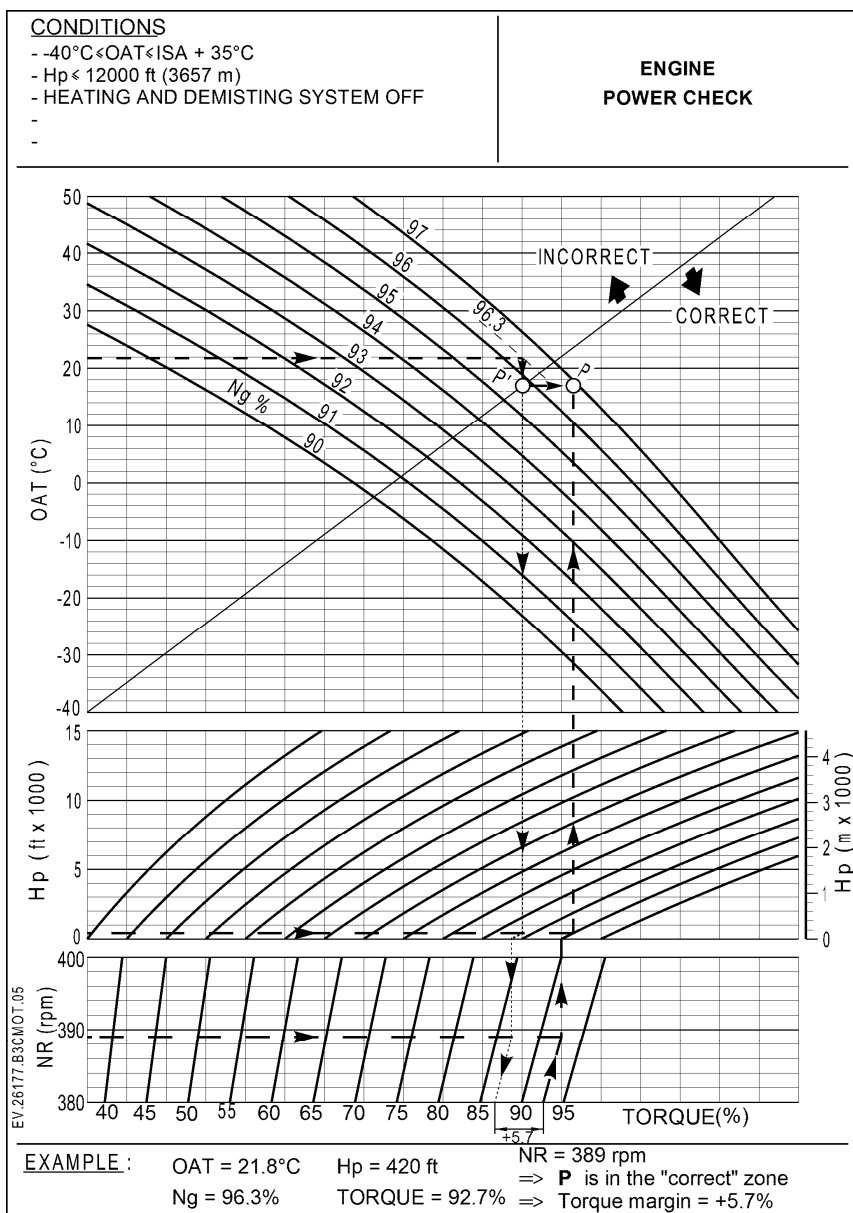


Figure 1

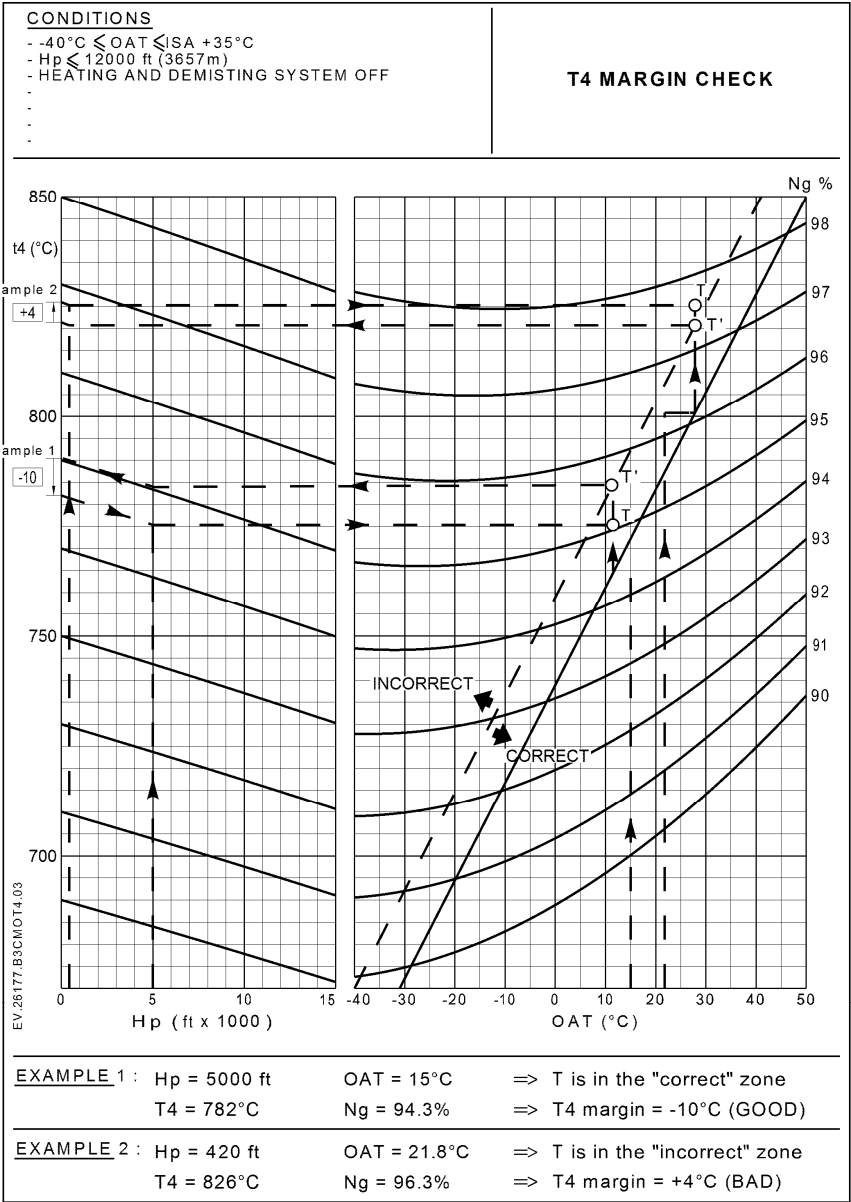


Figure 2

4 AIR DATA SYSTEM CALIBRATION

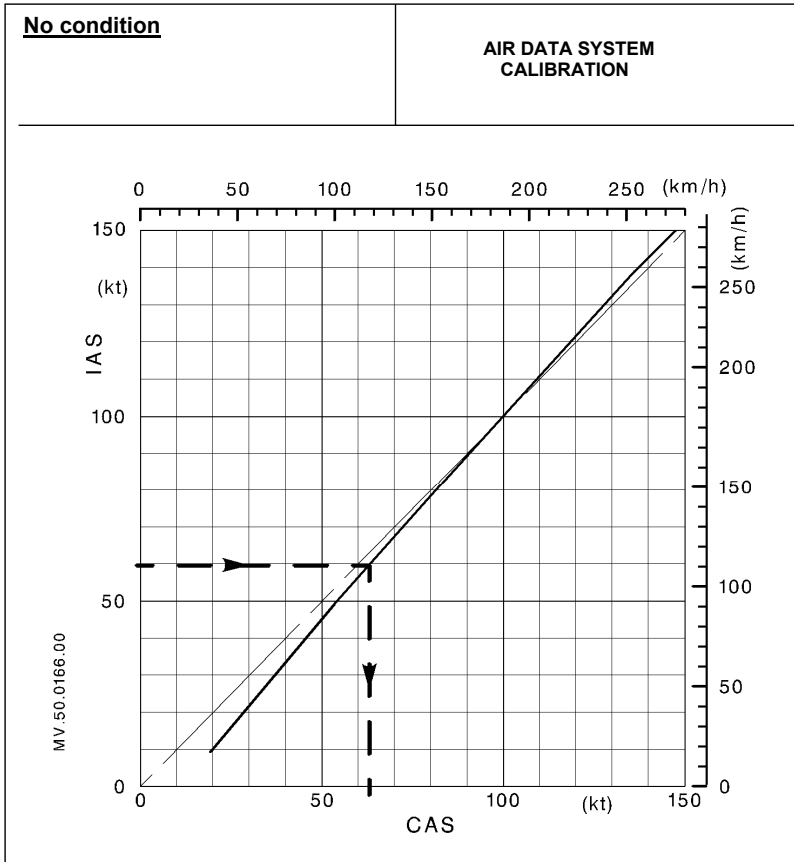


Figure 3

5 HEIGHT - VELOCITY DIAGRAM

The avoidance zone is defined by four points: A, B, C, D (refer to Figure 4).

- Point A: low hover point

Point A is at 8 ft (2.5 m) skid height at zero airspeed.

- Point B:

Point B is defined by:

- a constant height of 25 ft (8 m),
- a constant airspeed of 40 kt (74 km/h).

- Point C:

Point C is defined by:

- a constant height of 100 ft (30 m),
- a variable airspeed depending on the density altitude and on the aircraft weight as determined by line (C).

- Point D:

Point D is defined by:

- a variable height depending on the density altitude and on the aircraft weight as determined by line (D),
- a constant zero airspeed.

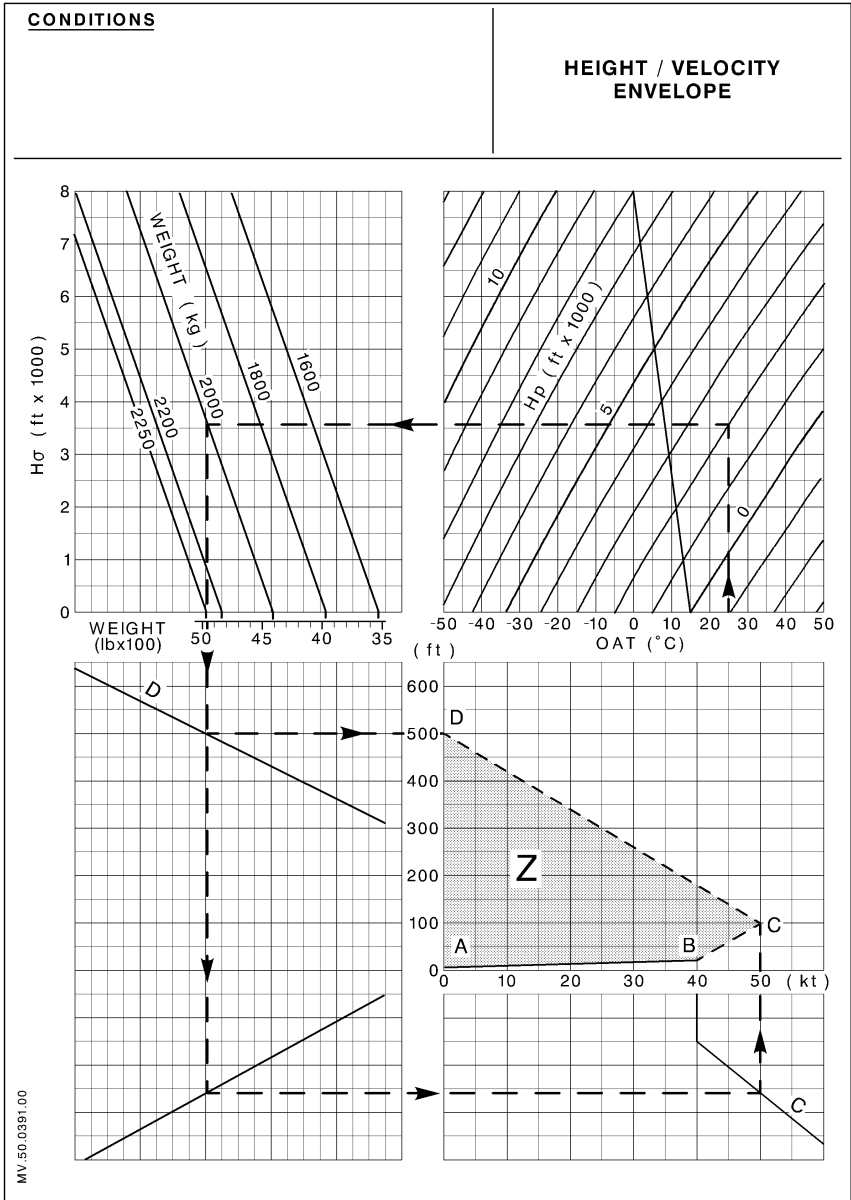


Figure 4

6 HOVER IN GROUND EFFECT

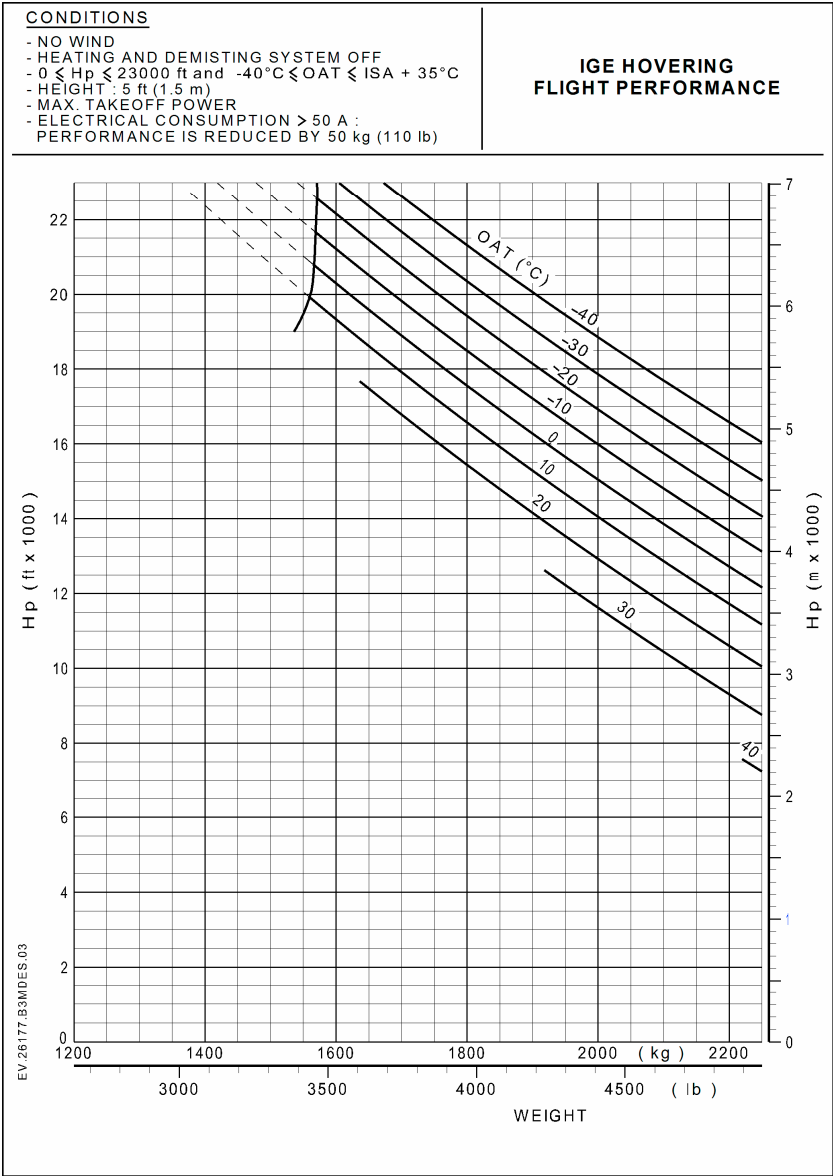


Figure 5

7 HOVER OUT OF GROUND EFFECT

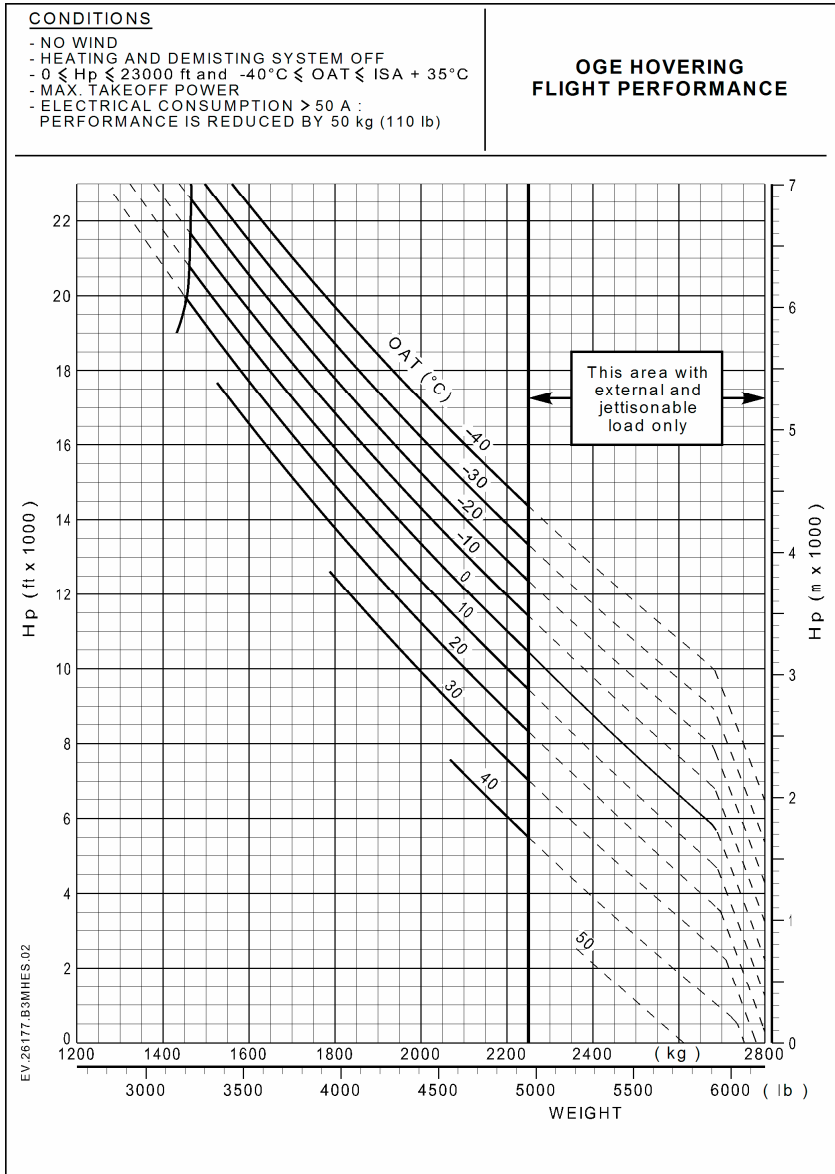


Figure 6

8 CORRECTED WEIGHT

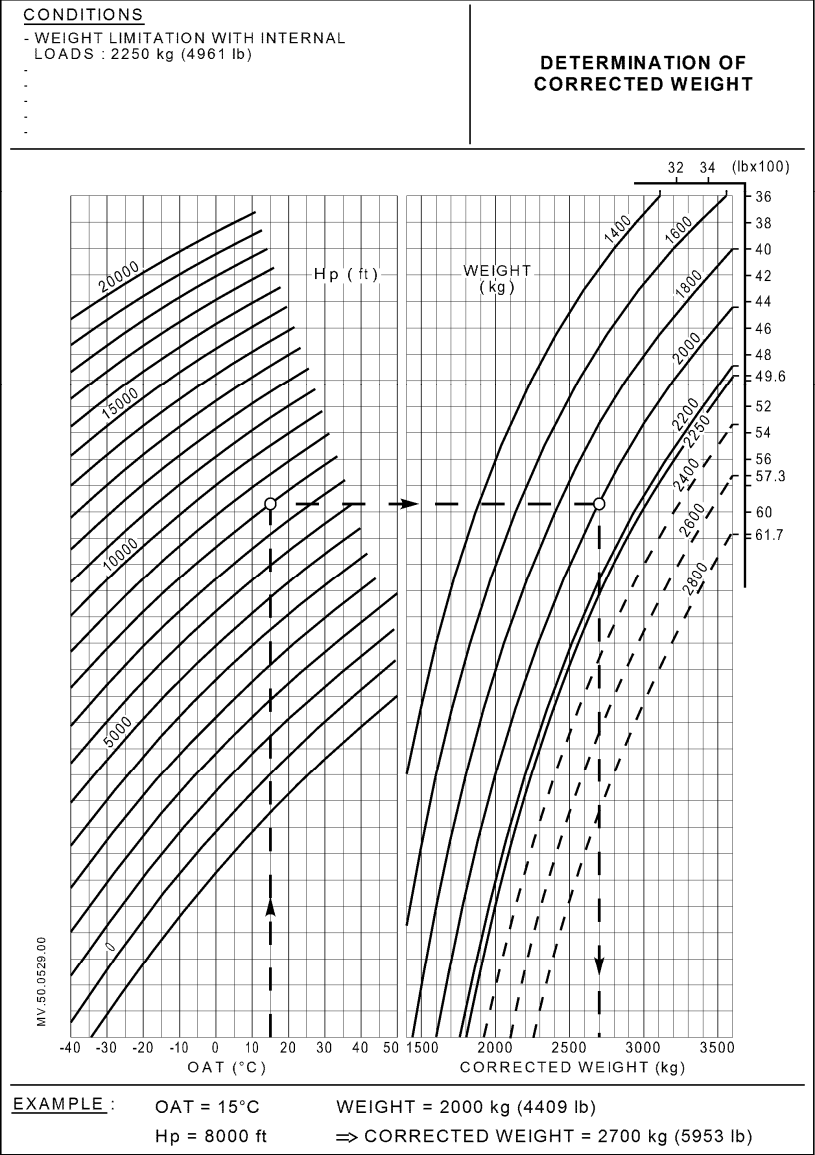


Figure 7

9 RATE OF CLIMB

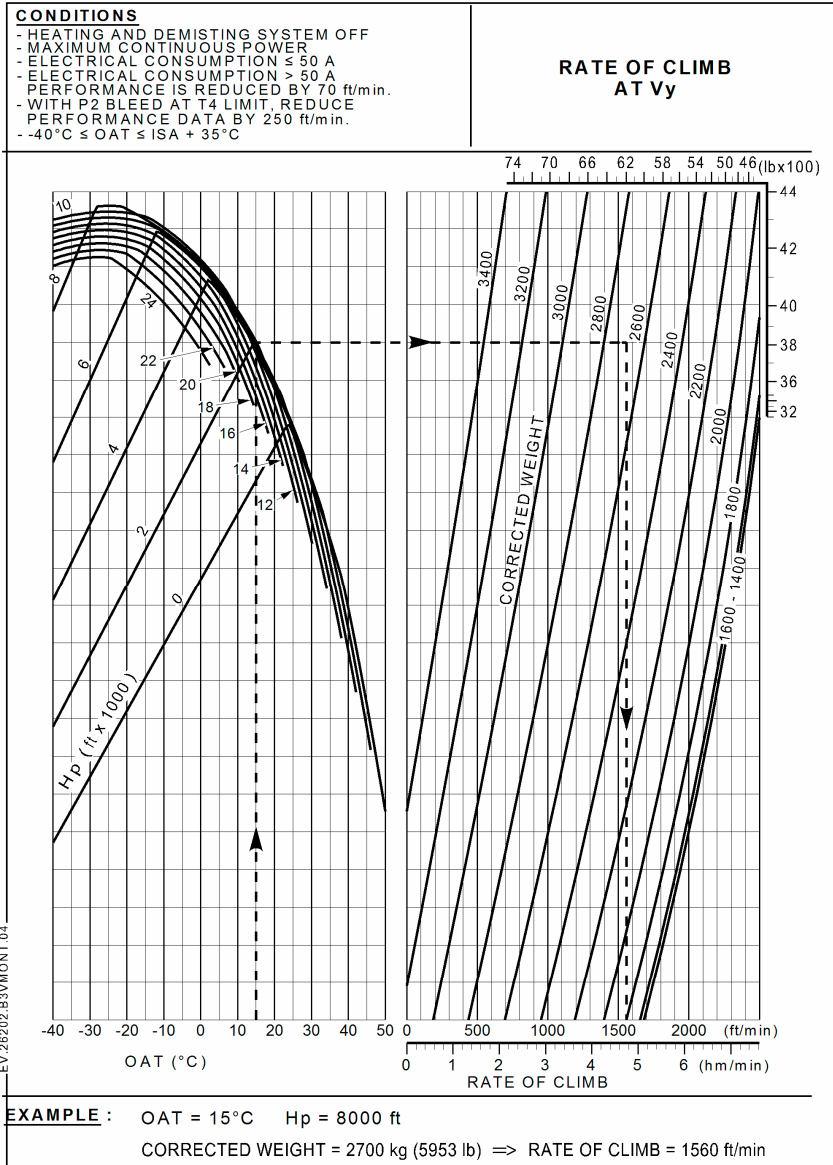


Figure 8

10 GLIDE DISTANCE IN AUTOROTATION

The distance flown in autorotation is:

0.54 NM per 1000 ft at IAS kt = 65 kt and NR $\cong$ 410 rpm

1000 m (distance) per 300 m (height) at IAS km/h = 120 km/h and NR $\cong$ 410 rpm

11 NOISE LEVEL

Noise characteristics as defined by the conditions of chapter 11 of the ICAO annex 16 are as follows:

Flight Phase	Noise Levels measured (EPNdB)	ICAO / EASA CS 36 Noise Limits (EPNdB)
Flyover	84.6	86.5



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

LIST OF SUPPLEMENTS

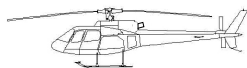
INCOMPATIBILITY OF USE

EFFECT ON PERFORMANCE DATA

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.



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Aéroport international Marseille-Provence 13725 Marignane Cedex - France

NOTE

Pages SUP.0.P3 and SUP.0.P4 concern the whole of the Supplements assigned to the helicopter mentioned on the title pages.

LIST OF SUPPLEMENTS

Some Supplements covering installations or procedures not used on this helicopter may be withdrawn from this manual. The complete list of Supplements appears on pages SUP.0.P2.

No.	TITLE
0	LIST OF SUPPLEMENTS - INCOMPATIBILITY OF USE - EFFECT ON PERFORMANCE DATA
1	RESERVED
2	RESERVED
3	RESERVED
4	INSTRUCTIONS FOR OPERATIONS IN COLD WEATHER
5	RESERVED
6	AUTOROTATION LANDING TRAINING PROCEDURE
7	HYDRAULIC FAILURE TRAINING PROCEDURE
8 to 11	RESERVED
11.1 to 11.2	RESERVED
12	TRANSPORT OF EXTERNAL LOADS: CARGO SLING 750 kg (1660 lb) "BREEZE EASTERN" (P/N 17149-1)
13	TRANSPORT OF EXTERNAL LOADS: CARGO SWING 1400 kg (3086 lb) with "SIREN" release unit (P/N AS21-5-7)
13.1	TRANSPORT OF EXTERNAL LOADS: CARGO SWING 1400 kg (3086 lb) with "SIREN" fixed release unit (P/N S1609)
13.2	TRANSPORT OF EXTERNAL LOADS: CARGO SWING 1400 kg (3086 lb) with "ON-BOARD" release unit (P/N 528-023-51)
14	SAND FILTER Reference: QB0550 / QB0777
15	RESERVED
16	AUTOMATIC FLIGHT CONTROL SYSTEM SFIM 85 T 31 (3-axis)

LIST OF SUPPLEMENTS (cond't)

No.	TITLE
17	EMERGENCY FLOATATION GEAR
18	HOIST INSTALLATION "BREEZE" or "AIR EQUIPMENT" Electric hoist 136 kg (300 lb)
19	HOIST INSTALLATION "BREEZE" Electric hoist 204 kg (450 lb)
19.1	HOIST INSTALLATION "BREEZE" Electric hoist 204 kg (450 lb), grip with support bracket
20	HYDRAULIC PUMP DRIVE ON MGB
21	TWO-PASSENGERS FRONT SEAT
22	LONG AND SHORT FOOTSTEPS Long footsteps P/N D.350-591-111-A Short footsteps P/N D.350-591-113
23	DUAL HYDRAULIC SYSTEM
24	LOUDSPEAKER INSTALLATION Optional OP 1810 and OP 1811
25 to 27	RESERVED
28	MAXIMUM INTERNAL GROSS WEIGHT Increased to 2370 kg (5225 lb)
29	200 A STARTER GENERATOR Optional OP 3821
30 to 39	RESERVED
40	SPECIAL COCKPIT LIGHTING
41 to 49	RESERVED
50	FERRY FLIGHT FUEL TANK
51	RESERVED
52	"BAMBI BUCKET" Model 2732S
53 to 55	RESERVED
55.1	RESERVED
55.2	GPS "GARMIN GNS 430/430W"
55.3	RESERVED
56	ABSEILING INSTALLATION

COMPOSITION

OF CONDITIONAL REVISIONS (RC)

This Manual assigned to the helicopter mentioned on the title page contains the following pink pages except those cancelled when the conditions are complied with.

CAUTION

The reader will have to insert the pink pages incorporating the paragraph(s) affected by the conditional revision so as the paragraph(s) cover(s) the paragraph(s) of the standard version or of the variant of standard definition.

(1) Paragraph Revision Code:

- **R** Revised, to be replaced
- **N** New, to be inserted

RC No.	SECTION or SUP.	PARAGRAPH	DATE CODE	Number of pages	(1)	Applicable before condition is met:
a	SUP.7	1.2.1.1 *RC*	19-40	1		MOD 07.3317
	SUP.7	4*RC*	19-40	1		
b	SUP.16	1.1 *RC*	15-45	1		MOD 07.3409
	SUP.16	3.3 *RC*	15-45	1		
	SUP.16	3.4 *RC*	15-45	1		
	SUP.16	3.5 *RC*	15-45	1		
	SUP.16	3.6 *RC*	15-45	1		
	SUP.16	3.7 *RC*	15-45	1		
	SUP.16	3.8.1 *RC*	15-45	1		
	SUP.16	4.1 *RC*	15-45	1		
	SUP.16	4.1.1 *RC*	15-45	1		
	SUP.16	4.1.2 *RC*	15-45	1		
	SUP.16	4.1.3 *RC*	15-45	1		
c	SUP.16	4.3.1.2 *RC*	15-45	1		MOD 07.3363
d	SUP.23	1 *RC*	21-02	1		MOD 07.4622
	SUP.23	3.2.2 *RC*	15-45	1		
	SUP.23	4 *RC*	23-40	1	R	

RC No.	SECTION or SUP.	PARAGRAPH	DATE CODE	Number of pages	(1)	Applicable before condition is met:
e	SUP.29	2 *RC*	15-45	1		Installation Starter generator 200 A Ref.: 515-030
f	SUP.13.2	1 *RC*	15-45	1		MOD 07.4614
	SUP.13.2	4.1 *RC*	15-45	1		
g	SUP.23	1 *RC*	21-02	2		MOD 07.4925

COMPOSITION
OF RUSH REVISIONS (RR)

The Supplements contain the following additional yellow page(s):

CAUTION

The reader will have to insert the yellow pages incorporating the paragraph(s) affected by the Rush Revision opposite the existing paragraph(s) of the standard version or of the variant of standard definition.

- (1) Paragraph Revision Code:
- **R** Revised, to be replaced
 - **N** New, to be inserted

RR No.	SECTION or SUP.	PARAGRAPHS	DATE CODE	Number of pages	(1)

LIST OF APPROVED EFFECTIVE PAGES - EASA CERTIFICATION

(1) AIRWORTHINESS EFFECTIVITY:

- Without indication..... Applicable to all aircraft
- A..... Specific to EASA

(2) VARIANT OF STANDARD DEFINITION EFFECTIVITY:

- Without indication..... Applicable to all aircraft
- XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.0.P1	1 to 1	15-45	A	
SUP.0.P2	1 to 2	22-22		
SUP.0.P3	1 to 2	23-40		
SUP.0.P4	1 to 1	15-45		
SUP.0.P5	1 to 3	23-40		
SUP.0	1 to 2	15-45		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

ISSUE 1: NR 0 to NR 15:

NORMAL REVISION 15 - JUNE 2014	Approved under the authority of EASA DOA No. 21J056 on September 03, 2015
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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved under the authority of EASA DOA No. 21J056 on May 31, 2016
Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 16-33		EASA approval No.10066154 on July 11, 2018
Title	Addition of a new Supplement: SUP.40 "SPECIAL COCKPIT LIGHTING"	
Revised information	SUP.0.P2, pages 1 and 2; SUP.0.P5, pages 1 and 2.	
Deleted information	None	
NORMAL REVISION 2 date code 19-40		Approved on October 23, 2020 under the authority of EASA DOA No. 21J700
Title	Data transfer from SUP.0.P3*RC* pink page to white page. SUP.0.P3*RC* is cancelled (no longer necessary).	
Revised information	SUP.0.P3 pages 1 and 2; SUP.0.P5 pages 1 and 2.	
Deleted information	None	
NORMAL REVISION 3 date code 21-02		Approved on July 09, 2021 under the authority of EASA DOA No. 21J700
Title	Hydraulic pump reversal (MOD 07- 4925)	
Revised information	SUP.0.P3 pages 1 and 2; SUP.0.P5 pages 1 and 2.	
Deleted information	None	

LOG OF APPROVED NORMAL REVISIONS**BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA**

NORMAL REVISION 4 date code 22-22		EASA approval No. 10080609 on November 15, 2022
Title	Modification of the title of SUP.17	
Revised information	SUP.0.P2 page 2; SUP.0.P5 pages 1 to 3.	
Deleted information	None	
NORMAL REVISION 5 date code 23-40		Approved on June 21, 2024 under the authority of EASA DOA No. 21J700
Title	Modification of SUP.0.P3	
Revised information	SUP.0.P3 page 1; SUP.0.P5 pages 1 to 3.	
Deleted information	None	

INCOMPATIBILITIES OF UTILIZATION BETWEEN OPTIONAL EQUIPMENT ITEMS

The following list is non-exhaustive and covers only those EASA-approved equipment items which are incompatible with one or several other items

NOTE

Incompatibility of installation between equipment items is stated in the Master Servicing Manual (MSM).

Item No.	Operation of the following installation:	Makes operation with the following optional equipment items impossible:	MANUAL SECTION No.
1	RESERVED		
2	Transport of external loads: "Cargo SLING"	7 - 8 - 9 - 12 - 13	SUP.12
3	Transport of external loads: "Cargo SWING"	7 - 8 - 9 - 12 - 13	SUP.13 or SUP.13.1 or SUP.13.2
4	RESERVED		
5	RESERVED		
6	Emergency floatation gear	7* - 12	SUP.17
7	Hoists installation	2 - 3 - 6* - 8 - 9 - 12 - 13	SUP.18 or SUP.19 or SUP.19.1
8	Two-passengers front seat	2 - 3 - 7	SUP.21
9	Ferry flight fuel tank	2 - 3 - 7	SUP.50
10	RESERVED		
11	RESERVED		
12	Abseiling installation	2 - 3 - 6 - 7	SUP.56
13	Air ambulance installation	2 - 3 - 7	9.14
14	Maximum internal gross weight increased to 2370 kg	15	SUP.28
15	Low landing gear installation	14	

(*) Hoisting remains possible when the floats are folded.

INFLUENCE OF OPTIONAL EQUIPMENT ITEMS ON PERFORMANCE DATA

When several optional equipment items are used simultaneously, the basic performance data must be reduced by the value corresponding to the influence of each optional item.

1 APPROVED PERFORMANCE DATA

- Takeoff weights:

When the installation of an optional equipment item modifies the takeoff weights specified in the basic Flight Manual SECTION 5.1, the relevant Supplement either provides the new takeoff weights by new charts or by a penalty relative to the basic flight performance.

- Rates of climb:

When the rates of climb in the basic Flight Manual SECTION 5.1 are modified, the relevant Supplement either provides a new chart or specifies a reduction with respect to the basic flight performance.

2 ADDITIONAL PERFORMANCE DATA

The influences of the optional equipment items are specified in SECTION 5.2 "Additional performance data".



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

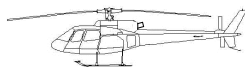
SUPPLEMENT

INSTRUCTIONS FOR OPERATIONS IN COLD WEATHER

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.



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Aéroport international Marseille-Provence 13725 Marignane Cedex - France

LIST OF APPROVED EFFECTIVE PAGES - EASA CERTIFICATION

- (1) AIRWORTHINESS EFFECTIVITY:
- Without indication..... Applicable to all aircraft
 - A..... Specific to EASA
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- Without indication..... Applicable to all aircraft
 - XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.4.P1	1 to 1	15-45	A	
SUP.4.P5	1 to 2	15-45		
SUP.4	1 to 4	15-45		

LOG OF APPROVED NORMAL REVISIONS**BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA**ISSUE 1: NR 0 to NR 1:

NORMAL REVISION 1 - MAY 2009	EASA approval No. 10027941 on November 18, 2009
------------------------------	--

ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	

1 GENERAL

This supplement details the procedures to be followed when the aircraft is operated in cold weather ($OAT \leq 0^{\circ}C$) and/or when aircraft is or could be exposed to falling or blowing snow.

Aircraft servicing does not require any special tools or routine replacement.

NOTE

Use of 3 to 3.9 cSt synthetic oil is recommended for low temperature operation.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

NOTE

Following an engine failure at light weight low Hp and low OAT, the stabilized NR may be below the audio warning threshold (360 rpm), the pilot can cut the horn using the [HORN] pushbutton.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

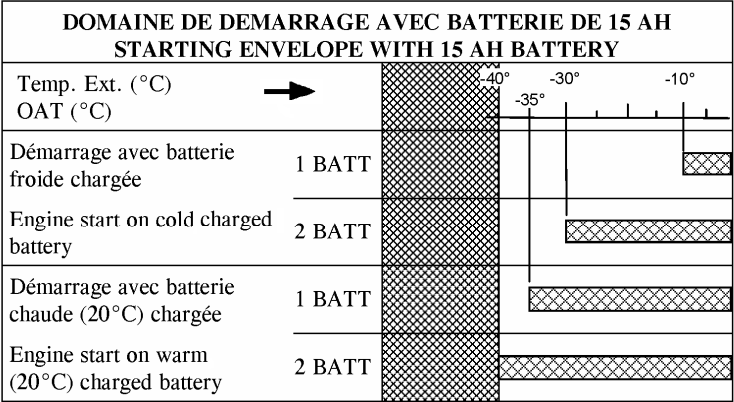
4.1 GENERAL RECOMMENDATIONS

For safe and rational operation of the aircraft in cold weather and snow, carry out the following basic operations:

- Remove ice or snow accumulations from the whole of the aircraft, particularly at hinges and on all parts of the dynamic and control systems (main rotor, rotor mast, tail rotor drive and tail rotor, flight controls, engine controls).
- When the aircraft has been subject to very low temperatures, it is recommended:
 - either to perform regular ground runs every two hours for temperatures of around $-20^{\circ}C$ or every hour for lower temperatures.
 - or to preheat the engine, transmission assemblies and cabin before engine starting (even if it is possible to start the engine at temperatures down to $-40^{\circ}C$).

4.2 USE OF BATTERIES FOR STARTING

During long periods on the ground it is recommended to store the battery in a warm area.
If a ground power unit is not available, startup may be carried out using the aircraft battery or two aircraft batteries connected in parallel.
The starting envelope is related to the temperature and is indicated in the following chart:



4.3 PREFLIGHT CHECK

WARNING

ICE OR SNOW ACCUMULATIONS THAT REMAIN IN OR AROUND THE ENGINE AIR INTAKE MAY BE INGESTED AND CAN CAUSE A SUDDEN IN-FLIGHT ENGINE FAILURE.

In addition to the inspections specified in the basic flight Manual, perform the following operations and inspections:

- Main rotor blades : Remove snow and ice.
- Main rotor hub and mast : Check for absence of ice on the swashplates, the scissors, the servo controls and the rotor head spring antivibrator.
- Engine :
 - Remove the engine air intake and exhaust nozzle blanking covers only after removal of snow from the aircraft surface.
 - Remove snow and ice accumulations around the air intake and on either side of the screen.
 - Check for absence of snow and ice accumulations inside the air intake.
- Drains and air pressure probes : Inspect fuel drain, check for absence of snow and ice on all ventilation and drain pipes as well as on static ports and pitot.
- Tail rotor :
 - Check for absence of ice on the tail rotor assembly.
 - Manually rotate the tail rotor so that the main rotor performs at least a complete turn:
 - Check free rotation,
 - Check freewheel operation.
- Cabin :
 - Remove the cabin cover just before engine starting to prevent windscreen icing.
 - Check that the windshield wiper is not stuck on the canopy.

NOTE

In falling or blowing snow conditions the engine air intake should be checked at the end of the exterior checks. The further checks before engine starting should then be performed without major delay.

4.4 RUN-UP CHECK

When the outside air temperature is below – 20°C, the run-up check is modified as follows:

- Hydraulic checks..... :
 - Proceed as for normal procedure and evaluate the loads when hydraulic switch is set to off. Complete the checks.
 - If operating loads were evaluated higher than at normal temperatures, move the cyclic stick 3-4 cm forward (nose-down) for 2 minutes to warm up the spherical thrust bearings.
- Yaw pedals : Move the pedals over approximately 50% of their travel range on either side of the mid-position.

4.5 AFTER LAST FLIGHT OF THE DAY

The normal procedures described in the basic flight manual are to be supplemented by the following:

- When the rotor stops turning, position the cyclic pitch stick close to the neutral position and the collective pitch lever locked at full low pitch, with tail rotor blades in the horizontal position.
- Do not leave doors open.
- Install the air intake cover and exhaust nozzle blank.
- When the aircraft is parked in an unsheltered area, it is recommended to apply anti-icing products and to carry out aircraft blanking and mooring.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

AUTOROTATION LANDING TRAINING PROCEDURE

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.



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LIST OF APPROVED EFFECTIVE PAGES - EASA CERTIFICATION

- (1) AIRWORTHINESS EFFECTIVITY:
- Without indication..... Applicable to all aircraft
 - A..... Specific to EASA
- (2) VARIANT OF STANDARD DEFINITION EFFECTIVITY:
- Without indication..... Applicable to all aircraft
 - XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.6.P1	1 to 1	15-45	A	
SUP.6.P5	1 to 2	15-45		
SUP.6	1 to 3	15-45		

LOG OF APPROVED NORMAL REVISIONS**BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA**ISSUE 1: NR 1 to 3:

NORMAL REVISION 3 - MAY 2009	EASA approval No. 10027941 on November 18, 2009
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ISSUE 2:

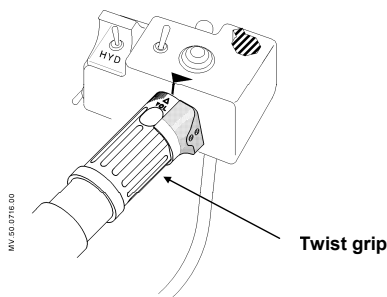
NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	

1 GENERAL

This procedure is used for training for autorotation landing with full touchdown or power recovery, with a simulated engine failure or loss of engine power.

In case of engine failure or sudden loss of power, the helicopter will yaw to the right, some red warnings may come on associated with the Gong audio warning, the NR will decay and the low NR audio warning will sound if NR goes below 360 rpm.

The procedure enables engine failure or loss of engine power to be simulated with the same symptoms by setting the twist grip to the IDLE position. Engine is thus set to idle.



2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable.

NOTE

Autorotation training shall be conducted within gliding distance of a suitable running landing area.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

NOTE

If necessary, it is possible to quickly turn the twist grip back to the FLIGHT position at any time and for any NR value.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

NOTE

No significant Nf transient is expected when switching from IDLE to FLIGHT position in autorotation configuration ($T_q \approx 0\%$) as no power is required from the engine.

4.1 FAILURE SIMULATION

- 1 Collective pitch.....REDUCE power
- 2. Twist gripIDLE position:
 - . **TWT GRIP**
 - . Gong sounds
 - . Engine is set to idle
 - . $N_g \cong 68\%$

then:

4.2 FULL TOUCHDOWN AUTOROTATION TRAINING PROCEDURE

- 1. Autorotation procedureAPPLY actions 1 to 10 of the procedure described in SECTION 3.2 § 1 of the basic flight manual

then:

After full stop landing:

- 2. Collective pitchREDUCE to full low pitch
- 3. Twist gripFLIGHT position:
 - . **TWT GRIP**
 - . Rotor speed increases to its normal governed value

4.3 POWER RECOVERY AUTOROTATION TRAINING PROCEDURE

1. Collective pitchAPPLY actions 1 to 4 of the procedure described in SECTION 3.2 § 1 of the basic flight manual

then:

At height $\geq$ 70 ft (21 m)

2. NR.....CHECK in normal operating range
3. Twist gripFLIGHT position:
 . **TWT GRIP**
 . Nf increases to its governed value
4. Collective pitchCONTROL to maintain NR in normal operating range
5. Cyclic.FLARE

At 20 - 25 ft (6/8 m) and at constant attitude

6. Collective pitch.....GRADUALLY INCREASE to reduce the rate of descent and forward speed
7. CyclicFORWARD slightly to adopt a landing attitude ($<10^\circ$)
8. Pedals.....ADJUST to cancel any side-slip tendency
9. Collective pitch.....INCREASE as necessary

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

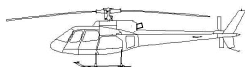
SUPPLEMENT

HYDRAULIC FAILURE TRAINING PROCEDURE

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.



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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.7.P1	1 to 1	15-45	A	
SUP.7.P5	1 to 2	19-40		
SUP.7	1 to 6	19-40		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

ISSUE 1: NR 0 to NR 7:

NORMAL REVISION 7 - JUNE 2014	Approved on September 03, 2015 under the authority of EASA DOA No. 21J056
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Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 19-40		Approved on October 23, 2020 under the authority of EASA DOA No. 21J700
Title	Hydraulic failure training procedure improvement	
Revised information	SUP.7.P5 pages 1 & 2; SUP.7 pages 1 to 6.	
Deleted information	None	

1 GENERAL

This procedure describes hydraulic failure training for an AS350 equipped with a single hydraulic system.

1.1 DESCRIPTION OF ACTUAL LOSS OF HYDRAULIC PRESSURE

- **HYDR** + "Gong"
- The main rotor servos are temporarily fed by the 3 accumulators giving hydraulic assistance for a limited operation of the main rotor flight controls allowing to reach the safety speed in case of hydraulic failure.
- Then the pilot must switch off the hydraulic cut-off switch on the RH collective grip (1) and apply the emergency procedure.
- The tail rotor control is assisted by a yaw load compensator with one rechargeable accumulator that provides continuous assistance on RH pedal.

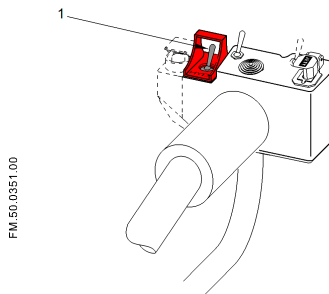


Figure 1: Hydraulic cut-off switch (RH collective grip)

Switching off only the hydraulic cut-off switch on the RH collective grip has the following effects:

- The hydraulic pump pressure is by-passed.
- The main rotor accumulators are immediately discharged, cyclic and collective control loads are significantly increased (it is necessary to discharge the main rotor accumulators to avoid non-homogenous main rotor cyclic flight control loads).
- RH pedal control loads remain low as the tail rotor load compensator remains pressurized.

1.2 DESCRIPTION OF HYDRAULIC FAILURE TRAINING PROCEDURE

- **The training for the loss of hydraulic pressure procedure is divided in two separate steps:**
 - First step: Hydraulic failure simulation by instructor and trainee initial reaction
 - Second step: Hydraulic failure emergency procedure handling and completion by trainee
- **Training procedure recommendations:**
 - It is recommended to avoid training with heavy aircraft weight as higher weight leads to higher control loads.
 - To minimize pilot's fatigue, the hydraulic training procedure should be performed close to an airfield that is suitable for a running landing.
 - In case of a go-around during hydraulic failure training procedure, it is recommended to abort the training and to reset the hydraulic cut-off switch to ON.

NOTE

When resetting the hydraulic cut-off switch to ON, be prepared for a significant decrease of cyclic and collective control loads.

1.2.1 First step: Hydraulic failure simulation by instructor and trainee initial reaction

NOTE

Upon instructor advice and training conditions and purposes, the first simulation step of the training procedure may be skipped.

In this step:

- The failure simulation is activated by the instructor pilot.
- The trainee recognizes the associated alarms and rejoins the hydraulic failure safety speed with the assistance of the main rotor accumulators.
- Once hydraulic failure safety speed is established, the instructor pilot restores hydraulic system to normal operation.

RC a

The paragraph 1.2.1.1 - **Hydraulic failure simulation procedure by instructor**, is modified as follows:

1.2.1.1 Hydraulic failure simulation procedure by instructor

To simulate a hydraulic pressure loss, the instructor press the [HYD TEST] pushbutton or [ACCU TST] (2) guarded pushbutton (SCU):

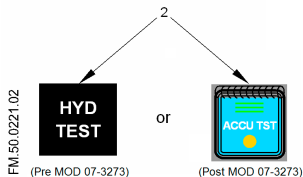


Figure 2: [HYD TEST] pushbutton or [ACCU TST] guarded pushbutton (SCU)

NOTE

The [HYD TEST] or [ACCU TST] action is not part of the hydraulic failure emergency procedure and should be performed by the instructor for simulation only.

Activating the [HYD TEST] or [ACCU TST] pushbutton has the following effects:

- **HYDR** + "Gong".
- The hydraulic pump pressure is by-passed (no more hydraulic pressure in the main rotor control circuit) simulating so the hydraulic pressure loss.
- The main rotor servos are temporarily fed by the 3 accumulators giving hydraulic assistance for a limited operation of the main rotor flight controls as in case of an actual pressure loss.

This simulation mean introduces an undesired side-effect not part of the actual hydraulic failure:

- On tail rotor control: Pressing the [HYD TEST] or [ACCU TST] removes all tail rotor control hydraulic assistance.
- On yaw pedals: Control loads increase immediately.

1.2.1.2 Hydraulic failure safety speed by trainee

- When **HYDR** lights up, trainee sets airspeed to hydraulic failure safety speed, between 40 and 60 KIAS (74 and 111 km/h).

1.2.1.3 Hydraulic system reconfiguration to normal mode by instructor

- Once hydraulic safety speed is established, instructor pilot restores hydraulic system to normal operations by resetting [HYD TEST] or [ACCU TST] to OFF position.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3317.

1.2.1.1 Hydraulic failure simulation procedure by instructor

To simulate a hydraulic pressure loss, the instructor press the [HYD TEST] pushbutton or [ACCU TST] (2) guarded pushbutton (SCU):

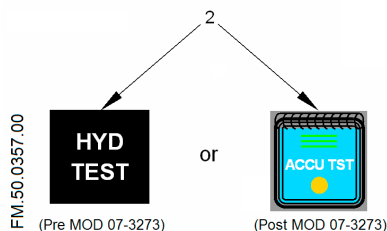


Figure 2: [HYD TEST] pushbutton or [ACCU TST] guarded pushbutton (SCU)

NOTE

The [HYD TEST] or [ACCU TST] action is not part of the hydraulic failure emergency procedure and should be performed by the instructor for simulation only.

Activating the [HYD TEST] or [ACCU TST] pushbutton has the following effects:

- **HYDR** flashes + "Gong" for each flash.
- The hydraulic pump pressure is by-passed (no more hydraulic pressure in the main rotor control circuit) simulating so the hydraulic pressure loss.
- The main rotor servos are temporarily fed by the 3 accumulators giving hydraulic assistance for a limited operation of the main rotor flight controls as in case of an actual pressure loss.

This simulation mean introduces an undesired side-effect not part of the actual hydraulic failure:

- On tail rotor control: Pressing the [HYD TEST] or [ACCU TST] removes all tail rotor control hydraulic assistance.
- On yaw pedals: Control loads increase immediately.

1.2.1.2 Hydraulic failure safety speed by trainee

- When **HYDR** lights up, trainee sets airspeed to hydraulic failure safety speed, between 40 and 60 KIAS (74 and 111 km/h).

1.2.1.3 Hydraulic system reconfiguration to normal mode by instructor

- Once hydraulic safety speed is established, instructor pilot restores hydraulic system to normal operations by resetting [HYD TEST] or [ACCU TST] to OFF position.

1.2.2 Second step: Hydraulic failure emergency procedure handling and completion by trainee

In this step:

- Trainee checks hydraulic failure safety speed is established.
- Trainee applies hydraulic failure procedure for landing.

2 LIMITATIONS

The limitations specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

2.1 CREW EXPERIENCE

Instructors shall be properly trained and with recent experience to perform hydraulic failure training exercises with trainees.

2.2 OPERATING LIMITATIONS

A thorough flight risk assessment taking in account trainee's experience and operational conditions (weather conditions, day/night conditions, ...) shall be performed before training.

Hover and low speed maneuvers are prohibited with hydraulic off.

Angle of bank more than 30° is prohibited with hydraulic off.

No passenger other than observers with training related duties shall be admitted on board.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable.

RC a

The paragraph 4 - **NORMAL TRAINING PROCEDURES, STEP1 : HYDRAULIC FAILURE SIMULATION** is modified as follows:

4 NORMAL TRAINING PROCEDURES

The normal procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

STEP 1: HYDRAULIC FAILURE SIMULATION

WARNING

THE [HYD TEST] or [ACCU TST] PUSHBUTTON SHALL NEVER BE ACTIVATED WHEN THE HYDRAULIC CUT-OFF SWITCH (RH COLLECTIVE GRIP) IS IN OFF POSITION.

In steady cruise flight conditions:

1. Instructor.....Activates [HYD TEST] or [ACCU TST]
Check **HYDR** + "Gong" and [ACCU TST] light on*

When **HYDR** lights up:

2. TraineeSet and maintain angle of bank below 30°
Avoid abrupt maneuvers
Set airspeed to hydraulic failure safety speed, between 40 and 60 KIAS (74 and 111 km/h)

Once hydraulic failure safety speed is established:

WARNING

IF THE [HYD TEST] or [ACCU TST] PUSHBUTTON IS NOT RESET TO OFF, HYDRAULIC ASSISTANCE CANNOT BE RESTORED.

3. Instructor.....Reset [HYD TEST] or [ACCU TST] to OFF position:
Check [ACCU TST] light off* and **HYDR** within 3 sec.

NOTE

Hydraulic assistance can be restored at any time during the simulation step by resetting [HYD TEST] or [ACCU TST] to OFF.

(*) Post MOD 07-3273

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3317.

4 NORMAL TRAINING PROCEDURES

The normal procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

STEP 1: HYDRAULIC FAILURE SIMULATION

WARNING

THE [HYD TEST] or [ACCU TST] PUSHBUTTON SHALL NEVER BE ACTIVATED WHEN THE HYDRAULIC CUT-OFF SWITCH (RH COLLECTIVE GRIP) IS IN OFF POSITION.

In steady cruise flight conditions:

1. Instructor.....Activates [HYD TEST] or [ACCU TST]
Check **HYDR** flashes + "Gong" and [ACCU TST] light on*

When **HYDR** lights up:

2. TraineeSet and maintain angle of bank below 30°
Avoid abrupt maneuvers
Set airspeed to hydraulic failure safety speed, between 40 and 60 KIAS (74 and 111 km/h)

Once hydraulic failure safety speed is established:

WARNING

IF THE [HYD TEST] or [ACCU TST] PUSHBUTTON IS NOT RESET TO OFF, HYDRAULIC ASSISTANCE CANNOT BE RESTORED.

3. Instructor.....Reset [HYD TEST] or [ACCU TST] to OFF position:
Check [ACCU TST] light off* and **HYDR** within 3 sec.

NOTE

Hydraulic assistance can be restored at any time during the simulation step by resetting [HYD TEST] or [ACCU TST] to OFF.

(*) Post MOD 07-3273

STEP 2: HYDRAULIC FAILURE PROCEDURE COMPLETION

WARNING

MAINTAIN ANGLE OF BANK BELOW 30° AND DO NOT ATTEMPT TO CARRY OUT HOVER FLIGHT OR ANY LOW SPEED MANEUVER BECAUSE THE INTENSITY AND DIRECTION OF THE CONTROL FEEDBACK FORCE WILL CHANGE RAPIDLY.

FAILURE TO COMPLY MAY RESULT IN POOR AIRCRAFT CONTROL AND POSSIBLE LOSS OF CONTROL.

AS CONTROL LOADS INCREASE, BE CAREFUL NOT TO INADVERTENTLY MOVE TWIST GRIP OUT OF FLIGHT POSITION (TWT GRIP LIGHT OFF).

1. Trainee Check airspeed set between 40 and 60 KIAS (74 and 111 km/h)

When ready for a significant cyclic and collective control loads increase:

2. Trainee Set hydraulic cut-off switch (RH collective grip) to OFF
Check **HYDR** + "Gong"
Control loads increase
3. Trainee Complete hydraulic failure emergency procedure

After landing:

4. Hydraulic cut-off switch..... Reset to ON (RH collective grip) to restore hydraulic assistance before subsequent takeoff or hovering
Check **HYDR** within 3 sec

NOTE

Instructor shall remain ready to restore hydraulic assistance as necessary.

Hydraulic assistance can be restored at any time during hydraulic failure procedure completion step by resetting the hydraulic cut-off switch to ON (RH collective grip).

5 PERFORMANCE DATA

The performance data specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

TRANSPORT OF EXTERNAL LOADS

CARGO SLING 750 kg (1660 lb) "BREEZE EASTERN" (P/N 17149-1)

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
THE EQUIPMENT MENTIONED ABOVE IS INSTALLED ON THE AIRCRAFT.



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LIST OF APPROVED EFFECTIVE PAGES - EASA CERTIFICATION

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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.12.P1	1 to 1	15-45	A	
SUP.12.P5	1 to 2	16-24		
SUP.12	1 to 7	16-24		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

ISSUE 1: NR 0 to NR 4:

NORMAL REVISION 4 - JULY 2009	EASA approval No. 10027529 on October 7, 2009
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Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 16-24		EASA approval No. 10059860 on October 24, 2016
Title	Modification of the operating limitation and normal procedure for the cargo hook	
Revised information	SUP.12.P5 pages 1 and 2, SUP.12 pages 4 and 6	
Deleted information	None	

1 GENERAL

The "CARGO SLING" external load installation (Figure 1) is composed of:

- A "BREEZE EASTERN" release unit and hook assembly (4) (P/N 17149-1) equipped with a load sensor unit (5) allowing:
 - the electrical hook opening,
 - the mechanical hook opening.
- A load indicator (1) with lighting (3), on the RH door pillar, with a zero setting control (2).

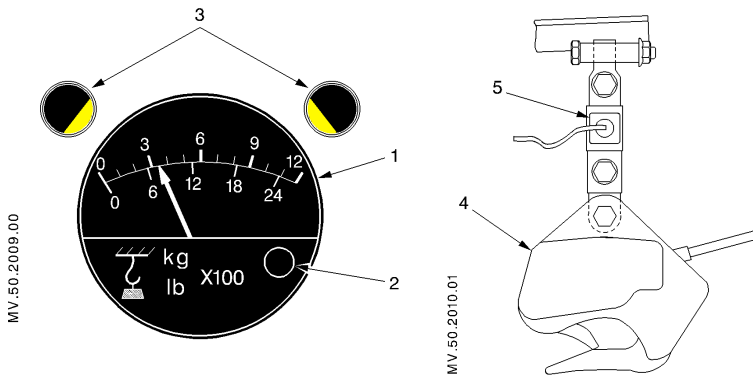


Figure 1: Cargo sling components

- A control system for the pilot (Figure 2), including:
 - a [SLING] pushbutton (4) located on the SCU (1), for powering-on the installation,
 - a release control (3) on the cyclic stick (electrical mode),
 - a release handle (2) located under the collective lever (mechanical mode).

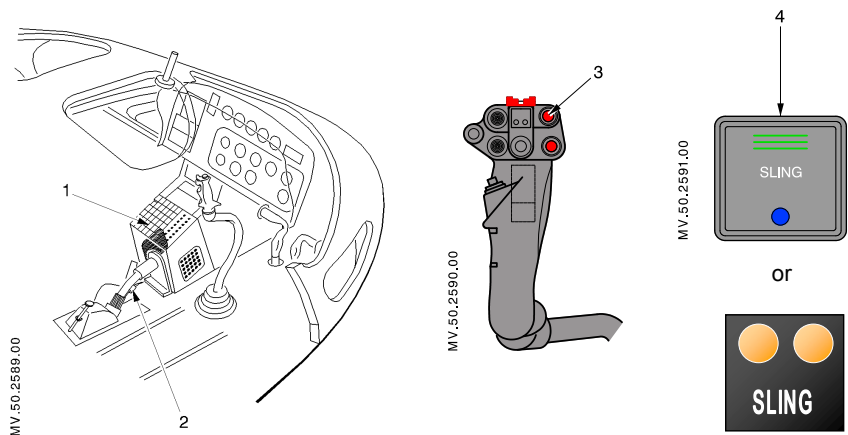


Figure 2: Cargo sling controls

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 WEIGHT LIMITATION

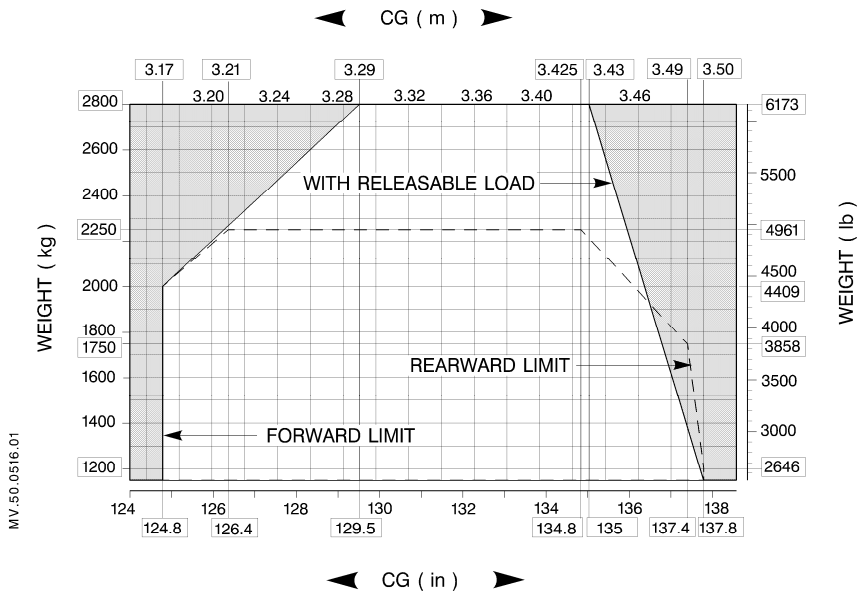
- Maximum authorized sling load weight: 750 kg (1660 lb).
- Maximum all up weight with an external load: 2800 kg (6173 lb)
or maximum authorized all up weight allowing hovering flight out of ground effect (the lowest of the two values).

CAUTION

The minimum and maximum weight without external load remains limited to the minimum and maximum weight specified in the limitations section of the basic flight manual.

2.2 LONGITUDINAL CG

With an external load, the longitudinal limits are defined according to the weight as per the graph below.



2.3 AIRSPEED LIMITATION

- Absolute maximum permissible indicated airspeed with external load 80 kt (148 km/h).

NOTE

The pilot is responsible for determining the limit speed according to the load and sling length. Particular care must be exercised when bulky loads are carried on the sling.

2.4 OPERATING LIMITATION

The cargo hook system is approved for lifting external load which is jettisonable and lifted free of land or water during rotorcraft operation.

Operation with an external load which remains in contact with land, water or any fixed structure is not demonstrated by the manufacturer. These operations shall not be conducted without approval from the responsible authority in accordance with the applicable operational regulations.

The external loads are limited to Non Human External Cargo (NHEC) only.

Flight with an empty net or unballasted sling as an external load is prohibited unless approved operational limits and procedures provided by the operator allow for such an operation.

Two placards visible to the ground operator and located on the lower fairing near to the hook indicate:

- the maximum sling load,
- the cargo hook rigging.

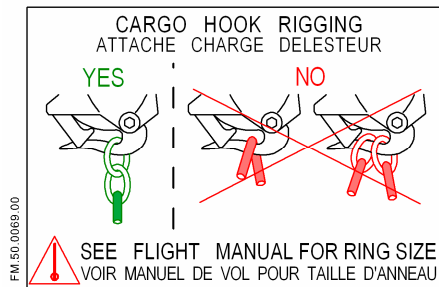


Figure 3: Cargo hook rigging placard

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

3.1 ENGINE FAILURE WITH EXTERNAL LOAD

- IN CRUISE FLIGHT
 1. Autorotation procedure: APPLY.
 2. External load RELEASE as soon as possible.
- IN HOVER
 1. Collective pitch REDUCE according to the height.
 2. External load RELEASE as soon as possible.
 3. Pedals Control yaw.
 4. Cyclic Forward to gain forward speed according to the height.
 5. Collective pitch INCREASE as needed to cushion touch-down.

NOTE

In case of a failure during the hooking phase, the pilot shall move the aircraft away to the right. Ground personnel are to be forewarned that in the event of an engine failure they have to move away to the left of the helicopter.

3.2 ELECTRICAL LOAD JETTISONING FAILURE

Collective pitch mechanical release ACTUATE

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Carrying heavy loads is a delicate operation due to the possible effects of a swinging load on the flight behavior of the helicopter. Consequently, pilots are advised to train with gradually increasing sling loads before undertaking heavy or bulky load carrying operations.
- The length of the sling cable must be determined in accordance with the type of mission. To carry a compact load, it is recommended to use the shortest possible cable.
- Operation with no or low load on a sling cable or in a net must be performed in such a way as to ensure that the trailing sling cable or net does not come close to the tail rotor.
- For permissible load attachment ring size refer to SECTION 9 of this Flight Manual.

WARNING

- 1- THE USE OF A LOAD ATTACHMENT RING WITH INCORRECT DIMENSIONS MAY LEAD TO LOSS OR JAMMING OF THE LOAD.
- 2- IN WET WEATHER, THE OPERATORS HANDLING THE HOOK AND LOADS SHOULD WEAR THICK RUBBER GLOVES. DISCHARGE STATIC ELECTRICITY BY PLACING AN ELECTRICAL CONDUCTOR CABLE OR TUBE BETWEEN THE GROUND AND THE CARGO RELEASE UNIT (HOOK).

4.1 CHECK OF THE INSTALLATION

- [SLING].....ON.
- Load indicatorZERO.
- Correct opening of the hook (both in electrical and mechanical control modes).....CHECK.
- Rotation of the retaining latchFREE.
- Operation of the retaining latch return spring.....CHECK.

4.2 TAKEOFF CHECK AND PROCEDURE WITH EXTERNAL LOAD

1. External loadHOOK and SECURE.
2. CollectiveIncrease very smoothly while maintaining the aircraft vertically above the load.
3. Cables tightenedDwell briefly before raising the load.
4. Lift the loadVertically.
5. Load indication.....CHECK.
6. Take-off path.....ADJUST to adopt an immediate forward climb attitude.
7. All parameters.....CHECK.

4.3 MANEUVERS

All control movements should be made very smoothly, with very gradual acceleration and deceleration, and only slightly banked turns.

4.4 APPROACH AND LANDING WITH EXTERNAL LOAD

Perform approach at minimum rate of descent.

- Establish zero translational ground speed sufficiently high to ensure that the load is not dragged along the ground.
- Then descend vertically until the load is set on the ground.
- Load.....RELEASE.
- Load release CHECK.
- All parameters.....CHECK.

NOTE

If the load is not released, actuate the mechanical release handle.

5 PERFORMANCE DATA

When no external load is carried on the hook, the performance data specified in the basic flight manual and in the flight manual supplements remain applicable.

- Hover out of ground effect performance is shown in Figure 6 of SECTION 5.1 § 7 of the basic flight manual.
- Hover and climb performance may be affected when carrying bulky loads.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

TRANSPORT OF EXTERNAL LOADS

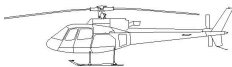
CARGO SWING 1400 kg (3086 lb) with "SIREN" release unit (P/N AS21-5-7)

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
THE EQUIPMENT MENTIONED ABOVE IS INSTALLED ON THE AIRCRAFT.



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LIST OF APPROVED EFFECTIVE PAGES - EASA CERTIFICATION

- (1) AIRWORTHINESS EFFECTIVITY:
- Without indication..... Applicable to all aircraft
 - A..... Specific to EASA
- (2) VARIANT OF STANDARD DEFINITION EFFECTIVITY:
- Without indication..... Applicable to all aircraft
 - XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.13.P1	1 to 1	15-45	A	
SUP.13.P5	1 to 2	16-24		
SUP.13	1 to 8	16-24		

LOG OF APPROVED NORMAL REVISIONS**BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA**ISSUE 1: NR 0 to NR 3:

NORMAL REVISION 3 - JULY 2009	EASA approval No. 10027529 on October 7, 2009
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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 16-24		EASA approval No. 10059860 on October 24, 2016
Title	Modification of the operating limitation and normal procedure for cargo hook	
Revised information	SUP.13.P5 pages 1 and 2, SUP.13 pages 4 and 6	
Deleted information	None	

1 GENERAL

The “CARGO SWING” external load installation (Figure 1) is composed of:

- A suspended pyramid frame (5) equipped with a free rotation hook “SIREN” release unit (4) (P/N AS21-5-7) allowing:
 - the electrical cargo hook opening,
 - the mechanical cargo hook opening.
- A load indicator, on the RH door pillar, with an indicator test pushbutton (1), a zero setting control (2) and a load-on indicator (3).

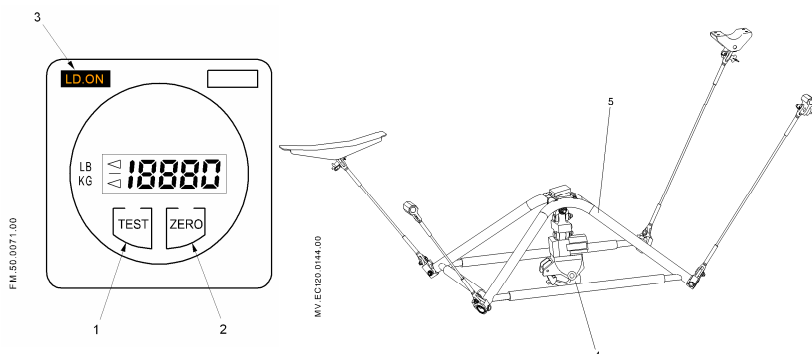


Figure 1: Cargo sling components

- A control system for the pilot (Figure 2), including:
 - a **[SLING]** pushbutton (4) located on the SCU (1), for powering-on the installation,
 - a release control (3) on the cyclic stick (electrical mode),
 - a release handle (2) located under the collective lever (mechanical mode).

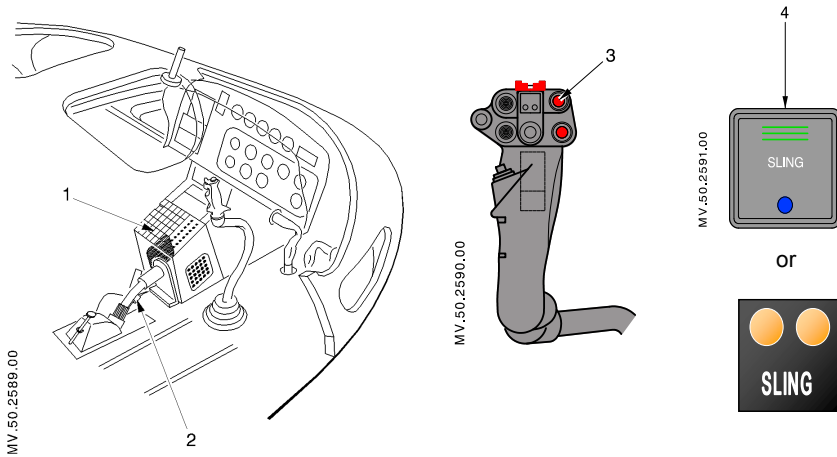


Figure 2: Cargo sling controls

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 WEIGHT LIMITATION

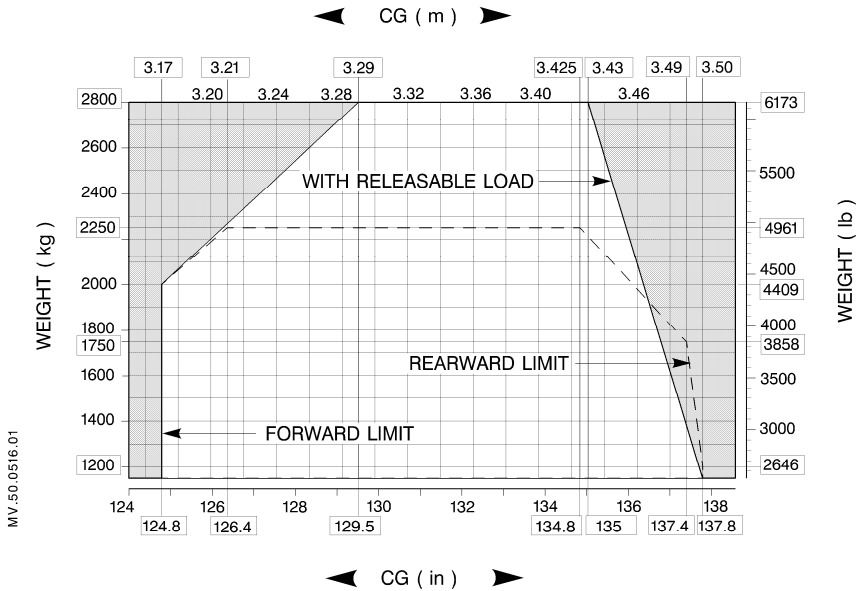
- Maximum authorized swing load weight: 1400 kg (3086 lb).
- Maximum all up weight with an external load: 2800 kg (6173 lb)
or maximum authorized all up weight allowing hovering flight out of ground effect
(the lowest of the two values).

CAUTION

The minimum and maximum weight without external load remains limited to the minimum and maximum weight specified in the limitations section of the basic flight manual.

2.2 LONGITUDINAL CG

With an external load, the longitudinal limits are defined according to the weight as per the graph below.



2.3 AIRSPEED LIMITATION

- Absolute maximum permissible indicated airspeed with external load 80 kt (148 km/h).

NOTE

The pilot is responsible for determining the limit speed according to the load and sling length. Particular care must be exercised when bulky loads are carried on the sling.

2.4 OPERATING LIMITATION

The cargo hook system is approved for lifting external load which is jettisonable and lifted free of land or water during rotorcraft operation.

Operation with an external load which remains in contact with land, water or any fixed structure is not demonstrated by the manufacturer. These operations shall not be conducted without approval from the responsible authority in accordance with the applicable operational regulations.

The external loads are limited to Non Human External Cargo (NHEC) only.

Flight with an empty net or unballasted sling as an external load is prohibited unless approved operational limits and procedures provided by the operator allow for such an operation.

Two placards visible to the ground operator and located on the lower fairing near to the hook indicate:

- The maximum sling load,
- The cargo hook rigging.

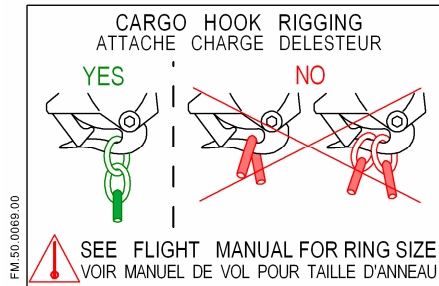


Figure 3: Cargo hook rigging placard

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

3.1 ENGINE FAILURE WITH EXTERNAL LOAD

- IN CRUISE FLIGHT
 1. Autorotation procedure: APPLY.
 2. External load RELEASE as soon as possible.
- IN HOVER
 1. Collective pitch REDUCE according to the height.
 2. External load RELEASE as soon as possible.
 3. Pedals CONTROL yaw.
 4. Cyclic FORWARD to gain forward speed according to the height.
 5. Collective pitch INCREASE as needed to cushion touch-down.

NOTE

In case of a failure during the hooking phase, the pilot shall move the aircraft away to the right. Ground personnel are to be forewarned that in the event of an engine failure they have to move away to the left of the helicopter.

3.2 ELECTRICAL LOAD JETTISONING FAILURE

Collective pitch mechanical release ACTUATE.

3.3 LOAD INDICATOR FAILURE

If the **LD ON** light comes on or goes out in an untimely manner:

- IN HOVER, during hooking or unhooking phase:
- Load RELEASE through electrical control.
- **LD ON** CHECK light status.



- IN CRUISE FLIGHT:
Perform a precautionary approach on the nearest helipad, then apply previous "IN HOVER" procedure.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Carrying heavy loads is a delicate operation due to the possible effects of a swinging load on the flight behavior of the helicopter. Consequently, pilots are advised to train with gradually increasing sling loads before undertaking heavy or bulky load carrying operations.
- The length of the sling cable must be determined in accordance with the type of mission. To carry a compact load, it is recommended to use the shortest possible cable.
- Operation with no or low load on a sling cable or in a net must be performed in such a way as to ensure that the trailing sling cable or net does not come close to the tail rotor.
- For permissible load attachment ring size refer to SECTION 9 of this Flight Manual.

WARNING

- 1- THE USE OF A LOAD ATTACHMENT RING WITH INCORRECT DIMENSIONS MAY LEAD TO LOSS OR JAMMING OF THE LOAD.
- 2- IN WET WEATHER, THE OPERATORS HANDLING THE HOOK AND LOADS SHOULD WEAR THICK RUBBER GLOVES. DISCHARGE STATIC ELECTRICITY BY PLACING AN ELECTRICAL CONDUCTOR CABLE OR TUBE BETWEEN THE GROUND AND THE CARGO RELEASE UNIT (HOOK).

4.1 CHECK OF THE INSTALLATION

- [SLING].....ON.

On the load indicator:

- **LD ON**.....CHECK **LD ON**.
- [ZERO].....PRESS, 0 is displayed,
- [TEST].....PRESS,

The test sequence is started; each display lasts for about 3 sec., followed by no display for approx. 3 sec.

- Display of: 
 - Display of the "calibration" value. This value should be the same as the one engraved on the transmitter fitted between the helicopter and the hook.
 - Display of ZERO offset (this value is the total offset which has been previously tared out by use of [ZERO]).
 - Display of two values separated by a dash: the first value, between 0 and 3, shows the filter selected and the second value shows the logic programmed in the computer.

"Cargo swing" check:

- Correct opening of the hook (both in electrical and mechanical control modes).....CHECK.
- Rotation of the retaining latchFREE.
- Operation of the retaining latch return spring.....CHECK.

4.2 TAKEOFF CHECK AND PROCEDURE WITH EXTERNAL LOAD

1. External load.....HOOK and SECURE.
2. CollectiveINCREASE very smoothly while maintaining the aircraft vertically above the load.
3. Cables tightenedDwell briefly before raising the load.
4. Lift the loadVertically.
5. Load indication.....CHECK **LD ON**.
6. Take-off path.....ADJUST to adopt an immediate forward climb attitude.
7. All parameters.....CHECK.

4.3 MANEUVERS

All control movements should be made very gently, with very gradual acceleration and deceleration, and only slightly banked turns.

4.4 APPROACH AND LANDING WITH EXTERNAL LOAD

Perform approach at minimum rate of descent.

- Establish zero translational ground speed sufficiently high to ensure that the load is not dragged along the ground.
- Then descend vertically until the load is on the ground.
- Load.....RELEASE.
- Effective load release.....CHECK **LD ON**.
- All parameters.....CHECK.

NOTE

If the load is not released, actuate the mechanical release handle.

5 PERFORMANCE DATA

When no external load is carried on the hook, the performance data specified in the basic flight manual and in the flight manual supplements remain applicable.

- Hover out of ground effect performance is shown in Figure 6 of SECTION 5.1 § 7 of the basic flight manual.
- Hover and climb performance may be affected when carrying bulky loads.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

TRANSPORT OF EXTERNAL LOADS

CARGO SWING 1400 kg (3086 lb) with "SIREN" fixed release unit (P/N S1609)

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.13.1.P1	1 to 1	15-45	A	
SUP.13.1.P5	1 to 2	16-24		
SUP.13.1	1 to 8	16-24		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

ISSUE 1: NR 0 to NR 3:

NORMAL REVISION 3 - JULY 2009	EASA approval No. 10027529 on October 7, 2009
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NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 16-24		EASA approval No. 10059860 on October 24, 2016
Title	Modification of the operating limitation and normal procedure for cargo hook	
Revised information	SUP.13.1.P5 pages 1 and 2, SUP.13.1 pages 4 and 6	
Deleted information	None	

1 GENERAL

The “CARGO SWING” external load installation (Figure 1) is composed of:

- A suspended pyramid frame (5) equipped with a fixed rotation hook “SIREN” release unit (4) (P/N S1609) allowing:
 - the electrical cargo hook opening,
 - the mechanical cargo hook opening.
- A load indicator, on the RH door pillar, with an indicator test pushbutton (1), a zero setting control (2) and a load-on indicator (3).

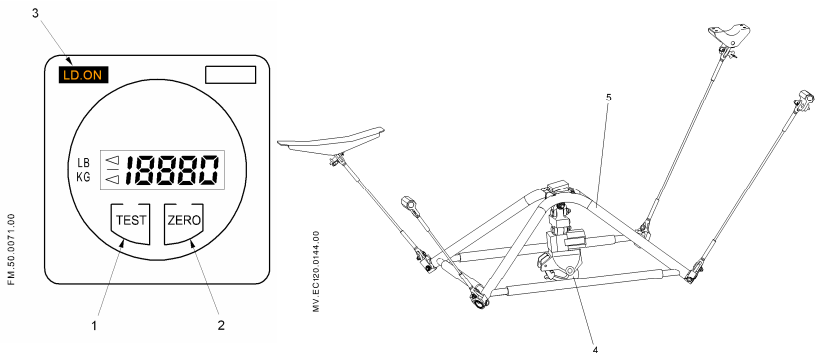


Figure 1: Cargo swing components

- A control system for the pilot (Figure 2), including:
 - a [SLING] pushbutton (4) located on the SCU (1), for powering-on the installation,
 - a release control (3) on the cyclic stick (electrical mode),
 - a release handle (2) located under the collective lever (mechanical mode).

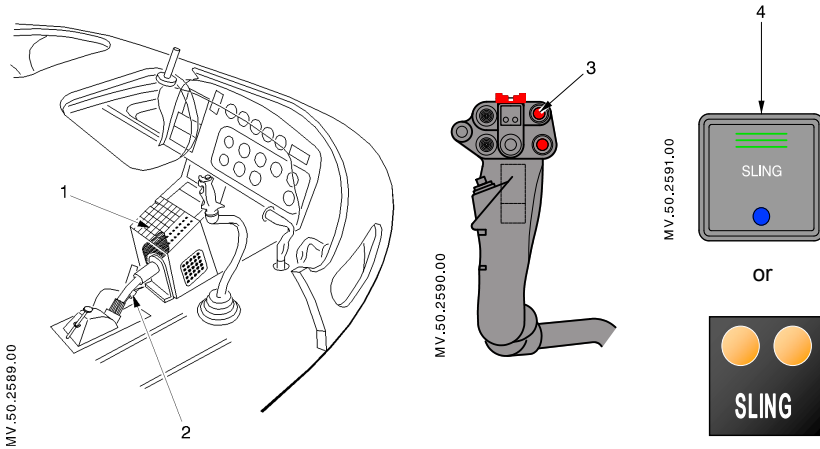


Figure 2: Cargo swing controls

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 WEIGHT LIMITATION

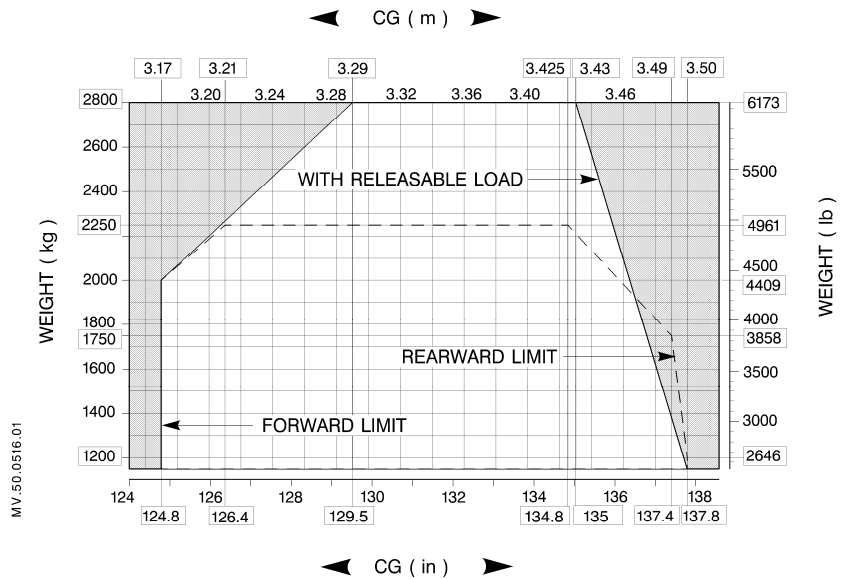
- Maximum authorized swing load weight: 1400 kg (3086 lb).
- Maximum all up weight with an external load: 2800 kg (6173 lb)
or maximum authorized all up weight allowing hovering flight out of ground effect
(the lowest of the two values).

CAUTION

The minimum and maximum weight without external load remains limited to the minimum and maximum weight specified in the limitations section of the basic flight manual.

2.2 LONGITUDINAL CG

With an external load, the longitudinal limits are defined according to the weight as per the graph below.



2.3 AIRSPEED LIMITATION

- Absolute maximum permissible indicated airspeed with external load 80 kt (148 km/h).

NOTE

The pilot is responsible for determining the limit speed according to the load and sling length. Particular care must be exercised when bulky loads are carried on the sling.

2.4 OPERATING LIMITATION

The cargo hook system is approved for lifting external load which is jettisonable and lifted free of land or water during rotorcraft operation.

Operation with an external load which remains in contact with land, water or any fixed structure is not demonstrated by the manufacturer. These operations shall not be conducted without approval from the responsible authority in accordance with the applicable operational regulations.

The external loads are limited to Non Human External Cargo (NHEC) only.

Flight with an empty net or unballasted sling as an external load is prohibited unless approved operational limits and procedures provided by the operator allow for such an operation.

Two placards visible to the ground operator and located on the lower fairing near to the hook indicate:

- The maximum sling load,
- The cargo hook rigging.

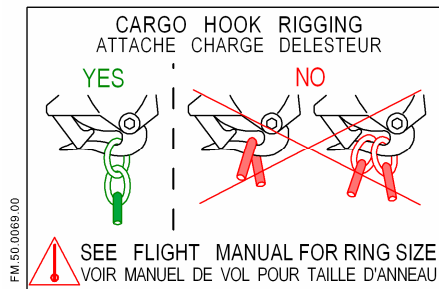


Figure 3: Cargo hook rigging placard

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

3.1 ENGINE FAILURE WITH EXTERNAL LOAD

- IN CRUISE FLIGHT
 - 1. Autorotation procedure: APPLY.
 - 2. External load RELEASE as soon as possible.
- IN HOVER
 - 1. Collective pitch REDUCE according to the height.
 - 2. External load RELEASE as soon as possible.
 - 3. Pedals CONTROL yaw.
 - 4. Cyclic FORWARD to gain forward speed according to the height.
 - 5. Collective pitch INCREASE as needed to cushion touch-down.

NOTE

In case of a failure during the hooking phase, the pilot shall move the aircraft away to the right. Ground personnel are to be forewarned that in the event of an engine failure they have to move away to the left of the helicopter.

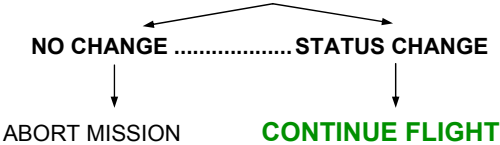
3.2 ELECTRICAL LOAD JETTISONING FAILURE

Collective pitch mechanical release ACTUATE.

3.3 LOAD INDICATOR FAILURE

If the **LD ON** light comes on or goes out in an untimely manner:

- IN HOVER, during hooking or unhooking phase:
 - Load..... RELEASE through electrical control.
 - **LD ON**..... CHECK light status.



- IN CRUISE FLIGHT:
Perform a precautionary approach on the nearest helipad, then apply previous "IN HOVER" procedure.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Carrying heavy loads is a delicate operation due to the possible effects of a swinging load on the flight behavior of the helicopter. Consequently, pilots are advised to train with gradually increasing sling loads before undertaking heavy or bulky load carrying operations.
- The length of the sling cable must be determined in accordance with the type of mission. To carry a compact load, it is recommended to use the shortest possible cable.
- Operation with no or low load on a sling cable or in a net must be performed in such a way as to ensure that the trailing sling cable or net does not come close to the tail rotor.
- For permissible load attachment ring size refer to section 9 of this Flight Manual.

WARNING

- 1- THE USE OF A LOAD ATTACHMENT RING WITH INCORRECT DIMENSIONS MAY LEAD TO LOSS OR JAMMING OF THE LOAD.
- 2- IN WET WEATHER, THE OPERATORS HANDLING THE HOOK AND LOADS SHOULD WEAR THICK RUBBER GLOVES. DISCHARGE STATIC ELECTRICITY BY PLACING AN ELECTRICAL CONDUCTOR CABLE OR TUBE BETWEEN THE GROUND AND THE CARGO RELEASE UNIT (HOOK).

4.1 CHECK OF THE INSTALLATION

- [SLING].....ON.

On the load indicator:

- **LD ON**.....CHECK **LD ON**.
- [ZERO].....PRESS, 0 is displayed,
- [TEST].....PRESS,

The test sequence is started; each display lasts for about 3 sec., followed by no display for approx. 3 sec.

- Display of: 
- Display of the "calibration" value. This value should be the same as the one engraved on the transmitter fitted between the helicopter and the hook.
- Display of ZERO offset (this value is the total offset which has been previously tared out by use of [ZERO]).
- Display of two values separated by a dash: the first value, between 0 and 3, shows the filter selected and the second value shows the logic programmed in the computer.

"Cargo swing" check:

- Correct opening of the hook (both in electrical and mechanical control modes).....CHECK.
- Rotation of the retaining latchFREE.
- Operation of the retaining latch return spring.....CHECK.

4.2 TAKEOFF CHECK AND PROCEDURE WITH EXTERNAL LOAD

1. External load.....HOOK and SECURE.
2. CollectiveINCREASE very smoothly while maintaining the aircraft vertically above the load.
3. Cables tightenedDwell briefly before raising the load.
4. Lift the loadVertically.
5. Load indication.....CHECK **LD ON**.
6. Take-off path.....ADJUST to adopt an immediate forward climb attitude.
7. All parameters.....CHECK.

4.3 MANEUVERS

All control movements should be made very gently, with very gradual acceleration and deceleration, and only slightly banked turns.

4.4 APPROACH AND LANDING WITH EXTERNAL LOAD

Perform approach at minimum rate of descent.

- Establish zero translational ground speed sufficiently high to ensure that the load is not dragged along the ground.
- Then descend vertically until the load is on the ground.
- Load.....RELEASE.
- Effective load release.....CHECK **LD ON**.
- All parameters.....CHECK.

NOTE

If the load is not released, actuate the mechanical release handle.

5 PERFORMANCE DATA

When no external load is carried on the hook, the performance data specified in the basic flight manual and in the flight manual supplements remain applicable.

- Hover out of ground effect performance is shown in Figure 6 of SECTION 5.1 § 7 of the basic flight manual.
- Hover and climb performance may be affected when carrying bulky loads.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

TRANSPORT OF EXTERNAL LOADS

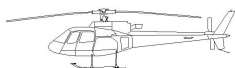
CARGO SWING 1400 kg (3086 lb) with "ON-BOARD" release unit (P/N 528-023-51)

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
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Aéroport international Marseille-Provence 13725 Marignane Cedex - France

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- Without indication..... Applicable to all aircraft
 - XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.13.2.P1	1 to 1	15-45	A	
SUP.13.2.P5	1 to 2	16-24		
SUP.13.2	1 to 8	16-24		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 16-24		EASA approval No. 10059860 on October 24, 2016
Title	Modification of the operating limitation and normal procedure for cargo hook	
Revised information	SUP13.2.P5 pages 1 and 2, SUP.13.2 pages 4 and 6.	
Deleted information	None	

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The paragraph 1, is modified as follows:

1 GENERAL

The “CARGO SWING” external load installation (Figure 1) is composed of:

- A suspended pyramid frame (6) held by two elastic straps (7) equipped with a bumper ring (P/N: 232-155-00) (5) and an “ON-BOARD” release unit (P/N 528-023-51) (4) allowing:
 - electrical cargo hook opening,
 - mechanical cargo hook opening.
- A load indicator, on the RH door pillar, with an indicator test pushbutton (1), a zero setting control (2) and a display window (3).

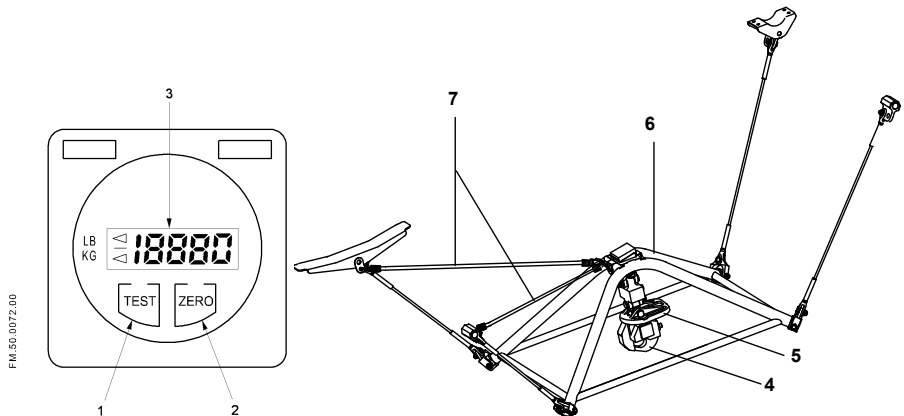


Figure 1: Cargo swing component

- A control system for the pilot (Figure 2), including:
 - a [SLING] pushbutton (4) located on the console (1), for powering-on the installation,
 - a release control (3) on the cyclic stick (electrical mode),
 - a release handle (2) located under the collective lever (mechanical mode).

The rest of the paragraph is unchanged.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.4614.

1 GENERAL

For detailed description of the load indicator and the “ON-BOARD” release unit, refer to their vendor manuals.

The “CARGO SWING” external load installation (Figure 1) is composed of:

- A suspended pyramid frame (3) held by two elastic straps (2) equipped with a bumper ring (P/N: 232-155-00) (4) and an “ON-BOARD” release unit (P/N 528-023-51) (5) allowing:
 - electrical cargo hook opening,
 - mechanical cargo hook opening.
- A load indicator (1), on the RH door pillar.

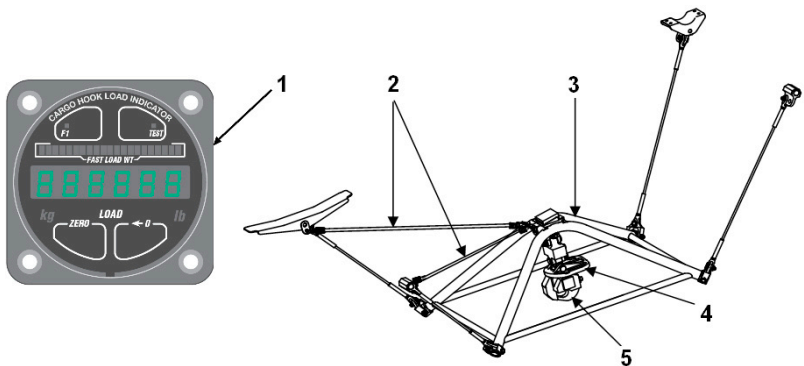


Figure 1: Cargo swing component

- A control system for the pilot (Figure 2), including:
 - a [SLING] pushbutton (4) located on the console (1), for powering-on the installation,
 - a release control (3) on the cyclic stick (electrical mode),
 - a release handle (2) located under the collective lever (mechanical mode).

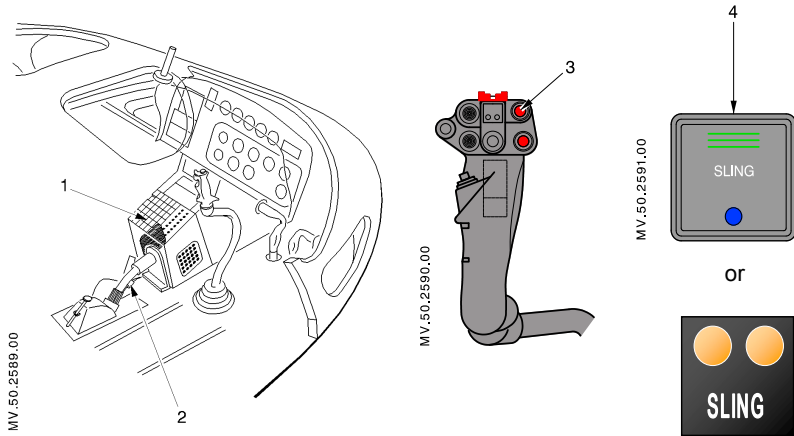


Figure 2: Cargo swing controls

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 WEIGHT LIMITATION

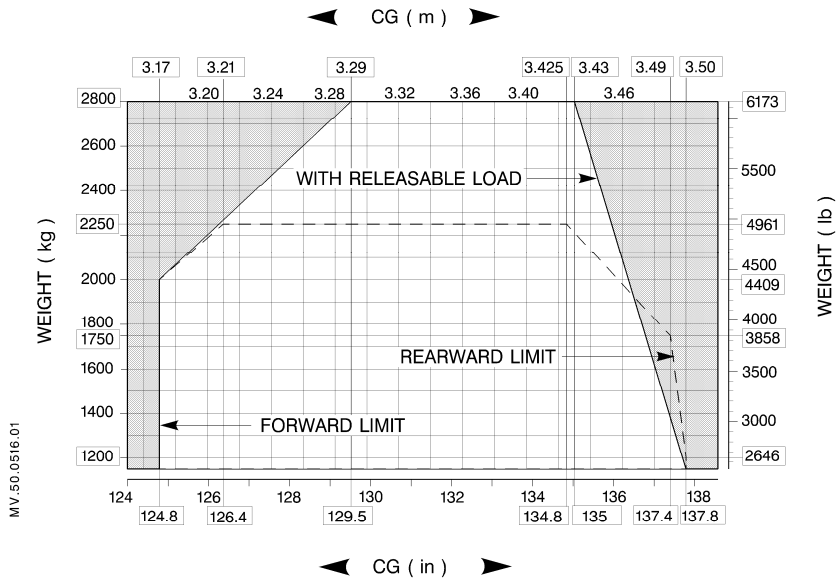
- Maximum authorized swing load weight: 1400 kg (3086 lb).
- Maximum all up weight with an external load: 2800 kg (6173 lb)
or maximum authorized all up weight allowing hovering flight out of ground effect (the lowest of the two values).

CAUTION

The minimum and maximum weight without external load remains limited to the minimum and maximum weight specified in the limitations section of the basic flight manual.

2.2 LONGITUDINAL CG

With an external load, the longitudinal limits are defined according to the weight as per the graph below.



2.3 AIRSPEED LIMITATION

- Absolute maximum permissible indicated airspeed with external load..... 80 kt (148 km/h).

NOTE

The pilot is responsible for determining the limit speed according to the load and sling length. Particular care must be exercised when bulky loads are carried on the sling.

2.4 OPERATING LIMITATION

The cargo hook system is approved for lifting external load which is jettisonable and lifted free of land or water during rotorcraft operation.

Operation with an external load which remains in contact with land, water or any fixed structure is not demonstrated by the manufacturer. These operations shall not be conducted without approval from the responsible authority in accordance with the applicable operational regulations.

The external loads are limited to Non Human External Cargo (NHEC) only.

Flight with an empty net or unballasted sling as an external load is prohibited unless approved operational limits and procedures provided by the operator allow for such an operation.

Two placards visible to the ground operator and located on the lower fairing near to the hook indicate:

- The maximum sling load,
- The cargo hook rigging.

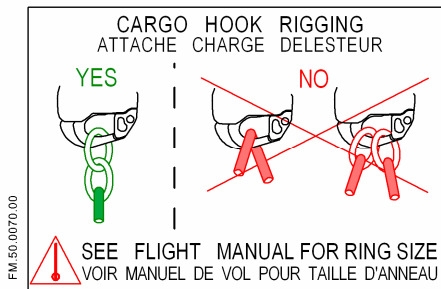


Figure 3: Cargo hook rigging placard

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

3.1 ENGINE FAILURE WITH EXTERNAL LOAD

- IN CRUISE FLIGHT
 - 1. Autorotation procedure APPLY.
 - 2. External load RELEASE as soon as possible.
- IN HOVER
 - 1. Collective pitch REDUCE according to the height.
 - 2. External load RELEASE as soon as possible.
 - 3. Pedals CONTROL yaw.
 - 4. Cyclic..... FORWARD to gain forward speed according to the height.
 - 5. Collective pitch INCREASE as needed to cushion touch-down.

NOTE

In case of a failure during the hooking phase, the pilot shall move the aircraft away to the right. Ground personnel are to be forewarned that in the event of an engine failure they have to move away to the left of the helicopter.

3.2 ELECTRICAL LOAD JETTISONING FAILURE

Collective pitch mechanical release ACTUATE.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are completed or modified by the following:

- Carrying heavy loads is a delicate operation due to the possible effects of a swinging load on the flight behavior of the helicopter. Consequently, pilots are advised to train with gradually increasing sling loads before undertaking heavy or bulky load carrying operations.
- The length of the sling cable must be determined in accordance with the type of mission. To carry a compact load, it is recommended to use the shortest possible cable.
- Operation with no or low load on a sling cable or in a net must be performed in such a way as to ensure that the trailing sling cable or net does not come close to the tail rotor.
- For permissible load attachment ring size refer to the section 9 of this Flight Manual.

WARNING

- 1- THE USE OF A LOAD ATTACHMENT RING WITH INCORRECT DIMENSIONS MAY LEAD TO LOSS OR JAMMING OF THE LOAD.**
- 2- IN WET WEATHER, THE OPERATORS HANDLING THE HOOK AND LOADS SHOULD WEAR THICK RUBBER GLOVES. DISCHARGE STATIC ELECTRICITY BY PLACING AN ELECTRICAL CONDUCTOR CABLE OR TUBE BETWEEN THE GROUND AND THE CARGO RELEASE UNIT (HOOK).**



The paragraph **4.1 - CHECK OF THE INSTALLATION**, is modified as follows:

4.1 CHECK OF THE INSTALLATION

- **[SLING]**.....ON.

On the load indicator:

- **[ZERO]**.....PRESS, 0 is displayed,
- **[TEST]**.....PRESS,

The test sequence is started; each display lasts for about 3 sec., followed by no display for approx. 3 sec.

- Display of 
- Display of the "calibration" value. This value should be the same as the one engraved on the transmitter fitted between the helicopter and the hook.
- Display of ZERO offset (this value is the total offset which has been previously tared out by use of **[ZERO]**).
- Display of two values separated by a dash: the first value, between 0 and 3, shows the filter selected and the second value shows the logic programmed in the computer.

Cargo swing check:

- Correct opening of the hook (both in electrical and mechanical control modes)CHECK.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.4614.

4.1 CHECK OF THE INSTALLATION

- [SLING]..... ON.

On the load indicator:

- [ZERO]..... PRESS, 0 is displayed,
- [TEST] PRESS,
The test sequence is started; each display lasts for about 1 sec.
 - Display of all digits and LEDs.
 - Display of the active software filter.
 - Display of the calibration value.

If a failure is detected, an error code will appear on the digital display.

NOTE

It is recommended to use the software filter "Hi-2" (very high filter) which presents the best indication stability.

Cargo swing check:

- Correct opening of the hook (both in electrical and mechanical control modes)..... CHECK.

4.2 TAKEOFF CHECK AND PROCEDURE WITH EXTERNAL LOAD

1. External load.....HOOK and SECURE.
2. CollectiveINCREASE very smoothly while maintaining the aircraft vertically above the load.
3. Cables tightened.....Dwell briefly before raising the load.
4. Lift the loadVertically.
5. Load indication.....CHECK.
6. Take-off pathADJUST to adopt an immediate forward climb attitude.
7. All parameters.....CHECK.

4.3 MANEUVERS

All control movements should be made very gently, with very gradual acceleration and deceleration, and only slightly banked turns.

4.4 APPROACH AND LANDING WITH EXTERNAL LOAD

Perform approach at minimum rate of descent.

- Establish zero translational ground speed sufficiently high to ensure that the load is not dragged along the ground.
- Then descend vertically until the load is on the ground.
- Load..... RELEASE.
- Effective load release CHECK.
- All parameters..... CHECK.

NOTE

If the load is not released, actuate the mechanical release handle.

5 PERFORMANCE DATA

When no external load is carried on the hook, the performance data specified in the basic flight manual and in the flight manual supplements remain applicable.

- Hover out of ground effect performance is shown in Figure 6 of SECTION 5.1 § 7 of the basic flight manual.
- Hover and climb performance may be affected when carrying bulky loads.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

SAND FILTER

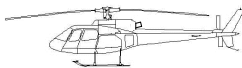
Reference: QB0550/QB0777

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
THE EQUIPMENT MENTIONED ABOVE IS INSTALLED ON THE AIRCRAFT.



Airbus Helicopters Direction Technique Support
Aéroport international Marseille-Provence 13725 Marignane Cedex - France

LIST OF APPROVED EFFECTIVE PAGES - EASA CERTIFICATION

- (1) AIRWORTHINESS EFFECTIVITY:
- Without indication..... Applicable to all aircraft
 - A..... Specific to EASA
- (2) VARIANT OF STANDARD DEFINITION EFFECTIVITY:
- Without indication..... Applicable to all aircraft
 - XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.14.P1	1 to 1	15-45	A	
SUP.14.P5	1 to 2	15-45		
SUP.14	1 to 9	15-45		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

ISSUE 1: NR 0 to NR 4:

NORMAL REVISION 4 - SEPTEMBER 2008	EASA Approval No. R.A.01582 on January 29, 2009
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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	

1 GENERAL

The sand filter installation is intended to protect the engine from sand ingestion, during hovering flight or when flying in sand-laden atmosphere.

This installation even when it does not use any P2 bleed air, is also designed to protect the air intake against any potential induction of snow in flight, in falling snow.

The installation consists essentially of:

- A filter fitted on the engine air intake, below the ice protection screen.
- A P2 air pressure supply system.
- An electrical control and monitoring system.

During engine operation, the ambient air flows through separator tubes which constitute the filter. The filtered air is forced towards the engine air intake. The sand is evacuated by scavenge tubes ventilated by P2 air.

The electrical circuit supplies an electric valve via **[SAND FILTER]** or **[SAND FILT]** push-button on the SCU. Opening and closing of the P2 air pressure circuit is controlled by the electric valve. A **P2** message on the FLI display of the VEMD indicates that the electric valve is fully opened.

The electrical circuit is protected by the SAND FILT fuse or breaker on the console side fuse or breaker panel.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- The heating and demisting systems must be switched off when sand filter is active.
- The flight under falling snow is permitted without any restrictions (refer to Section 2) when sand filter is fitted (active or not).

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

VEMD	CORRECTIVE ACTIONS
P2 Remains off. The " P2 " air valve fails to open.	Avoid flying the helicopter in sandy atmosphere.
P2 Remains on. The " P2 " air valve fails to close.	CONTINUE FLIGHT

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

WARNING

ICE OR SNOW ACCUMULATIONS THAT REMAIN IN OR AROUND THE ENGINE AIR INTAKE MAY BE INGESTED AND CAN CAUSE A SUDDEN IN-FLIGHT ENGINE FAILURE.

- Exterior checks:
 - ENG cowlng OPEN.
 - Engine air intake and sand filter Clean - no foreign objects or accumulations of ice or snow in or around the engine air intake and no stagnant water at the drain hole.
 - ENG cowlng CLOSED.
- Checks before starting the engine:
 - [SAND FILTER] or [SAND FILT] ON (**P2** on VEMD).
 - [SAND FILTER] or [SAND FILT] OFF (**P2** on VEMD).

- Flying in sandy atmosphere:
 - heating and demisting systems..... OFF.
 - [SAND FILTER] or [SAND FILT] ON (**P2** appears on VEMD).

NOTE

Operating the sand filter causes t4 temperature to rise by approximately 10°C.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

5.1 ENGINE POWER CHECK

- [SAND FILTER] or [SAND FILT] OFF.
 - Use the engine power check chart (Refer to Figure 1).

5.2 PERFORMANCE CALCULATION

- Refer to Figure 2 to Figure 5.

NOTE

VEMD engine power check and hovering performance are automatically modified if the sand filter is fitted.

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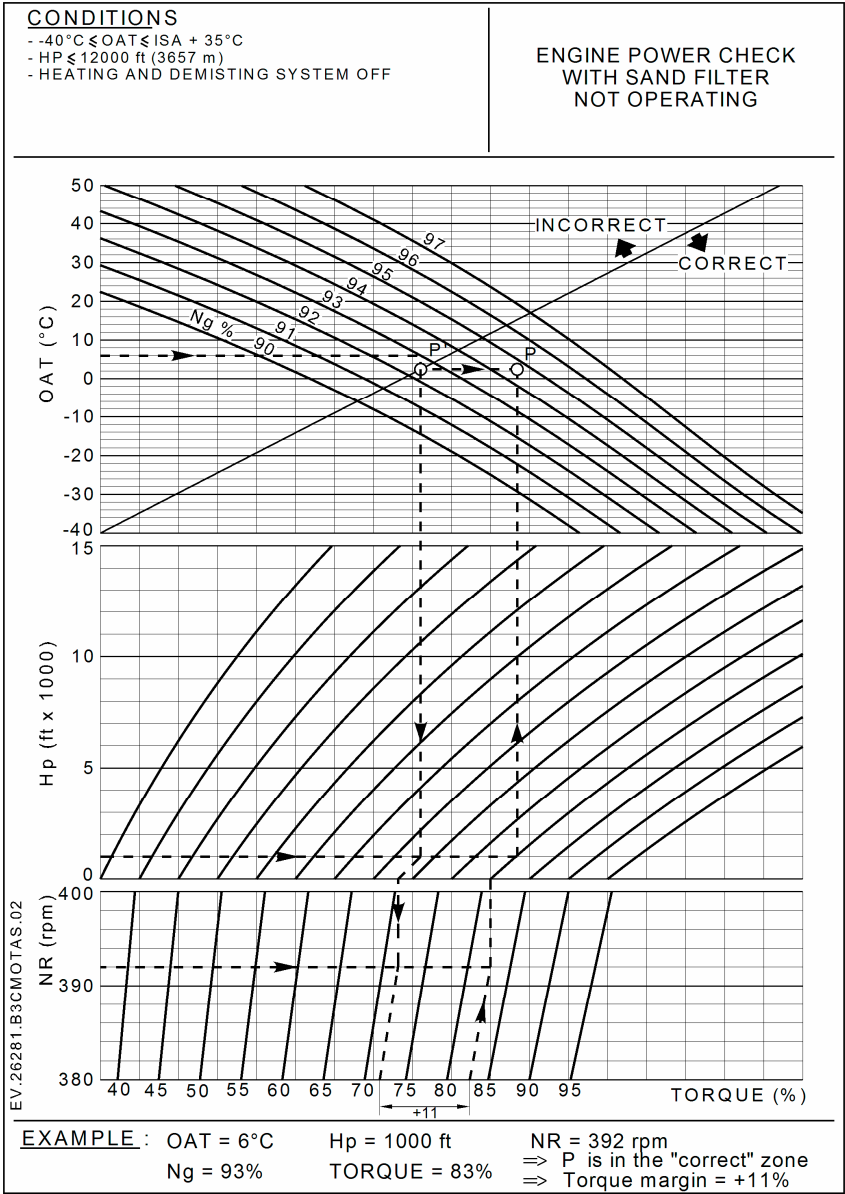


Figure 1

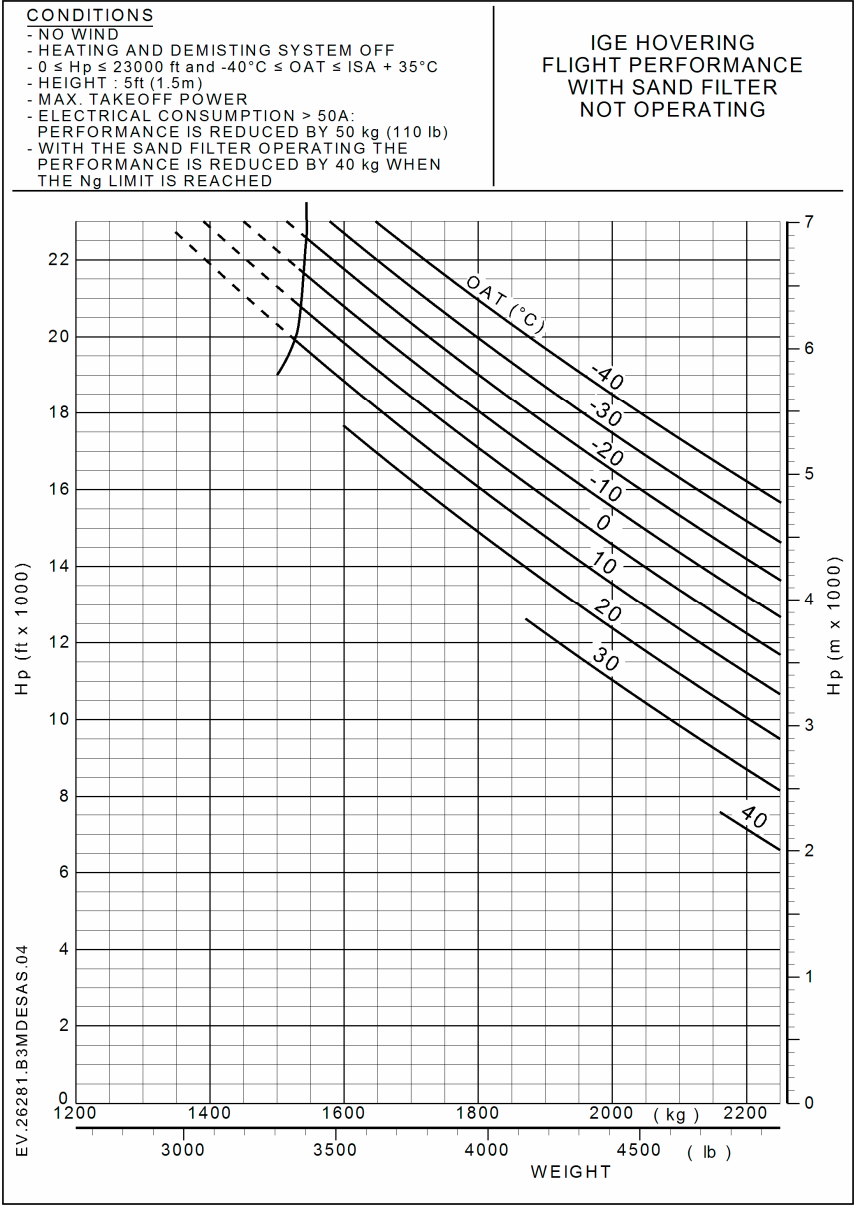


Figure 2

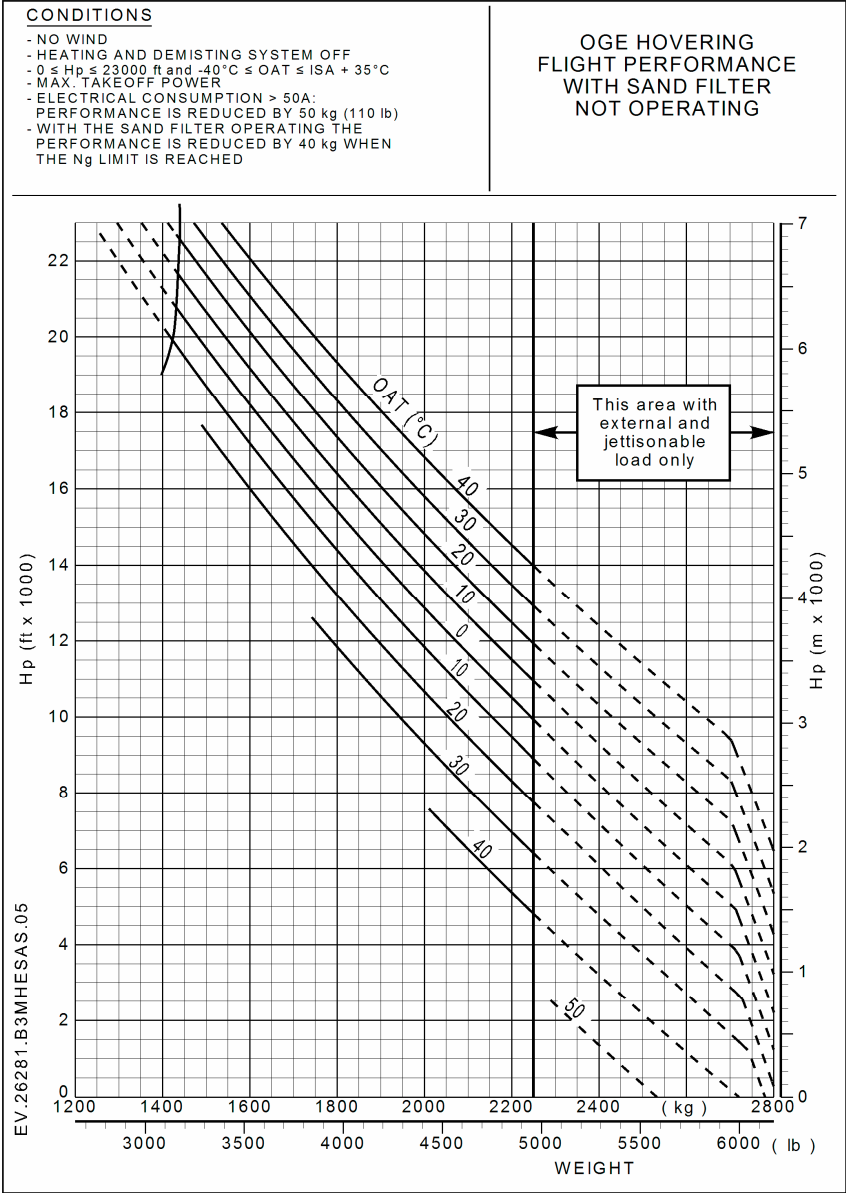
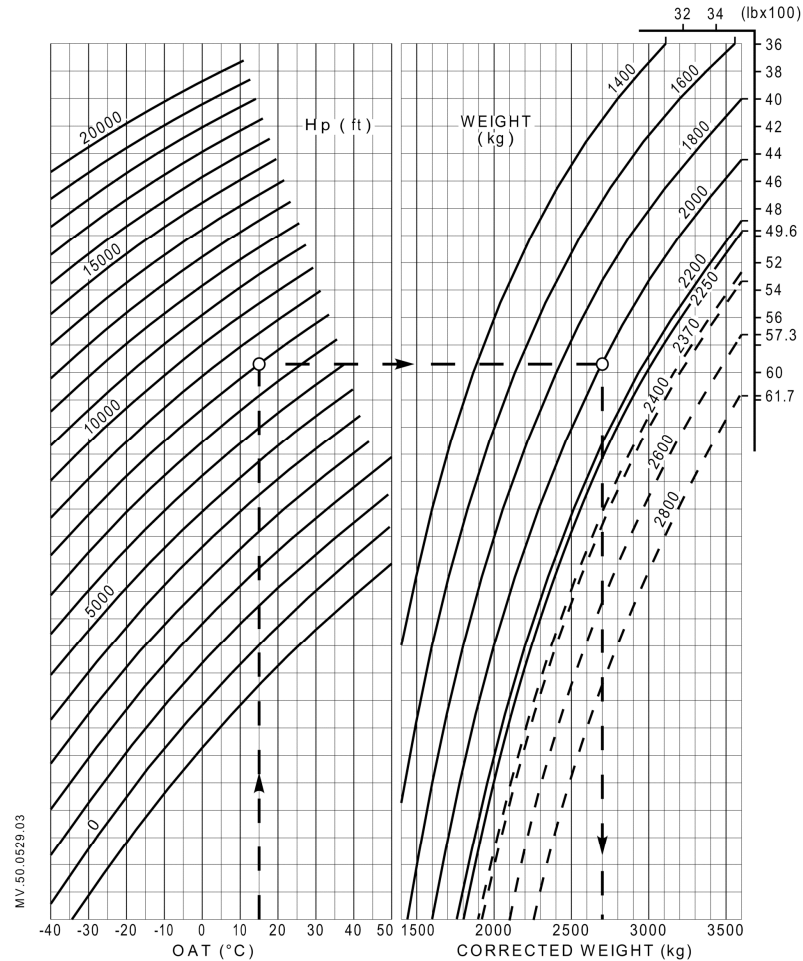


Figure 3

CONDITIONS

- WEIGHT LIMITATION WITH INTERNAL LOADS :
2250 kg (4961 lb)

DETERMINATION OF
CORRECTED WEIGHT



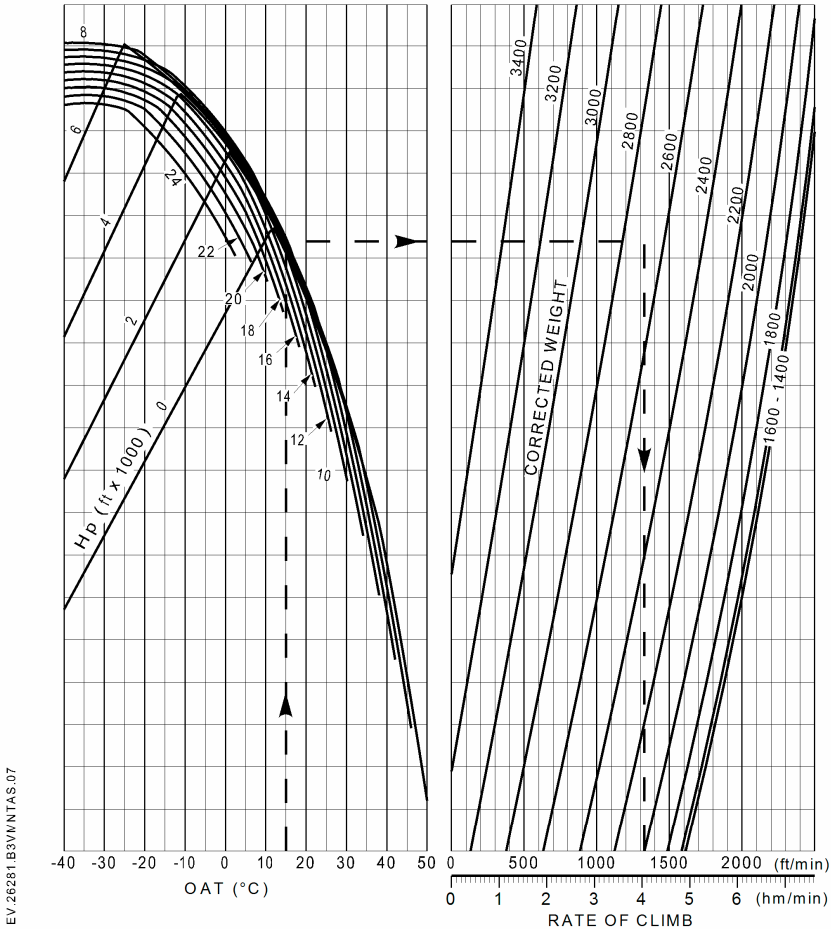
EXAMPLE : OAT = 15°C WEIGHT = 2000 kg (4409 lb)
 Hp = 8000 ft ⇒ CORRECTED WEIGHT = 2700 kg (5953 lb)

Figure 4

CONDITIONS

- HEATING AND DEMISTING SYSTEM OFF
- MAX CONTINUOUS POWER
- ELECTRICAL CONSUMPTION ≤ 50 A
- ELECTRICAL CONSUMPTION > 50 A:
PERFORMANCE IS REDUCED BY 70 ft/min
- WITH THE SAND FILTER OPERATING
THE PERFORMANCE IS REDUCED BY 80 ft/min.

RATE OF CLIMB
WITH SAND FILTER
NOT OPERATING
AT Vy



EXAMPLE : OAT = 15°C Hp = 8000 ft
CORRECTED WEIGHT = 2700 kg (5953 lb) ⇒ RATE OF CLIMB = 1330 ft/min

Figure 5



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

AUTOMATIC FLIGHT CONTROL SYSTEM 3 AXIS

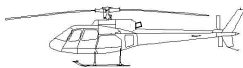
SFIM 85 T 31

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.16.P1	1 to 1	15-45	A	
SUP.16.P5	1 to 2	15-45		
SUP.16	1 to 11	15-45		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

ISSUE 1: NR 0 to NR 7:

NORMAL REVISION 7 - APRIL 2014	Approved under the authority of EASA DOA No. 21J056 on September 03, 2015
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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	

1 GENERAL

The SFIM 85 T 31 three-axis Automatic Flight Control System (AFCS) is intended to hold the attitudes (pitch, roll, yaw) and heading set by the pilot.

Additional modes:

- A/S : holds present airspeed.
- ALT : holds present pressure altitude.
- HDG : acquires and holds the heading selected on the HSI.
- T/C : automatic yaw/roll co-ordination during turns.

1.1 DESCRIPTION OF THE INSTALLATION

The AFCS or autopilot (AP) system includes the following items:

- An AFCS computer linked to following sensors:
 - Vertical gyro
 - Horizontal Situation Indicator (HSI)
 - Directional gyro and flux valve
 - Air data unit
 - Yaw pedal displacement detector
 - Lateral accelerometer (sideslip detector)

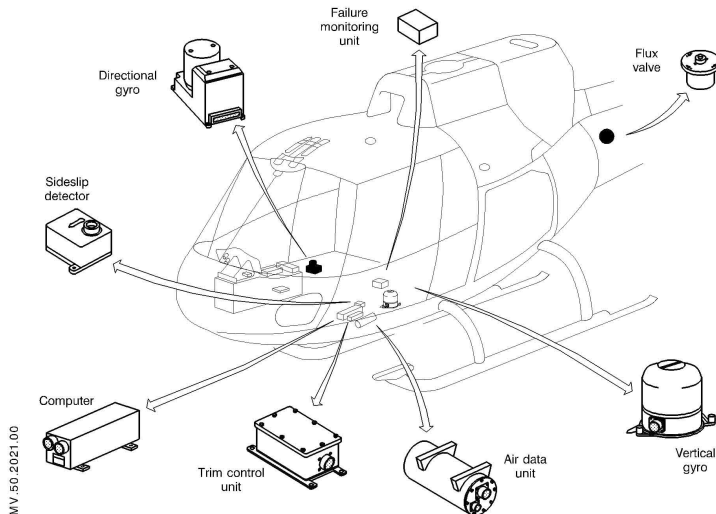


Figure 1: AFCS system components

- A yaw pedal friction adjustment system (in front of RH pilot's yaw pedals).
- Three series actuators (one per channel) associated with three galvanometers with "channel disengaged" indicator light:

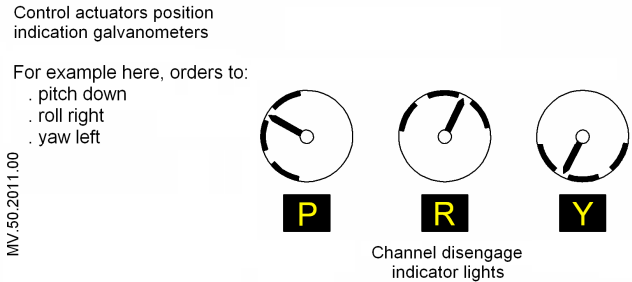


Figure 2: Serial actuators galvanometers (on instrument panel)

- Two parallel trim actuators (on pitch and roll channels) associated with the artificial load release system (trim release function):



Figure 3: Trims control pushbuttons (on SCU and cyclic grip)

- An AFCS warning and monitoring panel:

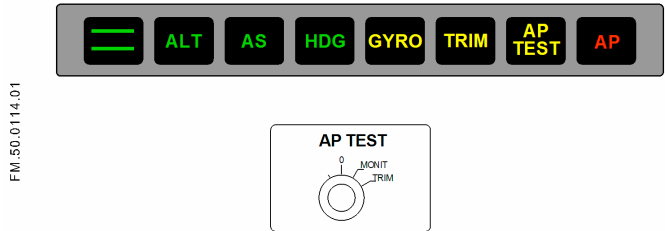


Figure 4: AFCS warning and monitoring panel (on instrument panel)

RC b

The paragraph 1.1 - **DESCRIPTION OF THE INSTALLATION**, is modified as follows:

- Two parallel trim actuators (on pitch and roll channels) associated to the artificial load release system (trim release function):

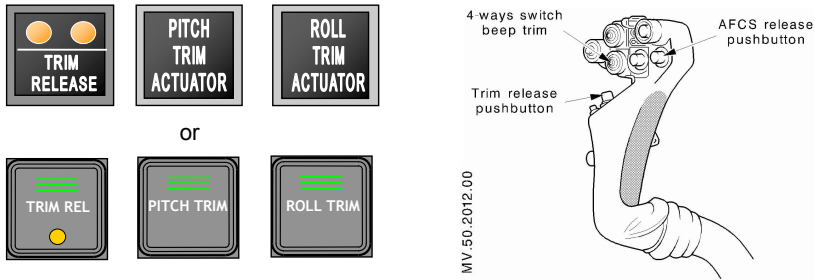


Figure 3: Trims control pushbuttons (on SCU and cyclic grip)

- An AFCS warning and monitoring panel:

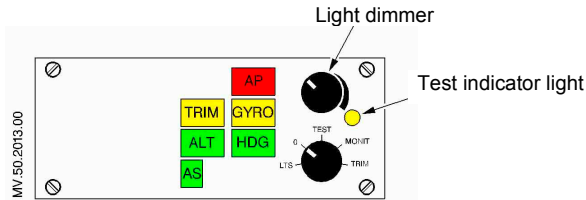


Figure 4: AFCS test and monitoring panel (on instrument panel)

The rest of the paragraph is unchanged.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3409.

- A failure monitoring unit:

This module monitors the operation of the pitch and roll channels at different levels by comparison of the output data from the vertical gyro and the instrument panel gyro horizon:

- Attitude sensors
- Control input generating system
- Control actuators

In case of discrepancy or abnormal operation, this unit warns the pilot and disengages the faulty channel. The monitoring unit is automatically switched on as soon as one pitch or roll channel is engaged.

- A control panel:

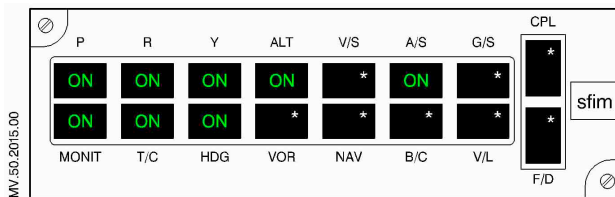


Figure 5: AFCS control panel

- [P] : pitch channel engage pushbutton
 [R] : roll channel engage pushbutton
 [Y] : yaw channel engage pushbutton
 [ALT] : altitude hold engage pushbutton
 [A/S] : airspeed hold engage pushbutton
 [HDG] : selected heading engage pushbutton
 [T/C] : coordinated turn engage pushbutton
 [MONIT] : failure monitoring unit and AP disengage pushbutton

(When channel is engaged, a green **ON** appears on the pushbutton)

- A static inverter (115 and 26 VAC, 400 Hz) for AC power supply.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 AFCS ENGAGEMENT

- When the aircraft is on ground, the AFCS must be disengaged except if checks are to be performed.
- Do not engage the AFCS before takeoff if trim test is not satisfactory.

2.2 HEIGHT LIMITATION

- Minimum height for AFCS hands-off operation: 400 ft (120 m)

2.3 WEIGHT LIMITATION

- Minimum all-up weight with AFCS in operation: 1300 kg (2866 lb).

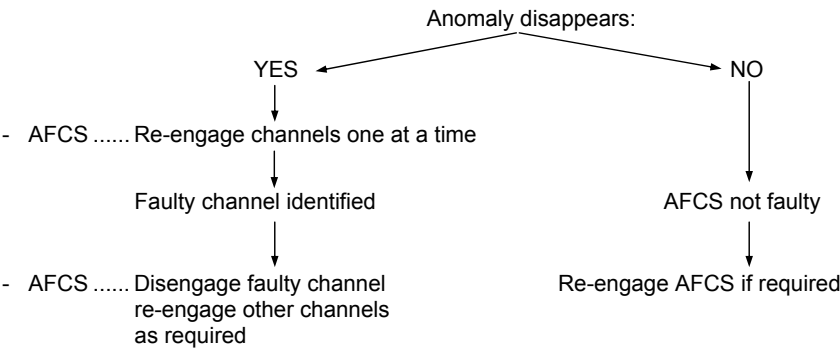
3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

3.1 UNCOMMANDED AFCS INPUTS

If jerks or sudden movements independent of air turbulence are felt during the flight with the AFCS engaged:

- AFCS Disengage



CONTINUE FLIGHT

RC b

3.2 HYDRAULIC SYSTEM FAILURE

- Apply hydraulic failure procedure (refer to SECTION 3).
- Disengage AFCS.

The paragraph 3.3 - VERTICAL GYRO, GYRO HORIZON OR MONITORING UNIT FAILURE, is superseded by the following:

WARNING	CORRECTIVE ACTIONS
AP Flashing for 10 sec. + GYRO or/and flag on gyro horizon + [P], [R], [MONIT] pushbutton lights go out + P R lights on. Monitoring unit failure	• Automatic disengagement of: . pitch channel. . roll channel. . monitoring unit. Cyclic and collective controlsHANDS ON. - HDG, ALT and A/S modes are inoperative. - Yaw channel remains operative. CONTINUE FLIGHT

The paragraph 3.4 - HARDOVER, is superseded by the following:

WARNING	CORRECTIVE ACTIONS
AP Flashing for 10 sec. + Automatic disengagement of faulty channel Hardover to the faulty channel	ControlsHANDS ON. NOTE Power reduction may be required to comply with the limitations. CONTINUE FLIGHT

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3409.

3.2 HYDRAULIC SYSTEM FAILURE

- Apply hydraulic failure procedure (refer to SECTION 3).
- Disengage AFCS.

3.3 VERTICAL GYRO, GYRO HORIZON OR MONITORING UNIT FAILURE

WARNING	CORRECTIVE ACTIONS
 Flashing for 10 sec. +  or/and flag on gyro horizon. + [P], [R], [MONIT] pushbutton lights go off   + lights on. Monitoring unit failure.	• Automatic disengagement of: . pitch channel, . roll channel, . monitoring unit. Cyclic and collective controls.....HANDS ON. - HDG, ALT and A/S modes are inoperative. - Yaw channel remains operative. CONTINUE FLIGHT

3.4 HARDOVER

WARNING	CORRECTIVE ACTIONS
 Flashing for 10 sec. + Automatic disengagement of faulty channel. Hardover on the faulty channel.	Controls HANDS ON. NOTE Power reduction may be required to comply with the limitations. CONTINUE FLIGHT

3.5 GYRO-COMPASS FAILURE

WARNING	CORRECTIVE ACTIONS
 Flashing for 10 sec. + HDG flag on HSI + [Y] and [T/C] pushbutton lights go off +  light comes on + [HDG] pushbutton +  lights go off (if HDG engaged).	- Automatic disengagement of: - yaw channel, - selected HDG function. Pedals FEET ON. - Yaw control by the pilot. - HDG mode and yaw channel are inoperative. - Pitch and roll channel remain operative. CONTINUE FLIGHT

3.6 TRIM FAILURE

WARNING	CORRECTIVE ACTIONS
 Comes on for 10 sec. + Automatic disengagement of defective trim. Before action of the safety device (auto disengagement), cyclic tends to move in direction of the failure	Controls HANDS ON. Trim release (on cyclic)..... Press momentarily while: Depending on trim failure:  -  - } PRESS, to deactivate the faulty trim channel. AFCS Re-engage. NOTE The AFCS continues to operate without the faulty channel being trimmed. Faulty channel galvanometer Re-center with cyclic trim release button. CONTINUE FLIGHT

RC b

The paragraph **3.5 - GYRO-COMPASS FAILURE**, is superseded by the following:

WARNING	CORRECTIVE ACTIONS
AP Flashing for 10 sec. + HDG flag on HSI + [Y] and [T/C] pushbutton lights go out +  light comes on. + [HDG] + HDG lights go out (if HDG engaged)	- Automatic disengagement of: - yaw channel, - selected HDG function, Pedals FEET ON. - Yaw control by the pilot, - HDG mode and yaw channel are inoperative, - Pitch and roll channel remain operative. CONTINUE FLIGHT

The paragraph **3.6 - TRIM FAILURE**, is superseded by the following:

WARNING	CORRECTIVE ACTIONS
TRIM comes on for 10 sec. + Automatic disengagement of defective trim. Before action of the safety device (auto disengagement), cyclic tends to move in direction of the failure.	Controls.....HANDS ON Trim release (on cyclic)...Depress momentarily the time to: Depending trim failure: -  or  -  or  Press to deactivate the faulty trim channel. AFCS Re-engage NOTE The AFCS continues to operate without the faulty channel being trimmed. Faulty channel galvanometer Re-center by cyclic trim release button. CONTINUE FLIGHT

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3409.

RC b

The paragraph 3.7 - **ARTIFICIAL LOAD SYSTEM SEIZURE**, is superseded by the following:

WARNING	CORRECTIVE ACTIONS
Cyclic stick seizure One trim parallel actuator is seized	Cyclic trim releasePRESS. Seizure disappears NO YES Press On corresponding cyclic channel.....Apply a force of 10 kg (22 lb) to break the trim mechanical shear device. CONTINUE FLIGHT

The paragraph 3.8 - **ELECTRICAL POWER SUPPLY FAILURE**, is superseded by the following:
3.8.1 - AC power supply failure

WARNING	CORRECTIVE ACTIONS
AP Flashing for 10 sec. + INVER	Automatic disengagement of AFCS Controls HANDS ON Check ON. YES NO Press IN AC power failure ControlsHANDS ON. CONTINUE FLIGHT

3.8.2 Complete electrical power supply failure

In the event of total power supply failure, AFCS disengages automatically and cannot be re-engaged.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3409.

3.7 ARTIFICIAL LOAD SYSTEM SEIZURE

WARNING	CORRECTIVE ACTIONS
Cyclic stick seizure. One parallel trim actuator is seized.	Cyclic trim release PRESS Seizure disappears. NO On corresponding cyclic channel: Apply a force of 10 kg (22 lb) to break the trim mechanical shear device. CONTINUE FLIGHT YES  PRESS.

3.8 ELECTRICAL POWER SUPPLY FAILURE

3.8.1 AC power supply failure

WARNING	CORRECTIVE ACTIONS
 Flashing for 10 sec. + 	Automatic disengagement of AFCS. Controls HANDS ON.  Check ON. YES AC power failure. Controls HANDS ON. CONTINUE FLIGHT NO  PRESS. 

3.8.2 Complete electrical power supply failure

In the event of total power supply failure, AFCS disengages automatically and cannot be re-engaged.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

4.1 RUN-UP CHECK

ADD:

1. **[PITCH TRIM]**, **[ROLL TRIM]** CHECK in ON position (button not pressed).
2. **[TRIM REL]** CHECK in OFF position (button not pressed).
3. Cyclic friction APPLY a low friction. To reduce the friction do not completely unscrew the adjustment nut.
4. Yaw pedal friction APPLY a moderate high friction.
5. **[INV]** ON **INV**.
6. HSI No HDG flag.
7. Gyro horizon(s) No flag.

NOTE

A too high cyclic friction and/or a too low pedal friction can lead to a malfunction of the AFCS.

4.1.1 AFCS test

1. Test selector switch set to "0" All AFCS lights OFF.
2. **[W/L TST]** PRESS, check:
 - Warning and monitoring panel lights ON.
 - AFCS control panel lights ON.
3. Pitch, roll, yaw channels Engage:
 - **[P]**, **[R]**, **[Y]** and **[MONIT]** check **ON**.
 - galvanometer lights OFF.
4. Cyclic beep trim Check operation in the 4 directions and that relevant galvanometer pointer direction changes.
5. AFCS Switch OFF via pilot's cyclic.

AP

flashes for 10 sec., then

AP
6. Repeat (3.) to (5.) for copilot's cyclic (if installed).

RC b

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

The paragraph **4.1 - RUN-UP CHECK**, is superseded by the following:

ADD:

1.    or    Pressed OUT.
2. Cyclic friction.....Un-tightened.
3. Yaw pedals friction.....Tightened.
4.  or ON, **INVER**.
5. HSI.....No HDG flag.
6. Gyro horizon(s).....No flag.

The paragraph **4.1.1 - AFCS TEST**, superseded by the following:

1. Test selector switch.....Set to "LTS", check:
(Fig. 4)
 - Control panel lights ON,
 - Monitoring panel lights ON,
(2.5 sec. Time delay)
 - Test light ON.
2. Test selector switch set to "0".....All AFCS lights OFF.
3. Pitch, roll, yaw channelsEngage:
 - **[P]**, **[R]**, **[Y]** and **[MONIT]**
Check: **ON**.
 - galvanometers lights OFF.
4. Cyclic beep trimCheck action in the 4 directions and that relevant galvanometer pointer direction changes.
5. AFCS.....Switch OFF via pilot's cyclic.
AP blinks for 10 sec, check **ON**.
6. Repeat (3.) to (5.) for copilot cyclic (if installed).

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3409.

RC b

The paragraph **4.1.2 - FAILURE MONITORING UNIT TEST**, is superseded by the following:

- Pitch channel test:
 1. Pitch channel Engage :
[P], [MONIT] check **ON**.
 2. Cyclic beep trim Actuate in pitch :
 - cyclic moves in relevant direction.
 3. Test selector switch Set to "MONIT":
(Fig. 4)
 - **AP** flashes for 10 sec.,
 - **GYRO** ,
 - cyclic stops moving,
 - galvanometers re-centers, light comes on.
 - [MONIT] **ON** light flashes then goes off.
 - [P] pushbutton check **ON** .
 4. Test selector switch Set to "0"
- Roll channel test:
Same procedure as pitch channel.

The paragraph **4.1.3 - TRIM TEST**, is superseded by the following:

CAUTION

Do not operate the AFCS if trim light does not come on or remains on during trim tests.

- Pitch channel test:
 1. Pitch channel Engage:
[P], [MONIT] check **ON** ,
 2. Test selector switch Set to "TRIM":
(Fig. 4)
 - **TRIM** lights up after 2 sec. delay then,
 - cyclic moves backward in alternately with **TRIM** light flashing.
 3. Test selector switch Set to "0"
 4. Pitch channel Disengage :
 - [P] pushbutton check **ON** .

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3409.

4.1.2 Failure monitoring unit test

- Pitch channel test:
 1. Pitch channel Engage:
[P], [MONIT], check **ON**.
 2. Cyclic beep trim Actuate in pitch:
 - cyclic moves in relevant direction.Set to "MONIT":
 - **AP TEST** lights on,
 - **GYRO** lights on and **AP** flashes for 10 sec., then **GYRO** and **AP**,
 - cyclic stops moving,
 - galvanometers re-centers, light comes on,
 - [MONIT] **ON** flashes then goes off,
 - [P] pushbutton check **ON**.
 4. Test selector switch Set to "0", check **AP TEST**.
- Roll channel test: Same procedure as pitch channel.

4.1.3 Trim test

CAUTION

Do not operate the AFCS if trim light does not come on or remains on during trim tests.

- Pitch channel test:
 1. Pitch channel Engage:
[P], [MONIT] check **ON**.
 2. Test selector switch Set to "TRIM":
(Fig. 4)
 - **AP TEST** lights on,
 - **TRIM** lights up after 2 sec. delaythen,
 - cyclic moves backward,alternately with **TRIM** light flashing,
 3. Test selector switch Set to "0", check **AP TEST**, and **TRIM**.
 4. Pitch channel Disengage:
 - [P] pushbutton, check **ON**.
- Roll channel test: Same procedure as pitch channel with left cyclic movements during test.

4.2 BEFORE TAKEOFF CHECK

ADD:

CAUTION

Do not keep the AFCS on with helicopter on the ground as trim may unwind, thus causing the cyclic stick to move up to the stops.

- Pitch, roll, yaw channels Engage.
 - [P], [R], [Y] and [MONIT] check **ON**.
 - galvanometer lights OFF.
- Test selector switch Check set to "0".

4.3 IN FLIGHT AFCS OPERATION

4.3.1 Basic modes

Pitch, roll, yaw and turn coordination: [P], [R], [Y], [T/C] engaged.

4.3.1.1 Pitch and roll channel flying modes

- "Hands Off" AFCS holds attitudes
- Cyclic stick trim released AFCS operates as a damper
- Through artificial loads..... Trim actuators are inhibited
series actuators counteract within
limits of their authority
- Through cyclic beep trim..... Attitude reference is changed

4.3.1.2 Yaw channel flying modes

- "Feet Off " AFCS holds present heading within
its authority.
- Operating yaw pedals Heading reference is altered.
- Heading change..... Pilot must bring the helicopter to the
desired heading, when rate of turn is
< 1.5 °/sec., present heading is
captured. Then the pilot may fly "feet
off" so that yaw channel can hold
new heading.

RC c

4.2 BEFORE TAKEOFF CHECK

ADD:

CAUTION

Do not keep the AFCS on with helicopter on the ground as trim may unwind, thus causing the cyclic stick to move up to the stops.

- Pitch, roll, yaw channels Engage.
 - [P], [R], [Y] and [MONIT]
 - check **ON**.
 - galvanometer lights OFF.
- Test selector switch Check set to "0".

4.3 IN FLIGHT AFCS OPERATION

4.3.1 Basic modes

Pitch, roll, yaw and turn coordination: [P], [R], [Y], [T/C] engaged.

4.3.1.1 Pitch and roll channel flying modes

- "Hands Off" AFCS holds attitudes
- Cyclic stick trim released AFCS operates as a damper
- Through artificial loads Trim actuators are inhibited
series actuators counteract within limits
of their authority
- Through cyclic beep trim Attitude reference is changed

The paragraph 4.3.1.2 - Yaw channel flying modes, is superseded by the following:

- "Feet Off " AFCS holds present heading within its authority.
- Operating yaw pedals Heading reference is altered.
- Heading change Pilot must bring the helicopter to the desired heading, when rate of turn is < 1.5 °/sec., present heading is captured. Then the pilot may fly "feet off" so that yaw channel can hold new heading.

NOTE 1

Collective pitch to yaw coupling is efficient when some yaw friction is applied to the pedals. It is therefore recommended to apply some friction.

NOTE 2

When the RH yaw pedal is near RH stop (e.g. with high LH crosswind), moving the collective pitch fully upward requires a greater effort than usual value due to the coupling spring rod.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.3363.

4.3.1.3 Coordinated turns, T/C mode

MODE	OPERATING ENVELOPE
T/C Yaw channel coordinates the turn.	- Airspeed greater than 50 KIAS (92 km/h – 57 MPH) - Angle of bank > 7°.

4.3.2 Additional modes

MODE	OPERATING ENVELOPE
HDG HSI selected heading capture and hold via roll channel	- Airspeed greater than 50 KIAS (92 km/h - 57 MPH).
ALT Altitude hold via pitch channel	- Airspeed greater than 60 KIAS (111 km/h – 69 MPH). - Engagement with R/C < 1000 ft/min.
A/S Airspeed hold via pitch channel	- Airspeed greater than 50 KIAS (92 km/h - 57 MPH).

4.4 AFTER LANDING

Disengage AFCS via the cyclic stick AFCS OFF pushbutton.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

EMERGENCY FLOATATION GEAR

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
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(1) AIRWORTHINESS EFFECTIVITY:

- Without indication..... Applicable to all aircraft
- A..... Specific to EASA

(2) VARIANT OF STANDARD DEFINITION EFFECTIVITY:

- Without indication..... Applicable to all aircraft
- XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
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Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 22-22		EASA approval No. 10080609 on November 15, 2022
Title	Suppression of "AERAZUR" in the title of SUP.17 Relocation of "FLOAT ARM" and "FLOAT FIRE" pushbuttons (MOD 07.20175)	
Revised information	SUP.17.P1 page 1; SUP.17.P5 pages 1 and 2; SUP.17 pages 1, 3 and 5	
Deleted information	None	
NORMAL REVISION 2 date code 23-50		EASA approval No. 10084221 on March 27, 2024
Title	Addition of performance information: demonstrated wave height (sea state)	
Revised information	SUP.17.P5 pages 1 and 2; SUP.17 pages 1, 2 and 5	
Deleted information	None	

1 GENERAL

The emergency floatation gear is approved for emergency use (not for ditching according to JAR 27) i.e. to aid in keeping rotorcraft sufficiently upright and in adequate trim to permit safe and orderly evacuation in case of emergency landing on water.

The emergency floatation gear consists of a landing gear assembly fitted with:

- Two floatation units mounted parallel along each skid of the aircraft (1).
- A system for inflating the floats from 2 cylinders (one for each float) (2). The cylinders are fitted with a pressure indicator.
- An electrical control system with a **[FLOAT ARM]** pushbutton (4) on the SCU or on the instrument panel.
- A **[FLOAT FIRING]** or **[FLOAT FIRE]** pushbutton (3) on the SCU or on the instrument panel.

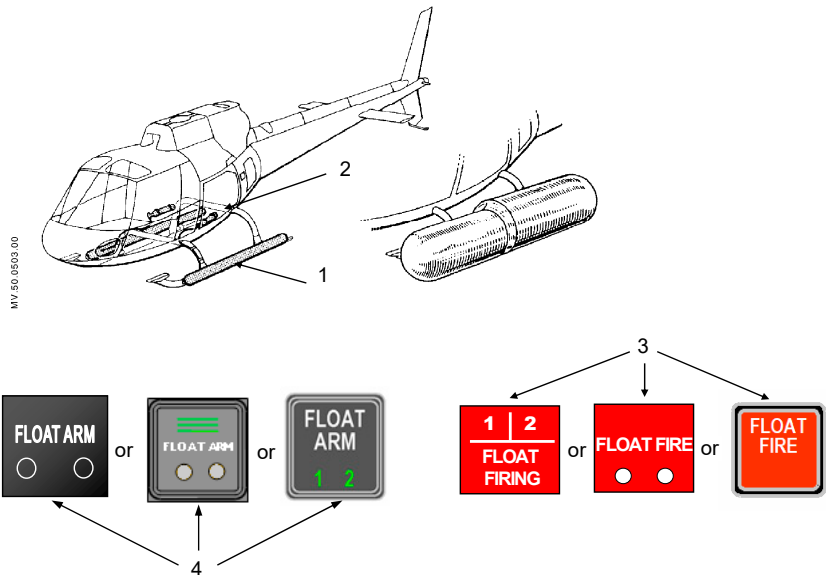


Figure 1: Emergency floatation gear

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- The emergency floatation system shall only be used for emergency landing on water.
- Takeoff after emergency landing on water is prohibited.
- Minimum weight with OAT < 0°C: 1480 kg (3263 lb) to comply with the minimum rotor rpm in case of engine failure.
- Emergency floatation gear stowed - system armed or floats inflated:
 - maximum indicated airspeed, power on: 135 kt (250 km/h)
 - maximum indicated airspeed, with Torque < 40 %: 100 kt (185 km/h)
- Maximum altitude for float inflation: 6600 ft (2000 m).
- When flying at less than 400 ft (122 m) above water, the emergency floatation gear must always be armed.
- The limit values for the pressure in the inflation cylinder, provided by the following tables, are applicable if the flight requires the emergency floatation gear to be armed.

MV 50.0657.00

TEMPERATURE °C	-45	-40	-30	-20	-10	0	10	20	30	40	50	60	70
FAHRENHEIT DEGREE	-49	-40	-22	-4	14	32	50	68	86	104	122	140	158
PRESSION MAX BARS	242	247	257	268	278	288	298	308	318	328	339	349	362
MAX PRESSURE PSI	3501	3582	3728	3887	4032	4177	4322	4467	4612	4757	4917	5062	5250
PRESSION MINI BARS	224	229	239	249	259	269	279	289	299	309	319	329	339
MINI PRESSURE PSI	3249	3321	3466	3611	3757	3902	4047	4192	4337	4482	4627	4772	4917

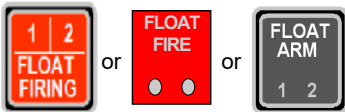
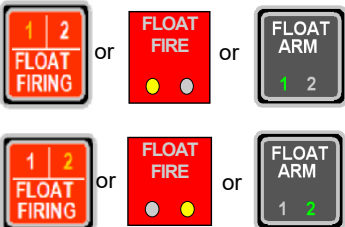
NOTE

The placard located adjacent to the cylinder provides the limit values.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- With **[FLOAT ARM]** set to arm:

	CORRECTIVE ACTIONS
 Floatation system failure	• Float inflation not possible. CONTINUE FLIGHT Flight path should remain within gliding distance from the coastline. NOTE The float inflation is not possible when [MASTER SW] or [EMER SW] is set to off position.
	• Float inflation remains possible for the both floatation units: CONTINUE FLIGHT

- INFLATION PROCEDURE

In the event of engine failure or other urgent requirement to alight on water, check the rotor speed then apply the following procedure:

[FLOAT ARM]..... PRESS.

[FLOAT FIRING] or

[FLOAT FIRE] or **[FLOAT ARM]** lights 1 and 2..... CHECK ON (at least one is ON).

[FLOAT FIRING] or

[FLOAT FIRE]..... PRESS (recommended firing IAS: below 80 kt - 148 km/h).

NOTE

A deceleration with a pitch down movement can occur when firing the floatation gear at a speed greater than 80 kt (148 km/h).

- **AUTORATION PROCEDURE OVER WATER WITH EMERGENCY FLOATATION GEAR**

1. Collective pitch**REDUCE** to maintain NR in normal operating range.
2. IAS.....**Vy**.
 - If relighting impossible or after tail rotor failure.
3. Engine starting selectorOFF position.
4. Maneuver to head the aircraft equally between the wind and wave direction on final approach.
- At height $\cong$ 70 ft (21 m)
5. CyclicFlare.
- At 20-25 ft (6/8 m) at constant attitude
6. Collective pitch.....GRADUALLY INCREASE
to reduce the rate of descent and forward speed.
7. CyclicFORWARD slightly to adopt attitude of 10° nose-up and a forward speed less than 10 kt (19 km/h) on touch-down.
8. Pedals.....ADJUST
to cancel any side-slip tendency.
9. Collective pitch.....INCREASE
to cushion touch down with minimum speed.
- After touch-down
10. Collective pitch.....Gradually decrease to fully down.
11. Rotor brakeAPPLY.
12. Evacuate aircraft once the rotor has stopped.

CAUTION

When the helicopter is afloat, the forward doors must be opened by actuating the door jettison handles.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Exterior check:
 - Floatation units..... LOCK in the lowered position.
 - Protective covers condition CHECK.
 - Pressure in the two inflation cylinders..... CHECK.

NOTE

The placard located adjacent to the cylinder provides the limit values.

- Interior check:
 - Arming of the emergency floatation gear:
 - [FLOAT ARM] ON.
 - [FLOAT FIRING] or
 - [FLOAT FIRE] or [FLOAT ARM] lights 1 and 2..... CHECK ON.
 - Disarming of the emergency floatation gear:
 - [FLOAT ARM] RESET in OFF position.
 - [FLOAT FIRING] or
 - [FLOAT FIRE] or [FLOAT ARM] lights 1 and 2..... CHECK OFF.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Stability after emergency landing on water was demonstrated up to a significant wave height of 2.5 m (up to sea state 4 as described on the World Meteorological Organization standards).
- The rate of climb must be reduced by 50 ft/min (0.25 m/s) (SECTION 5.1 Regulatory performance data) at Vy.
- The cruising speed is reduced by approximately 2 kt (3.7 km/h) (SECTION 5.2 Additional performance data).



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

HOIST INSTALLATION

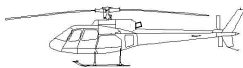
"BREEZE" or "AIR EQUIPMENT" electric hoist 136 kg (300 lb)

IMPORTANT NOTE

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Revised information	All	
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NORMAL REVISION 1 date code 19-47		Approved on May 28, 2020, under the authority of EASA DOA No. 21J700
Title	Improvement of "Warning" about static electricity (electrostatic device)	
Revised information	SUP.18.P5 pages 1 and 2; SUP.18 page 6	
Deleted information	None	

1 GENERAL

The hoist installation is designed to raise or lower loads while the aircraft is hovering.

The helicopter can be fitted with either of the following two electric hoists:

- "BREEZE" BL 16600
- "AIR EQUIPMENT" 76370

1.1 DESCRIPTION OF THE INSTALLATION

The hoist installation comprises:

- A pivoting jib (2) equipped with a locking device (3) (ferrying and hoisting position), mounted on the left side of the helicopter.
- A hoist (1) driven by a constant speed electric motor and fitted with:
 - A 33.5 m (110 ft) cable for the BREEZE hoist
 - A 40 m (130 ft) cable for the AIR EQUIPEMENT hoist
- A safety belt (5) for the hoist operator.
- A snap hook mounted on a pulley-block tackle (6) (if available),
- An electrical power supply and control circuit,
- two cable protectors (8) secured to the LH landing gear skid,
- A hand operated cable cutter stowed on the rear wall which allows the hoist operator to sever the cable, if necessary,
- An "Up travel limit" detector which cuts the electrical power supply to the motor when the hook is fully up,
- A "Down travel limit" which cuts the electrical power supply to the motor when 4 m of cable are left on the drum,
- A mechanical safety pin stopping the hoist drum when 3 m of cable are left in case of electrical "Down travel limit" failure.

The electrical system is protected by:

- One 100 A fuse located in the Electrical Master Box,
- One 60 A circuit-breaker (4) located on the aft wall near the hoist operator's grip support. This fuse and circuit breaker power the hoist,
- Two 2.5 A fuses or breakers protecting the "emergency release" circuit,
- One 2.5 A fuse or breaker protecting the "up-down" circuit.

NOTE

The hoist is fitted with a pyrotechnic cable cutter.

1.2 CONTROLS AND INDICATORS AVAILABLE TO THE CREW

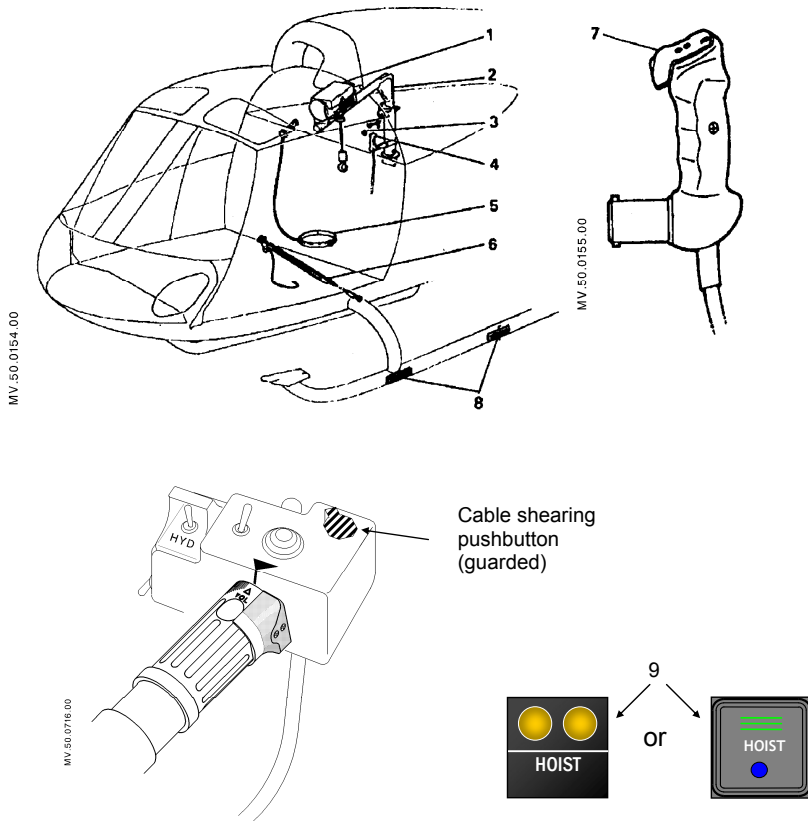


Figure 1: Hoist controls

- Controls available to the pilot:
 - A **[HOIST]** pushbutton (9) located on the console is used for switching on the installation.
 - A cable shear guarded pushbutton located on the collective pitch lever which is used to sever the cable in case of an emergency.
- Controls and indicators available to the hoist operator:
 - A rocker switch (7) located on the hoist operator's control grip use to raise, lower or stop the cable.
 - One green light on the hoist for "Hoist moving UP".
 - One red light on the hoist for "UP limit" microswitch test.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 MINIMUM CREW

One pilot and one hoist operator.

2.2 WEIGHT LIMITATION

Maximum permissible load on hoist cable: 136 kg (300 lb)

2.3 OPERATING LIMITATION

- AIR EQUIPMENT hoist:
 - To avoid overheating of hoist motor, never exceed 6 consecutive hoisting operations plus one descent with maximum load, and maximum cable reel-out or equivalent.
- BREEZE hoist:
 - After each operation (lowering or raising), wait 30 sec.

NOTE

After 3 complete cycles (one descent at max. load, the two following descents with no load, combined with three raising operations at max. load) it is recommended to stop the hoist for 40 min.

2.4 HAND CABLE CUTTER

A hand operated cable cutter must be readily available to the hoist operator.

2.5 LANDING

Landing with a load suspended on the hoist cable is prohibited.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

3.1 HOIST FAILURE

- According to the load carried and the circumstances, the crew can either:
- Manually pull the cable inside the aircraft if possible or,
 - Cut the cable or,
 - Return to base while slightly inclining the hoist cable, with a VNE of 60 kt (111 km/h).

3.2 ENGINE FAILURE

- IN HOVER:
 1. Collective pitch. REDUCE according to the height.
 2. Load RELEASE as soon as possible
(Emergency cable-shearing pushbutton on collective grip).
 3. Pedals..... Control yaw.
 4. Cyclic..... Forward to gain forward speed
according to the height.
 5. Collective pitch. INCREASE as necessary to cushion
touch-down.

NOTE

In case of an engine failure during a hoisting phase, the pilot should move away to the right. Ground crew if any must be forewarned that in the event of engine failure they must escape to the left of the helicopter.

3.3 ICS FAILURE

- According to the circumstances:
- Complete present hoisting operation,
 - Abort hoisting mission.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

4.1 EXTERIOR CHECK

ADD:

- Cable protectors on landing gear Check on and secured.
- Hoist jib and jib locking device Check.
- Hoist and cable general condition Check.
- If aircraft not fitted with LH sliding door:
 - LH cabin door and sub-door Removed.

4.2 INTERIOR CHECK

ADD:

- ICS system Check.
- Hoist operator control grip Installed.
- Hoist operator safety belt Installed and secured.
- Hoist operator's glove On board.
- Manual cable cutter In place.
- Check that the anticipated load on the hoist will permit the lateral CG to stay within limits.

4.3 RUN-UP CHECK

ADD:

1. [HOIST] ON.
2. Correct UP and DOWN operation CHECK.
3. [HOIST] OFF.

4.4 HOISTING OPERATION

4.4.1 Before hoisting

- Hoist operator
 1. Glove On the hand which guides the cable.
 2. Safety belt FASTEN and SECURED.

- Pilot
 1. Pilot/Hoist operator ICS CHECK operative.
 2. Door opening ORDER when IAS $\leq$ 60 kt (111 km/h).
 3. Heating and demisting OFF.
 4. Power margin available CHECK.
 5. All parameters CHECK.

4.4.2 Hoisting

WARNING

TO DISCHARGE POSSIBLE STATIC ELECTRICITY FROM THE AIRCRAFT, THE HOOK SHALL HAVE GROUND CONTACT BEFORE THE CABLE IS GRASPED.
IF AN ELECTROSTATIC DISCHARGE DEVICE SUCH AS A CABLE IS USED, IT HAS TO BE REMOVED AFTER THE HOISTING PHASE.
IF THE HOOK COMES INTO CONTACT FORCEFULLY WITH THE UPPER MECHANICAL STOP (FAILURE OF THE HOISTING AUTOMATIC STOP SYSTEM), ABORT THE HOISTING MISSION.

CAUTION

Take all precautions to prevent the load being hoisted from hitting any part of the helicopter.

- When helicopter stabilized above the hoisting site:
 1. Hoist jib SET in hoisting position.
 2. [HOIST] ON.

Hoist operator controls the hoist. To bring the load into the cabin, unlock the jib and pivot it inwards. The snap-hook (if available) can be used to hold the load while the load is unhooked.
The hoist may be operated by the pilot, using the rocker switch (if installed) on the cyclic grip.

4.4.3 After hoisting operation

1. Hoist jib SET in ferrying position.
2. [HOIST] OFF.
3. Door closing As required (check VNE with door open SECTION 2).

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable when the cable is fully reeled in, the hoist jib folded and no external load carried.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

HOIST INSTALLATION

"BREEZE" electric hoist 204 kg (450 lb)

IMPORTANT NOTE

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SUP.19.P1	1 to 1	15-45	A	
SUP.19.P5	1 to 2	19-47		
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Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 19-47		Approved on May 28, 2020, under the authority of EASA DOA No. 21J700
Title	Improvement of "Warning" about static electricity (electrostatic device)	
Revised information	SUP.19.P5 pages 1 and 2; SUP.19 page 6	
Deleted information	None	

1 GENERAL

The hoist installation is designed to raise or lower loads while the aircraft is hovering.

1.1 DESCRIPTION OF THE INSTALLATION

The hoist installation comprises:

- A pivoting jib (2) equipped with a locking device (3) (ferrying and hoisting position), mounted on the left side of the helicopter.
- A hoist (1) driven by a variable speed electric motor and fitted with a 50 meter cable (164 ft).
- A safety belt (6) for the hoist operator.
- A snap hook mounted on a pulley-block tackle (if available) (7),
- An electrical power supply and control circuit,
- Two cable protectors (8) secured to the LH landing gear skid,
- A hand operated cable cutter stowed on the rear wall allows the hoist operator to sever the cable, if necessary,
- An electronic control unit secured inside the LH side cargo hold providing:
 - Control of the hoist up/down speed,
 - Automatic deceleration before reaching the high and low travel limits,
 - Automatic stopping at travel limits and in the event of incorrect winding of the cable.

The system is protected by:

- One 130 A fuse located in the Electrical Master Box,
- Two 7.5 A circuit-breakers protecting the "emergency release" circuit located on the aft wall (4),
- One 100 A circuit-breaker on the hoist supply system for the resetable cut-off circuit located on the aft wall (4),
- One 2.5 A and one 6.3 A fuse or breaker for the control circuit located on the center console.

NOTE

The hoist is fitted with a pyrotechnic cable cutter.

1.2 CONTROLS AND INDICATORS AVAILABLE TO THE CREW

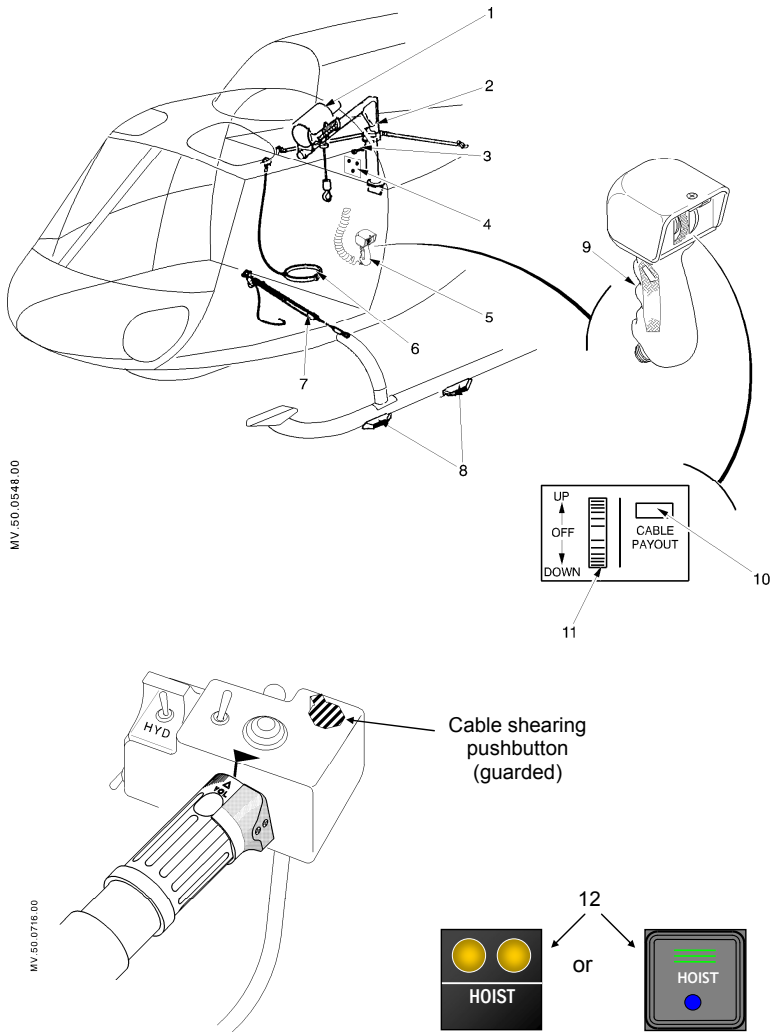


Figure 1: Hoist controls

- Controls available to the pilot:
 - A [HOIST] pushbutton (12) located on the console is used for switching on the installation.
 - A cable shear guarded pushbutton located on the collective pitch lever which is used to sever the cable in case of an emergency.
- Controls and indicators available to the hoist operator on the hoist operator's control grip (5):
 - An intercom push-to-talk switch (9),
 - A knurled knob (11) to control raising, lowering and stopping of the cable. The cable winding speed is proportional to the displacement of the knurled knob (11),
 - An indicator (10) shows the length of cable unwound in meters.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 MINIMUM CREW

One pilot and one hoist operator.

2.2 WEIGHT LIMITATION

Maximum permissible load on hoist cable: 204 kg (450 lb).

2.3 HAND CABLE CUTTER

A hand operated cable cutter must be within reach of the hoist operator.

2.4 LANDING

Landing with a load suspended on the hoist cable is prohibited.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

3.1 HOIST FAILURE

- According to the load carried and the circumstances, the crew can either:
- Manually pull the cable inside the aircraft if possible or,
 - Cut the cable or,
 - Return to the base with a slightly inclined hoist cable with a VNE of 60 kt (111 km/h).

3.2 ENGINE FAILURE

- IN HOVER:
 1. Collective pitch. REDUCE according to the height.
 2. Load RELEASE as soon as possible
(Emergency cable-shearing pushbutton on collective grip).
 3. Pedals..... Control yaw.
 4. Cyclic..... Forward to gain forward speed
according to the height.
 5. Collective pitch. INCREASE as necessary to cushion
touch-down.

NOTE

In case of an engine failure during a hoisting phase, the pilot should move away to the right. Ground crew if any must be forewarned that in the event of engine failure they must escape to the left of the helicopter.

3.3 ICS FAILURE

- According to the circumstances:
- Complete present hoisting operation,
 - Abort hoisting mission.

RR 2a

The paragraphs **4.1 Exterior check** and **4.3 Run-up check** are superseded by the following paragraphs:

4.1 EXTERIOR CHECK

ADD:

- Cable protectors on landing gear Check on and secured.
- Hoist jib and jib locking device Check.
- Hoist and cable general condition Check.
- Hoist hook Condition, free rotation, latch functioning and UP position (on STOP).
- If aircraft not fitted with LH sliding door:
 - LH cabin door and sub-door Removed.

4.3 RUN-UP CHECK

ADD:

1. [HOIST] ON.
2. Correct UP and DOWN operation CHECK.
3. Cable condition CHECK, where crimped to the bottom cup of the hook assembly.
4. Hoist hook UP position (on STOP).
5. [HOIST] OFF.

4 NORMAL PROCEDURES

The normal procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

4.1 EXTERIOR CHECK

ADD:

- Cable protectors on landing gear Check on and secured.
- Hoist jib and jib locking device Check.
- Hoist and cable general condition Check.
- If aircraft not fitted with LH sliding door:
 - LH cabin door and sub-door Removed.

4.2 INTERIOR CHECK

ADD:

- ICS system Check.
- Hoist operator control grip Installed.
- Hoist operator safety belt Installed and secured.
- Hoist operator's glove On board.
- Manual cable cutter In place.
- Check that the anticipated load on the hoist will permit the lateral CG to stay within limits.

4.3 RUN-UP CHECK

ADD:

1. [HOIST] ON.
2. Correct UP and DOWN operation CHECK.
3. [HOIST] OFF.

4.4 HOISTING OPERATION

4.4.1 Before hoisting

- Hoist operator
 1. Glove On the hand which guides the cable.
 2. Safety belt FASTEN and SECURED.

- Pilot
 1. Pilot/Hoist operator ICS CHECK operative.
 2. Door opening ORDER when IAS $\leq$ 60 kt (111 km/h).
 3. Heating and demisting OFF.
 4. Power margin available CHECK.
 5. All parameters CHECK.

4.4.2 Hoisting

WARNING

TO DISCHARGE POSSIBLE STATIC ELECTRICITY FROM THE AIRCRAFT, THE HOOK SHALL HAVE GROUND CONTACT BEFORE THE CABLE IS GRASPED.

IF AN ELECTROSTATIC DISCHARGE DEVICE SUCH AS A CABLE IS USED, IT HAS TO BE REMOVED AFTER THE HOISTING PHASE.

IF THE HOOK COMES INTO CONTACT FORCEFULLY WITH THE UPPER MECHANICAL STOP (FAILURE OF THE HOISTING AUTOMATIC STOP SYSTEM), ABORT THE HOISTING MISSION.

CAUTION

Take all precautions to prevent the load being hoisted from hitting any part of the helicopter.

- When helicopter stabilized above the hoisting site:
 1. Hoist jib SET in hoisting position.
 2. [HOIST] ON.

Hoist operator controls the hoist. To bring the load into the cabin, unlock the jib and pivot it inwards. The snap-hook (if available) can be used to hold the load while the load is unhooked.

The hoist may be operated by the pilot, using the rocker switch (if installed) on the cyclic grip.

4.4.3 After hoisting operation

1. Hoist jib SET in ferrying position.
2. [HOIST] RESET in OFF position.
3. Door closing As required (check VNE with door open SECTION 2).

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable when the cable is fully reeled in, the hoist jib folded and no external load carried.

RR 2a

The paragraph **4.4.3 After hoisting operation** is superseded by the following paragraph:

4.4.3 After hoisting operation

- 1. Hoist hook at the end of the hoisting operation and throughout cruise flight.....UP position (on STOP).
- 2. Hoist jibSET in ferrying position.
- 3. [HOIST].....OFF.
- 4. Door closing.....As required (check VNE with door open SECTION 2).



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

HOIST INSTALLATION

"BREEZE" electric hoist 204 kg (450 lb)

Grip with support bracket

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.19.1.P1	1 to 1	15-45	A	
SUP.19.1.P5	1 to 2	19-47		
SUP.19.1	1 to 6	19-47		

LOG OF APPROVED NORMAL REVISIONS**BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA**ISSUE 1: NR 0 to NR 5:

NORMAL REVISION 5 - MARCH 2015	Approved under the authority of EASA DOA No. 21J056 on June 03, 2015
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ISSUE 2:

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Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 19-47		Approved on May 28, 2020, under the authority of EASA DOA No. 21J700
Title	Improvement of "Warning" about static electricity (electrostatic device)	
Revised information	SUP.19.1.P5 pages 1 and 2; SUP.19.1 page 6	
Deleted information	None	

1 GENERAL

The hoist installation is designed to raise or lower loads while the aircraft is hovering.

1.1 DESCRIPTION OF THE INSTALLATION

The hoist installation comprises:

- A pivoting jib (2) equipped with a locking device (3) (ferrying and hoisting position), mounted on the left side of the helicopter.
- A hoist (1) driven by a variable speed electric motor and fitted with a 50 meter cable (164 ft).
- A safety belt (6) for the hoist operator.
- A snap hook mounted on a pulley-block tackle (7) (if available),
- An electrical power supply and control circuit,
- Two cable protectors (8) secured to the LH landing gear skid,
- A hand operated cable cutter stowed on the rear wall which allows the hoist operator to sever the cable, if necessary,
- An electronic control unit secured inside the LH side cargo hold providing:
 - Control of the hoist up/down speed,
 - Automatic deceleration before reaching the high and low travel limits,
 - Automatic stopping at travel limits and in the event of incorrect winding of the cable.

The system is protected by:

- One 130 A fuse located in the Electrical Master Box,
- Two 7.5 A circuit-breakers protecting the "emergency release" circuit located on the aft wall (4),
- One 100 A circuit-breaker on the hoist supply system for the resettable cut-off circuit located on the aft wall (4),
- One 2.5 A and one 6.3 A fuse or breaker for the control circuit located on the center console.

NOTE

The hoist is fitted with a pyrotechnic cable cutter.

1.2 CONTROLS AND INDICATORS AVAILABLE TO THE CREW

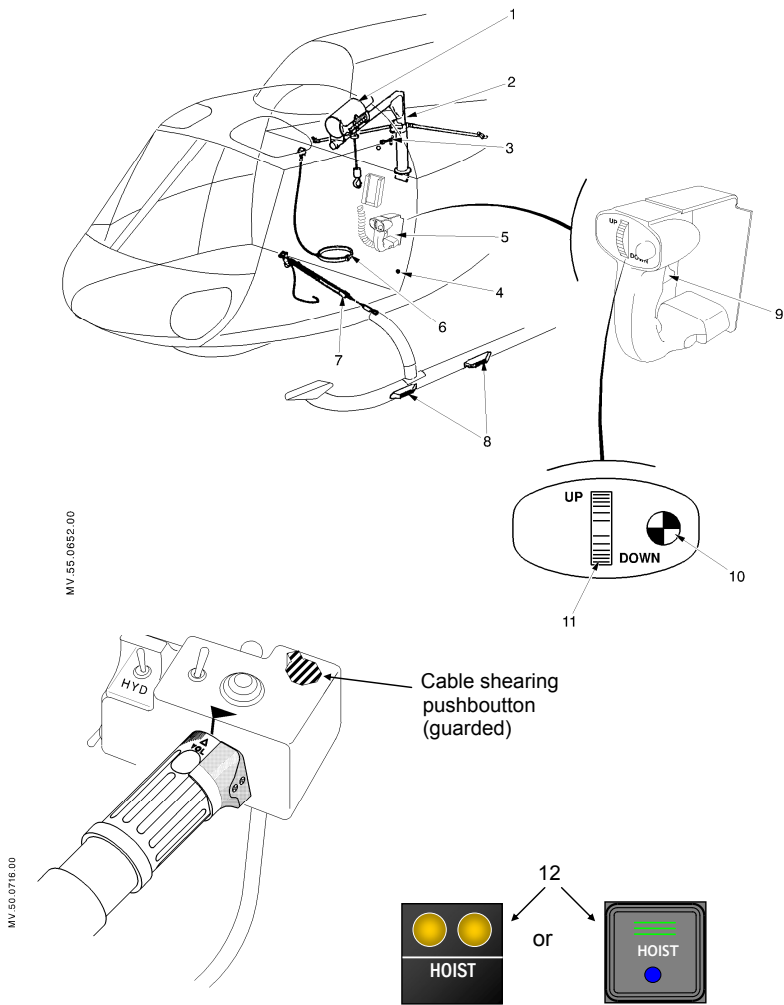


Figure 1: Hoist controls

- Controls available to the pilot:
 - A [HOIST] pushbutton (12) located on the console is used for switching on the installation.
 - A cable shear guarded pushbutton located on the collective pitch lever which is used to sever the cable in case of an emergency.
- Controls and indicators available to the hoist operator on the hoist operator's control grip (5):
 - An intercom push-to-talk switch (9),
 - A knurled knob (11) to control raising, lowering and stopping of the cable. The cable winding speed is proportional to the displacement of the knurled knob (11)
 - A white indicator light (10) shows that the hoist installation is energized.

2 LIMITATIONS

The limitations specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

2.1 MINIMUM CREW

One pilot and one hoist operator.

2.2 WEIGHT LIMITATION

Maximum permissible load on hoist cable: 204 kg (450 lb).

2.3 HAND CABLE CUTTER

A hand operated cable cutter must be within reach of the hoist operator.

2.4 LANDING

Landing with a load suspended on the hoist cable is prohibited.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

3.1 HOIST FAILURE

According to the load carried and the circumstances, the crew can either:

- Manually pull the cable inside the aircraft if possible or,
- Cut the cable or,
- Return to base with a slightly inclined hoist cable with a VNE of 60 kt (111 km/h)

3.2 ENGINE FAILURE

- IN HOVER:
 1. Collective pitch. REDUCE according to the height.
 2. Load RELEASE as soon as possible
(Emergency cable-shearing pushbutton on collective grip).
 3. Pedals..... Control yaw.
 4. Cyclic..... Forward to gain forward speed according to the height.
 5. Collective pitch. INCREASE as necessary to cushion touch-down.

NOTE

In case of an engine failure during a hoisting phase, the pilot should move away to the right. Ground crew if any must be forewarned that in the event of engine failure they must escape to the left of the helicopter.

3.3 ICS FAILURE

- According to the circumstances:
- Complete present hoisting operation,
 - Abort hoisting mission.

RR 2a

The paragraphs **4.1 Exterior check** and **4.3 Run-up check** are superseded by the following paragraphs:

4.1 EXTERIOR CHECK

ADD:

- Cable protectors on landing gear Check on and secured.
- Hoist jib and jib locking device Check.
- Hoist and cable general condition Check.
- Hoist hook Condition, free rotation, latch functioning and UP position (on STOP).
- .
- If aircraft not fitted with LH sliding door:
 - LH cabin door and sub-door Removed.

4.3 RUN-UP CHECK

ADD:

1. [HOIST] ON.
2. Correct UP and DOWN operation CHECK.
3. Cable condition CHECK, where crimped to the bottom cup of the hook assembly.
4. Hoist hook UP position (on STOP).
5. [HOIST] OFF.

4 NORMAL PROCEDURES

The normal procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

4.1 EXTERIOR CHECK

ADD:

- Cable protectors on landing gear Check on and secured.
- Hoist jib and jib locking device Check.
- Hoist and cable general condition Check.
- If aircraft not fitted with LH sliding door:
 - LH cabin door and sub-door Removed.

4.2 INTERIOR CHECK

ADD:

- ICS system Check.
- Hoist operator control grip Installed.
- Hoist operator safety belt Installed and secured.
- Hoist operator's glove On board.
- Manual cable cutter In place.
- Check that the anticipated load on the hoist will permit the lateral CG to stay within limits.

4.3 RUN-UP CHECK

ADD:

1. [HOIST] ON.
2. Correct UP and DOWN operation CHECK.
3. [HOIST] OFF.

4.4 HOISTING OPERATION

4.4.1 Before hoisting

- Hoist operator
 1. Glove On the hand which guides the cable.
 2. Safety belt FASTEN and SECURED.

- Pilot
 1. Pilot/Hoist operator ICS CHECK operative.
 2. Door opening ORDER when IAS $\leq$ 60 kt (111 km/h).
 3. Heating and demisting OFF.
 4. Power margin available CHECK.
 5. All parameters CHECK.

4.4.2 Hoisting

WARNING

TO DISCHARGE POSSIBLE STATIC ELECTRICITY FROM THE AIRCRAFT, THE HOOK SHALL HAVE GROUND CONTACT BEFORE THE CABLE IS GRASPED.

IF AN ELECTROSTATIC DISCHARGE DEVICE SUCH AS A CABLE IS USED, IT HAS TO BE REMOVED AFTER THE HOISTING PHASE.

IF THE HOOK COMES INTO CONTACT FORCEFULLY WITH THE UPPER MECHANICAL STOP (FAILURE OF THE HOISTING AUTOMATIC STOP SYSTEM), ABORT THE HOISTING MISSION.

CAUTION

Take all precautions to prevent the load being hoisted from hitting any part of the helicopter.

- When helicopter stabilized above the hoisting site:
 1. Hoist jib SET in hoisting position.
 2. [HOIST] ON.

Hoist operator controls the hoist. To bring the load into the cabin, unlock the jib and pivot it inwards. The snap-hook (if available) can be used to hold the load while the load is unhooked.

The hoist may be operated by the pilot, using the rocker switch (if installed) on the cyclic grip.

4.4.3 After hoisting operation

1. Hoist jib SET in ferrying position
2. [HOIST] OFF
3. Door closing As required (check VNE with door open SECTION 2)

5 PERFORMANCE DATA

The performance data specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable when the cable is fully reeled in, the hoist jib folded and no external load carried.

RR 2a

The paragraph **4.4.3 After hoisting operation** is superseded by the following paragraph:

4.4.3 After hoisting operation

- 1. Hoist hook at the end of the hoisting operation and throughout cruise flightUP position (on STOP).
- 2. Hoist jibSET in ferrying position.
- 3. [HOIST].....OFF.
- 4. Door closing.....As required (check VNE with door open SECTION 2).



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

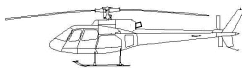
HYDRAULIC PUMP DRIVE ON MGB

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
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SUP.20.P5	1 to 2	15-45		
SUP.20	1 to 1	15-45		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

ISSUE 1: NR 0 to NR 1:

NORMAL REVISION 1 - DECEMBER 2005	EASA approval R.A. 01107 on April 07, 2006
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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	

1 GENERAL

An optional mechanical power output on the MGB enables the installation of a supplementary hydraulic pump for specific use.

This output is driven by the MGB spiral bevel gear wheel with a speed of 6000 rpm for a corresponding NR of 386 rpm.

When the hydraulic pump is running off load (no specific equipment is using this specific hydraulic power), the power taken from the MGB must be less than 3 kW so as not to affect the helicopter's performance.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Maximum power consumed by hydraulic pump on load..... 32 kW,
- Maximum power consumed by hydraulic pump off load..... 3 kW,
- The use of hydraulic power is prohibited:
 - in hover,
 - in climb,
 - in flight at IAS < 30 kt (56 km/h) and IAS > 80 kt (148 km/h).

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- In case of engine failure or loss of power, IMMEDIATELY shut-down the specific hydraulically powered system.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

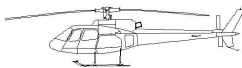
TWO-PASSENGERS FRONT SEAT

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic Flight Manual and/or the Supplements listed in section Supplement 0.

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SUP.21.P5	1 to 2	17-23		
SUP.21	1 to 1	17-23		

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NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 17-23		Approved EASA No. 10064298 on January 11, 2018
Title	- Deletion of the weight limitation for optional two-place seat, - Wording improvement.	
Revised information	All	
Deleted information	None	

1 GENERAL

Installed in place of the copilot's seat, the two-passenger front seat increases the total seating capacity from six to seven people, including the pilot.

2 LIMITATIONS

The limitations specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

- Maximum number of occupants
 (including flight crew) Seven,
- Dual controls (optional) shall be removed to install the two-passenger seat.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable.

4 NORMAL PROCEDURES

The normal procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable. Special attention shall be paid to CG computation.

NOTE

Flying with seven people on board then with one pilot only presents a very significant change in the CG position. Each configuration must be carefully checked.

5 PERFORMANCE DATA

The performance data specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

LONG AND SHORT FOOTSTEPS

Long footsteps P/N D.350-591-111-A

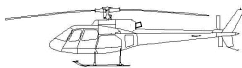
Short footsteps P/N D.350-591-113

IMPORTANT NOTE

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SUP.22.P5	1 to 2	15-45		
SUP.22	1 to 1	15-45		

LOG OF APPROVED NORMAL REVISIONS

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ISSUE 2:

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Title	New issue	
Revised information	All	
Deleted information	None	

1 GENERAL

The long footsteps (P/N D. 350-591-111-a) on the high landing gear facilitate cabin access and inspection of the transmission and engine decks.

The short footsteps (P/N D. 350-591-113) on the high landing gear facilitate cabin access.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Rate of climb: reduce by 2.5 %.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

DUAL HYDRAULIC SYSTEM

IMPORTANT NOTE

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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
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SUP.23.P5	1 to 2	23-40	A	
SUP.23	1 to 8	23-40		

LOG OF APPROVED NORMAL REVISIONS**BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA**ISSUE 1: NR 0 to NR 9:

NORMAL REVISION 9 - JUNE 2014	Approved under the authority of EASA DOA No. 21J056 on September 03, 2015
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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016 under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	
NORMAL REVISION 1 date code 21-02		Approved on July 09, 2021 under the authority of EASA DOA No. 21J700
Title	Hydraulic pump reversal (MOD 07- 4925), procedure improvement and minor corrections.	
Revised information	SUP.23.P5 pages 1 and 2, SUP.23 pages 1, 2, 4, 5 and 7	
Deleted information	None	
NORMAL REVISION 2 date code 23-40		Approved on June 21, 2024 under the authority of EASA DOA No. 21J700
Title	- Deletion of servo control fluid limitations (described in section 2.5) and hydraulic pressure values. - Improvement of the procedures	
Revised information	SUP.23.P5 pages 1 to 2, SUP.23 pages 4 and 7.	
Deleted information	None	

RC g

The paragraph **1 - GENERAL** , is modified as follows:

1 GENERAL

The hydraulic system reduces the pilot's workload by providing hydraulic assistance to actuate the main rotor controls. This optional dual hydraulic system is designed to enhance safety and reliability.

- System description:

Two independent circuits, with the same architecture for the main rotor:

- A separate reservoir secured on the top of the MGB,
- A constant flowrate gear-pump generates the hydraulic power,
- A regulator valve/filter unit delivers the pressure at 35 bars (507 psi),
- A distribution system which comprises flexible pressure and return hoses, supplies the three dual-body servo-controls for the main rotor,

The RH circuit also supplies the accumulator, the load compensator and the single body servo-control for the yaw control.

The two pumps are driven differently:

- The LH circuit pump is supported and gear-driven by the MGB (MGB front accessory mechanical output),
- The RH circuit pump is belt-driven by the engine power drive shaft at the MGB power input,

The total volume of fluid in each system is 2 liters (0.53 US gal – 0.44 UK gal) when the reservoir is at max. level line.

- Commands and controls:

- one yaw servo hydraulic switch on the collective switch to isolate the yaw hydraulic circuit,
- One [SERVO TEST] or [SERVO TST] pushbutton to check servo distributor jamming,
- One [HYD TEST] or [ACCU TST] pushbutton on the SCU to perform the yaw servo accumulator test,
- Pre MOD 07-4622: one **HYDR** caution light which comes on if the pressure in one of the hydraulic systems drops below 20 bars $\pm$ 2 bars (290 psi $\pm$ 29 psi),
- Post MOD 07-4622: two caution lights **HYD1** and **HYD2** (one per circuit) which come on if the pressure in the corresponding circuit of the hydraulic system drops below 20 bars $\pm$ 2 bars (290 psi $\pm$ 29 psi),
HYD2 flashes if the yaw hydraulic cut-off switch on the collective grip is OFF.
- One **SERVO** caution light which comes on in case of servo distributor seizure,
- One **LIMIT** caution light which comes on if the maximum lateral servo load is reached.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.4925.

1 GENERAL

The hydraulic system reduces the pilot's workload by providing hydraulic assistance to actuate the main rotor controls. This optional dual hydraulic system is designed to enhance safety and reliability.

- System description:

Two independent circuits, with the same architecture for the main rotor:

- A separate reservoir secured on the top of the MGB,
- A constant flowrate gear-pump generates the hydraulic power,
- A regulator valve/filter unit delivers the pressure at 35 bars (507 psi),
- A distribution system which comprises flexible pressure and return hoses, supplies the three dual-body servo-controls for the main rotor,

The RH circuit also supplies the accumulator, the load compensator and the single body servo-control for the yaw control.

The two pumps are driven differently:

- The LH circuit pump is belt-driven by the engine power drive shaft at the MGB power input,
- The RH circuit pump is supported and gear-driven by the MGB (MGB front accessory mechanical output),

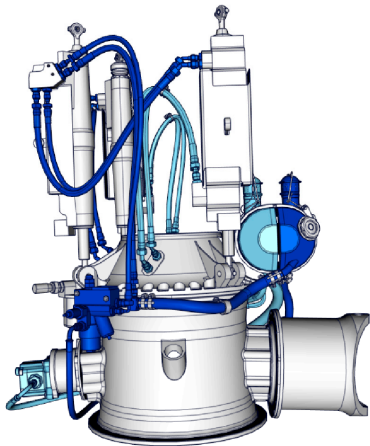
The total volume of fluid in each system is 2 liters (0.53 US gal – 0.44 UK gal) when the reservoir is at max. level line.

- Commands and controls:

- one yaw servo hydraulic switch on the collective switch to isolate the yaw hydraulic circuit,
- One [SERVO TEST] or [SERVO TST] pushbutton to check servo distributor jamming,
- One [HYD TEST] or [ACCU TST] pushbutton on the SCU to perform the yaw servo accumulator test,
- Pre MOD 07-4622: one **HYDR** caution light which comes on if the pressure in one of the hydraulic systems drops below 20 bars $\pm$ 2 bars (290 psi $\pm$ 29 psi),
- Post MOD 07-4622: two caution lights **HYD1** and **HYD2** (one per circuit) which come on if the pressure in the corresponding circuit of the hydraulic system drops below 20 bars $\pm$ 2 bars (290 psi $\pm$ 29 psi),
HYD2 flashes if the yaw hydraulic cut-off switch on the collective grip is OFF.
- One **SERVO** caution light which comes on in case of servo distributor seizure,
- One **LIMIT** caution light which comes on if the maximum lateral servo load is reached.

View from LH side

FM.50.0368.00



View from RH side

FM.50.0369.00

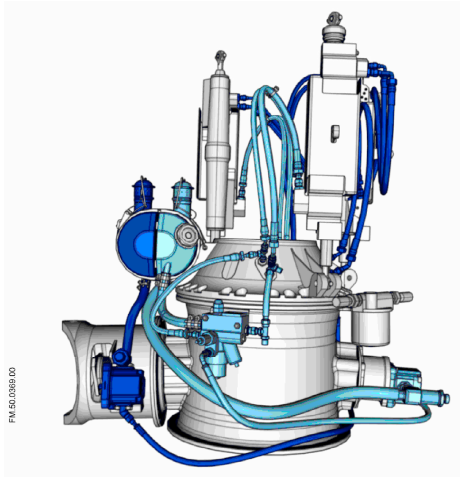
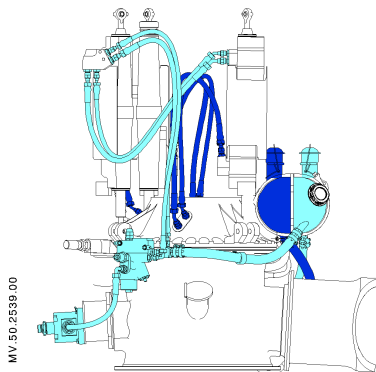


Figure 1: Hydraulic system circuits

RC g

The paragraph **1 - GENERAL (cont'd)**, is modified as follows:
The Figure 1 is superseded by:

View from LH side



View from RH side

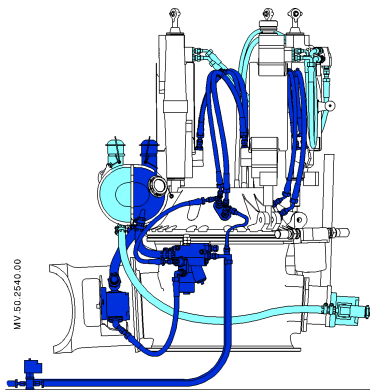


Figure 1: Hydraulic system circuits

The rest of the paragraph is unchanged.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.4925.

RC d

The paragraph 1 - **GENERAL**, is modified as follows:

The Figure 2 is superseded by:

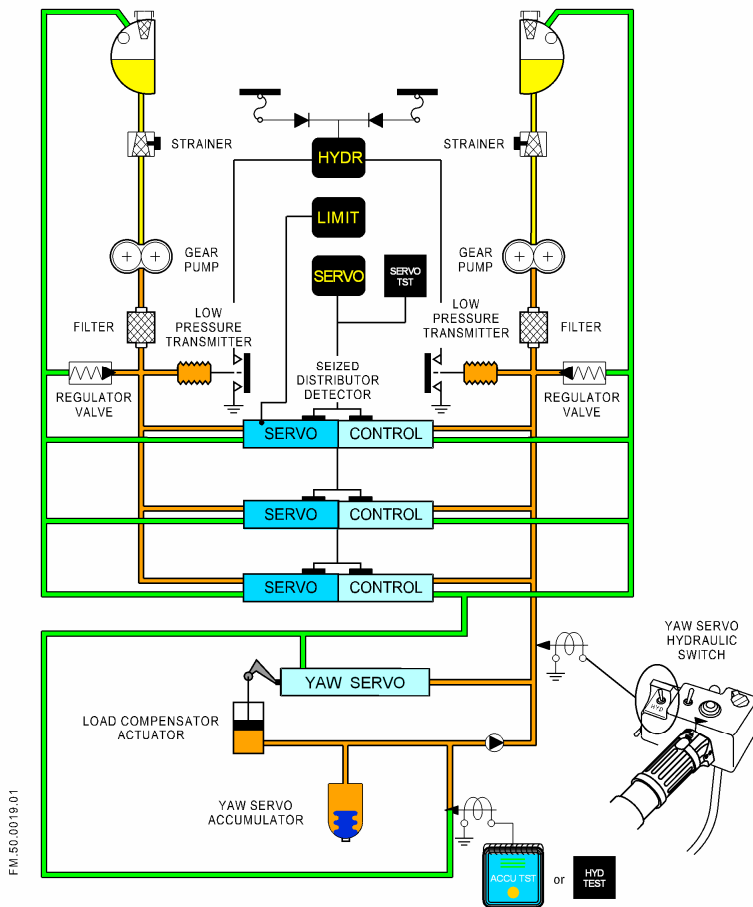


Figure 2: Hydraulic system block-diagram

CAUTION

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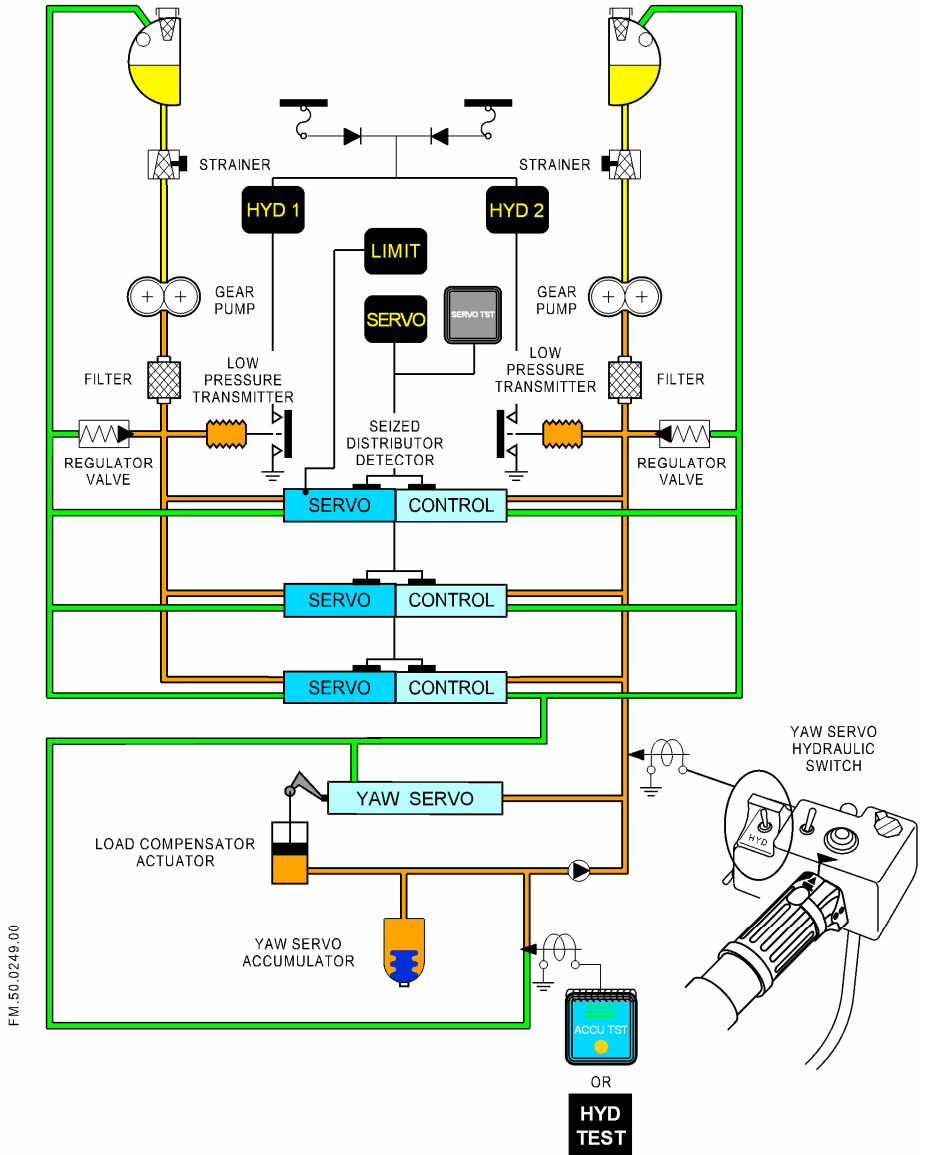


Figure 2: Hydraulic system block-diagram

2 LIMITATIONS

The limitations specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

2.1 MANEUVERING LIMITATIONS

Do not exceed a load factor that causes **LIMIT** light to come on.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

3.1 LOSS OF TAIL ROTOR CONTROL

Symptom: Jamming of pedals or loss of effectiveness of the pedals. These conditions make it impossible to control tail rotor thrust with the pedals.

WARNING

WHEN AIRSPEED IS LESS THAN 20 KT (37 KM/H) AND PARTICULARLY NEAR TO THE GROUND, GO-AROUND MAY BE IMPOSSIBLE DUE TO LOSS OF VERTICAL FIN EFFICIENCY.

- 1. Cyclic and collective..... ADJUST to set IAS to 70 kt (130 km/h) in level flight
- 2. Hydraulic cut-off switch (collective grip)..... OFF
- 3. [HYD TEST] or Pre MOD 07.4719 [ACCU TST] ON (load compensator depressurizes)

Post MOD 07.4719 (applicable only for aircraft equipped with dual hydraulic system):

- 3. [ACCU TST]..... PRESS for 2 sec. (load compensator depressurizes)

On a suitable area for a running landing procedure:

Make a shallow approach with a slight left sideslip. Perform a running landing, the sideslip will be reduced progressively as airspeed is reduced and collective is applied to cushion the landing.

NOTE

Landing is made easier with a RH wind component.

3.2 HYDRAULIC SYSTEM FAILURES

3.2.1 Yaw servo slide-valve seizure

- IN HOVER:

- If no movement about yaw axis: Hydraulic cut-off switch (collective grip) OFF

LAND normally

NOTE

Yaw load compensator assists with heavier pedal forces.

- If yaw axis rotation: Hydraulic cut-off switch (collective grip) OFF

Control yaw, LAND normally

NOTE

Yaw load compensator assists with heavier pedal forces.

- IN CRUISE FLIGHT:

1. Airspeed REDUCE IAS to between 40 and 60 kt (74 and 111 km/h) in level flight, enter sideslip if necessary
2. Hydraulic cut-off switch (collective grip) OFF
3. Normal approach and landing

NOTE

Yaw load compensator assists with heavier pedal forces.

3.2.2 Caution and warning panel hydraulic alarms

WARNING PANEL	CORRECTIVE ACTIONS
HYD1 or HYD2 + SERVO Loss of hydraulic pressure on corresponding circuit.	CAUTION During high load factor maneuvers, servocontrol reversibility may be encountered. Keep aircraft in a more or less level attitude. Avoid abrupt maneuvers. Maintain angle of bank less than 30°. Maintain IAS below 110 kt (204 km/h) (or VNE if less). NOTE One remaining circuit allows continued safe flight and landing. HYD1 ↓ Loss of LH circuit LIMIT may come on HYD2 ↓ Loss of RH circuit Tail rotor control hydraulic power lost CAUTION Do not press the [HYD TEST] or [ACCU TST] pushbutton as this would discharge the yaw load compensator and right pedal loads may be significant. LAND AS SOON AS PRACTICABLE For all cases, normal approach and landing
SERVO Jamming of a distributor valve on main servo unit.	LAND AS SOON AS PRACTICABLE
LIMIT Main servo unit max. load reached.	In high speed cruise flight or steep maneuvers: 1. Collective Reduce power 2. Cyclic Reduce speed or load factor CONTINUE FLIGHT NOTE LIMIT may appear associated with HYD1 in case of hydraulic failure.

RC d

The paragraph 3.2.2 - Caution and warning panel hydraulic alarms, is superseded by:

3.2.2 Caution and warning panel hydraulic alarms

WARNING PANEL	CORRECTIVE ACTIONS
HYDR + SERVO Loss of hydraulic pressure in one circuit.	CAUTION During high load factor maneuvers, servocontrol reversibility may be encountered. Keep aircraft in a more or less level attitude. Avoid abrupt maneuvers. Maintain angle of bank less than 30°. Maintain IAS below 110 kt (204 km/h) (or VNE if less). • If loads and friction are felt on yaw pedals: RH circuit failure. CAUTION Do not press the [HYD TEST] or [ACCU TST] pushbutton as this would discharge the yaw load compensator and right pedal loads may be significant. • No loads are felt on yaw pedals: LH circuit failure. NOTE LIMIT light may be on if LH circuit failed or during load factor maneuvers. The remaining circuit allows continued safe flight and landing. LAND AS SOON AS PRACTICABLE For all cases, normal approach and landing.
SERVO Jamming of a distributor valve on main servo unit.	LAND AS SOON AS PRACTICABLE
LIMIT Main servo unit max. load reached.	In high speed cruise flight or steep maneuvers: 1. Collective Reduce power. 2. Cyclic..... Reduce speed or load factor. CONTINUE FLIGHT NOTE LIMIT may appear associated with HYDR in case of hydraulic failure.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.4622.

RC d

The paragraph **4 - NORMAL PROCEDURES** is modified as follows:

4 NORMAL PROCEDURES

The normal procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

- **ENGINE PRESTART CHECK** (SECTION 4.3 § 1 of the basic Flight Manual):

Replace items 14 to 20 by the following:

14. CWP lights:

- With battery power.....CHECK



- With EPU powerCHECK:
Same lights as above +

BATT

15. [SERVO TEST] or [SERVO TST]PRESS:

SERVO

16. VEMDCHECK:

- . 3-data page: no message
- . Vehicle page: no message
- . Battery voltage > 22 V
- . Check fuel quantity
- .  (Bleed valve open)

17. Control pedals.....Free travel, then left pedal 2 cm (0.8 in) forward

18. Cyclic pitch.....CENTER, friction adjusted

19. Collective pitch.....LOCK, friction adjusted

20. Heating, demisting,
air conditioning (if fitted).....OFF

- **ENGINE STARTING** (SECTION 4.3 § 2 of the basic Flight Manual):

Replace item 7 by the following:

7. CWP :

SERVO

ENG P

HYDR

MGB P

The rest of paragraph is unchanged.

CAUTION

THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.4622.

4 NORMAL PROCEDURES

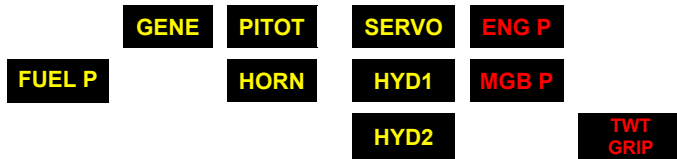
The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- **ENGINE PRESTART CHECK** (SECTION 4.3 § 1 of the basic flight manual):

Replace items 14 to 19 by the following:

14. CWP lights:

- With battery power..... CHECK



- With EPU power CHECK:

Same lights as above +

BATT

15. [SERVO TEST] or [SERVO TST] PRESS:

SERVO

16. VEMD CHECK:

- . 3-data page: no message
- . Vehicle page: no message
- . Battery voltage > 22 V
- . Check fuel quantity
- .  (Bleed valve open)

17. Control pedals..... Free travel, then left pedal 2 cm (0.8 in) forward

18. Cyclic pitch..... CENTER, friction adjusted

19. Collective pitch..... LOCK, friction adjusted

20. Heating, demisting,
air conditioning (if fitted)..... OFF

- **ENGINE STARTING** (SECTION 4.3 § 2 of the basic flight manual):

Replace item 7 by the following:

7. CWP :



- **RUN-UP CHECK** (SECTION 4.3 § 3 of the basic flight manual):

Replace item 1 by the following:

1. Hydraulic checks:

- Servo distributors seizure check:

- [SERVO TEST] or [SERVO TST] PRESS: **SERVO**

The rest of the basic flight manual procedure remains applicable.

5 PERFORMANCE DATA

The performance data specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

LOUDSPEAKER INSTALLATION

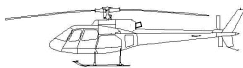
Optional OP 1810 and OP 1811

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
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 - XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.24.P1	1 to 1	15-45	A	
SUP.24.P5	1 to 2	15-45		
SUP.24	1 to 1	15-45		

LOG OF APPROVED NORMAL REVISIONS**BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA**ISSUE 1: NR 0 to NR 2:

NORMAL REVISION 2 - MARCH 2006	EASA Approval R.A. 01126 on May 23, 2006
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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	

1 GENERAL

The loudspeaker installation is designed to transmit external high power audio messages. The installation comprises:

- An amplifier unit incorporating an audio adjust potentiometer, located on the center console,
- A microphone,
- Two loudspeakers secured to the landing gear aft cross member.

The installation is protected by two fuses or breakers and can be controlled by the [LS] pushbutton on the SCU.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- VNE power on: 135 kt or basic helicopter VNE, the lower of the two.
- VNE power off: same as basic helicopter.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

Exterior check:

ADD:

Loudspeaker installation.....Check attachment.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Rate of climb: reduce by 10 %.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

MAXIMUM INTERNAL GROSS WEIGHT

Increased to 2370 kg (5225 lb)

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.28.P1	1 to 1	15-45	A	
SUP.28.P5	1 to 2	15-45		
SUP.28	1 to 12	15-45		

LOG OF APPROVED NORMAL REVISIONS

BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA

ISSUE 1: NR 0 to NR 3:

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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	

1 GENERAL

This supplement (OP 3369) allows an increased maximum internal weight when aircraft is equipped with:

- The high landing gear,
- The dual hydraulic system (refer to SUP.23).

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 TYPE OF OPERATIONS

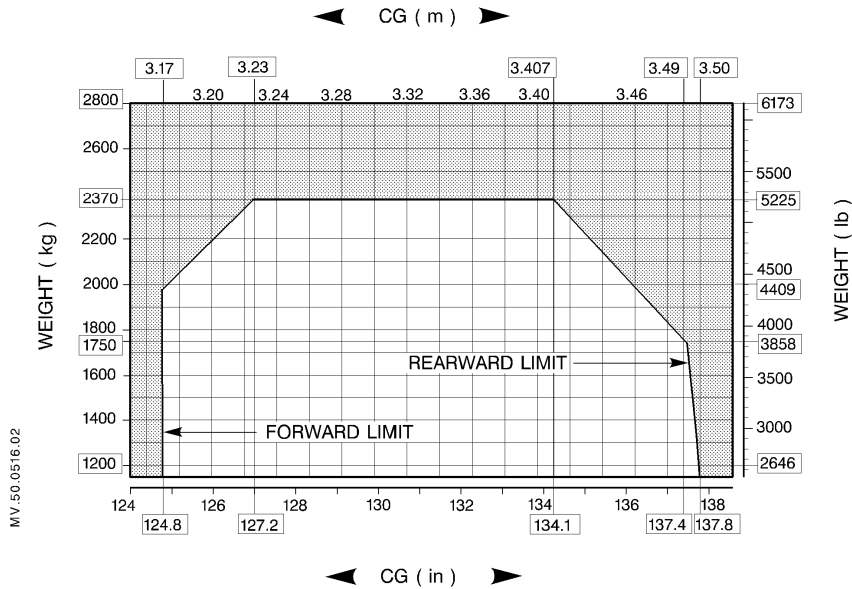
The following is forbidden:

- Operation with jettisonable external load if the internal weight is higher than 2250 kg (4961 lb).

2.2 WEIGHT LIMITATION

- Maximum internal weight in flight..... 2370 kg (5225 lb),
- Maximum internal weight for IGE,
take-off and landing 2370 kg (5225 lb).

2.3 LONGITUDINAL CG



2.4 LATERAL CG

2.4.1 Lateral CG in flight

The lateral CG limits specified in the basic flight manual and in the flight manual supplements remain applicable.

2.4.2 Lateral CG at landing

- Internal gross weight $\leq$ 2250 kg (4961 lb):
Maximum left CG 0.18 m (7.08 in).
Maximum right CG 0.14 m (5.51 in).
- 2250 kg (4961 lb) < Internal gross weight $\leq$ 2370 kg (5225 lb):
Maximum left CG 0.08 m (3.15 in).
Maximum right CG 0.08 m (3.15 in).

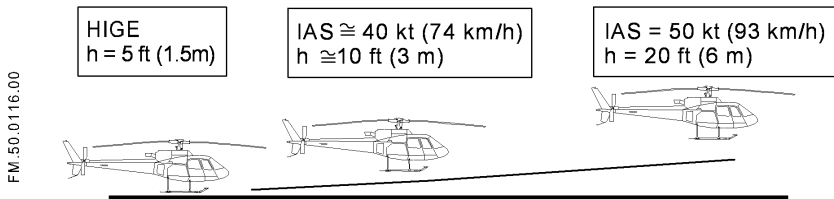
3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Increase airspeed with HIGE power until IAS = 40 kt (74 km/h), then begin to climb so as to clear 20 ft (6 m) at IAS = 50 kt (93 km/h).



5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following.

5.1 HEIGHT - VELOCITY DIAGRAM

- Refer to pages SUP.28 pages 4 and 5 (Figure 1).

5.2 PERFORMANCE CALCULATION

- Refer to pages SUP.28 pages 6 to 11 (Figures 2 to 7).

CAUTION

Pilot shall limit the flight envelope and weight displayed on VEMD performance pages to the relevant limitations of SUP.28 § 2.2.

5.3 HEIGHT - VELOCITY DIAGRAM

5.3.1 Internal gross weight ≤ 2250 kg (4961 lb)

Refer to SECTION 5.1 Figure 4.

5.3.2 2250 kg (4961 lb) < Internal gross weight ≤ 2370 kg (5225 lb)

The avoidance zone is defined by five points: A, B, C, D, E (refer to Figure 1).

- Point A: low hover point.

Point A is at 7 ft (2.1 m) skid height at zero airspeed.

- Point B:

Point B is defined by:

- a constant height of 30 ft (9 m),
- a constant airspeed of 50 kt (93 km/h).

- Point C:

Point C is defined by:

- a constant height of 100 ft (30 m),
- a constant airspeed of 60 kt (111 km/h).

- Point D:

Point D is defined by:

- a variable height ($700 \text{ ft} \leq \text{height} \leq 1000 \text{ ft}$) depending on the altitude and on the aircraft weight as determined by line (E D),
- a constant airspeed of 20 kt (37 km/h).

- Point E :

Point E is defined by:

- a variable height ($700 \text{ ft} \leq \text{height} \leq 1000 \text{ ft}$) depending on the altitude and on the aircraft weight as determined by line (E D),
- a constant zero airspeed.

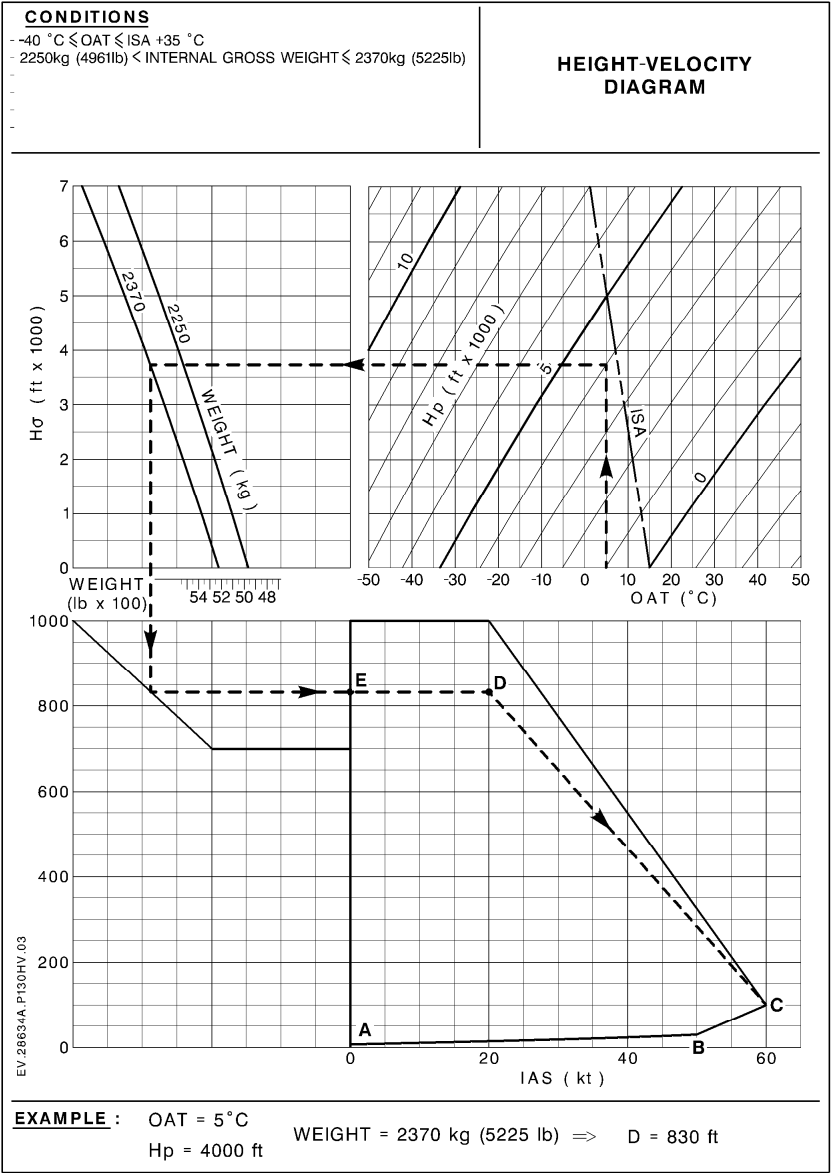


Figure 1

5.4 HOVER IN GROUND EFFECT

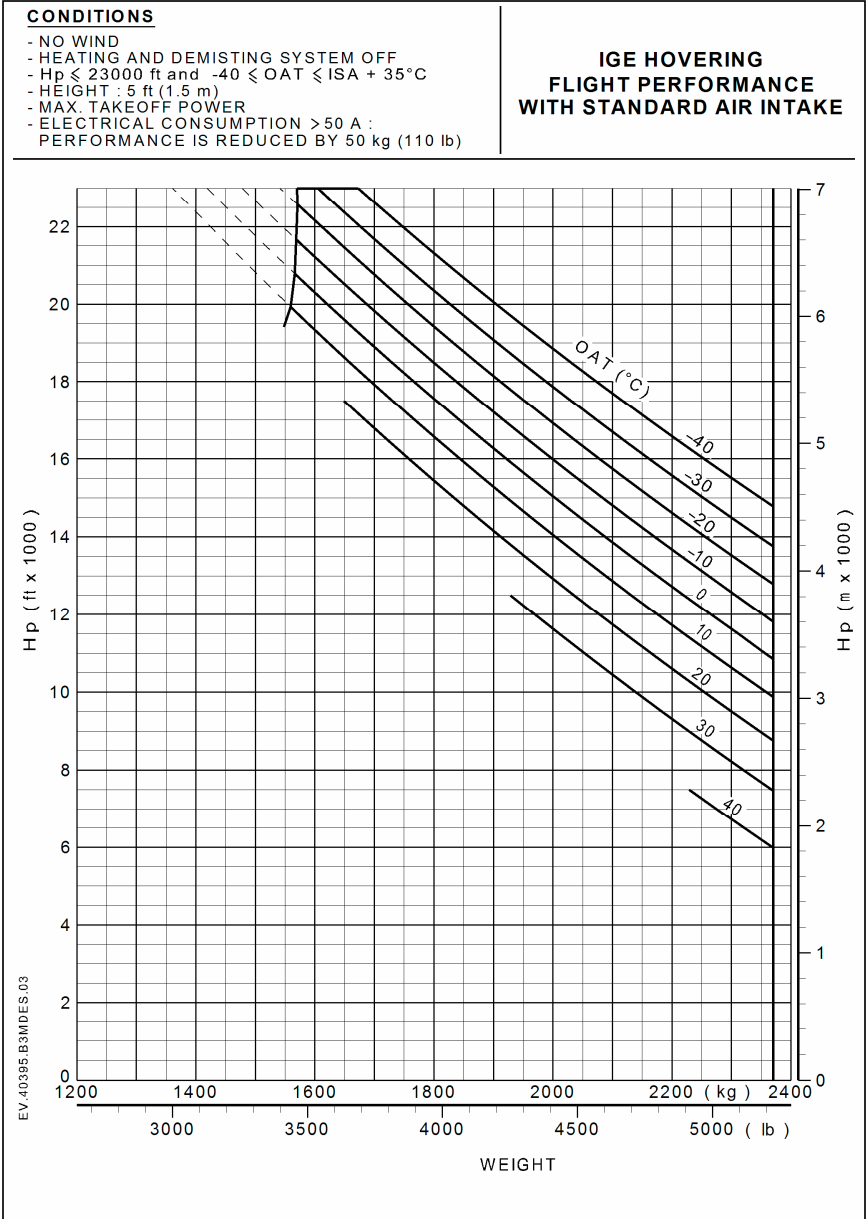


Figure 2

CONDITIONS

- NO WIND
- HEATING AND DEMISTING SYSTEM OFF
- $H_p \leq 23000$ ft and $-40^{\circ}\text{C} \leq \text{OAT} \leq \text{ISA} + 35^{\circ}\text{C}$
- HEIGHT : 5ft (1.5m)
- MAX. TAKEOFF POWER
- ELECTRICAL CONSUMPTION $> 50\text{A}$:
PERFORMANCE IS REDUCED BY 50 kg (110 lb)
- WITH THE SAND FILTER OPERATING THE
PERFORMANCE IS REDUCED BY 40 kg WHEN
THE Ng LIMIT IS REACHED

**IGE HOVERING
FLIGHT PERFORMANCE
WITH SAND FILTER
NOT OPERATING**

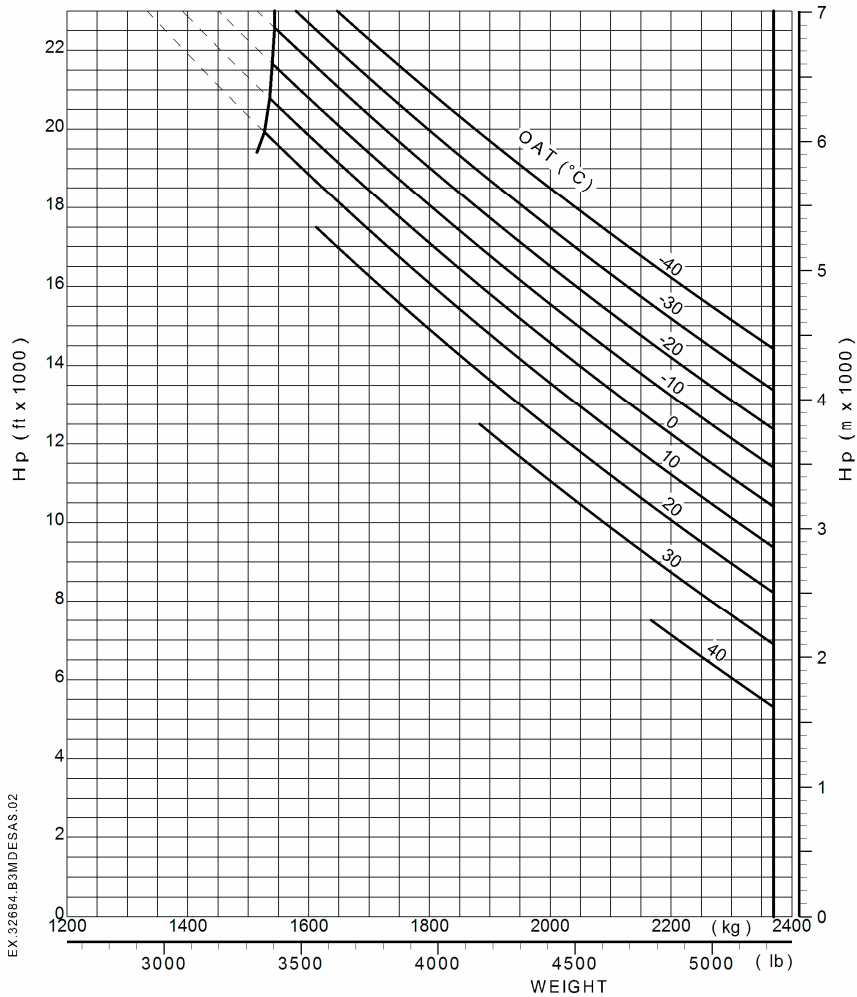


Figure 3

5.5 HOVER OUT OF GROUND EFFECT

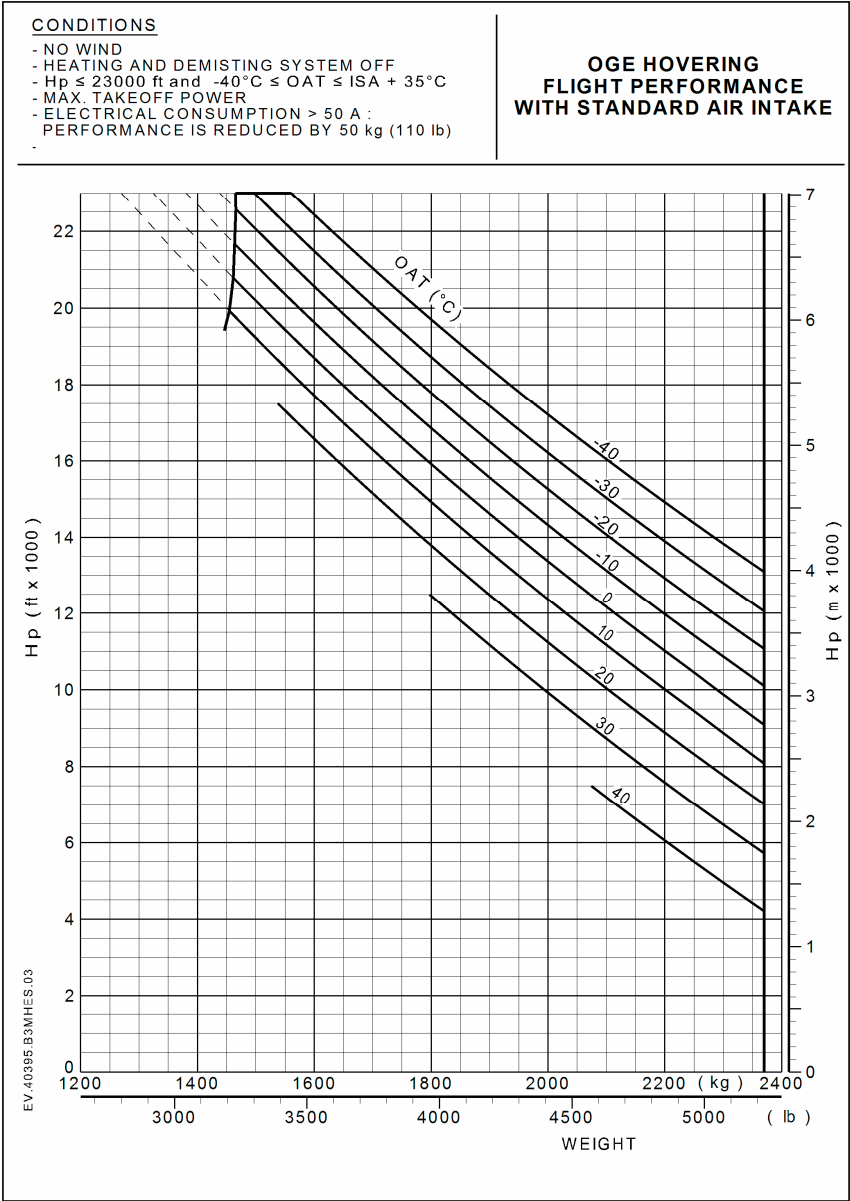


Figure 4

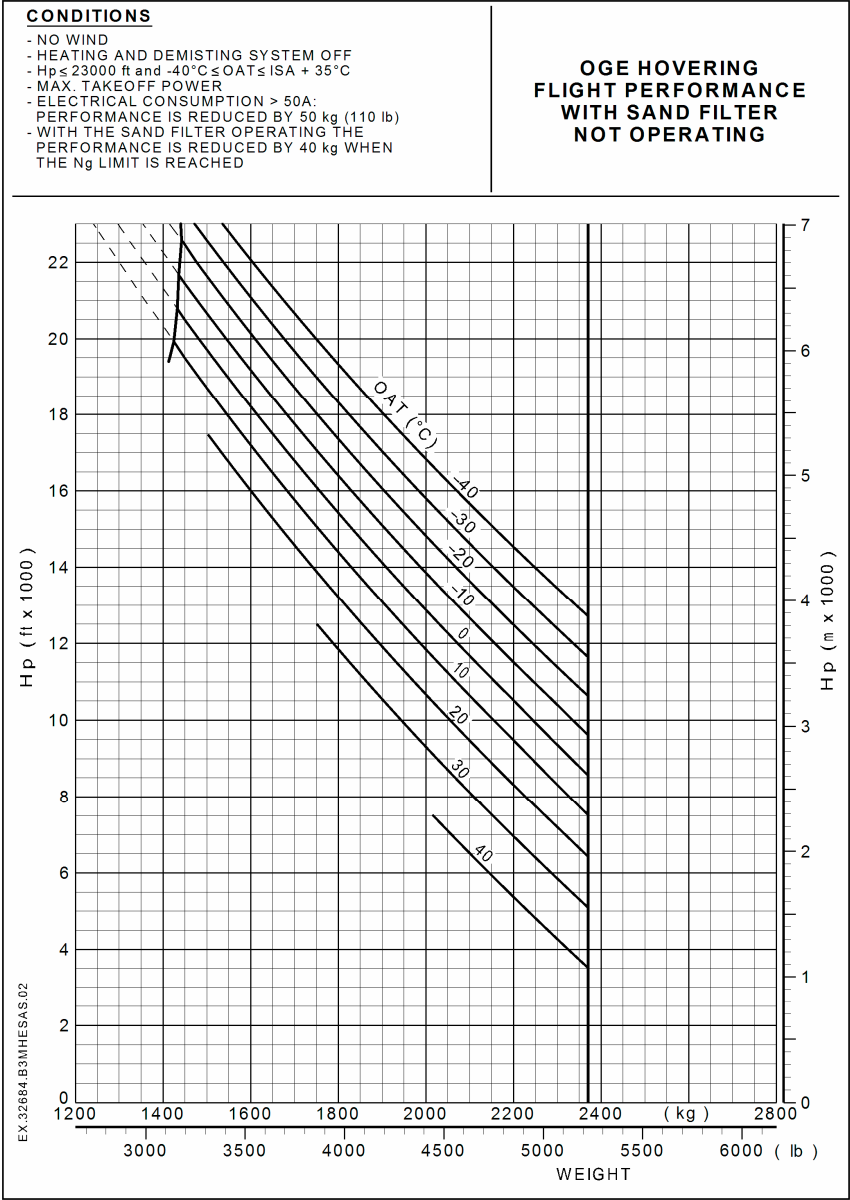


Figure 5

5.7 RATE OF CLIMB

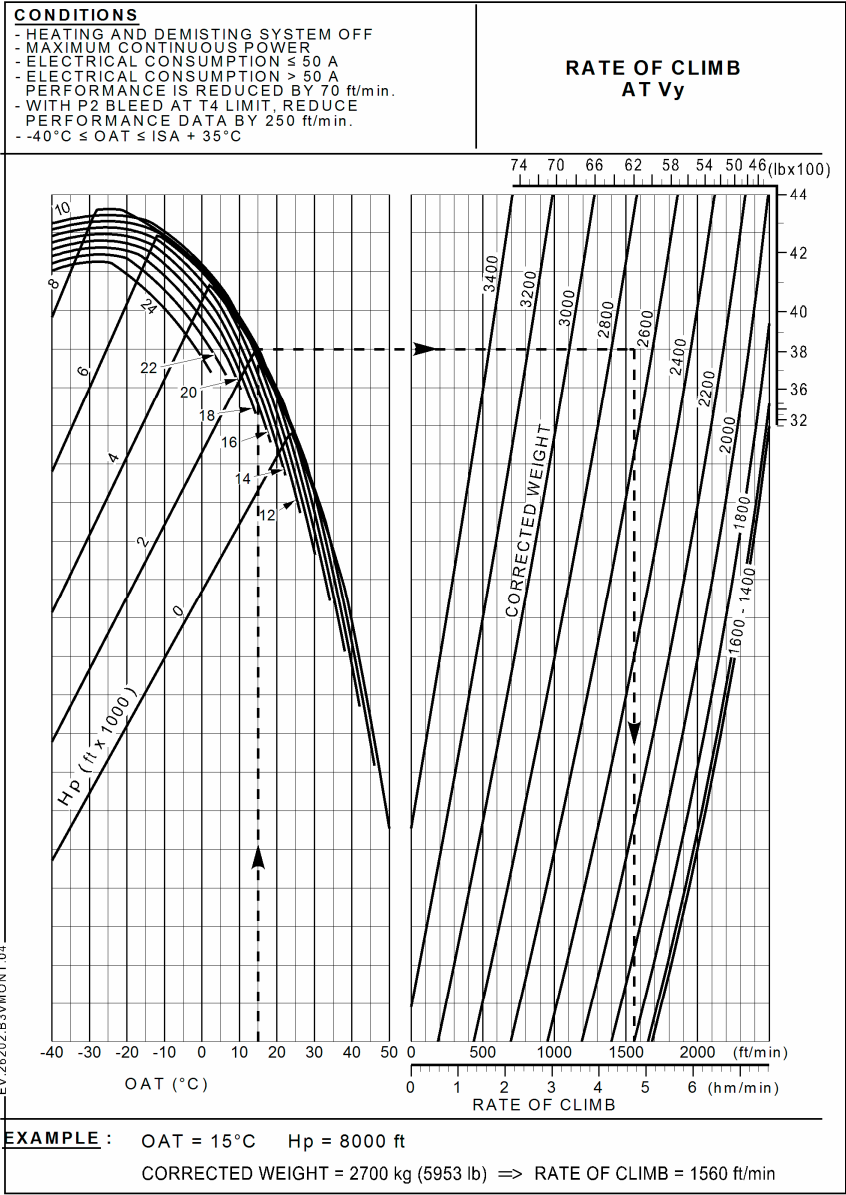


Figure 7

5.8 NOISE LEVEL

Noise characteristic as defined by the conditions of chapter 11 of the ICAO annex 16 are as follows:

Noise Level Measured dB(A)	ICAO / JAR 36 Noise Limits dB(A)
84.1	86.8

The measured noise level corresponds to a reference V_H (maximum horizontal speed) of 132 kt (244 km/h) TAS.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

200 A STARTER GENERATOR

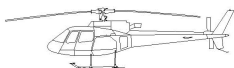
Optional OP 3821

IMPORTANT NOTE

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- (1) AIRWORTHINESS EFFECTIVITY:
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- Without indication..... Applicable to all aircraft
 - XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
SUP.29.P1	1 to 1	15-45	A	
SUP.29.P5	1 to 2	15-45		
SUP.29	1 to 1	15-45		

LOG OF APPROVED NORMAL REVISIONS**BASIC RFM REVISIONS - EFFECTIVITY (1) (2) EASA**ISSUE 1: NR 0 to NR 1:

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ISSUE 2:

NORMAL REVISION 0 date code 15-45		Approved on May 31, 2016, under the authority of EASA DOA No. 21J056
Title	New issue	
Revised information	All	
Deleted information	None	

RC e

The paragraph **2 - LIMITATIONS**, is superseded by the following:

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Maximum continuous current:180 A
- An associated placard located under the VEMD (Pre MOD 07-3354 only)

CHARGE MAXIMUM CONTINUE GENERATEUR = 180 A

or

MAXIMUM CONTINUOUS GENERATOR LOAD = 180 A

CAUTION
THIS PAGE MUST ONLY BE REMOVED FROM THE MANUAL AFTER THE AIRCRAFT IS EQUIPPED WITH THE STARTER GENERATOR 200 A (REFERENCE: 515-030).

1 GENERAL

The starter generator 200 A is an optional installation allowing increasing electrical capacity of the aircraft by replacement of the basic starter generator 150 A.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Maximum continuous current 200 A.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

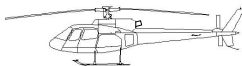
SPECIAL COCKPIT LIGHTING

IMPORTANT NOTE

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- Without indication..... Applicable to all aircraft
 - XXX..... Specific to aircraft equipped with XXX

SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
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Deleted information	None	

1.GENERAL

The information contained herein supplements the information of the basic Flight Manual.

It has been shown that the installation of the special cockpit lighting system does not affect the aircraft capability for VFR day or night operations.

No compliance demonstration has been performed to show that the installed system complies with the applicable regulations for NVIS. No changes have been introduced to comply with the applicable regulations.

1.1 SYSTEM DESCRIPTION

Refer to SECTION 7.6 of the complementary Flight Manual.

2.LIMITATIONS

The limitations specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following limitation:

CAUTION

If NVG operations are envisaged, an airworthiness and subsequent operational approval by the competent authority are necessary.

3.EMERGENCY PROCEDURES

The emergency procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

NOTE

If the lighting mode can not be reselected to DAY mode during a transition from night to day flight, the VEMD and NR indications become progressively unreadable.

Failure of light mode selection

If NR/VEMD unreadableApply "VEMD SCREEN FAILURE, Failure of both screens" emergency procedure.

4 NORMAL PROCEDURES

The normal procedures specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable and are supplemented or modified by the following:

For night VFR:

- Selector [DAY/NIGHT HI/NIGHT LO]..... "NIGHT HI" or "NIGHT LO" as required.
- Brightness of displays Adjust as required.

CAUTION

When the lighting mode is switched from "NIGHT HI" to "NIGHT LO" the red arcs/symbols on the VEMD are initially difficult to see. After approximately 3 min., the color stabilizes and is readable with the appropriate brightness level set.

If switching from "NIGHT HI" to "NIGHT LO" mode:

- Power..... Set to ensure power margin available.
- Flight path..... Maintain steady level flight for approximately 3 min., until the red arcs are visible.
- VEMD brightness..... Adjust as required.

5 PERFORMANCE

The performance data specified in the basic Flight Manual and in the Flight Manual Supplements remain applicable.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

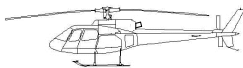
FERRY FLIGHT FUEL TANK

IMPORTANT NOTE

The information contained herein supplements or supersedes the information given in the basic flight manual and/or the supplements listed in section supplement 0.

The effectivity of the manual at the latest revision is specified on the list of effective pages.

THIS SUPPLEMENT MUST BE INCLUDED IN THE FLIGHT MANUAL WHEN
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Revised information	All	
Deleted information	None	

1 GENERAL

The use of this installation is subject to the approval of the operational authorities concerned.

To increase the maximum range of the helicopter, a specific ferry flight fuel tank can be installed transversally in the rear cabin area.

The installation consists essentially of:

- A 475 liter (125 US gal – 104.5 UK gal) removable tank with negligible unusable fuel quantity.
- A vent line.
- A fuel transfer pipe with a manually actuated valve between the ferry tank and helicopter fuel tank.

The fuel is transferred by gravity into the helicopter fuel tank.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 OPERATIONS AUTHORIZED

This installation is to be used only for ferry flight, provided such operation is permitted by the appropriate aviation authorities of the country concerned.

2.2 CREW LIMITATION

Only persons essential to the mission shall be allowed on board the aircraft.

2.3 FERRY TANK FUEL WEIGHT LIMITATION

The weight of fuel that be carried in the ferry tank will depend on the overall loading of the helicopter. Refer to SECTION 6 of the basic flight manual to determine aircraft CG.

CAUTION

The maximum ferry tank fuel weight must be determined carefully to remain within the forward CG limit given by Section 2 of the basic flight manual specifically with a full main tank.

2.4 SPECIFIC INSTRUCTIONS

Smoking in cabin is prohibited.

Maximum ventilation of the cabin is to be ensured.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Fuel leak in the cabin:

LAND AS SOON AS POSSIBLE

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Checks before filling ferry tank:
 - Ferry tank Condition, attachment,
 - Ferry tank vent line Installed and correctly secured,
 - Transfer valve. Closed,
 - Max. ferry tank fuel quantity Determined, refer to SECTION 6 of the basic flight manual to determine aircraft CG.
- Filling procedure:
 - Main fuel tank Fill,
 - Ferry tank Fill with quantity previously determined.
- In flight transfer procedure:

Takeoff and cruise flight:

 - Transfer valve Maintain closed until main fuel gauge reads 80 %.

When fuel gauge reads 80 %:

 - Transfer valve Open,
 - Fuel transfer Check effective (fuel gauge indicator is moving).

CAUTION

If fuel transfer is not operative, land before the fuel gauge indicator reads 60%. Failure to land above 60% may result in CG exceeding the approved limits.

The fuel level in the two tanks will be equal when the ferry tank quantity is approximately 300 l (79.2 US gal – 66.0 UK gal).
If there is a difference in fuel level, fuel will be transferred and the levels will be balanced within 10 min.

When the fuel levels are balanced in the two tanks, the quantity corresponding to the gauge reading is:

Fuel gauge reading (%)		90	80	70	60	50	40	30	20
V O L U M E	Liters	805	705	605	505	405	305	205	105
	US gal	212	186	159	133	107	80	54	27
	UK gal	176	154	132	110	88	66	45	23

When the gauge reads 20 % the ferry tank is empty and the quantity of fuel remaining in the main tank is 105 l (27 US gal – 23 UK gal).

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable.



FLIGHT MANUAL

AS 350 B3

Arriel 2B1

SUPPLEMENT

"BAMBI BUCKET"

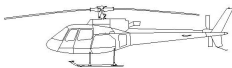
Model 2732S

IMPORTANT NOTE

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1 GENERAL

The use of this installation is subject to the approval of the operational authorities concerned.

This supplement does not constitute an operational authorization.

The "BAMBI BUCKET" model 2732S installation is a fire-fighting system used to carry and jettison water or other liquids in flight.

Maximum capacity: 1225 litres.

The installation is composed of a bucket secured to the external load carrying device via a shackle (refer to SUP.13.1 or SUP.13.2).

The bucket is filled by plunging it into water (swimming pool, sea, lake, etc.).

The "BAMBI BUCKET" enables the water load to be released in a single drop.

A foam injection system "SACKSAFOAM" (P/N: SF02-2044) can be installed as an option.

1.1 GENERAL DESCRIPTION

The set of controls consist of:

- A [SLING] pushbutton located on the console for switching on the installation.
- A pushbutton (1) located on the cyclic grip controls enables the release of the contents of the bucket.
- A pushbutton (2) located on the cyclic grip controls provides an electrical control to release the entire load suspended from the sling hook.
- A mechanical control (3) on the collective grip provides a mechanical release of the load as a whole.

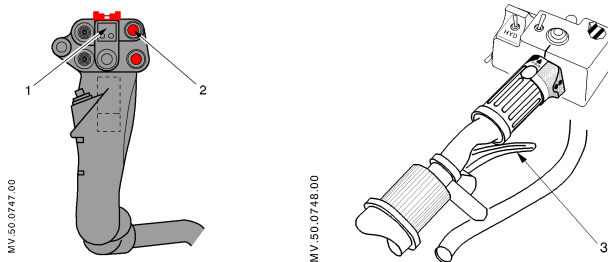


Figure 1: Release controls

- A load indicator is fitted to indicate the weight suspended from the hook (refer to SUP.13.1 or SUP.13.2),
- A control unit located on the instrument panel or the console enables the control of the optional foam injection system.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

Speed limitations:

- Empty bucket VNE = 90 kt (167 km/h - 104 MPH),
- Bucket full or partially filled..... VNE = 80 kt (148 km/h - 92 MPH).

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

3.1 WATER EMERGENCY JETTISON

If it is impossible to jettison the contents of the "BAMBI BUCKET" via the electrical control on the cyclic grip, the pilot may place the bucket (filled or empty) on the ground as he would do with a load on the cargo sling.

The ground staff may then unhook the bucket via the load release device.

3.2 BUCKET EMERGENCY JETTISON

If there is a problem on the helicopter requiring bucket jettison, the bucket (filled or empty) may be released via the cargo sling electrical or mechanical release controls.

In order to release the bucket, it is recommended to enter a right hand turn with a slight load factor.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

4.1 CHECK BEFORE TAKEOFF

- Check that the bucket is correctly secured,
- Check that the manufacturer's plate on shackle and the ballast inside the bucket are oriented in the forward direction.

5 PERFORMANCE DATA

For weights with external load > maximum weight specified in the limitations section of the basic flight manual, refer to SECTION 5.1 "REGULATORY PERFORMANCE DATA" of the basic flight manual.

The curves are plotted in dotted lines on Figure 6.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

GPS

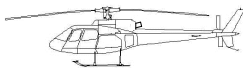
GARMIN GNS 430/430W

IMPORTANT NOTE

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SECTION or SUP.	PAGES	DATE CODE	(1)	(2)
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1 GENERAL

The use of this installation is subject to the approval of the operational authorities concerned.

The “GNS 430/430W” includes a VHF COM transceiver, a VOR/ILS receiver and a GPS navigation system.

The GARMIN “GNS 430/430W” GPS system complies with the requirements as a supplement to VFR navigation.

Abbreviations

- BRG : Bearing to waypoint.
- CDI : Course Deviation Indicator.
- DTK : Desired Track.
- DIS : Distance to waypoint.
- ETE : Estimated Time En-route.
- GS : Ground Speed.
- HSI : Horizontal Situation Indicator.
- OBS : Omni Bearing Selector.
- RAIM : Receiver Autonomous Integrity Monitoring.
- TRK : Track.
- XTK : Cross Track error, the cross track error has a manual or an automatic adjustable scale on the CDI.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

2.1 PILOT'S GUIDE

2.1.1 GNS 430 pilot's guide

The Pilot's Guide operating manual (Part number 190-00140-00, Rev.F dated July 2000 or later revision) must be on board the aircraft and readily available to the pilot.

2.1.2 GNS 430W pilot's guide

The Pilot's Guide operating manual (Part number 190-00356-00, Rev.C dated March 2008 or later revision) must be on board the aircraft and readily available to the pilot.

2.2 SOFTWARE VERSIONS

2.2.1 GNS 430 software versions

The GNS 430 must be loaded with the following or later approved software versions:

GNS 430 sub-system	Software version
Main	2.25
GPS	2.11
COMM	5.00
VOR/LOC	3.01
G/S	2.03

The main software version is displayed on the GNS 430 self-test page 5 seconds after power-on. The other system software versions can be checked on the AUX group sub-page 2:

SOFTWARE/DATABASE Ver.

From main software version 5.01, a TAWS (TERRAIN) function has been added to the GNS 430. USING THE TAWS FUNCTION OF THE GPS IS PROHIBITED. Consequently, this function is de-activated by configuration and shall remain so.

2.2.2 GNS 430W software versions

The GNS 430W must be loaded with the following or later approved software versions:

GNS 430W sub-system	Software version
Main	3.10
GPS	3.1
COMM	7.00
VOR/LOC	5.02
G/S	4.00

The main software version is displayed on the GNS 430W self-test page 5 seconds after power-on. The other system software versions can be checked on the AUX group sub-page 2:

SOFTWARE/DATABASE Ver.

USING THE TAWS FUNCTION OF THE GPS IS PROHIBITED. Consequently, this function is de-activated by configuration and shall remain so.

The GPS receiver is capable of tracking SBAS (WAAS, EGNOS) satellites. USING THE SBAS MODE OF THE GPS IS PROHIBITED. Consequently, SBAS (WAAS, EGNOS) operation is de-activated in set-up sub-group page 2 and shall remain so.

2.3 OPERATION

The use of the GPS is restricted to VFR flight only.

All the navigation means required for each route phase of the intended flight must be available and serviceable.

As the database is not guaranteed, the crew must check, before the flight if possible, the validity and the accuracy of the database information by reference to the official documentation.

Before starting navigation, the crew must read the GNS 430/430W self-test messages to check that all necessary validities are present.

2.4 PLACARDS

GPS OPERABLE IN VFR
CONDITIONS ONLY

GPS UTILISABLE EN VFR
UNIQUEMENT

Location: Next to the GNS 430/430W on the instrument panel.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

3.1 GPS FAILURES

FAILURES	CORRECTIVE ACTIONS
NAV Flag on HSI (or external CDI). + MSG	[MSG] key Press and check message. ↓ 1. GPS navigation data not available or invalid. 2. RAIM POSITION WARNING. In both cases, use remaining operational means of navigation (GNS 430/430W VOR or any other available means). CONTINUE THE FLIGHT
MSG + INTEG	[MSG] key Press and check message. ↓ RAIM IS NOT AVAILABLE Revert to other operational means of navigation (GNS 430/430W VOR or any other available means) approved for the route and flight phase. During En-route phase, GPS navigation can still be used provided the position can be checked with other means of navigation at least every 15 min. CONTINUE THE FLIGHT

NOTE

Bottom row key [MSG] is used on GNS 430/430W to display the message.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

4.1 OPERATING PROCEDURES

The detailed operating procedures are described in the Pilot's Guide referenced in paragraph 2.1 of this supplement.

NOTE 1

Use of the VHF frequencies listed below may degrade GPS receiver operation after 10 to 15 seconds of transmission time, returning to normal operation a few seconds after transmission ends.

Frequencies = 121.150 / 121.175 / 121.200 / 131.200 / 131.250 / 131.275 and 131.300 MHz.

NOTE 2

Correct operation of the GPS is not guaranteed for cabin temperatures below - 20°C.

4.2 CONTROLS AND INDICATORS

The GNS 430/430W GPS navigation system provides the following information to the pilot's HSI or external CDI:

- XTK.
- TO / FROM.
- Validity flag.

NOTE 1

XTK full scale deviation is the same for the HSI, or external CDI, and the GNS 430/430W integrated CDI. Default setting is 5 NM (meaning that full deviation is achieved when XTK reaches 5 NM) except within 30 NM range of the departure/destination airfield. Within 30 NM of the destination airfield, the full scale deviation gradually ramps from 5 to 1 NM. Likewise, upon departure, default setting is 1 NM gradually increasing up to 5 NM beyond 30 NM from the departure airfield.

XTK scale is also selectable by the pilot. However, the GNS 430/430W will automatically select the lowest value between the default setting and the value selected by the pilot. Current selected scale is displayed on either side of the GNS 430/430W's CDI. Recommended full-scale value for helicopter "En-route" navigation is 1 NM.

NOTE 2

The HSI or CDI course is not automatically slaved to the desired track (DTK). Consequently, when GPS navigation is selected, (**GPS** on GNS 430/430W screen) as HSI or external CDI navigation source, the course pointer on the HSI or course selector on the external CDI must be manually set to the DTK indicated by the GNS 430/430W. Particular attention is required during automatic navigation leg changes and subsequent change of DTK. However, if the course selected on pilot's HSI or external CDI differs from the DTK by more than 10°, the **MSG** annunciator will flash and the message **Set course to xxx** will be displayed on the GNS 430/430W "MSG" page.

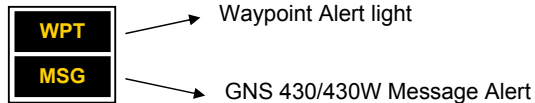
NOTE 3

Pressing the [CDI] key on the GNS 430/430W toggles HSI or external CDI navigation source between GPS and VOR/ILS (**GPS** or **VLOC** displayed above [CDI] key).

NOTE 4

Pressing the [OBS] key on the GNS 430/430W toggles between manual mode (OBS mode) and automatic sequencing of waypoints. Activating OBS mode, indicated by the **OBS** annunciator above the [OBS] key, holds current active waypoint as the navigation destination and prevents the GNS 430/430W from sequencing to the next waypoint. In OBS mode, the DTK to/from the active waypoint is controlled via the pilot's HSI course pointer or external CDI course selector.

The "GNS 430/430W" GPS navigation system is also associated with a two-label indicator on the pilot's instrument panel (If installed):



5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable.



FLIGHT MANUAL

AS 350 B3 Arriel 2B1

SUPPLEMENT

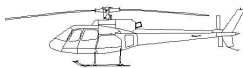
ABSEILING INSTALLATION

IMPORTANT NOTE

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1 GENERAL

The use of this installation is subject to the approval of the operational authorities concerned.

This installation allows trained personnel to perform abseiling.

It consists of two rings secured to the cabin floor in front of the passenger seats and of a protection for the lower rail of each sliding door.

2 LIMITATIONS

The limitations specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Abseiling is limited to hover flight.
After completion of the abseiling operation, transition to forward flight or landing is prohibited with the ropes extended.
- The load on the abseiling installation is limited to 120 kg (265 lb) per ring.
A placard affixed close to each ring indicates the maximum load.

3 EMERGENCY PROCEDURES

The emergency procedures specified in the basic flight manual and in the flight manual supplements remain applicable.

4 NORMAL PROCEDURES

The normal procedures specified in the basic flight manual and in the flight manual supplements remain applicable and are supplemented or modified by the following:

- Before takeoff, determine the weight and CG conditions which will prevail during the mission, knowing that the load on the abseil ropes is located at:
 - 2.24 m (88 in) from the longitudinal datum.
 - 1.09 m (43 in) from the aircraft centerline.

5 PERFORMANCE DATA

The performance data specified in the basic flight manual and in the flight manual supplements remain applicable.

SECTION 5.2

ADDITIONAL PERFORMANCE DATA

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SECTION 5.2

ADDITIONAL PERFORMANCE DATA

1 EFFECT OF EQUIPMENT ITEMS ON PERFORMANCE DATA

Equipment installed	Fast cruise				Recommended cruise			
	Airspeed km/h	kt	Hourly fuel consumption	Range	Airspeed km/h	kt	Hourly fuel consumption	Range
Heating and demisting systems ON*	- 8	- 4	+ 4%	- 4%	- 8	- 4	+ 4%	- 4%
Electrical consumption > 50A	- 4	- 2	+ 1%	- 2%	-	-	+ 2%	- 2%
High landing gear	- 4	- 2	-	- 1.5%	- 4	- 2	-	- 1.5%
Skis	- 2	- 1	-	- 1%	- 2	- 1	-	- 1%
Hoist	- 6	- 3	-	- 2%	- 6	- 3	-	- 2%
Emergency floatation gear	- 4	- 2	-	- 1.5%	- 4	- 2	-	- 1.5%
Long footsteps	- 4	- 2	-	- 1.5%	- 4	- 2	-	- 1.5%
Hailers (OP 1810 and 1811)	- 4%		-	- 4%	- 4%		-	- 4%

* No fast or recommended cruise performance penalty is to be taken into account when power limitation is the maximum continuous torque.

A speed performance penalty of – 4 kt (8 km/h) is to be taken in account if the first power limitation is the maximum continuous t4.

2 DETERMINATION OF CORRECTED WEIGHT

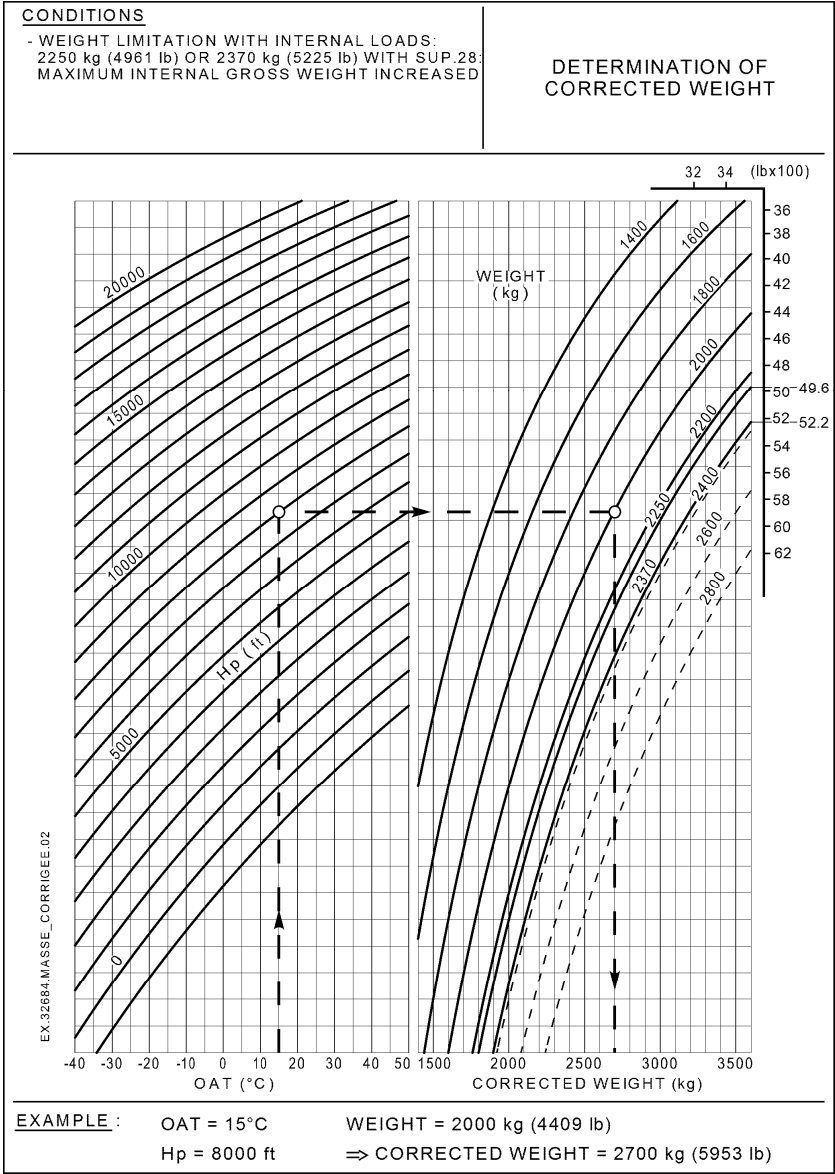


Figure 1

3 TAS/CAS IN FAST CRUISE

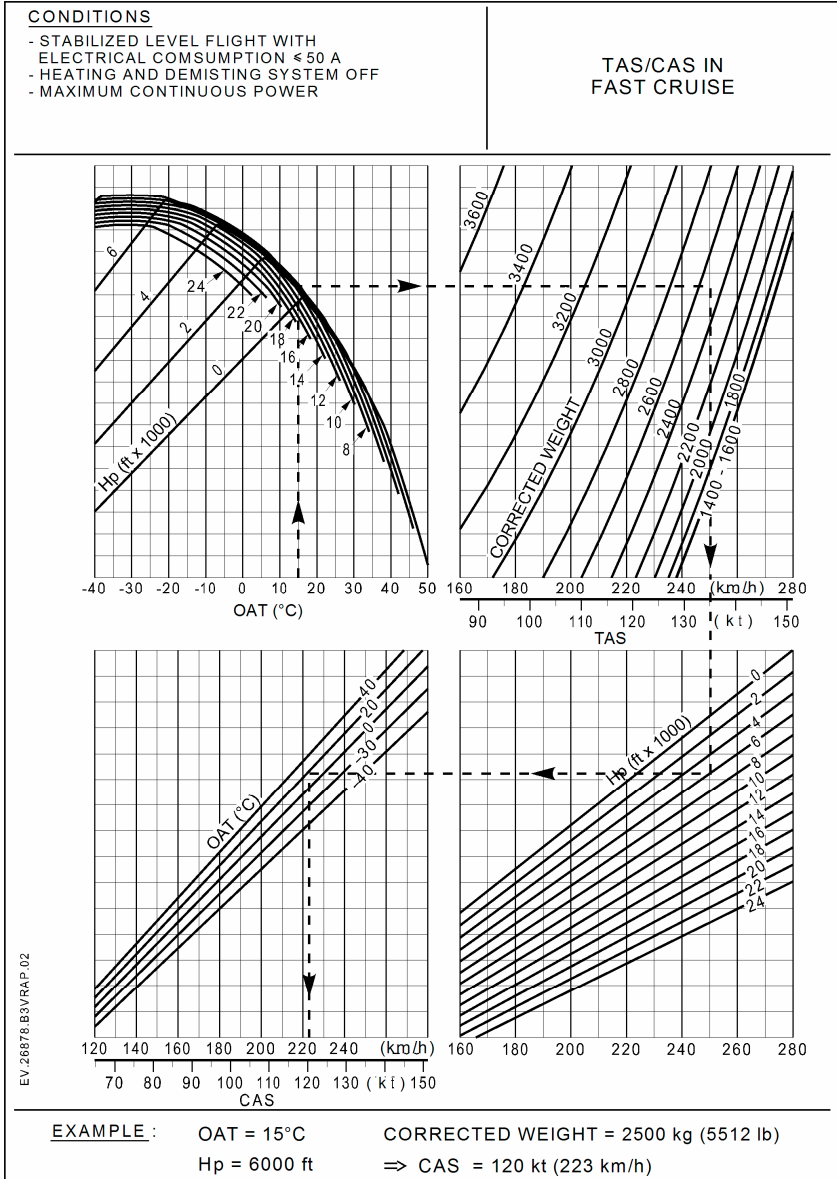


Figure 2

4 DETERMINATION OF CORRECTED WEIGHT

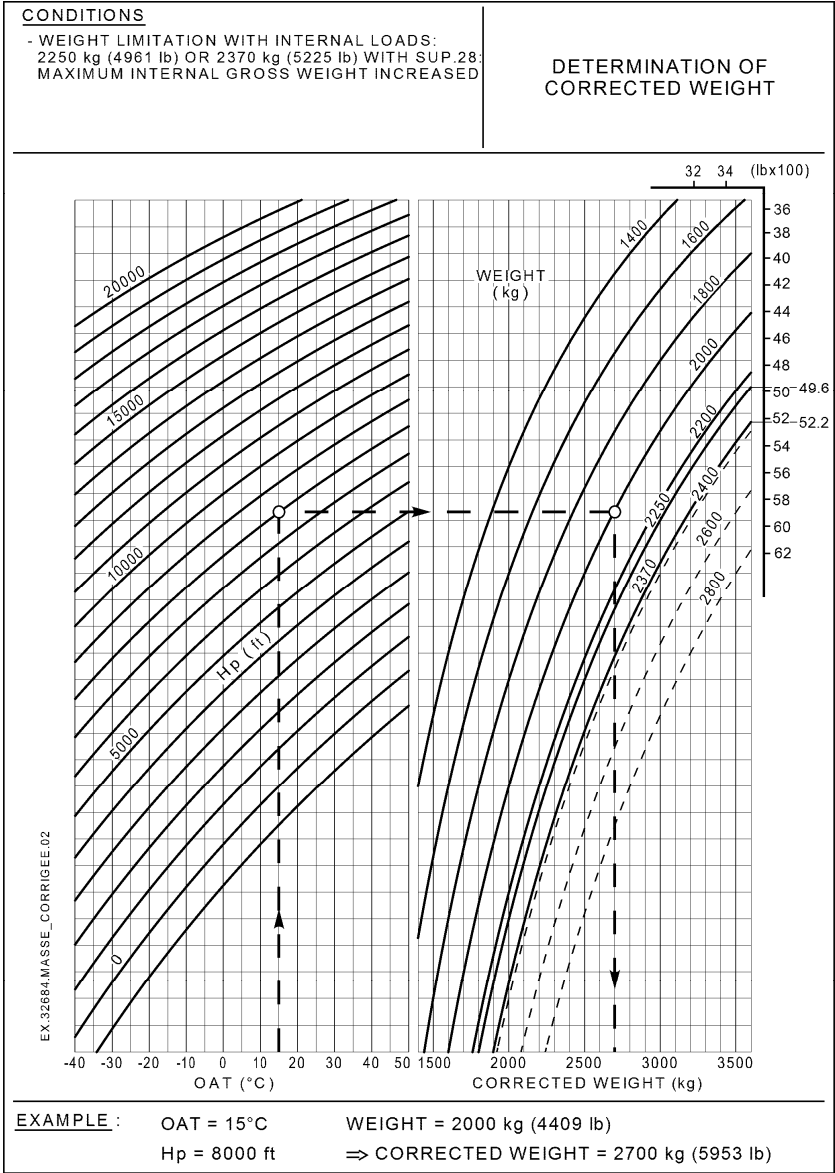


Figure 3

5 TAS/CAS IN RECOMMENDED CRUISE

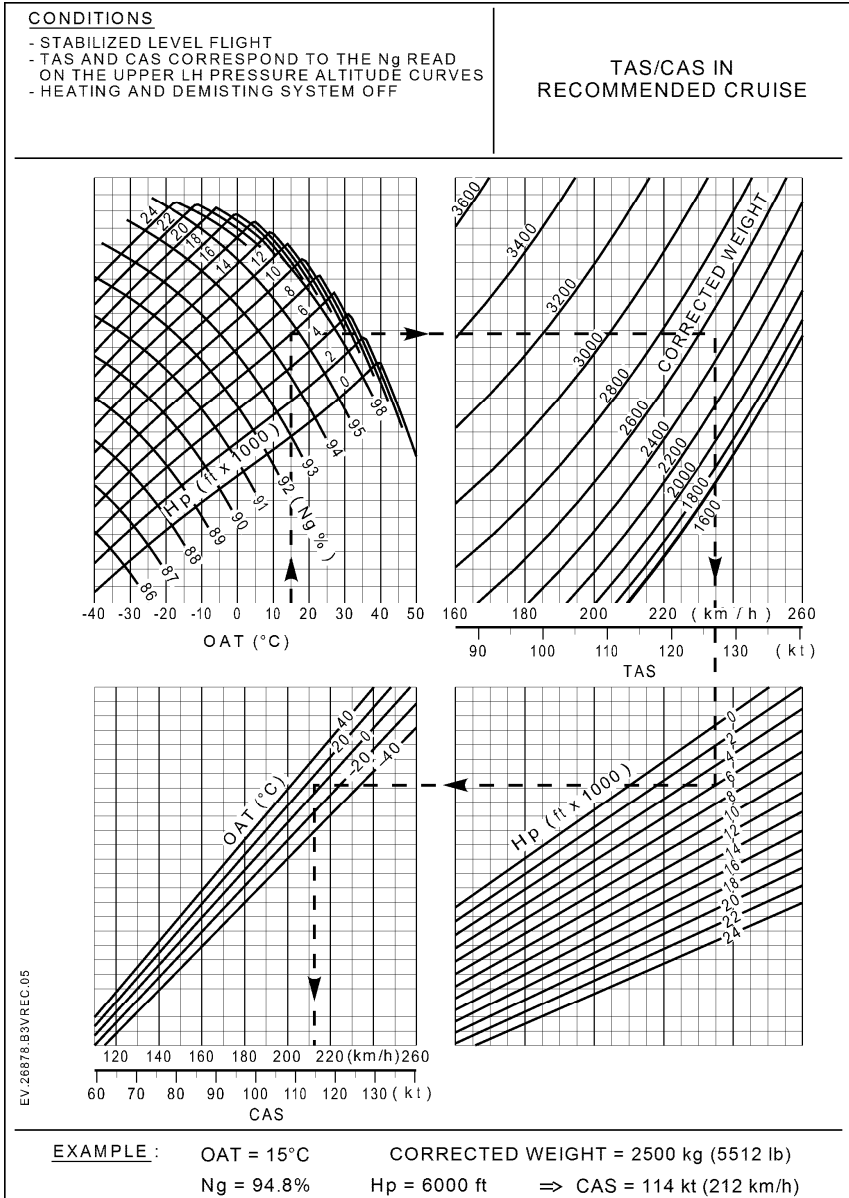
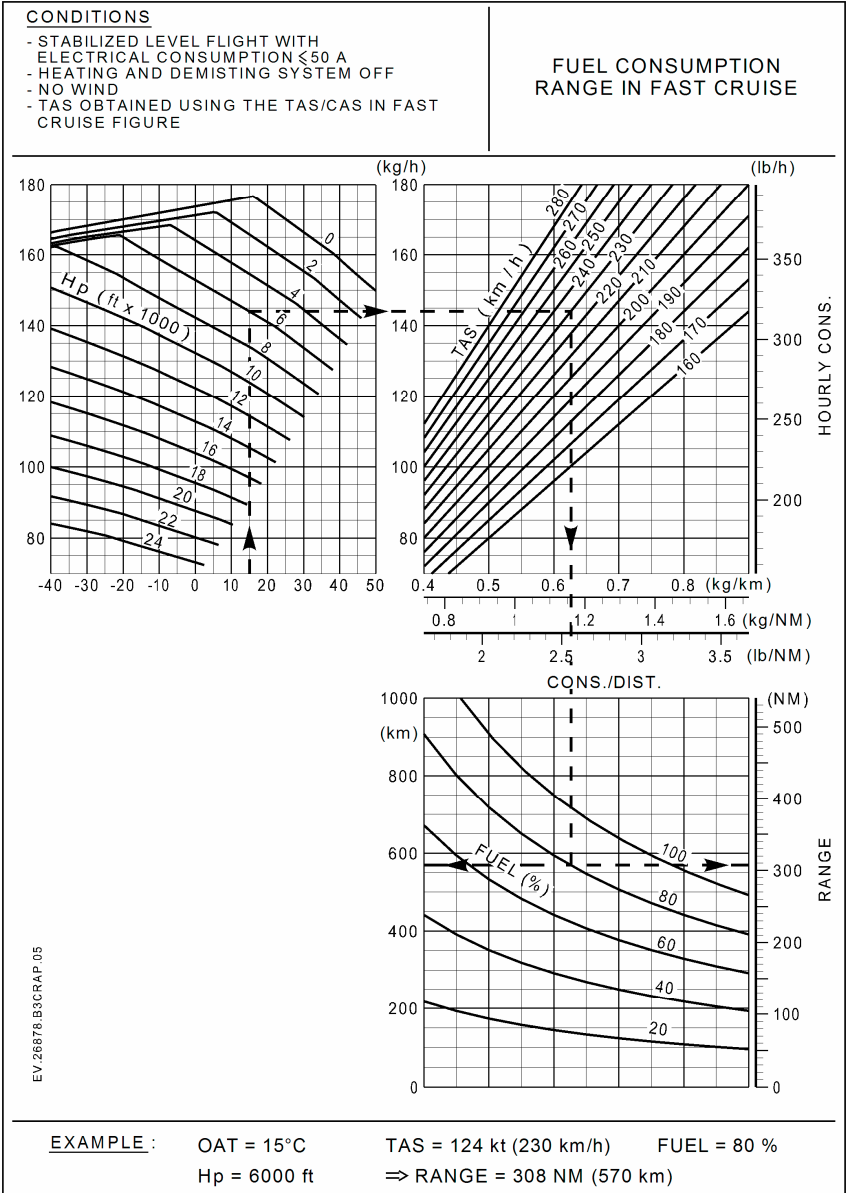


Figure 4

6 FUEL CONSUMPTION-RANGE IN FAST CRUISE



EV 26878-B3CRAP_05

Figure 5

7 FUEL CONSUMPTION-ENDURANCE IN RECOMMENDED CRUISE

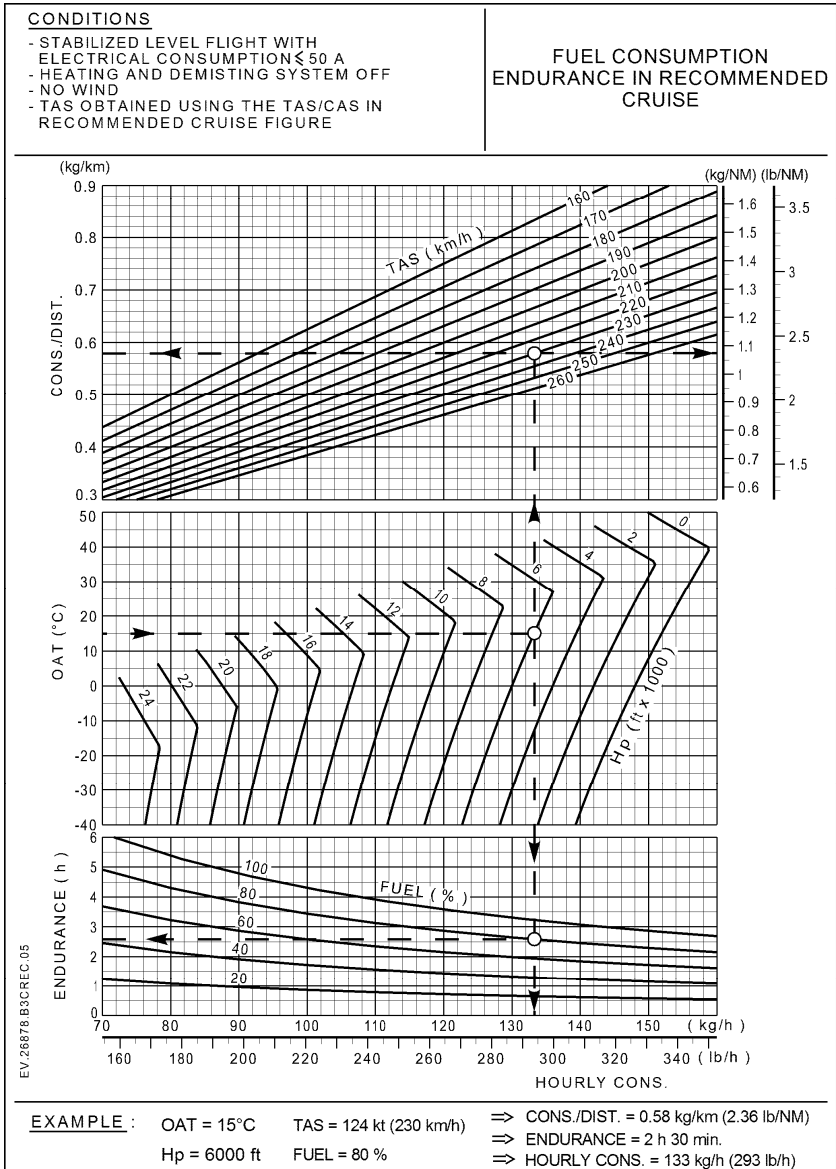


Figure 6

8 RANGE IN RECOMMENDED CRUISE

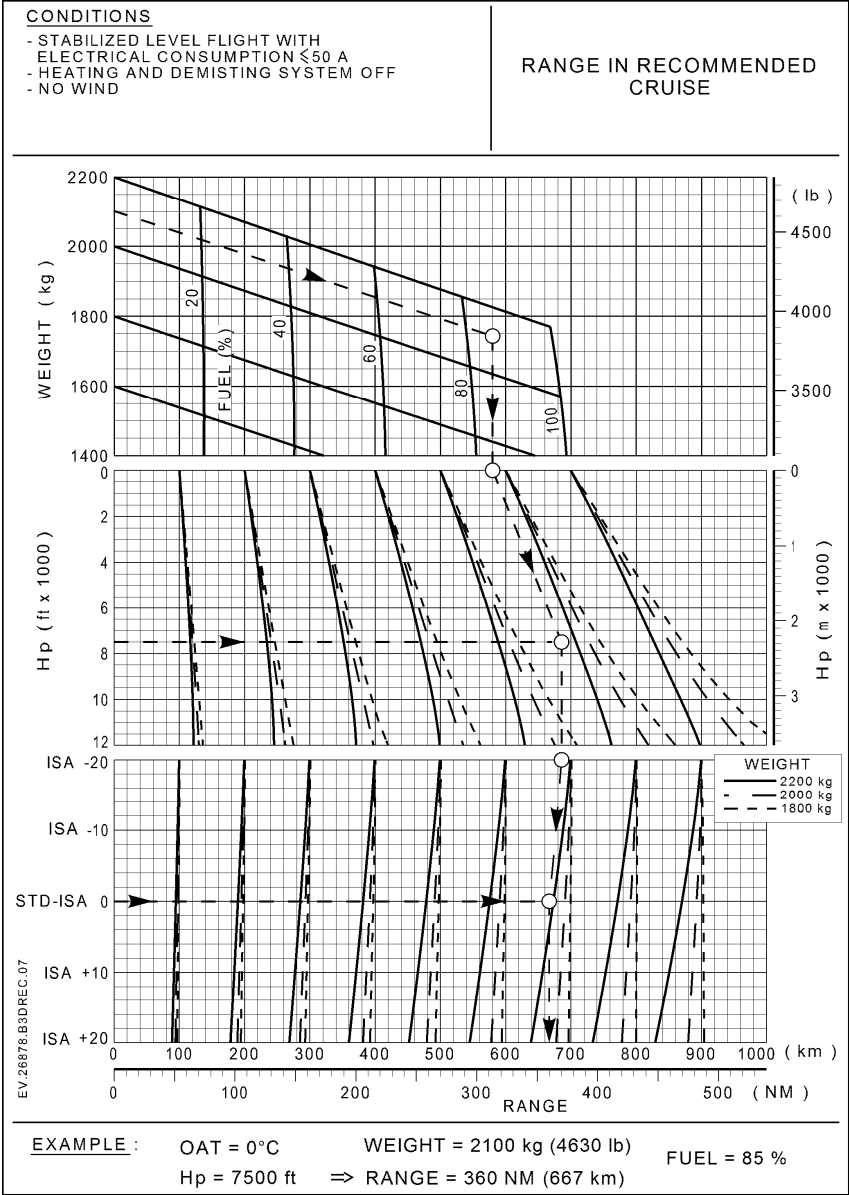


Figure 7

9 FUEL CONSUMPTION - ENDURANCE IN CRUISE AT MINIMUM HOURLY FUEL CONSUMPTION

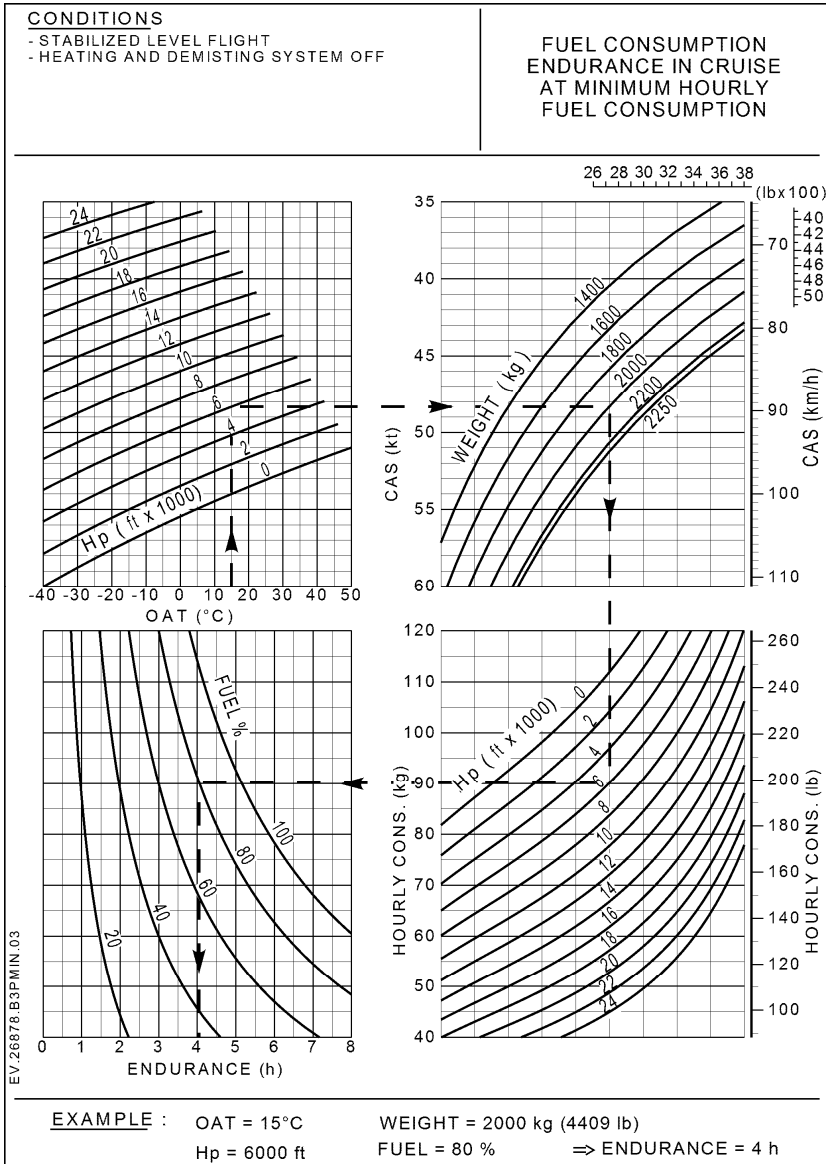


Figure 8

10 LEVEL FLIGHT PERFORMANCE WITH SAND FILTER OPERATING

The level flight performances are modified as follow:

- First limitation : torque limit

PERFORMANCE DATA	FUEL CONSUMPTION	RANGE
Fuel consumption-Range in fast cruise (Figure 13)	+ 1 %	- 1 %

- First limitation : Ng or t4 limit

PERFORMANCE DATA	OAT	
	OAT < ISA + 15 °C	OAT ≥ ISA +15 °C
TAS/CAS in fast cruise (Figure 10)	- 1,1 kt (- 2 km/h)	- 2,7 kt (- 5 km/h)
TAS/CAS in recommended cruise (Figure 12)	0 kt (0 km/h)	- 2,7 kt (- 5 km/h)
Fuel consumption in fast cruise (Figure 13)	0 %	- 1 %
Fuel consumption in recommended cruise (Figure 14)	+ 1 %	- 1 %
Range in recommended cruise (Figure 15)	- 1 %	- 3 %

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12 TAS/CAS IN FAST CRUISE

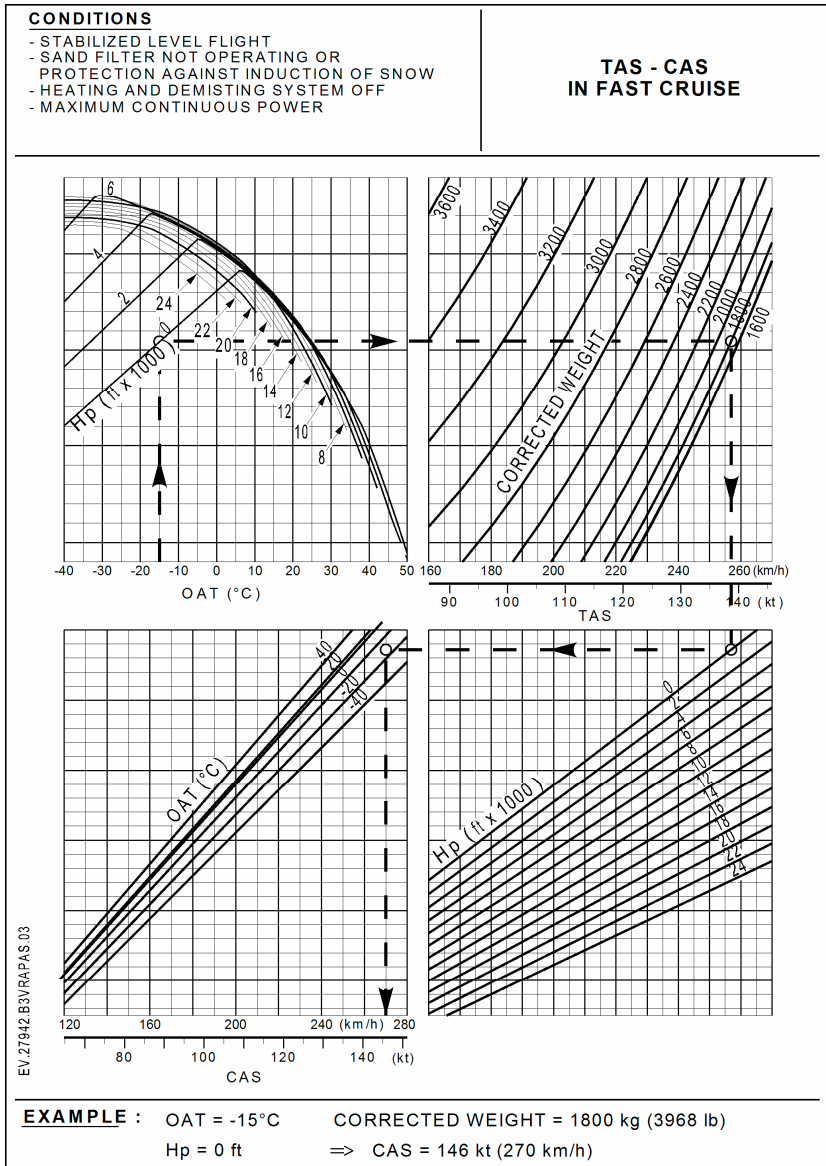


Figure 10

14 TAS/CAS IN RECOMMENDED CRUISE

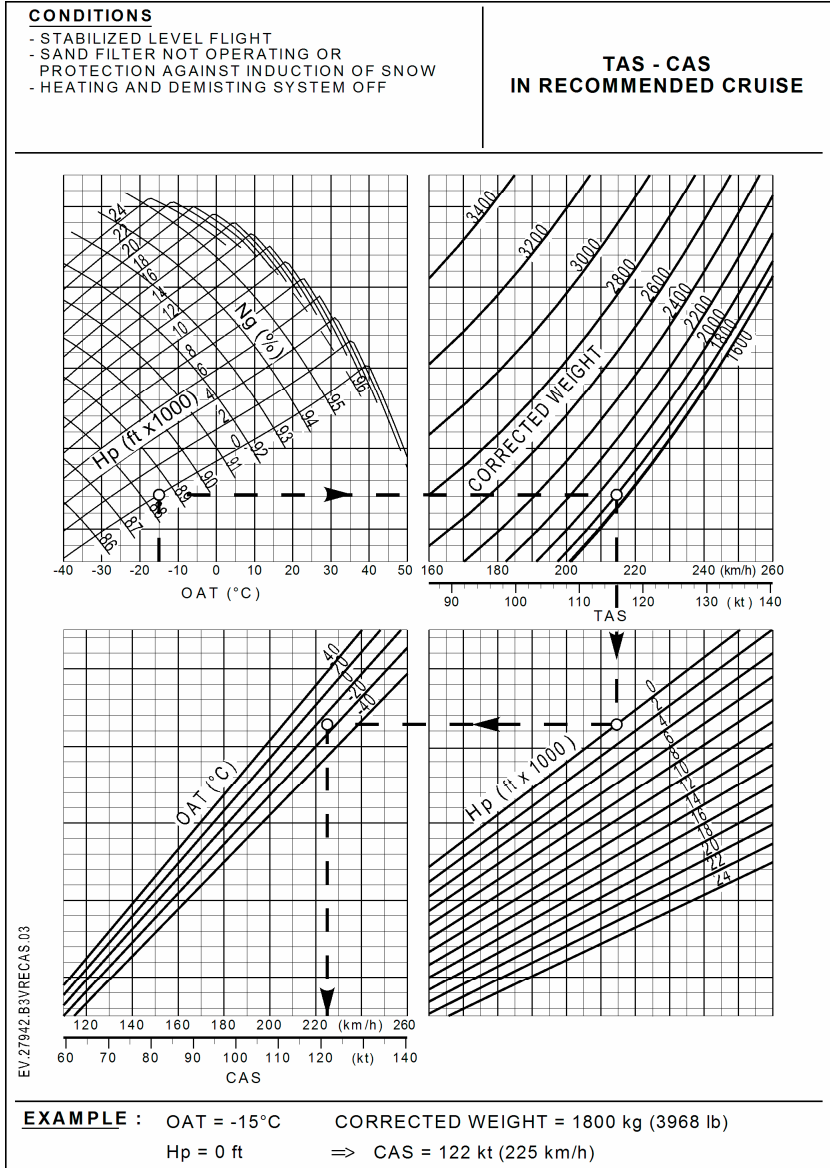
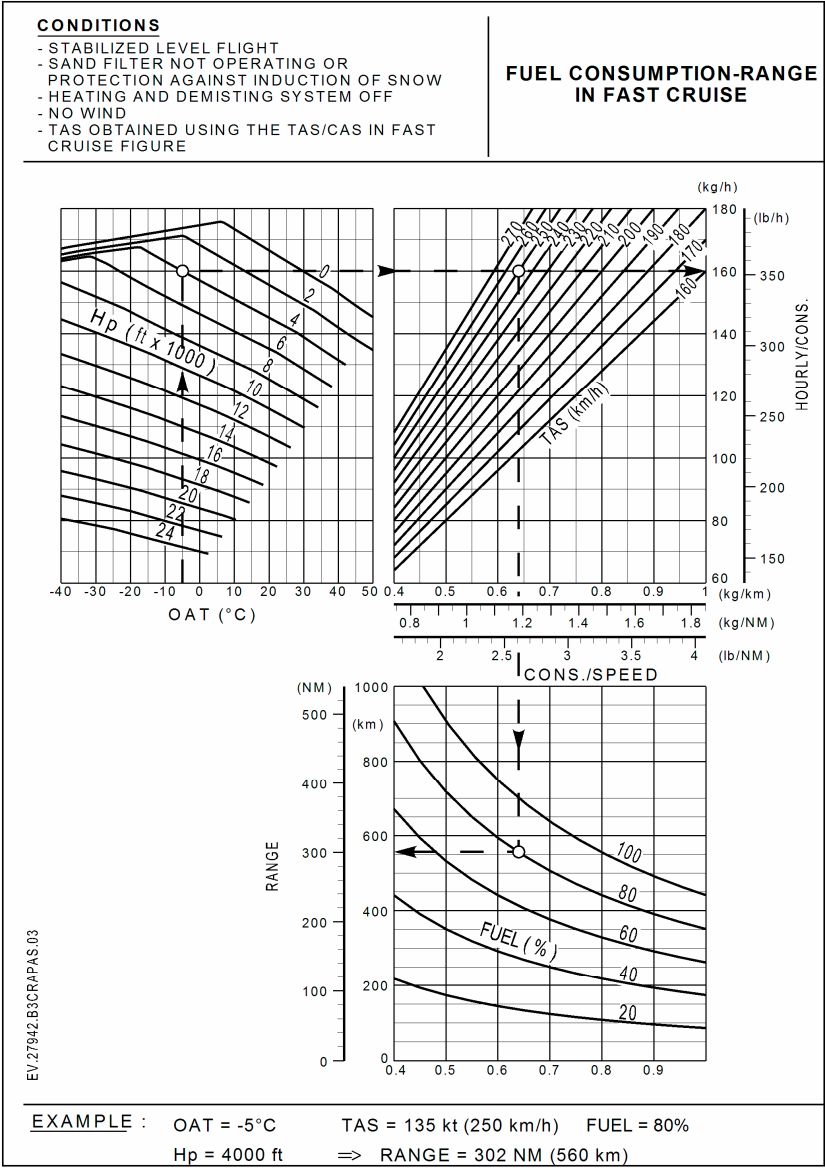


Figure 12

15 FUEL CONSUMPTION - RANGE IN FAST CRUISE



EV 27942.B3CRAPAS.03

Figure 13

16 FUEL CONSUMPTION – ENDURANCE IN RECOMMENDED CRUISE

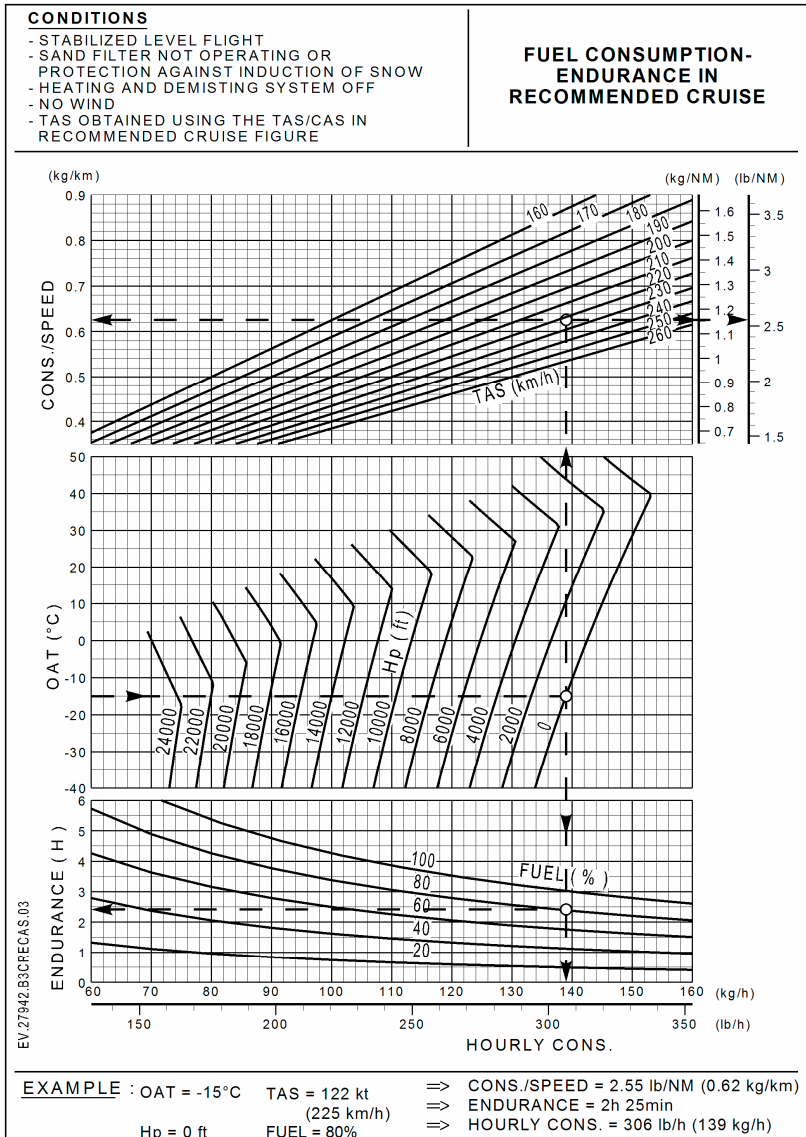


Figure 14

17 RANGE IN RECOMMENDED CRUISE

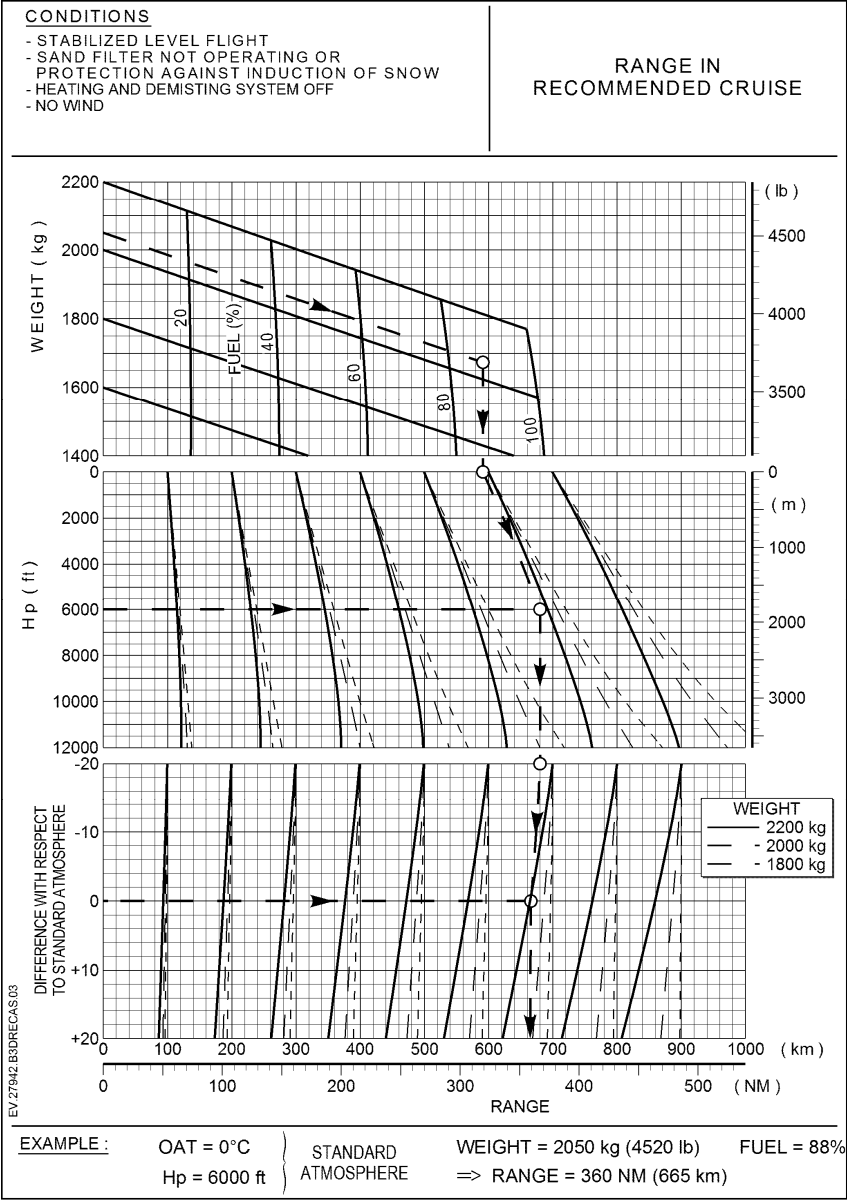


Figure 15

SECTION 6

WEIGHT AND BALANCE

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SECTION 6.1

WEIGHT AND BALANCE

1 GENERAL

The purpose of this section is to provide data for use when evaluating a proposed loading configuration or calculating the weight and center of gravity of an aircraft in service.

2 WEIGHT AND BALANCE

2.1 WEIGHT - STANDARD DEFINITIONS

- Empty Weight (EW)
This corresponds to the sum of the weights of the permanent assemblies and equipment:
 - The vehicle and its power plant.
 - Equipment common to all missions.
 - Lubricants and hydraulic fluids.
 - Unusable fuel.EW then is constant for a given aircraft.
- Equipped Empty Weight (EEW)
This is the sum of:
 - Empty weight (EW).
 - Weight of the specific operational or mission equipment.EEW varies according to the proposed mission.
- All-up Weight (AUW)
This is the sum of:
 - Equipped empty weight (EEW).
 - Crew weight.
 - Payload.
 - Usable fuel weight.
- Maximum Weight
Refer to limitations (SECTION 2).

2.2 CENTER OF GRAVITY CONVENTIONAL TERMS

- The center of gravity is defined by dimensions measured perpendicular to the three basic datum planes. These planes are as follows:
 - A horizontal plane parallel to the cabin floor datum, the Z datum plane, located 2.60 m (102.3 in) above this datum.
 - A vertical plane perpendicular to the cabin floor datum. This Y datum plane is the aircraft plane of symmetry. Dimensions to the left (port) are negative, dimensions to the right (starboard) are positive.
 - A vertical plane perpendicular to the two mentioned above, situated 3.40 m (133.8 in) forward of the center of the main rotor. This is the X datum plane, from which the longitudinal reference stations and CG positions are measured.

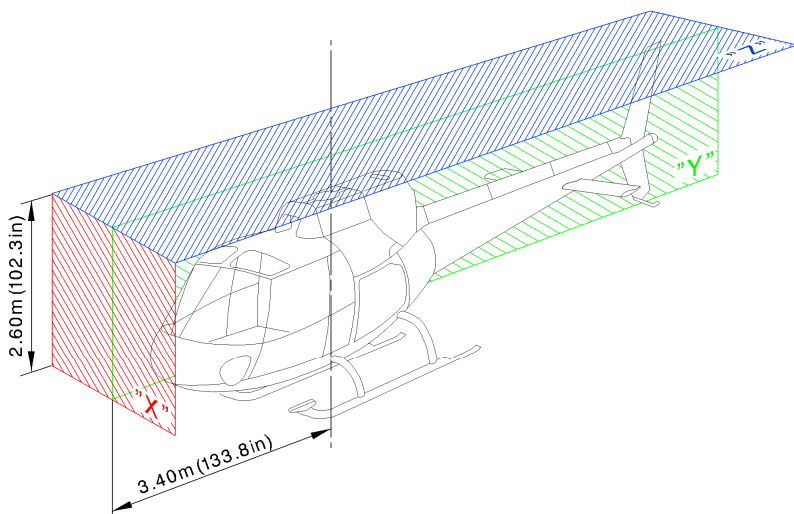


Figure 1: Basic datum planes

NOTE

CG location limits must not be exceeded. Refer to SECTION 2 Limitations.

CAUTION

A CG location which is correct on takeoff may vary during the mission, due to fuel weight reduction or loading variation and therefore exceed acceptable limits.

- Longitudinal CG must be monitored more closely.
- Lateral CG need be considered only in very asymmetrical loading configurations.

SECTION 6.2

LONGITUDINAL CG LOCATION

1 DETERMINATION OF LONGITUDINAL CG LOCATION

- Procedure

The distance from the aircraft center of gravity to the datum plane is obtained using the formula:

$$\frac{\text{Sum of moments}}{\text{Sum of weights}} = \text{CG ready for flight.}$$

- Example: Analysis for a passenger transport mission

• Before takeoff

- 1) Determine the maximum permissible takeoff weight.
- 2) Note the equipped empty weight and the moment.
- 3) Refer to the tables given below to determine loading conditions; totalize weights and moments.
- 4) Calculate the CG location.
- 5) Check that CG falls within permissible limits.

Example:

	kg	m.kg	lb	in.lb
EEW	1200	4272	2646	370837
Front seats	175	271	386	23554
Rear seats	150	381	331	33097
Rear cabin freight	50	112	110	9744
Side cargo hold	100	320	220	27716
Rear cargo hold	50	230	110	19921
Fuel	427	1484	941	128738
TOTAL	2152	7070	4744	613607

CG = $\frac{7070}{2152}$ = 3.285 m	CG = $\frac{613607}{4744}$ = 129.3 in
------------------------------------	---------------------------------------

Longitudinal CG is within the permissible limits.

2 LOADING DATA

- Crew and passengers

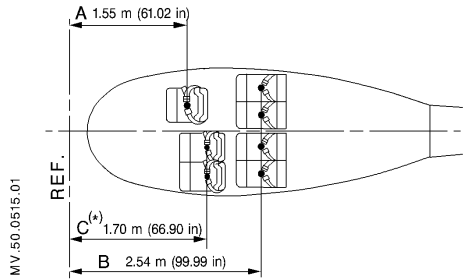
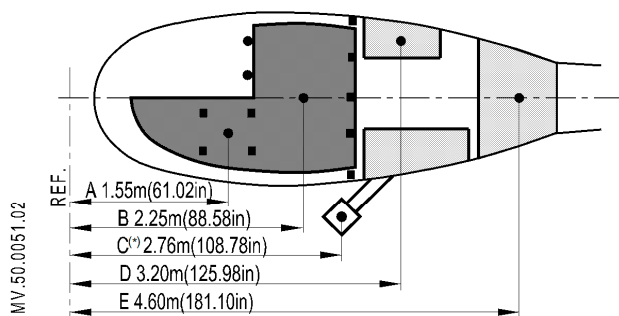


Figure 1: Longitudinal location of seats

METRIC UNITS				OTHER UNITS			
WEIGHT	MOMENT : m.kg			WEIGHT	MOMENT : in.lb		
(kg)	(A)	(B)	(C)*	(lb)	(A)	(B)	(C)*
60	93	152	102	100	6102	9999	6693
80	124	203	136	150	9153	14999	10040
100	155	254	170	200	12204	19998	13386
120	186	305	204	250	15255	24998	16733
140	217	356	238	300	18306	29997	20079
154	239	391	262	340	20717	33948	22713
160	248	406		350	21357	34997	
180	279	457		400	24408	39996	
200	310	508		450	27459	44996	
220	341	559		500	30510	49995	
240	372	610		550	33561	54995	
260		660		600		59994	
280		711		650		64994	
300		762		700		69993	
320		813		750		74993	
340		864		800		79992	
360		914		850		84992	
380		965		900		89991	
400		1016		950		94991	

(*) Optional

- Freight and baggage transport



Eligible area for cabin freight and baggage (Refer to SDS 25-20-00,01 §B.4. Cargo)

Figure 2: Longitudinal location of loads

NOTE

The Cabin floor rings are certified as provisions only. There is no restraint system, cabin freight and baggage configuration certified by Airbus Helicopters.

METRIC UNITS					
WEIGHT (kg)	MOMENT : m.kg				
	(A)	(B)	(C)*	(D)	(E)
10	15.5	22.5	27.6	32	46
20	31	45	55.2	64	92
50	77.5	112.5	138	160	230
70	108.5	157.5	193.2	224	322
80	124	180	220.8	256	368
100	155	225	276	320	
120	186	270	331.2	384	
150	232.5	337.5	414		
200		450	552		
250		562.5			
300		675			
310		697.5			

(*) Optional

- Freight and baggage transport (Con't)

OTHER UNITS					
WEIGHT (lb)	MOMENT: in.lb				
	(A)	(B)	(C)*	(D)	(E)
50	3051	4429	5439	6299	9055
100	6102	8858	10878	12598	18110
150	9153	13287	16317	18897	27165
176	10740	15590	19145	22172	31874
200	12204	17716	21756	25196	
220	13424	19488	23932	27716	
250	15255	22145	27195	31495	
264	16109	23385	28718	33259	
300	18306	26574	32634		
330	20137	29231	35897		
400		35432	43512		
500		44290	54390		
600		53148			

(*) Optional

- Fuel

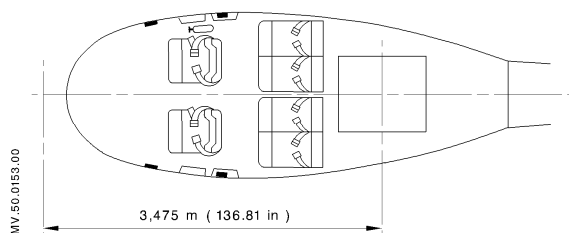


Figure 3: Longitudinal location of fuel

NOTE**Fuel specific gravity: 0.79**

METRIC UNITS										
litre	kg	m. kg		litre	kg	m.kg		litre	kg	m. Kg
25.32	20	69.5		227.88	180	625.5		430.38	340	1181.5
50.64	40	139		253.20	200	695		455.70	360	1251
75.96	60	208.5		278.52	220	764.5		481.01	380	1320.5
101.28	80	278		303.84	240	834		506.33	400	1390
126.60	100	347.5		329.16	260	903.5		531.64	420	1459.5
151.92	120	417		354.48	280	973		540	427	1484
177.24	140	486.5		379.8	300	1042.5				
202.56	160	556		405.06	320	1112				

OTHER UNITS								
US gal	UK gal	lb	in.lb		US gal	UK gal	lb	in.lb
7.58	6.32	50	6841		83.43	69.47	550	75246
15.17	12.63	100	13681		91.02	75.79	600	82086
22.75	18.95	150	20522		98.60	82.11	650	88927
30.34	25.26	200	27362		106.19	88.42	700	95767
37.92	31.58	250	34203		113.77	94.73	750	102608
45.51	37.89	300	41043		121.36	101.05	800	109448
53.09	44.21	350	47884		128.94	107.37	850	116289
60.68	50.52	400	54724		136.53	113.68	900	123129
68.26	56.84	450	61565		142.60	118.74	940	128601
75.85	63.15	500	68405					

3 CG CHARTS

The following charts (metric and other units) are used to easily determine the aircraft center of gravity. When the point obtained is close to the limits, it should be confirmed by calculations.

Example:

		Item on chart
- The weighing operation locates the CG at 3.56 m (140.15 in) for an EEW of 1200 kg (2646 lb)	:	①
- 2 front seats used	: 175 kg (386 lb)	: ②
- 2 rear seats used	: 150 kg (331 lb)	: ③
- Freight on the rear seat	: 50 kg (110 lb)	: ④
- Freight in the lateral hold	: 100 kg (220 lb)	: ⑤
- Freight in the rear hold	: 50 kg (110 lb)	: ⑥
- Zero fuel weight	: 1725 kg (3803 lb)	: ⑦
- Fuel	: 427 kg (941 lb)	
- TOTAL WEIGHT	: 2152 kg (4744 lb)	: ⑧

These charts are designed so that the variations in fuel weight make the CG move along a vertical line.

The takeoff weight is 2152 kg (4744 lb) (item ⑧) with a center of gravity at 3.285 m (129.3 in).

The longitudinal CG is within the permissible limits.

- During the flight, after consuming 300 kg (661 lb) of fuel (item ⑨), the center of gravity will be 3.255 m (128.1 in).

The weight and CG limits are given in LIMITATIONS (SECTION 2) and may be modified by the Supplements corresponding to the optional items installed.

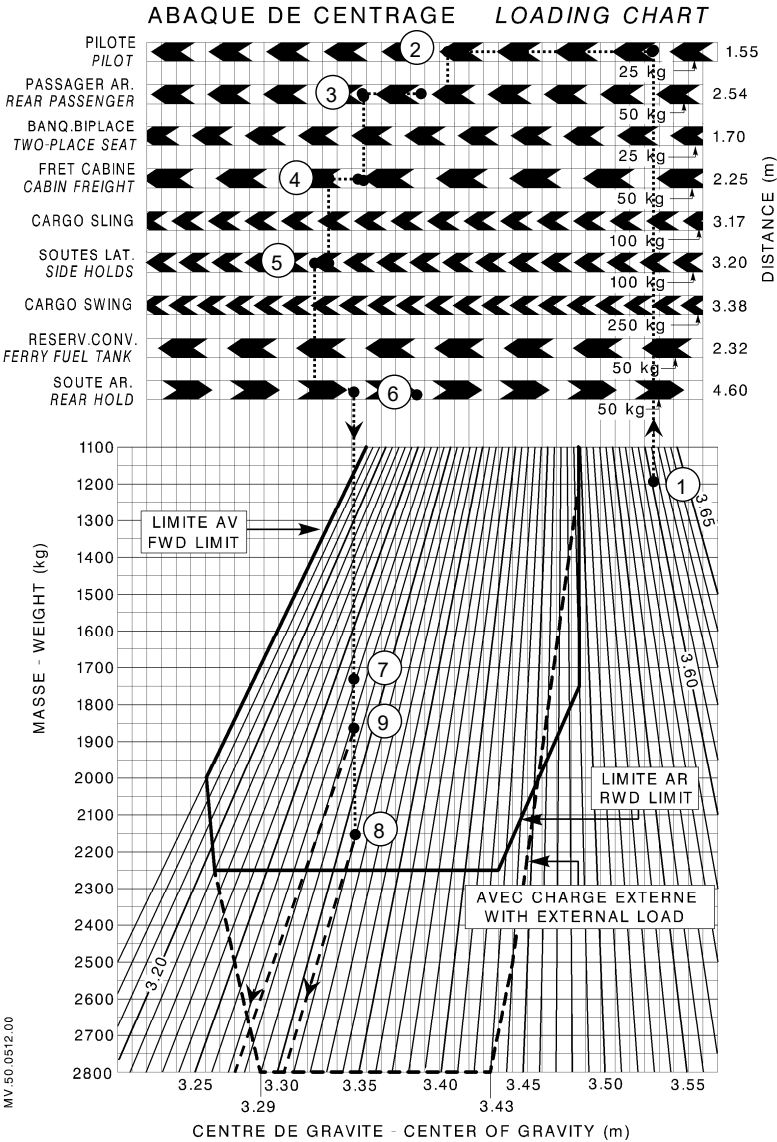


Figure 4: Center of gravity

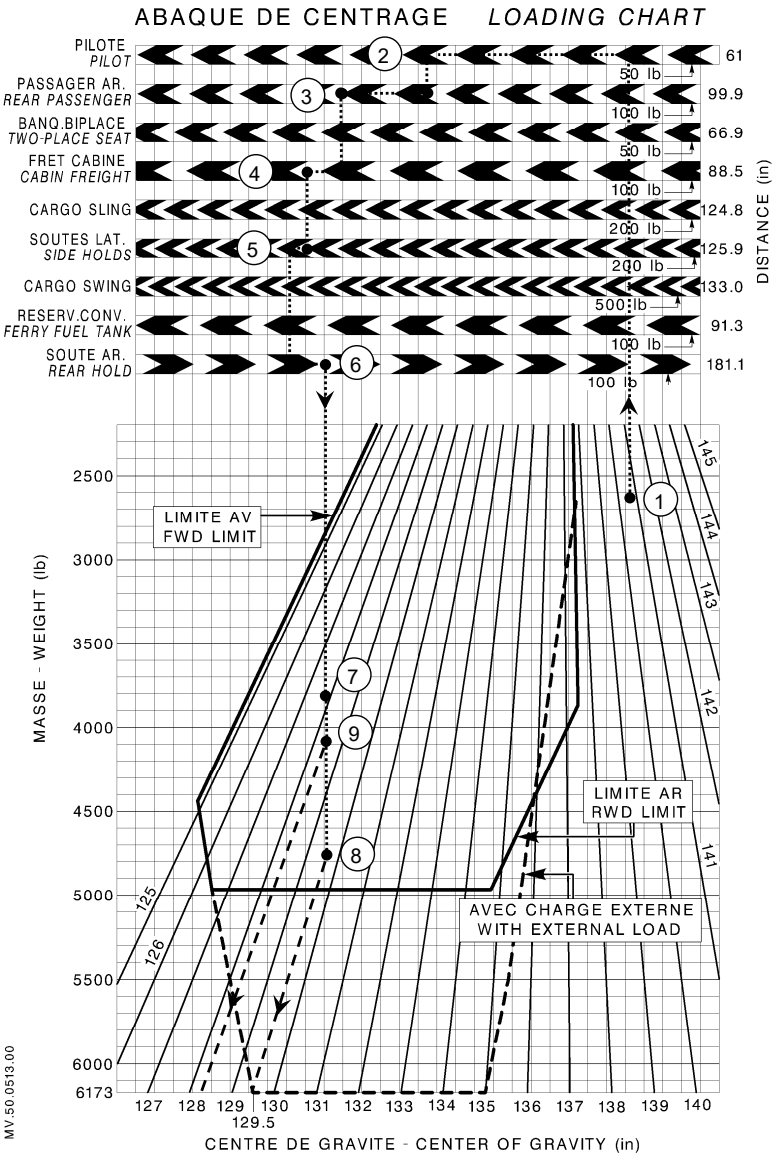


Figure 5: Center of gravity (cont'd)

SECTION 6.3

LATERAL CG LOCATION

The tables below give the lateral CG positions for different weights and their moments with respect to the Y plane (positive dimensions on the right, negative dimensions on the left).

1 DETERMINATION OF LATERAL CG LOCATION

The computation method is the same as that used for determining the longitudinal CG location (SECTION 6.2 paragraph 1).

Add weights and moments to the aircraft empty weight and moment referring to the preceding pages.

Lateral CG location values during the mission shall fall within the permissible limits.

Example:

- Before flight:

	kg	m.kg	lb	in.lb
EEW	1200	9.70	2646	842
Front seats	175	0	386	0
Middle right rear seat	75	15.5	165	1345
Full right rear seat	75	46.5	165	4034
Left rear cabin freight	50	- 21	110	- 1819
LH side cargo hold	80	- 45	176	- 3880
RH side cargo hold	20	11	45	970
Rear cargo hold	50	0	110	0
Fuel	427	0	941	0
TOTAL	2152	16.7	4744	1492

CG = $\frac{16.7}{2152}$ = 0.0078 m	CG = $\frac{1492}{4744}$ = 0.32 in
-------------------------------------	------------------------------------

Lateral CG is within the permissible limits.

2 LOADING DATA

- Crew and passengers

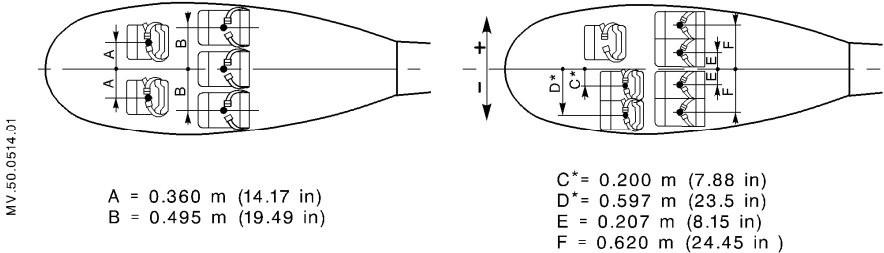


Figure 1: Lateral location of seats

WEIGHT (kg)	METRIC UNITS					
	MOMENT: m.kg					
	(A)	(B)	(C)*	(D)*	(E)	(F)
50	18	25	10	30	10	31
60	22	30	12	36	12	37
70	25	35	14	42	14	43
80	29	40	16	48	17	50
90	32	45	18	54	19	56
100	36	50	20	60	21	62
120	43	59	24	72	25	74

WEIGHT (lb)	OTHER UNITS					
	MOMENT: in.lb					
	(A)	(B)	(C)*	(D)*	(E)	(F)
100	1417	1949	788	2350	815	2445
120	1700	2339	946	2820	978	2934
140	1984	2729	1103	3290	1141	3423
160	2267	3118	1261	3760	1304	3912
180	2551	3508	1418	4230	1467	4401
200	2834	3898	1576	4700	1630	4890
220	3117	4288	1734	5170	1793	5379
240	3401	4678	1891	5640	1956	5868
260	3684	5067	2049	6110	2119	6357

(*) Optional

- Baggage compartment and stretchers

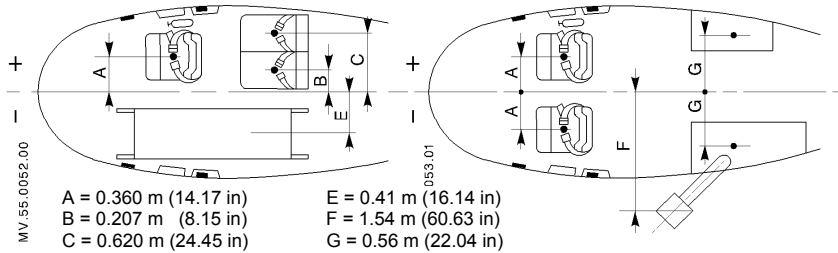


Figure 2: Lateral location of stretchers and loads

METRIC UNITS						
WEIGHT	MOMENT: m.kg					
(kg)	(A)	(B)	(C)	(E)*	(F)*	(G)
50	18	10	31	21	77	28
60	22	12	37	25	92	34
70	25	14	43	29	108	39
80	29	17	50	33	123	45
90	32	19	56	37	139	50
100	36	21	62	41	154	56
120	43	25	74	49	185	67
136	49	28	84	56	209	
160					246	
180					277	
204					314	

OTHER UNITS						
WEIGHT	MOMENT: in.lb					
(lb)	(A)	(B)	(C)	(E)*	(F)*	(G)
100	1417	815	2445	1634	6079	2189
120	1700	978	2934	1961	7295	2627
140	1984	1141	3423	2288	8511	3065
160	2267	1304	3912	2614	9726	3502
180	2551	1467	4401	2941	10942	3940
200	2834	1630	4890	3268	12158	4378
220	3117	1793	5379	3595	13374	4816
240	3401	1956	5868	3922	14590	5254
260	3684	2119	6357	4248	15805	5691
300	4251	2445	7335	4902	18237	
350					21277	
400					24316	
450					27356	

(*) Optional

SECTION 6.4

WEIGHT AND MOMENT OF EQUIPMENT ITEMS

The following list covers the equipment items. It gives the approximate weight and moment of the removable components.

EQUIPMENT ITEM	WEIGHT		ARM		MOMENT	
	kg	lb	m	in	m.kg	in.lb
Aircraft ground tool kit • Twin handling wheels • Helicopter hoisting ring	39.5	87.8	-	-	-	-
Aircraft tool kit • Static pressure port cover • Pitot tube cover • Mooring ring • Document bag • Storage bag • Main rotor blade socks • Engine tail pipe cover • Air intake cover • Tail rotor blade locking tool	7.1	15.7	-	-	-	-
Air ambulance system • Lower (with stretcher) • upper (with stretcher) • Stretcher	15.61	34.41	1.77	69.68	27.63	2397.69
	17.2	37.92	1.91	75.2	32.85	2851.58
	15.1	33.29	1.77	69.68	26.73	2319.76
Arriel engine flushing system	0.65	1.43	4.08	160.63	2.65	230.18
Axe	1.18	2.6	1.75	68.9	2.07	179.23
Battery • 1st battery • 2nd battery (very-cold weather starting system)	17.3	38.1	3.99	157.09	69.0	5985.13
	14.38	31.7	3.99	157.09	57.38	4979.75
Cabin fire extinguisher	1.96	4.32	1.7	66.93	3.33	289.2
Cable cutter	6.95	15.32	1.03	40.55	7.16	621.23
Cargo sling 750 kg	3.19	7.03	3.45	135.83	11.01	954.88
Cargo swing 1400 kg • Release unit • Fixed release unit in rotation • "ON BOARD" release unit	17.51	38.6	3.45	135.83	60.41	5243.04
	16.85	37.15	3.45	135.83	58.13	5046.08
	13.38	29.5	3.36	132.28	44.96	3902.26
"DATCON" elapsed time indicator	0.14	0.31	1.1	43.31	0.15	13.43

EQUIPMENT ITEM	WEIGHT		ARM		MOMENT	
	kg	lb	m	in	m.kg	in.lb
Doors						
Standard door and small-sized door on LH side	14.62	32.23	2.38	93.7	34.8	3019.95
• Standard door without improved visibility	9.98	22	2.38	93.7	23.75	2061.4
• Small-sized door + window	4.64	10.23	2.38	93.7	11.04	958.55
Sliding door and reduced door on RH side	16.31	35.96	2.38	93.7	38.82	3369.45
• Sliding door with fixed window	8.46	18.65	2.38	93.7	20.13	1747.51
• Reduced door + windows	7.85	17.31	2.38	93.7	18.68	1621.95
Dual controls	3.44	7.58	1.66	65.35	5.71	495.63
Emergency floatation gear	64.14	141.4	3.36	132.28	215.51	18704.39
• Floatation gear kit	51.22	112.92	3.36	132.28	172.1	14937.06
• Cylinder (*2)	12.92	28.48	3.36	132.28	43.41	3767.33
Emergency locator transmitter KANAD 121 AF-H						
• ELT fixed parts	1.86	4.1	2.68	105.51	4.98	432.65
• ELT removable parts	1.09	2.4	4.28	168.31	4.66	404.44
KANAD 406 AF-H						
• ELT fixed parts	1.86	4.1	2.68	105.51	4.98	432.65
• ELT removable parts	1.19	2.62	4.28	168.31	5.09	441.55
External mirror						
RH external mirrors	2.83	6.24	0.46	18.11	1.3	112.99
• Fixed part installation	0.2	0.44	0.56	22.05	0.11	9.72
• Removable part installation	2.63	5.8	0.46	18.11	1.21	105.04
RH deiced external mirrors	2.25	4.96	0.48	18.9	1.08	93.74
• Fixed part installation	0.45	0.99	0.56	22.05	0.25	21.87
• Removable part installation	2.25	4.96	0.48	18.9	1.08	93.74
Ferry fuel tank	27.62	60.89	2.34	92.13	64.63	5609.8
• Tank	26.01	57.34	2.34	92.13	60.86	5282.73
• Complement for installation	1.61	3.55	2.34	92.13	3.77	327.06
First aid kit						
• On pilot seat	3.32	7.32	1.65	64.96	5.48	475.46
• Under passenger rear seat	2.82	6.22	2.45	96.46	6.91	599.66
Footstep "DART"						
• On high landing gear	8.94	19.71	2.24	88.19	20.03	1738.11
• On low landing gear	8.54	18.83	2.24	88.19	19.13	1660.34

EQUIPMENT ITEM	WEIGHT		ARM		MOMENT	
	kg	lb	m	in	m.kg	in.lb
Hoist						
Electric hoist "AIR EQUIPMENT" (136 kg – 300 lb) with jib, hoist, grip, pulley-block, safety belt, hand cable cutter.	38.22	84.26	2.8	110.24	107.02	9288.4
Electric hoist "BREEZE" (204 kg – 450 lb) with jib, faired hoist, grip, pulley-block, safety belt, hand cable cutter.	51.89	114.4	2.8	110.24	145.29	12610.55
Drip pan	11.27	24.85	2.86	112.6	32.33	2797.58
Landing gear						
• Low landing gear	47.11	103.86	3.34	131.5	157.35	13656.89
• High landing gear	57.77	127.36	3.34	131.5	192.95	16747.16
• Wear plates	1.77	3.9	3.18	125.2	5.63	488.53
Lights						
• "LOCATOR" search light	10.1	22.27	0.92	36.22	9.29	806.49
• Swivelling and retractable landing light	5.25	11.75	1.9	74.68	9.98	865.44
Sand filter						
• Removable parts	8.84	19.49	4.17	164.17	36.86	3199.67
Seats						
• High front seat	10.6	23.37	1.55	61.02	16.43	1426.03
• Low front seat	7.3	16.09	1.55	61.02	11.32	982.08
• Comfort front seat with energy absorption (one seat)	11.44	25.22	1.78	70.08	20.36	1767.41
• Standard front seat with energy absorption (one seat)	12.55	27.67	1.78	70.08	22.34	1939
• Two-passengers front seat	13.33	29.39	1.7	66.93	22.66	1966.85
• Four-passengers rear seats including cushions	21.1	46.52	2.54	100	53.59	4651.67
• Three-passengers rear seat (complete with armrests)	26.2	57.76	2.54	100	66.55	5776
Skis "SURFAIR"	26.88	59.26	4.23	166.54	113.7	9869

SECTION 6.5

WEIGHING

1 PRELIMINARY ACTIONS

- The weighing operation must be carried out in a closed shelter to avoid any errors caused by the wind.
- Clean the aircraft carefully before weighing.
- If the weighing operation is used to determine CG location, level the aircraft before weighing (transmission deck set to 2°, nose down).
- In principle, all equipment items included in the aircraft's empty weight must be installed. Draw up a brief inventory of those equipment items and include it in the weighing record.
- All weighing instruments must be checked for correct "zero" setting before use. It is important that the weighing instruments are placed on suitably levelled ground for correct measurement.
- Unless otherwise specified, the fuel cell must be drained.

NOTE

A certain quantity of fuel remaining in the system is defined by the Airworthiness Regulations as the "unusable fuel", i.e. the quantity of fuel below which anomalies in fuel supply begin to appear in certain aircraft attitudes and/or flying maneuvers.

The weight (and moment) of the "unusable fuel" is indicated in the specifications, airworthiness sheets, etc.

2 WEIGHING PROCEDURES

The aircraft is weighed and the CG location is determined as follows:

- After the inventory has been made and the checks have been performed, level the aircraft by means of the appropriate markings and using a clinometer, with the landing gear off the ground.
- Check that the fuel cell is drained
- The distances of the jacking points are defined by the manufacturer when the aircraft is assembled on the jig.
- Record the weight measured at each jacking point.
- Compute the moment by multiplying the weight by the distance of the corresponding jacking point.
- Calculate the sum of the moments.
- Divide the total moment by the total weight to obtain the aircraft CG location.

The empty weight (and CG) must include the weight of unusable fuel.

As a general rule, these values are calculated from the "aircraft dry" weight.

WEIGHING RECORD SHEET

AIRCRAFT /VERSION			
SERIAL No			
CUSTOMER No			
DATE:			
SIGNATURE		REMARKS	
JACK POINTS	WEIGHT (kg) or (lb)	ARM (m) or (in) X Y	
LH FORWARD (A1)		2.31 m 90.94 in	-0.3985 m -15.689 in
RH FORWARD (A2)		2.31 m 90.94 in	+0.3985 m +15.689 in
TOTAL LH.FWD + RH FWD (A)			
AFT (A3)		5.072 m 199.68 in	0
TOTAL WEIGHT			MX= MY=
LONGITUDINAL CG..... $X = \frac{MX}{\text{TOTAL WEIGHT}} =$			
LATERAL CG $Y = \frac{MY}{\text{TOTAL WEIGHT}} =$			
WEIGHING CORRECTIONS AND LONGITUDINAL CG CORRECTIONS			
EQUIPMENT ITEMS	WEIGHT (± kg) or (± lb)	ARM (m) or (in)	MOMENT (± m.kg) or (± lb.in)
EMPTY WEIGHT OF EQUIPPED AIRCRAFT			
CORRECTED WEIGHT			
CORRECTED MOMENT			
CORRECTED LONGITUDINAL CG			

INVENTORY AT TIME OF AIRCRAFT WEIGHING			
FURNISHINGS		RADIO COMMUNICATION	
Aircraft tool kit		RADIO NAVIGATION	
First aid kit		AVIONICS	
Flight manual		VHF/1	
Fire extinguisher		VHF/2	
OPTIONAL EQUIPMENT		VHF/FM	
Enlarged instrument panel		UHF	
Retractable right landing light		VOR/1	
Swivelling left landing light		VOR/2	
Battery		Homing	
2 nd battery		Marker	
Cabin heating system		ADF/1 Auto. Direction Finder	
Dual controls		Transponder	
Sand filter		ICS 2 lanes	
Hailer installation		Public Address System	
Fuel flowmeter		Encoding altimeter	
Emergency floatation gear		Radio altimeter	
Cargo sling		Gyro. horizon	
Cargo swing		Stand-by horizon	
Rear-view mirror		Gyro compass	
Hoist		DME	
Sliding door (LH)		HF/SSB	
Sliding door (RH)		Emergency locator	
Ambulance hooks		Radar	
Cabin trimming (standard)		Automatic pilot	
Cabin trimming (comfort)		Global position system (GPS)	
Passenger seats 4 places		Searchlight LOCATOR/SPECTROLAB	
Carpet		Forward high seat + harness	
Two-place bench seat		NADIR	
Skis		DOPPLER	
Cable-cutter		TACAN	
High type landing gear		IFF	
Low type landing gear		Coupleur	
Ballast for balancing			
3 main blades			
Stowage net bay			
Pilot windshield wiper			
Copilot windshield wiper		MISCELLANEOUS	
Engine washing installation		Fuel	
Air conditioning system			
Hourmeter			

(0) : not fitted

(1) : fitted

(FP) : Fixed provisions

SECTION 7

DESCRIPTION AND SYSTEMS

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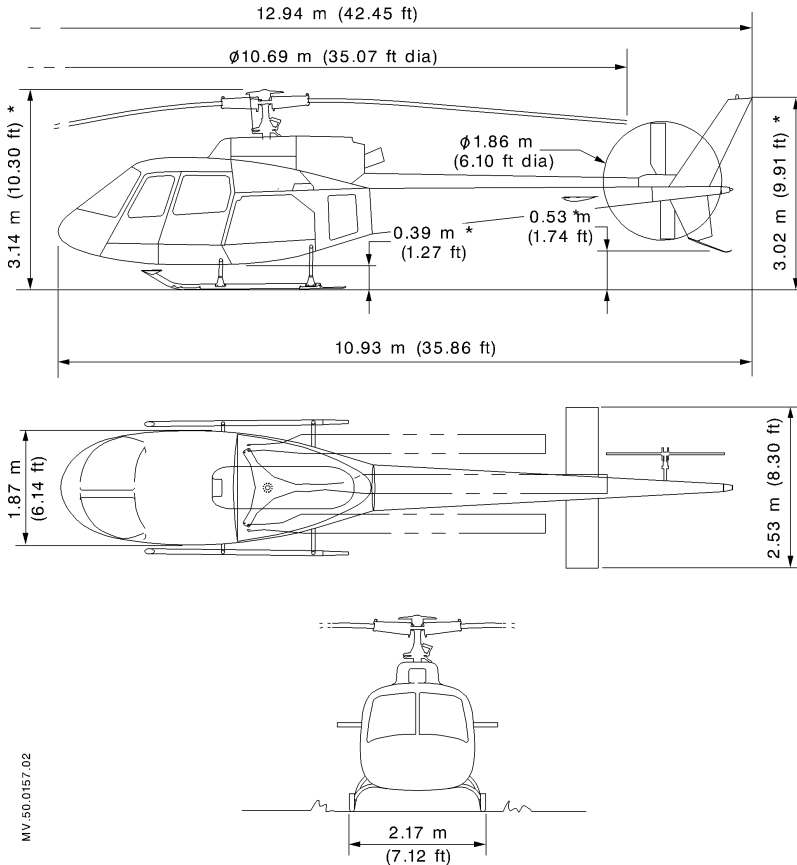
7.14 APPAREO VISION 1000

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SECTION 7.1

MAIN AIRCRAFT DIMENSIONS

1 AIRCRAFT DIMENSIONS



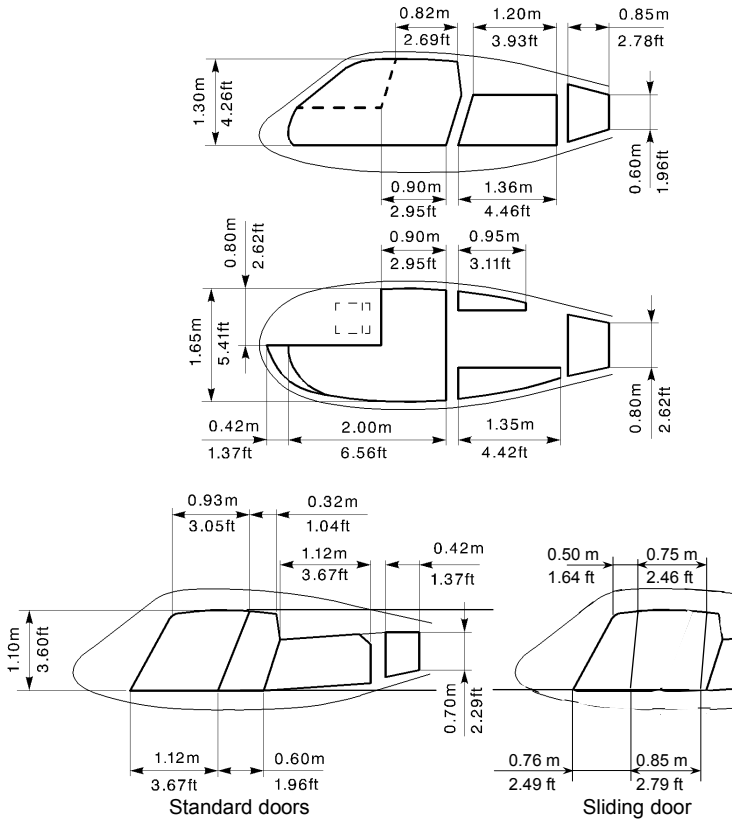
(*) Add 0.20 m (0.65 ft) when aircraft is fitted with high landing gear.

NOTE

The values which vary according to weight are given at the maximum weight.

Figure 1: Three-view drawing

MV.55.0046.01



LH HOLD
Area 0.43 m ² (4.62 ft ²)
Volume 0.235 m ³ (8.29 ft ³)

CABIN
Area 2.60 m ² (27.98 ft ²)
Volume 3.00 m ³ (105.94 ft ³)

REAR HOLD
Area 0.55 m ² (5.92 ft ²)
Volume 0.565 m ³ (19.94 ft ³)

RH HOLD
Area 0.35 m ² (3.76 ft ²)
Volume 0.200 m ³ (7.06 ft ³)

Figure 2: Access doors and compartments

2 DESCRIPTIVE DATA

2.1 ENGINE

- Number	: 1	- Minimum shaft available power (ISA, at sea level):
- Manufacturer	: TURBOMECA	- Max. takeoff power rating (MTP): 557 kW (747 SHP)
- Model	: ARRIEL	- Max. continuous power rating (MCP): 543 kW (728 SHP)
- Type	: 2B1	

2.2 MAIN ROTOR

- Type	: STARFLEX	- Diameter	: 10.69 m (35.07 ft)
		- Number of blades	: 3
		- Nominal rotor speed	: 386 rpm

2.3 TAIL ROTOR

- Type	: Flexible seesaw	- Diameter	: 1.86 m (6.10 ft)
		- Number of blades	: 2
		- Nominal tail rotor speed	: 2044 rpm

2.4 FUEL

- Total capacity	: 540 l (427 kg) (142.7 US gal)	- Usable fuel:	
		• Standard fuel tank	: 538.7 l (425.6 kg) (142.3 US gal, 938 lb)
		• Crash resistant fuel tank	: 538 l (425 kg) (142.1 US gal, 937 lb)

2.5 OIL

- MGB oil capacity including filter	: 6.5 l (1.7 US gal.)	- Engine oil capacity	: 6.2 l (1.64 US gal.)
- TGB oil capacity	: 0.33 l (0.08 US gal.)	- Servo control fluid capacity per reservoir:	3 l max (0.79 US gal.)

SECTION 7.2

COCKPIT

1 INSTRUMENT PANEL AND CONSOLE

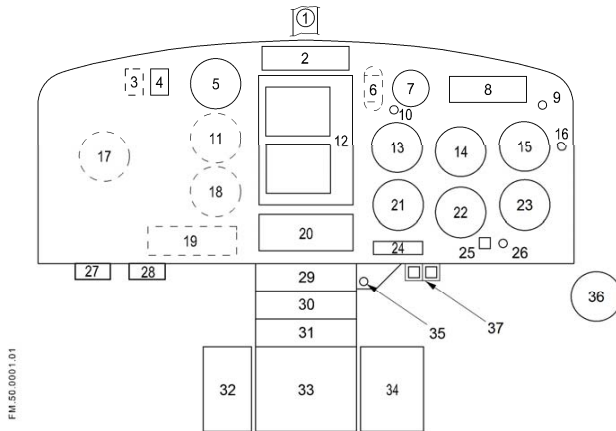


Figure 1: Instrument panel and console

- | | |
|--|---|
| 1 - Compass | 18 - (Spare) or Clock |
| 2 - Lighting potentiometers | 19 - ADF (*) or HF SSB (*) or Hailers control unit (*) |
| 3 - Gyro slaving control (*) | 20 - VHF1-NAV1 (VOR/ILS/GPS) |
| 4 - ELT control switch | 21 - Turn and bank indicator or Radio altimeter indicator (*) |
| 5 - CDI or Clock | 22 - Directional gyro or HSI (*) |
| 6 - "EMERG. SW" switch (post MOD 07 3273) | 23 - Vertical speed indicator |
| 7 - NR/Nf rpm indicator | 24 - DME indicator (*) |
| 8 - Caution and Warning Panel | 25 - NAV1-NAV2 indicator |
| 9 - "LAND LT" indicator light (*) | 26 - NAV1-NAV2 selector |
| 10 - NR/Nf lighting potentiometer | 27 - Air conditioning breakers(*) |
| 11 - (Spare) or CDI or Clock or Turn and bank indicator or RMI (*) | 28 - Air conditioning control unit (*) |
| 12 - VEMD | 29 - ICS |
| 13 - Airspeed indicator | 30 - VHF2-NAV2 (VOR/ILS) |
| 14 - Horizon | 31 - Transponder |
| 15 - Altimeter | 32 - Fuses or breakers panel |
| 16 - "FUEL PUMP" indicator light (Pre MOD 07 3273) | 33 - SCU console and fuses or breakers panel |
| 17 - (Spare) or Turn and bank indicator or standby horizon (*) | 34 - Fuses or breakers panel |
| | 35 - Cabin ventilation control |
| | 36 - Sling load indicator (*) |
| | 37 - Emergency floatation gear pushbuttons (*) |
- (*) If installed

- Post MOD 07-3368:

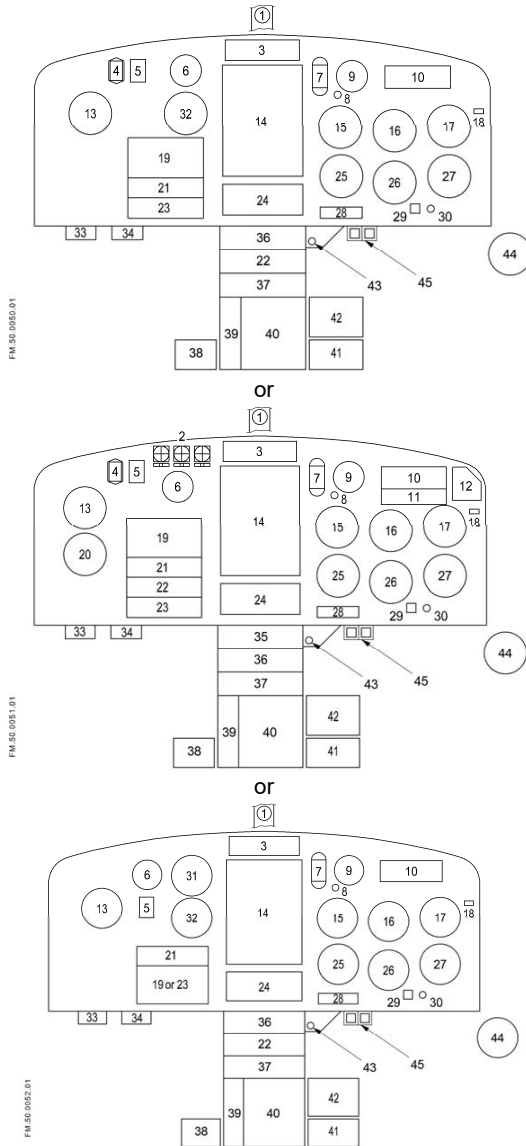


Figure 1.1: Instrument panel and console

- | | |
|---|---|
| 1 – Compass | 23 – Hailers control unit* |
| 2 – AP galvanometers* | 24 – VHF1 - NAV 1 (VOR/ILS/GPS) |
| 3 – Lighting potentiometers | 25 – Turn and bank indicator or
radio altimeter indicator* |
| 4 – Gyro slaving control* | 26 – Directional gyro or HSI* |
| 5 – ELT control switch | 27 – Vertical speed indicator |
| 6 – Clock | 28 – DME indicator* |
| 7 – "EMERG. SW" switch | 29 – NAV 1 - NAV 2 indicator |
| 8 – NR/Nf lighting potentiometers | 30 – NAV 1 - NAV 2 selector |
| 9 – NR/Nf rpm indicator | 31 – CDI |
| 10 – Caution and Warning Panel | 32 – RMI* |
| 11 – AP Caution and Warning Panel* | 33 – Air conditioning breakers* |
| 12 – AP test selector* | 34 – Air conditioning control unit* |
| 13 – (Spare) or turn and bank indicator*
or standby horizon* | 35 – AP control unit* |
| 14 – VEMD | 36 – ICS |
| 15 – Airspeed indicator | 37 – VHF2 - NAV 2 (VOR/ILS) |
| 16 – Horizon | 38 – Direct battery/ 16α breakers panel |
| 17 – Altimeter | 39 – 30α breakers panel |
| 18 – "LIGHT" indicator light | 40 – SCU |
| 19 – HF SSB* | 41 – 44α breakers panel |
| 20 – ADF indicator* | 42 – 31/32α breakers panel |
| 21 – ADF* | 43 – Cabin ventilation control |
| 22 – Transponder | 44 – Sling load indicator* |
| | 45 – Emergency floatation gear
pushbuttons* |

(*) If installed

SECTION 7.3

CENTRAL WARNING AND ANCILLARY SYSTEMS

1 DESCRIPTION

The caution and warning panel (CWP) comprises the following components:

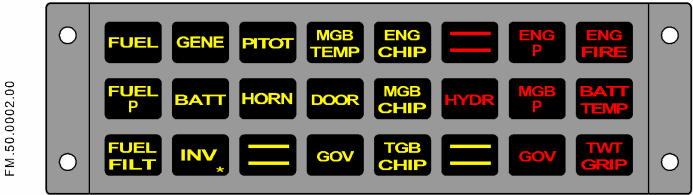
- Red warning lights for alarms which require immediate action,
- Amber caution lights for alarms requiring action which can be delayed.

Audio alarms are generated through the intercommunication system. The audio warning system is activated when [HORN] on the central console is set to ON position. In this case, **HORN** on the caution and warning panel.

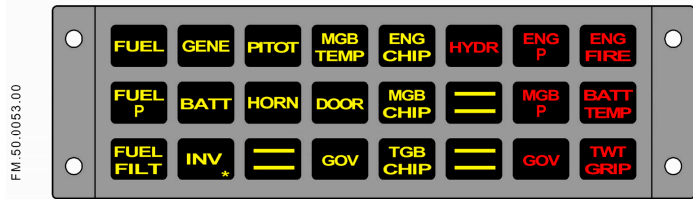
2 CHARACTERISTICS

The caution and warning panel is supplied by a dual 28 VDC power supply and protected by fuses or circuit breakers.

(Pre MOD OP3346) Single hydraulic system



or



(Post MOD OP3346) Dual hydraulic system



or



(*) If installed

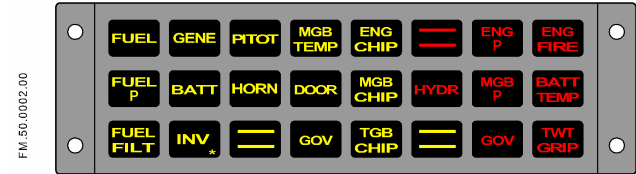
Figure 1: Caution and warning panel

RC a

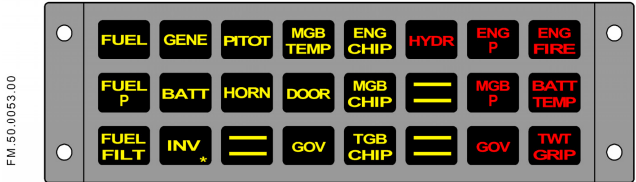
The paragraph 2 - **CHARACTERISTICS**, is modified as follows:

The caution and warning panel is supplied by a dual 28 VDC power supply and protected by fuses or circuit breakers.

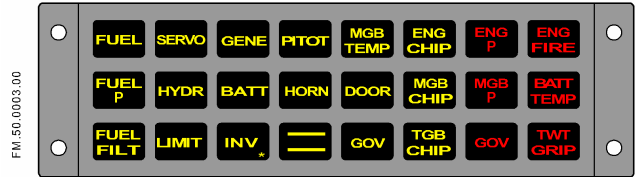
(Pre MOD OP3346) Single hydraulic system



or



(Post MOD OP3346) Dual hydraulic system



or



(*) If installed

Figure 2 : Caution and warning panel

CAUTION
THIS PAGE MUST NOT BE REMOVED FROM THE MANUAL UNTIL EMBODIMENT OF MODIFICATION 07.4622.

3 ANCILLARY SYSTEMS

- General

The ancillary systems are composed of 3 Ancillary Systems Unit (ASU) cards to perform the ancillary service functions of the helicopter:

- Management of all audio warnings, some visual warnings, and the processing of specific electrical signals,
- Management of the engine fuel control back-up system.

- Characteristics

The ASU cards are supplied with a dual 28 VDC power supply and are protected by fuses.

- Description

ASU No.1 performs the following functions:

- Management of the **ENG FIRE** warning light,
- Generation of the audio high and low NR warnings,
- Management of other audio warnings:
 - Due to red alarms: "Gong",
 - Due to MTOP overlimit: continuous low tone.

ASU No.2 performs the following functions:

- Management of the **ENG CHIP** **MGB CHIP** **TGB CHIP** caution lights,
- Management of the **BATT TEMP** warning light,
- Processing and filtering of VEMD electrical supply.

ASU No.3 is part of the Engine Back-up Control Ancillary Unit (EBCAU) and performs the following functions:

- Engine Nf indicator signal acquisition,
- Computing and signal processing for back-up fuel metering valve,
- "Back to neutral" function of back-up fuel metering valve (safety device or necessary after system maintenance test).

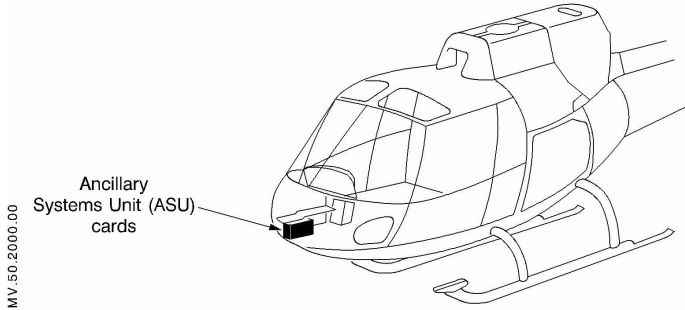


Figure 2: ASU cards location

SECTION 7.4

VEHICLE AND ENGINE MANAGEMENT DISPLAY (VEMD)

1 GENERAL

The VEMD is a duplex indicator equipped with two matrix liquid crystal displays. It is located in the center of the instrument panel. The VEMD displays all necessary engine and vehicle parameters. The VEMD comprises 3 modules:

- Two processing modules: LANE 1 and LANE 2,
- One display module which includes two screens and the control pushbuttons.

2 CHARACTERISTICS

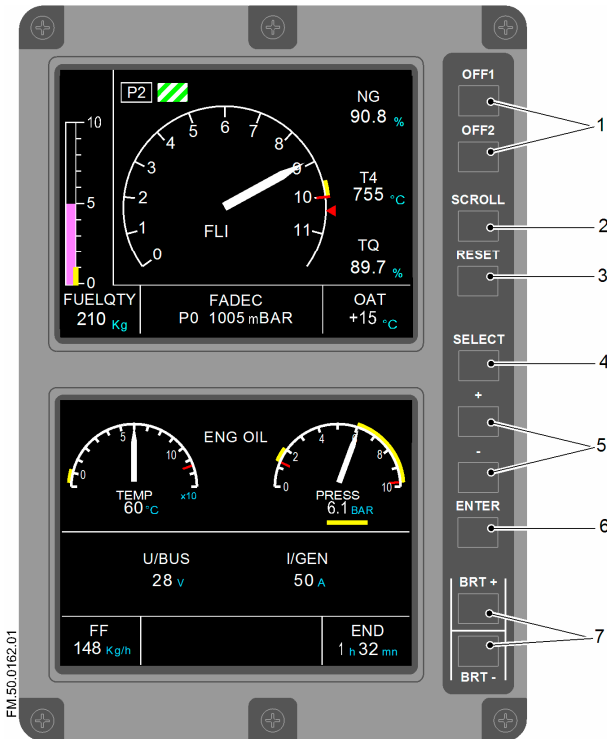
The VEMD is supplied with a dual 28 VDC power supply and is protected by fuses or circuit breakers.

3 OPERATING MODES

Three operating modes are accessible:

- **"OPERATIONAL" mode:** accessible in ground and flight condition, this mode constitutes the main operating mode of the equipment. It contains the ENGINE, VEHICLE, FLI, FLIGHT REPORT and ENGINE POWER CHECK pages,
- **"CONFIGURATION" mode:** only accessible in ground condition, this mode allows configuration of the VEMD.
 1. [OFF1] and [OFF2]PRESS to switch OFF the VEMD,
 2. [SELECT] and [ENTER].....PRESS and HOLD,
 3. [OFF1] and [OFF2]PRESS to switch ON the VEMD,
 4. Hold until message **RELEASE KEY** appears on both screens.
- **"MAINTENANCE" mode:** only accessible in ground condition. This mode allows the selection of the different maintenance functions: Flight report, Failure report, Overlimits, Engine Health Check, Operating times, EECU data and Data loading. To access this mode use the same procedure as "CONFIGURATION" mode except item 2, replace by the following.
 2. [SCROLL] and [RESET]PRESS and HOLD,

4 VEMD CONTROLS



- 1 - **[OFF1 / OFF 2]** pushbuttons:
 . Switch ON/OFF processing module 1/2 and the upper or lower screen.
- 2 - **[SCROLL]** pushbutton:
 . Scrolling through the pages.
- 3 - **[RESET]** pushbutton:
 . Return to nominal display configuration.
- 4 - **[SELECT]** pushbutton:
 . Select a data field.
- 5 - **[+ / -]** pushbuttons:
 . Increase/decrease the numerical values of the selected data.
- 6 - **[ENTER]** pushbutton:
 . Validate the selected data.
 . Go through a list of available data.
- 7 - **[BRT +/-]** pushbutton:
 . Screen brightness control.

Figure 1: VEMD controls

5 OPERATION

The VEMD is automatically powered up when [EXT PWR/BAT] or [BAT/EPU] is switched ON.

The equipment performs an initialization test which checks correct operation of each of the two lines. During the test, the following message is displayed.

TEST IN PROGRESS

If the test is faulty, the following is displayed:

MIN 50.2533.00

"LANE 1 FAILED"
"PRESS OFF 1"

or

"LANE 2 FAILED"
"PRESS OFF 2"

The line concerned can be cut-off by pressing the associated pushbutton (OFF1 or OFF2). This validates the initialization tests and switches the remaining line to operating mode.

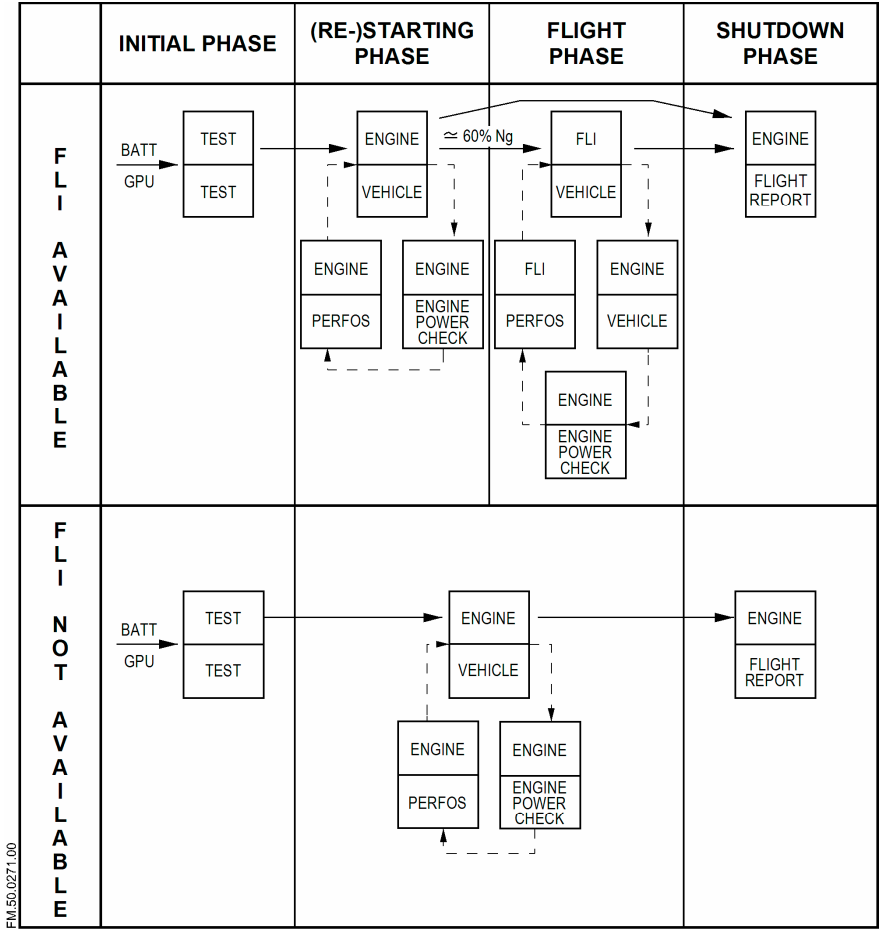
If the test is successful, the VEMD automatically goes to "OPERATIONAL" mode.

6 OPERATIONAL MODE

This mode is displayed by default, when no other mode is selected.

The [SCROLL] pushbutton is used to scroll the pages as shown on the following diagrams (Figures 2 and 3).

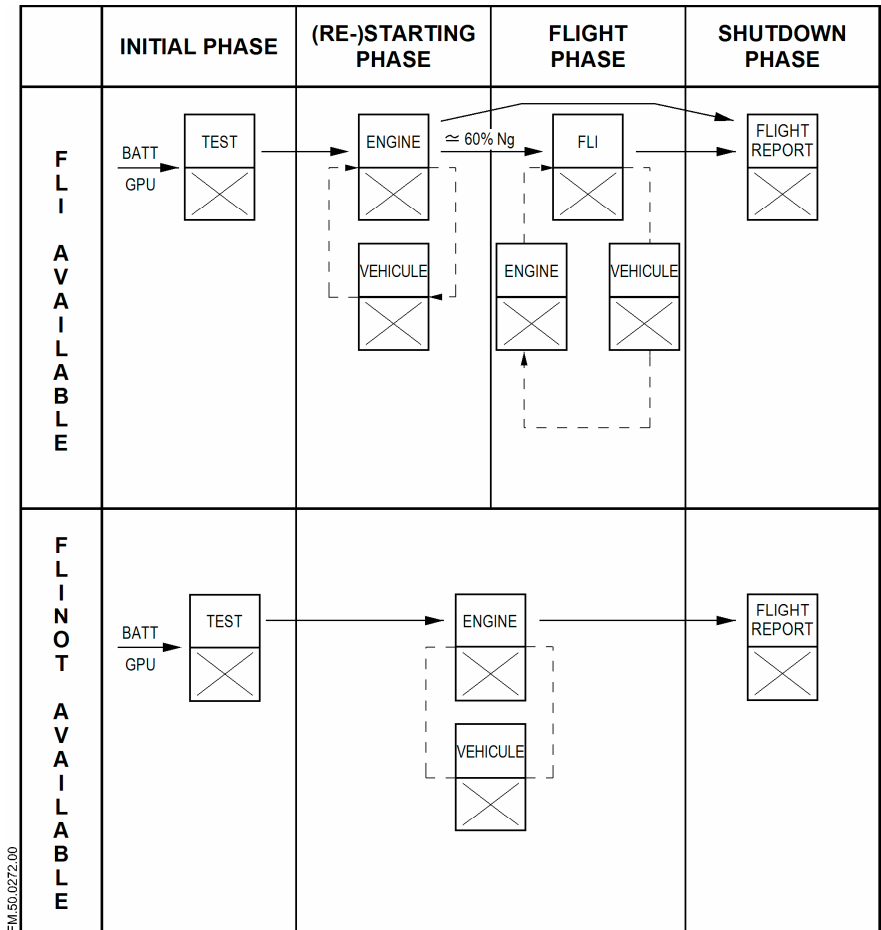
6.1 MANAGEMENT OF PAGES IN NORMAL FLIGHT MODE



- ➔ Automatic change-over at end of phase
- - - ➔ Page selected manually by pressing "SCROLL"

Figure 2: VEMD management in normal FLIGHT mode

6.2 MANAGEMENT OF PAGES IN DEGRADED DISPLAY FLIGHT MODE



- > Automatic change-over at end of phase
 - - - - -> Page selected manually by pressing "SCROLL"

Figure 3: VEMD management in degraded display mode

6.3 FIRST LIMITATION INDICATOR (FLI) PAGE

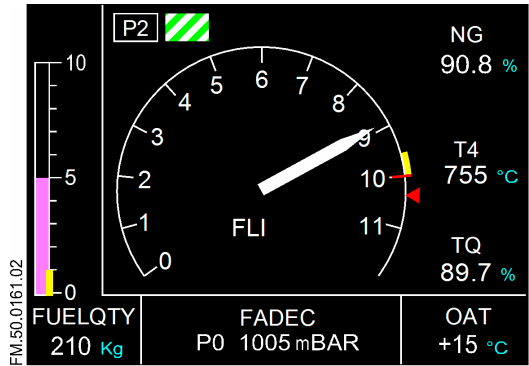


Figure 4: FLI page (Values given as an example)

NOTE

If one of the parameters on the FLI page becomes invalid, the ENGINE page is automatically displayed; the parameters can then be read on independent scales.

6.4 ENGINE PAGE

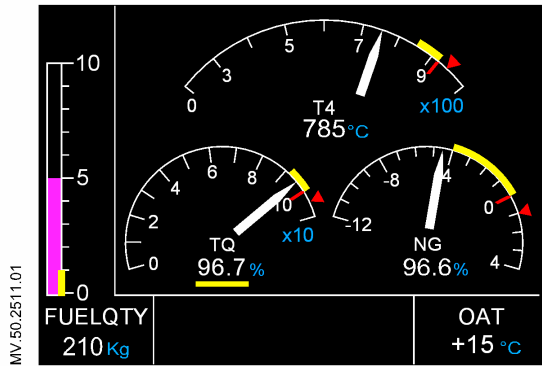


Figure 5: ENGINE page

6.5 VEHICLE PAGE

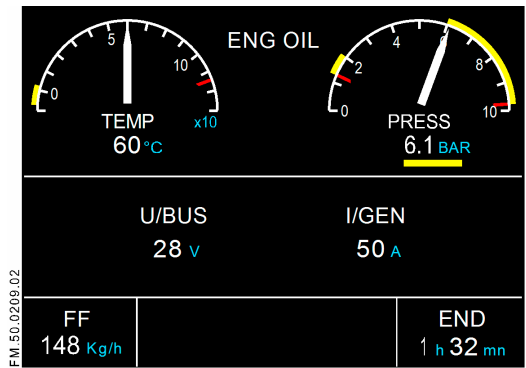


Figure 6: VEHICLE page

6.6 ENGINE POWER CHECK (EPC) PAGE

The first page displays the conditions of compliance, where applicable, when the EPC is requested, in order to obtain a correct engine power check. The check is broken down into three phases:

- A value stabilization phase,
- A more restrictive stabilization phase,
- A margin computation phase.

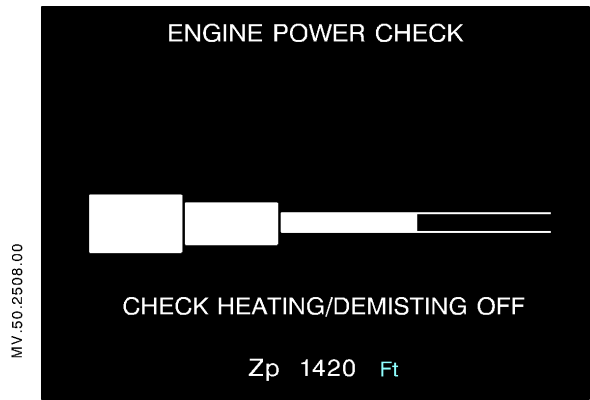


Figure 7: First page of the EPC

The second page displays the result of the EPC according to 6 parameters (Ng, Nf, t4, Hp (Zp), Tq, OAT) and the positive or negative differences in t4 and torque.

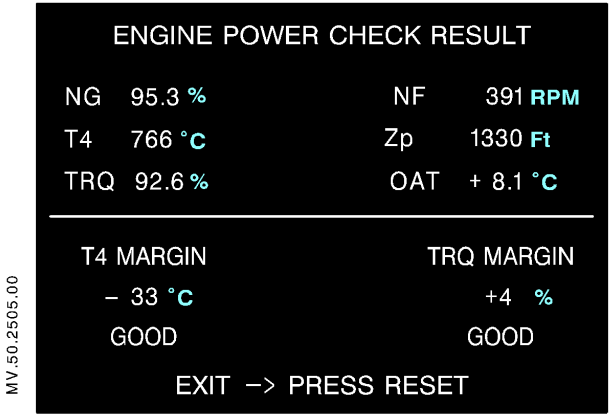


Figure 8: Second page of the EPC

6.7 PERFORMANCE PAGE

This page is used to calculate aircraft weight and performance in the form of takeoff weights, in and out of ground effect.

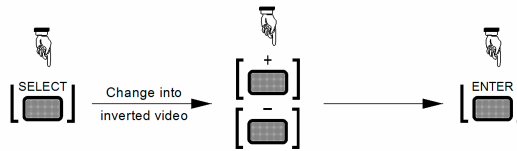
The following parameters must be entered:

- The equipped empty weight of the aircraft (EEW). At power up, the value displayed is the value from the previous flight, it needs to be updated if the aircraft configuration has been changed.
- The weight of the crew (CREW). At power up, the default value is 80 kg (176 lb).
- The weight of the payload (PAYLOAD). At power up, the PAYLOAD value is automatically set to match the maximum internal takeoff weight.

Fuel and external parameters Hp and OAT are taken into account automatically.

For mission planning purposes, Hp (Zp) and OAT can be modified.

When Hp (Zp) is modified, the OAT decreases in accordance with the standard atmosphere law. When the page is changed or another parameter is selected, the VEMD takes into account the actual Hp (Zp) and OAT values. To set or modify the parameters, apply the following procedure.



Use of + / - Keys

	EEW / CREW / PAYLOAD	Zp	OAT
Press > 5 sec.	± 100 kg (200 lb)	± 500 ft (150 m)	± 5°C (10°F)
Press < 5 sec.	± 2 kg (4 lb)	± 100 ft (30 m)	± 1°C (2°F)

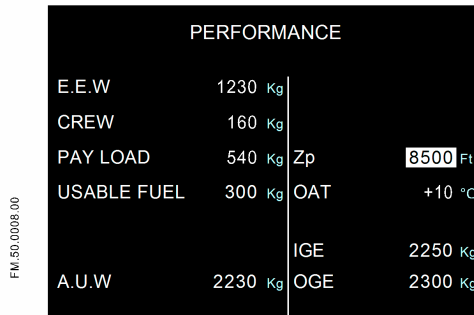


Figure 9: PERFORMANCE page

NOTE

When the IGE and OGE values are less than the aircraft All-Up Weight, they are displayed in yellow.

If the actual hover performance is higher than the demonstrated hover performance (dashed lines beyond the max. demonstrated H₀ in the HIGE / HOG performance charts) IGE / OGE is displayed in white and the corresponding weight in yellow.

6.8 FLIGHT REPORT PAGE

The purpose of this page is to provide the crew with a synthetic report of the last flight performed. The "FLIGHT REPORT PAGE" automatically replaces the "VEHICLE" page when the VEMD detects the engine "shutdown" and NR below 70 rpm status.



- 1 - Flight number, which is incremented automatically.
- 2 - Flight time (from Ng > 60% after start to Ng < 50% at engine shutdown).
- 3 - Generator cycles / Total cycles.
- 4 - Free turbine cycles / Total cycles.
- 5 - Message area (in yellow) if a discrepancy is detected during the flight.

If a message appears, refer to the "MAINTENANCE" mode in the systems description manual.

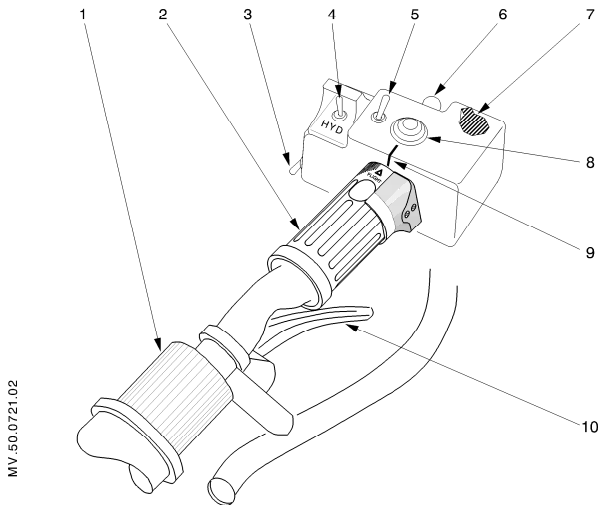
To exit this page, press the [RESET] key.

Figure 10: FLIGHT REPORT page

SECTION 7.5

FLIGHT CONTROLS

1 COLLECTIVE LEVER GRIP

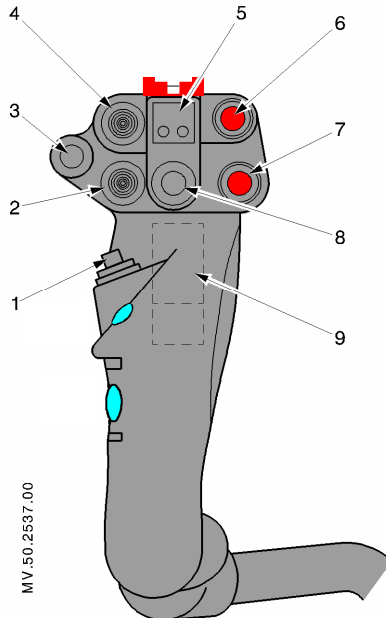


- | | |
|--|--|
| 1 - Collective lever friction ring | 7 - Hoist cable shearing pushbutton* |
| 2 - Twist grip | 8 - Swiveling landing light directional control* |
| 3 - VEMD page scroll | 9 - Twist grip position index |
| 4 - Hydraulic pressure cut-off switch | 10 - External load mechanical release* |
| 5 - Swiveling landing light ON/OFF switch* | |
| 6 - Collective pitch locking device | |

(*) If installed

Figure 1: Collective lever grip

2 CYCLIC STICK GRIP

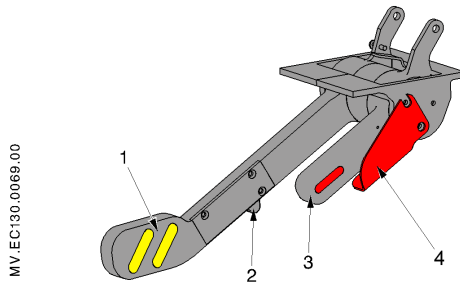


- | | |
|---|--|
| 1 - AFCS trim release (*) | 5 - Flares (*) / Sling load lower hook release (*) |
| 2 - Radio frequency control / AFCS trim (*) | 6 - Sling load release (*) |
| 3 - (Spare) | 7 - AFCS release (*) |
| 4 - Mirror orientation control (*) | 8 - Camera (*) |
| | 9 - Radio/ICS push to talk switch |

(*) Optional

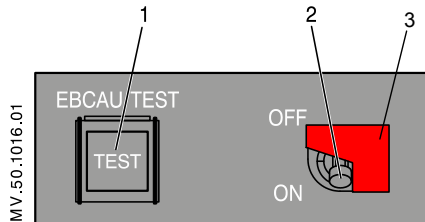
Figure 2: Cyclic stick grip

3 OTHER CONTROLS



- 1 - Rotor brake
- 2 - Rotor brake safety device
- 3 - Fuel shut-off control
- 4 - Fuel shut-off control safety device

Figure 3: Rotor brake - Fuel emergency shut-off



- 1 - Engine backup control test switch (guarded)
- 2 - Engine starting selector: OFF-ON
- 3 - Engine starting selector safety device

Figure 4: EBCAU test switch - Starting selector

SECTION 7.6

LIGHTING SYSTEM

1 INTERIOR LIGHTING

1.1 SYSTEM DESCRIPTION

1.1.1 Cabin lighting

The cabin lighting is provided by:

- One adjustable spot light at the front for the crew.
- Two adjustable reading lights at the rear for passengers lighting.

The forward adjustable spot light has its own switch and potentiometer.

It is supplied directly by the battery in case of emergency. The spot light is used as an emergency light in case of total lighting failure.

Two adjustable reading lights at the rear for passenger lighting has its own switch and potentiometer.

An adjustable glareshield (if installed) can be set to minimize reflections from the instrument panel and console equipment on the windshield.

Alternate configuration:

- One adjustable spot light at the front for the crew.
- Two swiveling dome lights at the front for the crew.
- Two swiveling dome lights at the rear for passengers lighting.

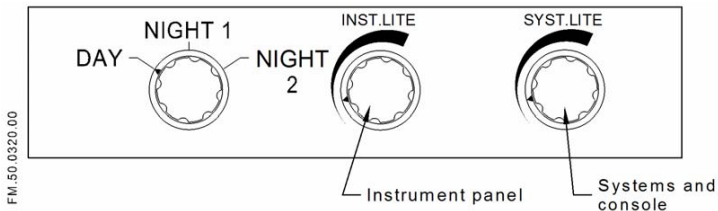
1.1.2 Cockpit lighting

The cockpit lighting is provided by:

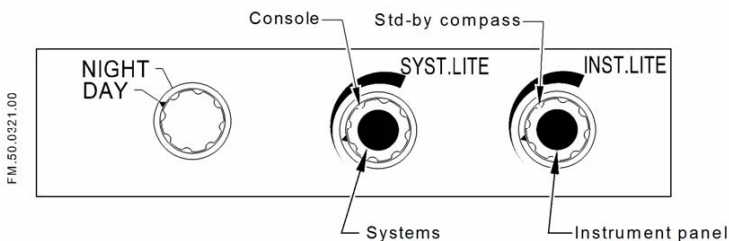
- The instrument integral lighting system for normal and back-up lighting,
- Integral lighting for the SCU, VEMD controls, radio and radio-navigation controls,
- LCD or LED displays on VEMD, NR/Nf indicator, radio and radio-navigation systems,
- CWP and other warning and advisory lights,
- Interior light for the stand-by compass.

Pre MOD 07-3368 (Figure 1):

- Two pushbuttons on the SCU:
[INST LTs 1] or **[INST L1]**: Circuit 1 control for instrument panel lighting and the systems control unit (SCU).
[INST LTs 2] or **[INST L2]**: Circuit 2 control for the instrument panel lighting.
- A selector with 3 positions **[DAY/NIGHT1/NIGHT2]**
- Two potentiometers:
INST.LITE and **SYST.LITE** to adjust the instrument panel brightness and the systems brightness

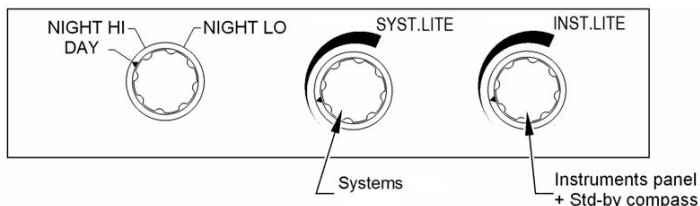
**Figure 1: Lighting system controls****Post MOD 07-3368 (Figure 2):**

- Two pushbuttons on the SCU:
[INST L1]: Circuit 1 control for the systems control unit (SCU).
[INST L2]: Circuit 2 control for instrument panel lighting.
- A selector 2 positions **[DAY/NIGHT]**
- Two dual potentiometers:
INST.LITE to adjust instrument panel and Stand-by compass brightness.
SYST.LITE to adjust the console and systems brightness.

**Figure 2: Lighting system controls**

Alternate configuration (Figure 3):

- Two pushbuttons on the SCU:
 - [INST L1]: Circuit 1 control for instrument panel and systems lighting (SCU),
 - [INST L2]: Circuit 2 control for instrument panel and systems lighting connected to [DCT/BAT],
- A selector with 3 positions [DAY/NIGHT HI/NIGHT LO]
- Two potentiometers:
 - INST.LITE to adjust instrument panel and Stand-by compass brightness
 - SYST.LITE to adjust systems brightness

**Figure 3: Lighting system controls****1.2 OPERATION**

The lighting in the cockpit is controlled using the lighting control on the instrument panel (Figures 1, 2 and 3). Some equipment has additional lighting level adjustments.

1.2.1 Operating modes

The following operating modes are possible:

Pre MOD 07-3368:

DAY: VEMD displays, caution, warning and advisory lights are at nominal brightness,

NIGHT 1: VEMD displays, caution, warning and advisory lights are dimmed,

NIGHT 2: Same as "NIGHT 1" but stand-by compass lighting is switched off.

The NR/Nf indicator has its own nearby potentiometer to adjust the brightness of LCD displays when the lighting selector is on "NIGHT1" or "NIGHT2" position.

Post MOD 07-3368:

DAY: VEMD displays, caution, warning and advisory lights are at nominal brightness,

NIGHT: VEMD displays, caution, warning and advisory lights are dimmed.

The NR/Nf indicator has its own nearby potentiometer to adjust the brightness of LCD displays when the lighting selector is on **"NIGHT"** position.

Alternate configuration:

DAY: Swiveling dome lights, VEMD displays, **NAV1 - NAV2 - LIGHT** lights, pushbutton indicator lights on the SCU, caution, warning lights are at nominal brightness.

NIGHT HI: Swiveling dome lights, VEMD displays, **NAV1 - NAV2 - LIGHT** lights, pushbutton indicator lights on the SCU, caution, warning lights are dimmed.

NIGHT LO: VEMD displays, **NAV1 - NAV2 - LIGHT** lights, pushbutton indicator lights on the SCU, caution, warning lights are dimmed at minimum level. The internal lighting of the horizon is switched off. The swiveling dome lights (two for crew and two for passenger) are available in vario mode with the maximum lighting level reduced.

The brightness of the LCD displays on the NR/Nf indicator has a dedicated brightness potentiometer and can be adjusted when the 3 positions **[DAY/NIGHT HI/NIGHT LO]** selector is on **"NIGHT HI"** or **"NIGHT LO"** position.

1.2.2 Adjustment

Two potentiometers **INST.LITE** and **SYST.LITE** are used following the version to adjust the instrument panel, stand-by compass, SCU, VEMD and equipment brightness control.

The brightness of the adjustable dome lights can be set manually using their potentiometer.

The brightness of the VEMD screen can be adjusted manually using the **[BRT +/-]** pushbuttons.

The brightness of some avionics equipment * (e.g. GTX330, GNS430, ...) displays is automatically adjusted by light sensitive cells.

Alternate configuration:

The brightness of the swiveling dome lights can be set manually using their potentiometer.

The brightness of the VEMD screen can be adjusted manually using the **[BRT +/-]** pushbuttons.

The brightness of GMA340H, GTX330, GNS430 and KX165A displays is automatically adjusted by light sensitive cells.

(*) If installed

2 EXTERIOR LIGHTING

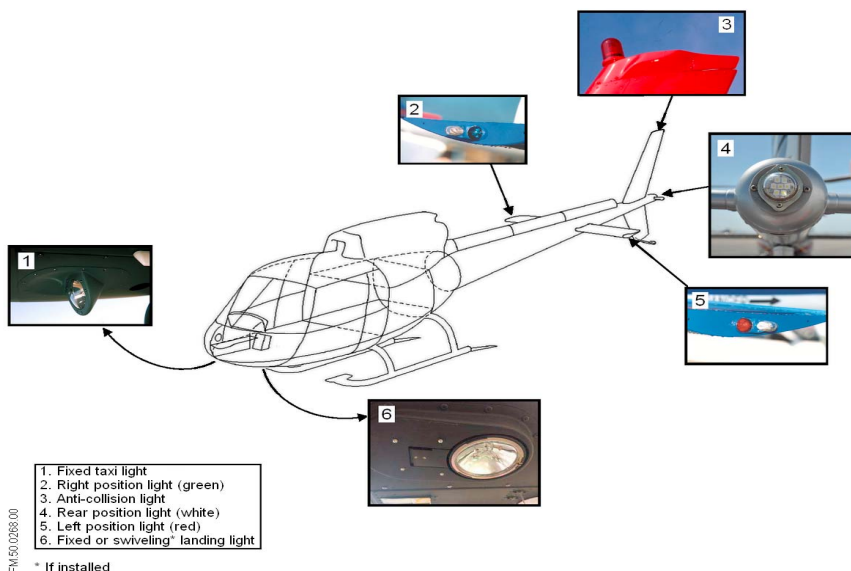
2.1 DESCRIPTION

WARNING

WHEN HOVER, LANDING AND/OR SEARCH LIGHTS ARE SWITCHED ON, MAKE SURE THAT THERE IS NO FLAMMABLE OBJECT WITHIN A DISTANCE OF 3 M (10 FT).

CAUTION

WITH AIRCRAFT ON GROUND, WHEN SWITCHED ON, AND FOR A FEW MINUTES AFTER THEY HAVE BEEN EXTINGUISHED, HOVER, LANDING AND SEARCH LIGHTS MAY SET FIRE TO DRY VEGETATION.



Exterior lighting includes:

- Three position lights,
- an anti-collision light,
- a fixed taxi light,
- a fixed or swiveling landing light *.

The position lights and anti-collision light are switched ON/OFF by the [POS.LT.] or [POS LT] and [A/COL LT] or [A/COL.LT] pushbuttons on the SCU.

(*) if installed

2.2 CONFIGURATION

- Standard configuration (one taxi light and one fixed landing light)

The taxi and the fixed landing lights are set ON/OFF by the following pushbuttons on the SCU: **[TAXI LT]** or **[TAXI LIGHT]** and **[LDG LT]** or **[LAND LIGHT]**.

The swiveling landing light* is switched ON/OFF using the **"ON/OFF/RET"** switch (**RET** position is unstable) on the collective grip. A directional control allows the orientation of the landing light.

Pre MOD 07-3368:

An advisory light "LAND LT" indicates that the swiveling landing light is on. If the swiveling landing light is not installed there is no indicator "LAND LT" on the instrument panel.

Post MOD 07-3368:

An advisory light **LIGHT** indicates that at least one of the taxi, fixed or swiveling landing lights is on.

- Alternate configuration (one fixed landing light and one swiveling landing light)

The fixed landing light is controlled by the pushbutton **[LDG LT]** on the SCU.

The swiveling landing light is controlled by the **"ON/OFF/RET"** switch (**RET** position is unstable) located on the collective grip. In **"NIGHT LO"** mode the swiveling landing light is available with the maximum lighting level reduced and the brightness adjustable by the **"BRT+/-"** switch located on the collective grip.

An advisory light **LIGHT** indicates that the swiveling landing light is on.

(*) If installed

SECTION 7.7

POWER PLANT

1 GENERAL

The TURBOMECA ARRIEL 2B1 is a 650 kW (870 hp) class turboshaft engine. It is located in a separate fireproof compartment aft of the MGB and above the rear cargo compartment. It is connected to the MGB by a shaft mounted between two flexible couplings. The engine is a modular free-wheel engine. It is composed of five independent modules:

- The Axial Compressor Module (single-stage with bleed valve),
- The Gas Generator Module:
 - A centrifugal compressor,
 - An annular combustion chamber,
 - A single-stage gas generator turbine.
- The Free Turbine Module,
- The Reduction Gear Module:
 - Reduces the free turbine speed from 39158 down to 6000 rpm.
- The Output Shaft Module:
 - Transmits power to the MGB and accessory drive couplings.

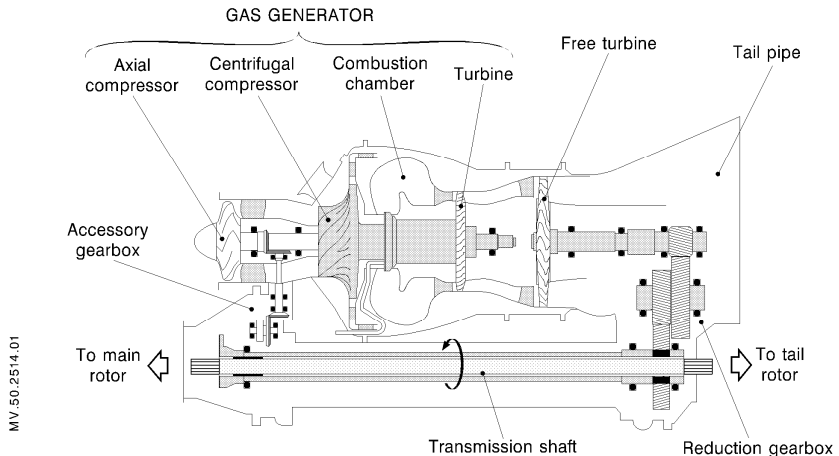


Figure 1: Engine view

2 ENGINE OIL SYSTEM

The engine oil system is divided into two systems:

- An external system installed in both the MGB and engine compartment. It includes one tank and one cooler. The hoses installed in the engine compartment are fireproof.
- An internal system integrated into the engine. It includes one pressure and three scavenge pumps, a filter with a by-pass valve and one electrical magnetic chip detector.

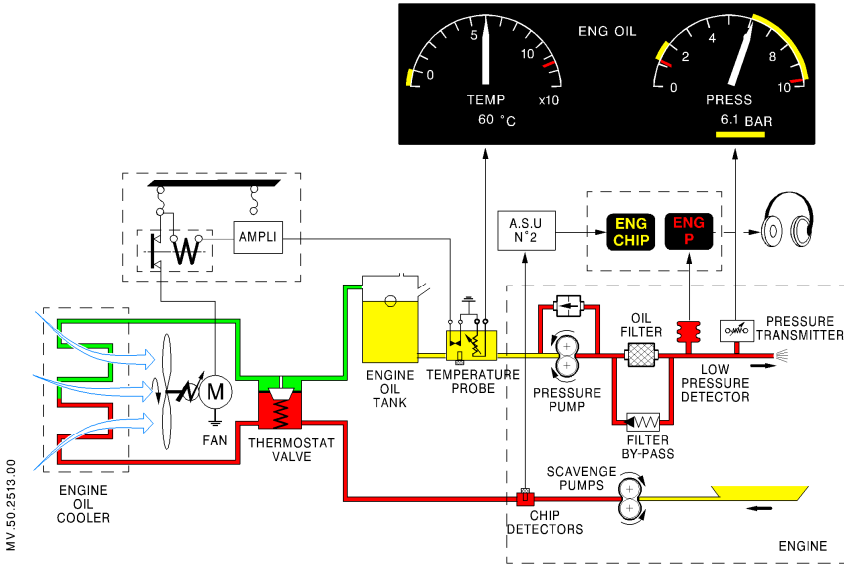


Figure 2: Lubrication system

SECTION 7.8

FUEL SYSTEM

1 GENERAL

The fuel system comprises a helicopter part and an engine part. The first part comprises a spin-moulded tank, a supply system, a gravity refueling filler, and a monitoring system. The second part comprises an LP pump, a fuel filter, an HP pump and a fuel control hydro-mechanical unit which are integral with the engine. The hydro-mechanical unit is controlled by a Full Authority Digital Engine Control system (FADEC) and a back-up system.

2 HELICOPTER SUPPLY SYSTEM

The fuel tank is located in the body structure beneath the transmission deck and is equipped with a fuel level transmitter. The tank also includes a starting priming pump and a decanting sump with a water drain valve. A venting device on the RH side and a filler on the LH side are fitted to the tank. The fuel is then supplied to the engine via a fire fuel shut-off valve. For starting and purging purposes, the tank booster pump primes the fuel line during starting sequence.

3 ENGINE FUEL SUPPLY SYSTEM

Drawn by the engine low pressure pump, the fuel is sucked up through the fuel filter by the high pressure pump. At the entry of the circuit, an ejector fed by the HP pump helps to prime the LP fuel pump. The filter is fitted with an electrical pre-clogging indicator, a by-pass valve with a visual indicator and a low pressure detector.

Heated by the combined fuel/oil engine filter assembly the fuel is then delivered to the hydro-mechanical unit of the FADEC. An engine fuel control back-up system allows to control the fuel flow via a back-up valve in case of a total FADEC system failure. The heat exchanger allows operation at -20°C without anti-ice additive in the fuel.

The fuel flows through a shut-off solenoid valve which enables the engine to be shut down, and flows through a pressurizing valve before entering the main injector and the injection wheel.

A three-way electrovalve distributes the fuel to two starting injectors during the starting phase. To prevent carbonization, they are supplied with P3 air once the starting sequence is over.

During shutdown sequence a purge valve drains the fuel remaining in the main injection system.

4 CONTROLS AND MONITORING

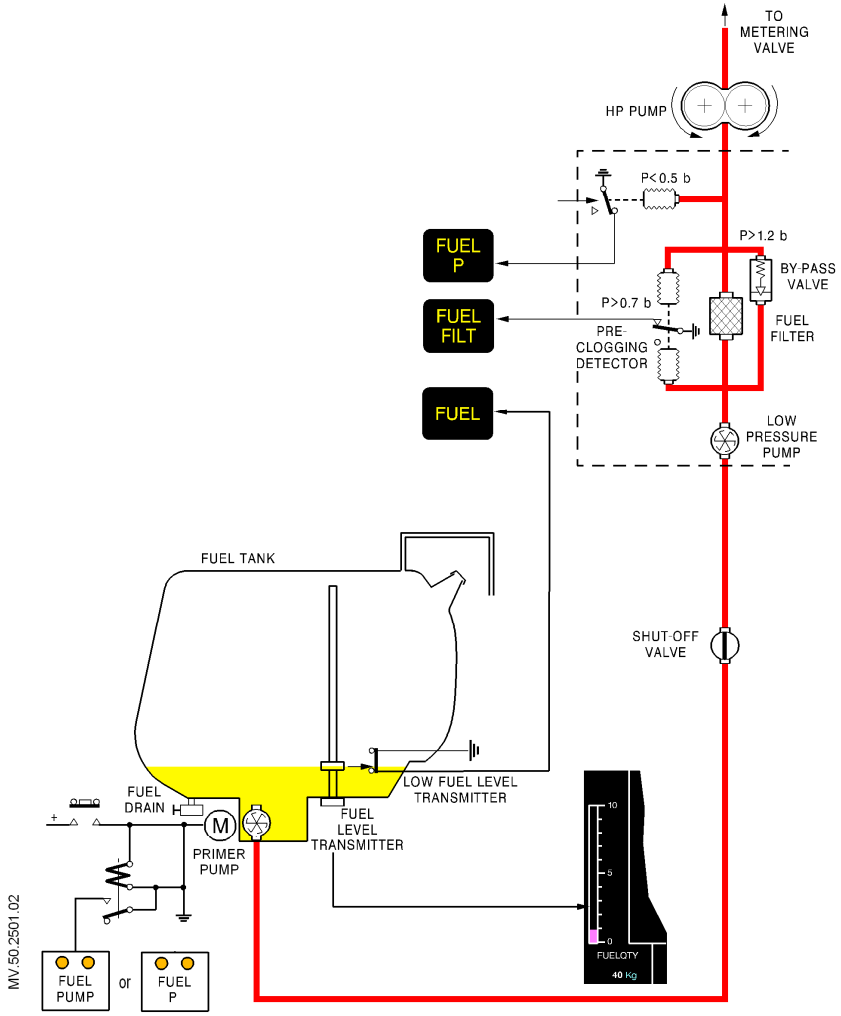


Figure 1: Fuel system

5 FUEL FLOW CONTROL

The fuel flow is managed by a FADEC system. Based on a dual channel digital engine control unit called Electronic Engine Control Unit (EECU) in normal flight mode, the principle is to control the power turbine speed, N_f , independently of the power drawn from the engine, by adjusting the gas generator speed (N_g). The EECU provides fuel flow modulation via a stepper motor which operates a metering valve in the engine's hydro-mechanical unit. A control channel device gives control to the channel with no major failure detected or to channel A "preferred channel" if both channels are sane.

- FADEC functions

Installed on the airframe, in the rear cargo bay, the EECU provides the following functions:

- Automatic engine starting preventing t_4 overlimit,
- Proportional power turbine speed control with rotor noise signature reduction feature,
- Protection against engine surge and flame out during transients,
- Bleed valve monitoring,
- Protection against N_g and torque overlimits,
- Failure detection and indication,
- Engine Power Check (EPC),
- N_g and N_f cycles counting,
- Post MOD 07-20060 and 07-20061: Automatic engine shutdown in the event of N_f overspeed ($N_f > 120\%$ (equivalent to $NR = 463 \text{ rpm}$)).

Post MOD 07-20060 and 07-20061: The FADEC features a function protecting the engine free turbine from overspeed events and then has the ability to automatically shutdown the engine by closing the main fuel metering valve. The engine restart capability is not impacted.

- FADEC power supplies

The FADEC system electrical power supply is designed in such way that normal engine operation is independent from the helicopter electrical system as soon as the gas generator speed is sufficient ($N_g > 60\%$). FADEC electrical supplies are:

- The helicopter 28 VDC system,
- An alternator with 2 independent channels driven by the gas generator.

Helicopter electrical power supply remains necessary in all operating conditions:

- For the starting sequence and the monitoring system,
- As a back-up source for the EECU fuel control section.

In case of a total loss of helicopter DC supply, even if DC is restored, the **GOV** light will remain on, IDLE is not usable (FADEC forced to FLIGHT regime), until a complete reset of the FADEC (engine off, battery off and on again on the ground).

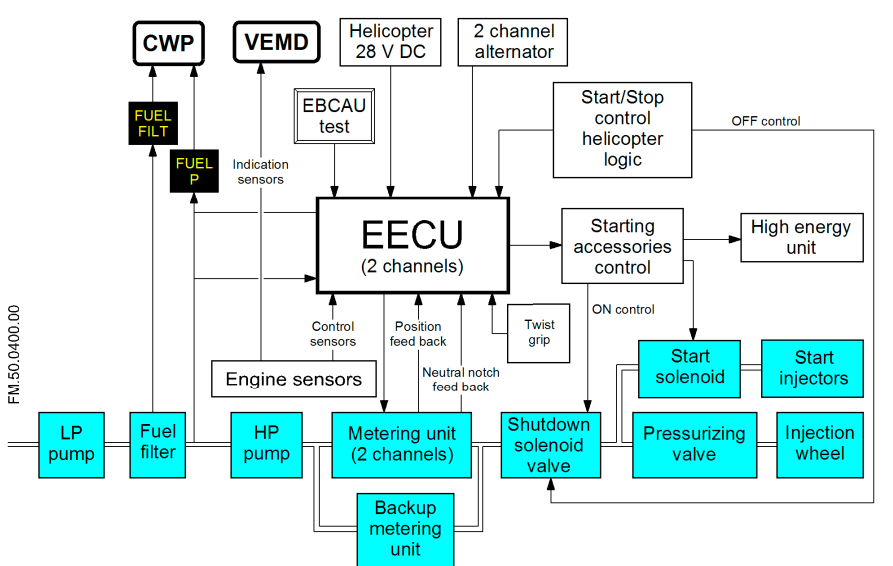


Figure 2: Fuel control block diagram

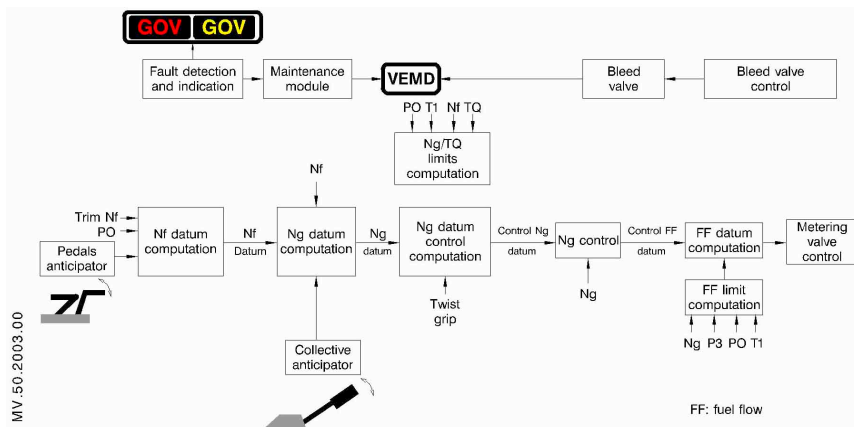


Figure 3: FADEC block diagram

- FADEC failures

FADEC system failures are classified in three levels:

- Level 1 (flashing **GOV** at idle, engine starting or stopped): No effect on engine control, redundancy loss.

Actions to be taken:

- Reset FADEC if before start (battery OFF/ON),
Refer to SECTION 3 of the Flight Manual.

- Level 2 (steady **GOV**): Failures leading to a degraded engine control or back-up governing system out of neutral position, loss of accuracy, engine control is degraded but protected, loss of power check.

Actions to be taken:

Refer to SECTION 3 of the Flight Manual.

- Level 3 (**GOV**): Total FADEC failure, FADEC main metering valve is frozen at last computed value, back-up system is automatically activated, automatic start impossible.

Actions to be taken:

Refer to SECTION 3 of the Flight Manual.

- Fuel governor back-up system (**GOV** and steady **GOV**)

- The back-up governor takes control of the engine fuel flow governing. It is based on an electronic computer, the Engine Back-up Control Ancillary Unit (EBCAU), independent from the FADEC. It controls the Nf between 388 and 400 rpm.
- The back-up governor is activated immediately after a total FADEC failure detection.
- The back-up governor may also be activated by the pilot on ground for maintenance purposes. At the end of the test, the back-up valve returns to neutral position.

Controlled by the EBCAU through a Nf control loop, a DC actuator operates the back-up metering valve which closes or opens slots partly in series and partly in parallel within the main metering valve of the engine's hydro-mechanical unit. This increases or decreases fuel flow, depending on the fixed fuel flow delivered by the frozen FADEC main metering valve.

When the FADEC is operative, the back-up system is maintained in the neutral position (safety device).

A maintenance test of the EBCAU is to be performed periodically on the ground to check the back-up system. At the end of the test, the EBCAU will automatically set the back-up valve back to the neutral position when the system is reset (EBCAU test switch back to OFF).

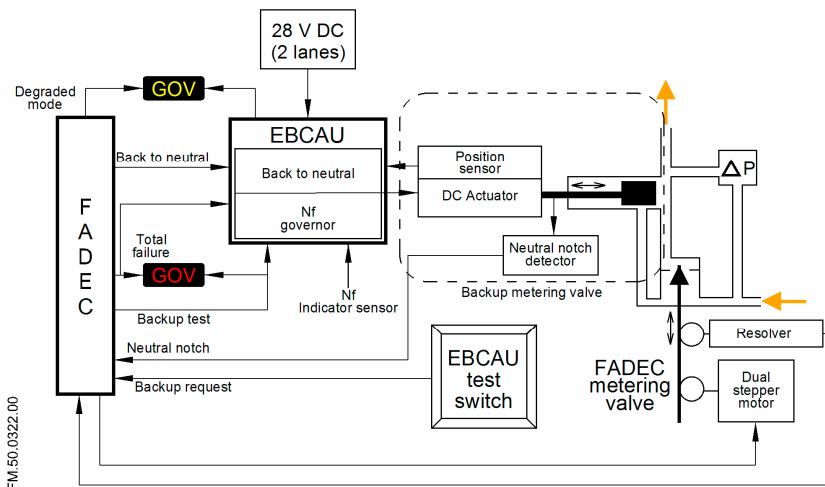


Figure 4: Engine back-up fuel control block diagram

SECTION 7.9

POWER TRANSMISSION SYSTEM AND ROTORS

1 POWER TRANSMISSION

The transmission system consists of:

- Engine / MGB coupling,
- Main gear box (MGB),
- Tail rotor drive shaft,
- Tail gear box (TGB).

- ENGINE / MGB COUPLING

The engine / MGB coupling transmits the engine power to the MGB.

It consists of:

- A drive shaft connecting the engine to the MGB through two flexible couplings at each end,
- A coupling tube around the drive shaft,
- A casing for connection to the MGB,
- A gimbal ring between the connecting casing and the coupling tube.

- MGB

- It transmits the power from the engine to the main rotor with a speed reduction,
- It drives and supports the hydraulic pump, the MGB lubricating pump and the rotor brake,
- It supports the servocontrols and suspension bar attachment fittings.

Made of three interchangeable modules, it includes its own lubricating system, monitoring systems and access for maintenance.

The lubricating pump sucks the oil up from the MGB sump through a strainer and delivers it through an oil cooler and a filter. The oil returns to the sump by gravity.

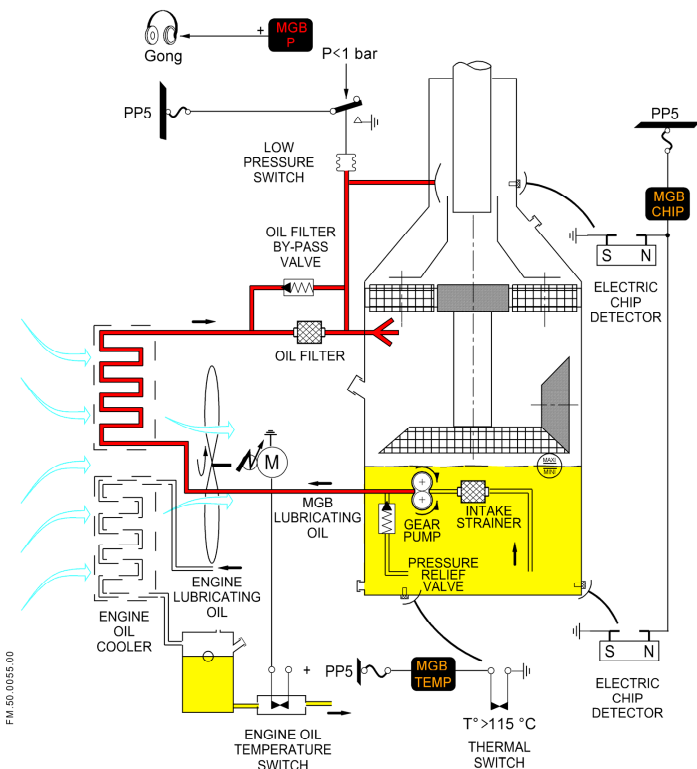


Figure 1: Main gear box oil system

- TAIL ROTOR DRIVE SHAFT AND TGB

The tail rotor drive shaft is composed of two shafts, a forward steel shaft, and a long light alloy shaft. The shafts are connected to each other, to the engine and to the tail gear box by 3 flexible couplings.

The long shaft is supported by five ball bearing/support assemblies mounted on elastomeric bushes which provide vibration damping.

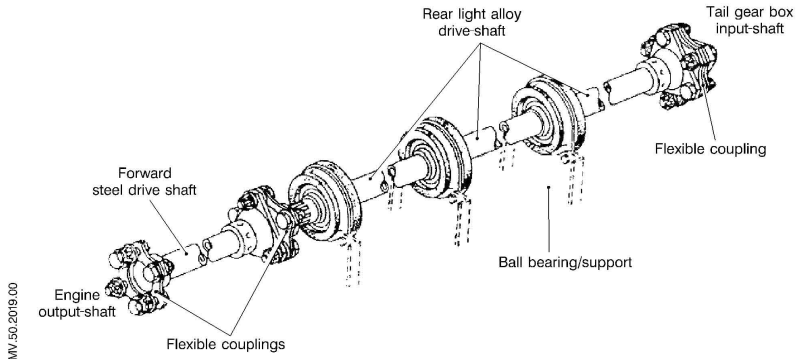


Figure 2: Tail rotor drive shaft

- TAIL GEAR BOX

The TGB is fitted to the rear end of the tail boom and it comprises power transmission and pitch control modules in the same casing.

The TGB is splash-lubricated and comprises a visual oil level indicator and a chip detection device

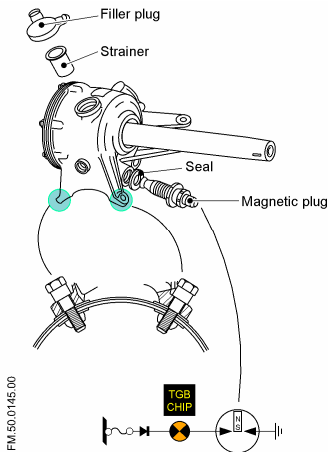


Figure 3: Tail gear box

2 ROTORS

- MAIN ROTOR

The main rotor is of semi-rigid design: the “STARFLEX” rotor hub has no ball bearings nor lubrication system. The rotor includes three blades of flexible glass-resin laminated construction. It rotates clockwise when viewed from above. Flapping is achieved by the “star” composite arms and lead-lag and pitch hinges are provided through distortion of elastomeric items.

- TAIL ROTOR

The two-blade tail rotor is see-saw mounted on the TGB.

The blades rotate counterclockwise when viewed from the RH side of the aircraft.

SECTION 7.10

HYDRAULIC SYSTEM

1 GENERAL

To reduce collective, cyclic and pedal control forces, the flight controls are hydraulically assisted. There are three main rotor servos, one longitudinal and two laterals and also a tail rotor servo for yaw control.

The hydraulic fluid used must comply with the approved specifications in SECTION 2 of the present Flight Manual.

Total system fluid volume is 3 liters (0.79 US gal or 0.66 UK gal) up to the maximum level mark on the reservoir.

2 SYSTEM DESCRIPTION

2.1 HYDRAULIC SYSTEM COMPONENTS

- A separate reservoir mounted on the top of the MGB,
- A single constant flow rate gear-pump generates the hydraulic power. The hydraulic pump is driven by a drive belt mounted between the pump pulley and a pulley on the engine power drive shaft close to the MGB power input,
- A regulator unit fitted with:
 - A pressure regulating valve set to 40 bars (580 psi),
 - A pressure switch,
 - A 3-micron filter with a clogging indicator, and,
 - A solenoid electro valve.
- A distribution system which comprises flexible pressure and return hoses, supplies the four single-body servos and the tail rotor load compensator,
- Three single-body servos to control the main rotor, which move the non-rotating swash plate. Each servo is fitted with a safety unit which consists of:
 - A hydraulic accumulator,
 - A non-return valve and,
 - A solenoid electro valve.

The safety units allow for continued hydraulic assistance for a limited time in the event of a hydraulic pressure loss in the system.

This limited time is sufficient to allow the pilot to reach a flight regime under which the control feedback forces are acceptable without hydraulic assistance,

- A single-body yaw servo,
- A load compensating system to reduce, in the event of a hydraulic pressure loss, the yaw pedal feedback loads for an indefinite period. The load compensator pressure can only be dumped by selecting the accumulator test switch to TEST position (down). This system consists of:
 - A hydraulic accumulator,
 - A non-return valve,
 - A pressure relief valve,
 - A pressure-drop solenoid electrovalve on the accumulator, and,
 - A load compensator actuator.
- Hydraulic system warnings:

In case of loss of pressure or low pressure, the following cockpit indications come on:

- A red **HYDR** light on the CWP,
and
- A Gong sounds once (the Gong sounds when any red warning light comes on).

2.2 SYSTEM CONTROLS AND MONITORING

The hydraulic system is controlled using two switches:

- The Hydraulic cut-off switch: Guarded switch mounted on the collective grip with two positions, ON and OFF.

Normally left in the ON position, allows the main-rotor servos to be powered when the hydraulic system is operating normally. Selected to OFF, during pre-flight checks, emergency procedures, and also when performing hydraulics-off training, the hydraulic system is then depressurized, the accumulators on the main rotor servos are depressurized simultaneously, the tail rotor load compensating system retains its assist function.

- The accumulator test pushbutton: **[HYD TEST]** or **[ACCU TST]** pushbutton switch mounted on the center console with two positions: TEST (down) and OFF (up). It is normally left in the OFF position.

Selected to the TEST position during pre-flight checks, emergency procedures and also when performing hydraulics-off training, it will result in the solenoid valve opening on the regulator unit, which depressurizes the hydraulic system. It will also open the tail rotor servo solenoid, depressurizing the tail rotor load compensating servo but allowing the main rotor servos to be powered by the accumulators in their respective safety units.

2.3 MAIN ROTOR SERVO

The AS 350 can be fitted with servos produced by two suppliers, SAMM and Dunlop. The servos are interchangeable, but have some differences.

SAMM Servos – SAMM servos are fitted with an input clearance locking system. The input clearance locking system was designed to reduce control system freeplay, and therefore enhance aircraft handling qualities when operating in the unpowered (bypass) mode. During system pressurization and with collective in locked position, as hydraulic pressure rises, if the input locking devices on the lateral servos unlock at different pressures, a large cyclic stick movement to the right or to the left may occur in normal operation. The pilot can prevent the cyclic from moving by firmly holding the cyclic with his hand and knees. The force required to prevent control movement is approximately 5 daN (11 lbf). The movement occurs because one lateral servo can become hydraulically assisted before the other due to:

- The main rotor control loads not being equally applied to both lateral servos and/or,
- The hydraulic pressure threshold necessary to activate unlocking of the servos (change from non-assisted mode to hydraulically assisted mode) can be slightly different for the RH and the LH lateral servos due to manufacturing tolerances.

In flight, a small cyclic movement may occur when switching from assisted to non-assisted (bypass) and vice versa.

Dunlop Servos – Dunlop main rotor lateral servos do not include an input locking device, and are not subject to asymmetric switching from unpowered (bypass) mode to powered mode. Small cyclic movement may occur when switching from powered to unpowered (bypass) and vice versa.

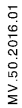


Figure 1: Hydraulic system block diagram

3 NORMAL OPERATION

At start-up, hydraulic pressure is zero **HYDR**.

The hydraulic pump operates as soon as the rotor is spinning.

When the pressure in the circuit is above minimum threshold, on the CWP: **HYDR**.

When the flight control system is operating normally there is no control force feedback to the pilot, except when reaching servo control reversibility when manoeuvring the aircraft under high load factors.

Before each flight, two hydraulic tests shall be performed as normal procedure:

- The hydraulic accumulator test [**HYD TEST**] or [**ACCU TST**] depressed enables the pilot to check that the accumulators still provide hydraulic assistance should the hydraulic power system fail (i.e. checks that there are no significant leaks in the accumulators).

HYDR (flashes post MOD 07-3317) and the Gong comes on to indicate a loss of hydraulic pressure.

- The hydraulic pressure cut-off test (hydraulic cut-off switch on collective pitch set to OFF) enables the pilot to check the electrovalves (dump valves) of the main servos for correct operation. These electrovalves are used to cut off the hydraulic power system in accordance with the AS 350 flight manual emergency procedures, in the event of a hydraulic power system failure or other flight control malfunctions. When the hydraulic cut-off switch is placed in the cut-off position, the accumulators are depressurized simultaneously by opening of the three electrovalves.

Non-related to hydraulic malfunction, the [**HYD TEST**] or [**ACCU TST**] switch is only used in flight to deplete the tail rotor accumulator and load compensator in case of tail rotor control failure in order to bring tail rotor to low pitch. On ground the [**HYD TEST**] or [**ACCU TST**] pushbutton is used to deplete the tail rotor accumulator and load compensator after engine shutdown.

- The accumulators will be repressurized when the hydraulic cut-off switch is returned to the ON position.

The accumulators are checked for correct nitrogen pressurization by measuring the time required for the hydraulic system pressure to return to its nominal level. It will take normally 3 seconds until pressure rises to the nominal operating level. The pilot must note the time from placing the hydraulic cut-off switch to the normal position

HYDR.

Expected cyclic movement is explained in the following paragraph.

4 ABNORMAL OPERATION

4.1 ACCUMULATOR MALFUNCTION

After completion of the hydraulic pressure cut-off test, the normal period for accumulator repressurization is 2 to 3 seconds, but it is reduced to 1 sec. or increased to more than 4 sec. if at least one of the accumulators is faulty. If a faulty accumulator is detected then maintenance action must be performed prior to flight.

4.2 HYDRAULIC PRESSURE LOSS

Usually, hydraulic system failure is caused by a loss of pressure or a low pressure of the hydraulic system, with normal functioning of the servos, accumulators, safety units, electro-valves and hydraulic cut-off switch.

Indications to pilot:

- **HYDR**.
- Gong sounds,
- Controls remain powered by the accumulators.

The pilot must perform the AS 350 flight manual hydraulic pressure loss (illumination of **HYDR** warning light) emergency procedure:

- The average time to reach the recommended safety speed range 40 to 60 kt (74 to 111 km/h), from VNE or hover, is less than 30 seconds. If the accumulators are properly serviced, they will power the flight controls throughout the manoeuvres required to reach the recommended safety speed range. If control load feedback is felt prior to reaching the safety speed range, then the pilot must immediately set the hydraulic cut-off switch to OFF.
- As soon as the recommended safety speed is reached the hydraulic system has to be cut-off. This is necessary even if the accumulators still provide some hydraulic assistance because it enables simultaneous depressurization of the three main rotor accumulators.
This prevents the situation where one of the two lateral accumulators depressurizes while the other is still operative. This condition would result in asymmetric control loads.
- The pilot must continuously exert the following forces in order to maintain aircraft attitude when at the recommended safety speed 40 to 60 kt (74 to 111 km/h) and with the collective in its neutral position (approximately 40 percent Torque):
 - Lateral Cyclic: force to push left, approximately 4 daN (9 lbf),
 - Longitudinal Cyclic: force to push forward, approximately 5 daN (11 lbf),

- The collective control will have a tendency to reach by itself the neutral position where zero force is required from the pilot while providing the appropriate power setting for the recommended safety speed in level flight, unless the pilot decides to change power from the neutral position.
- As the aircraft flies at recommended safety speed without significant control loads, this generally allows the pilot sufficient time to choose a landing area suitable for a running landing.
- If necessary, increase IAS, but the control load feedback will also increase in both cyclic axes as the speed is increased, and on the collective as the collective is moved from the neutral (zero force) position. The pilot is required to exert forces continuously in order to keep control of the aircraft and must be careful not to become excessively tired and unable to keep control of the aircraft.
- The recommended flat approach at low speed and the slight running landing can be performed with very little change to collective, which results in reduced cyclic force variations. During the running landing at around 10 kt (19 km/h), the pilot may have to exert a forward longitudinal force up to 17 daN (38 lbf) for less than 30 seconds with low lateral forces.
- If the aircraft is hovering, the control loads change in both direction and intensity as the pilot attempts to maintain a steady position. The pilot will have to exert longitudinal and lateral forces of up to 5 daN (11 lbf) which can change quickly in direction. This results in excessive pilot workload and controllability problems.
- Due to pilot fatigue, it is not recommended to prolong the flight after a hydraulic failure and the pilot should divert to the nearest suitable landing site.

4.3 OTHER HYDRAULIC SYSTEM MALFUNCTIONS

The hydraulic flight control system consists of many different components (hydraulic pump, servos, accumulators, check valves, electro-valves, the hydraulic cut-off switch, [HYD TEST] or [ACCU TST] pushbutton, circuit breakers, relays, electrical wires etc.), which can fail individually. Failures will be characterized by the following indications, some or all of which may be present, thus modifying the usual failure symptoms:

- **HYDR**,
- Gong sounds,
- Flight control load feedback on one or all of the controls (lateral cyclic, longitudinal cyclic, collective, pedals),
- Flight control loads may vary differently with airspeed than they normally do (control loads may increase with decreasing airspeed, or be almost constant with varying airspeed),

- Lateral cyclic loads may be to the **LEFT**, requiring the pilot to pull the cyclic **RIGHT** to maintain attitude,
- Flight control feedback loads may be felt immediately upon component failure. There may be little or no delay between first indication of failure and load feedback,
- The hydraulic cut-off switch may not be effective in opening all the electro-valves, and dumping all the pressure in the accumulators simultaneously,

Even if the usual feedback loads are not felt in the flight controls in case of failure, the pilot will be able to keep control of the aircraft and should immediately set the hydraulic cut-off switch to OFF. Once the switch has been set to OFF, the control loads should return to usual for hydraulics off.

The exception to this is if the hydraulic cut-off switch is inoperative due to loss of electrical power, broken wires, or a faulty switch. The operation of the switch is checked prior to each flight when performing the pre-flight checks. If the hydraulic cut-off switch is inoperative, the control loads should become normal (for hydraulics off) after all the accumulators have depressurized.

There may be some cases where the control forces remain unusual for the duration of the flight. The pilot should minimize the flight time and plan a shallow approach and a slow running landing into wind without hovering.

The maximum forces that the pilot will have to exert on the controls in order to maintain aircraft attitude are approximately:

- Lateral cyclic 15 daN (34 lbf) left or right,
- Longitudinal cyclic 17 daN (38 lbf) forward.

Although these forces are high, they are generally found at the extremes of the speed envelope. The pilot can reduce the required input forces by reaching the safety speed range of 40 to 60 kt (74 to 111 km/h).

SECTION 7.11

ELECTRICAL POWER SYSTEMS

1 DC ELECTRICAL POWER

1.1 GENERAL

The generating and distribution system supplies the electrical network with 28 VDC regulated voltage. The network is supplied by:

- A starter generator located on the engine accessory gear box,
- A 15 A/h battery located in the RH rear cargo bay,
- An optional second battery may be installed,
- A 28 V external power unit (EPU) receptacle on the right side (400 A max).

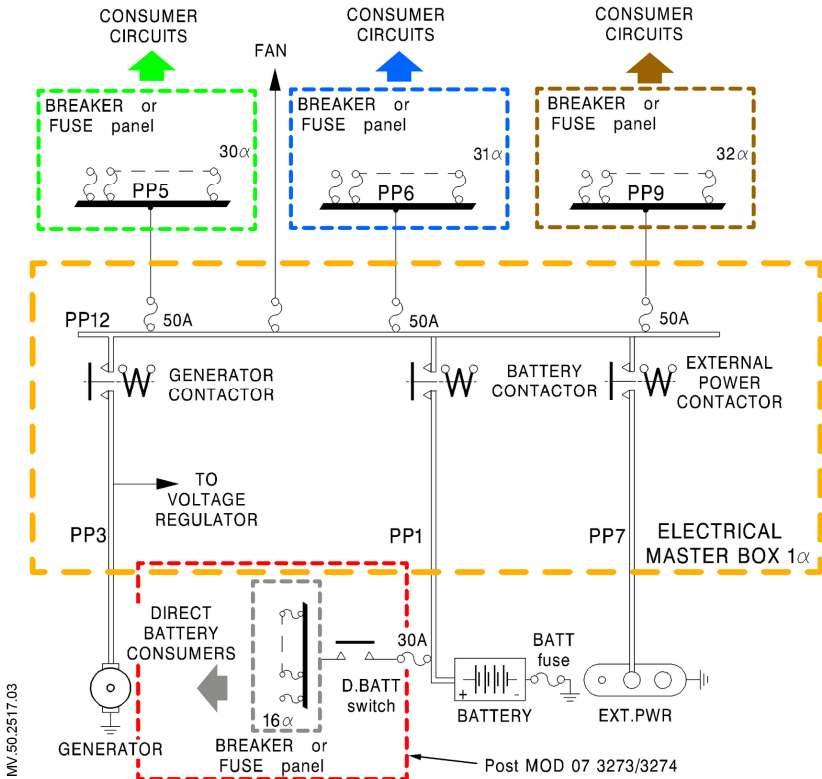


Figure 1: DC distribution description

1.2 DESCRIPTION AND DISTRIBUTION

NOTE

The modification 07-3273 introduces a new SCU panel and modification 07-3274 introduces DC distribution modifications, a direct battery bus-bar and breaker panels in replacement of fuse panels.

The three power sources are connected to the Electrical Master Box (EMB) which ensures the following functions:

- Regulation of the starter generator,
- Connection of the power supplies to the electrical network by line contactors,
- Electrical network distribution and protection against failure of power supplies,
- Interface between the generating and distribution systems and the indicating, control and monitoring system.

The DC distribution system includes:

- An electrical master box (EMB) (rear cargo bay),
- A direct battery bus-bar (PP25) for aircraft after MOD 07-3274,
- A primary bus-bar (PP12 in EMB) and secondary bus-bars (PP5, 6, 9, etc.),
- A cockpit compartment fuse or breaker panel,
- A Systems Control Unit (on the cockpit console),
- An ancillary connector in the cabin (rear of cockpit console).

- Cockpit fuse or breaker panel

For aircraft pre MOD 07-3273/3274:

The cockpit 31 α and 32 α fuse panel is installed on the right side of the cockpit console and the 30 α fuse panel on the right side of the Systems Control Unit.

For aircraft post MOD 07-3273/3274:

The cockpit breaker panels are installed on the left (16 α) and right side (31/32 α , 44 α) of the cockpit console. The 30 α breaker panel is on the left side of the Systems Control Unit.

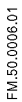
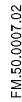
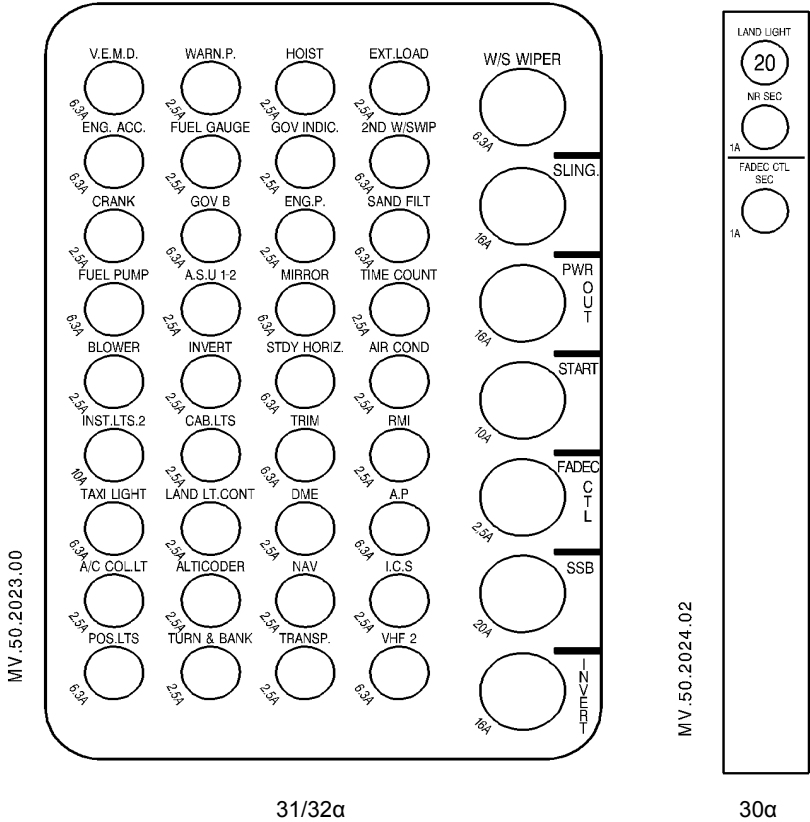
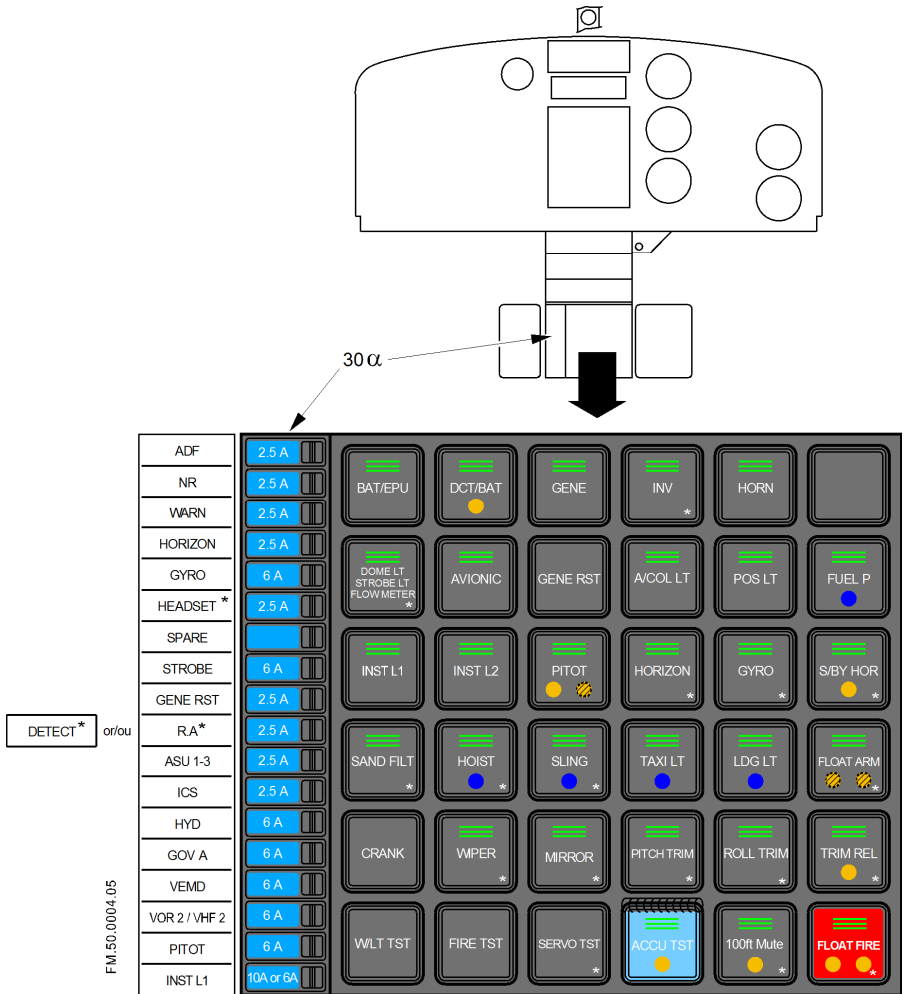
16 α 44 α

Figure 2: Cockpit breaker panels (post MOD 07-3274)



- **Systems control unit**



(⚡) Indicator light illuminated during warning light test only, without further function.

(*) If installed

**Figure 4: Systems Control Unit and 30α breaker panel
(Post MOD 07-3273/3274)**

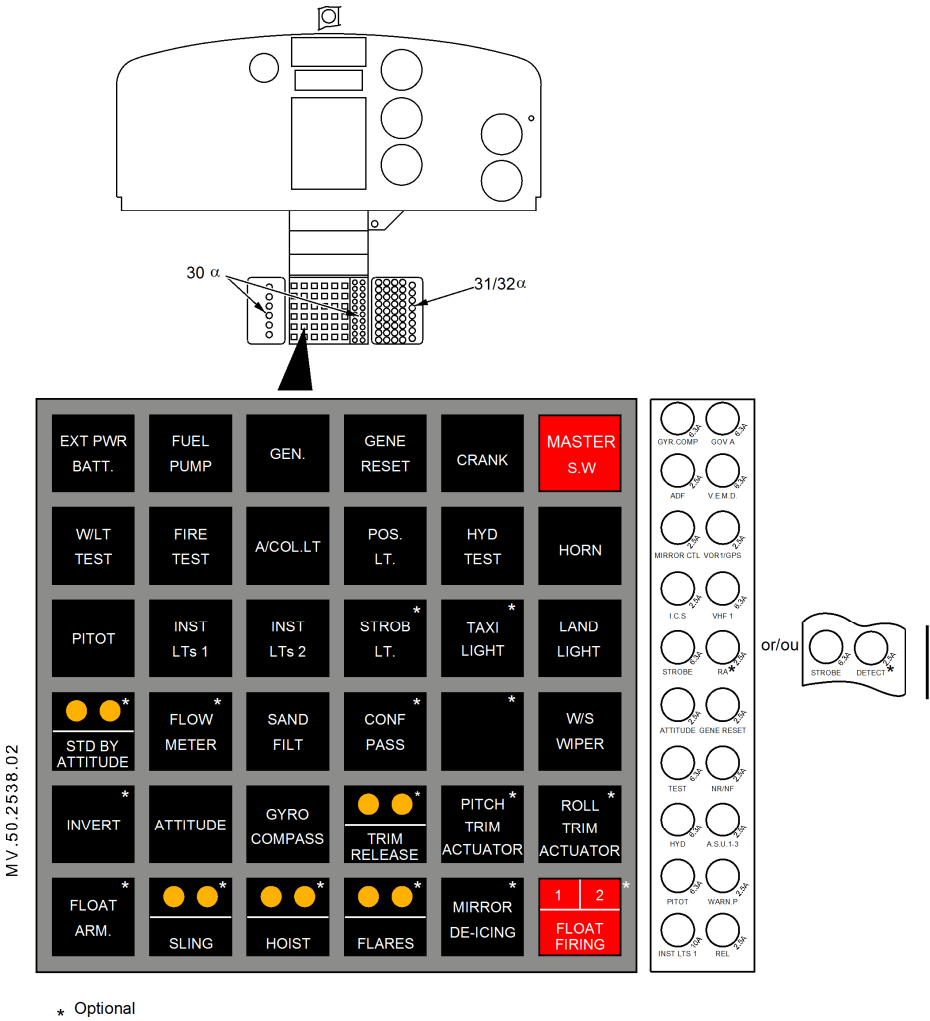


Figure 5: Systems Control Unit and 30 α fuse panel
(Pre MOD 07-3273/3274)

1.3 CONTROL AND MONITORING

- Controls

The following switches and pushbuttons control the DC power system; they are located on the SCU or on the instrument panel:

Pre MOD 07-3273/3274	Post MOD 07-3273/3274
[MASTER SW]	[EMER SW] (lockwired)
-	[DCT/BATT]
[EXT PWR BATT.]	[BAT/EPU]
[CRANK]	[CRANK]
[GEN]	[GENE]
[GENE RESET]	[GENE RST]

- Monitoring

The following warning/caution lights are included on the CWP:

- One red warning light: **BATT TEMP**
- Two amber lights : **GENE** and **BATT**.

The vehicle page on the VEMD displays the following parameters:

- DC bus voltage (U/BUS),
- Generator current (I/GEN),
- Starting current during engine start (I/START).

1.4 OPERATION

- External power system

The EPU is coupled to the primary distribution bus by means of its line contactor when the following conditions are met:

- 28 VDC power is available at EPU plug,
- [MASTER SW] on SCU or [EMER SW] on the instrument panel is in normal (ON) position,
- [EXT PWR BATT.] or [BAT/EPU], is pressed,

On the CWP: **GENE** and **BATT**.

The battery and generator are isolated from the DC system until the EPU is disconnected.

- Battery power system

For aircraft pre MOD 07-3273/3274:

The battery is coupled to the primary distribution bus by means of its line contactor when the following conditions are met:

- 28 VDC power is not available at EPU plug,
- **[MASTER SW]** on the SCU is in normal (ON) position,
- **[EXT PWR BATT.]** is pressed.

A specific direct battery linkage feeds some vital consumers in case of electric circuit failures or **[MASTER SW]** or **[EMER SW]** disconnection:

- Transponder,
- NR/Nf instrument,
- Instrument lighting (circuit n°2),
- Crew adjustable reading light,
- Start-stop engine switch to FADEC (via a relay box).

The battery may be isolated from DC system if either:

- Automatically by using EPU 28 VDC power,
- Manually by setting in OFF position the **[MASTER SW]** or **[EXT PWR BATT.]** pushbuttons.

For aircraft post MOD 07-3273/3274:

The battery is coupled to the primary distribution bus by means of its line contactor when the following conditions are met:

- 28 VDC power is not available at EPU plug,
- **[EMER SW]** on the instrument panel is in normal (ON) position,
- **[BAT/EPU]** is pressed.

A direct battery bus-bar feeds directly some equipments via the **[DCT/BATT]** pushbutton:

- Transponder,
- VHF1/NAV1,
- Instrument lighting (circuit n°2),
- One ICS power line (circuit n°2).

Thus on ground, powering the aircraft only via **[DCT/BATT]** allows the crew to monitor radio communications and to plan its navigation while saving battery power.

The battery may be isolated from DC system if either:

- Automatically by using EPU 28 VDC power,
- Manually by setting in OFF position the **[EMER SW]** or **[BAT/EPU]** and the **[DCT/BATT]** pushbuttons.

In case of an electrical emergency, after actuating the **[EMER SW]** in OFF position, all DC power is switched off except some vital consumers directly fed by the battery:

- NR/Nf instrument,
- Crew adjustable spot light,
- Start-stop engine switch to FADEC (via a relay box).

- Generator system

The 150 A (200 A optional) generator is coupled to the primary distribution bus by means of its line contactor if:

- Engine is running,
- 28 VDC power is not available at EPU plug,
- **[MASTER SW]** or **[EMER SW]** is in normal (ON) position,
- **[GEN]** or **[GENE]** is switched ON,
- Generator voltage exceeds battery voltage by at least 0.5 V.

The generator may be isolated from DC system:

- Manually:
 - By **[GEN]** or **[GENE]** to OFF,
 - For aircraft pre MOD 07-3273/3274 by pressing in **[MASTER SW]** or **[EXT PWR BATT.]** pushbuttons,
 - For aircraft post MOD 07-3273/3274 by setting **[EMER SW]** in OFF position or by switching off **[BAT/EPU]**,
- Automatically:
 - By using EPU 28 VDC power,
 - If a reverse current is detected from battery to generator,
 - If the generator voltage exceeds 31.5 V,
 - When **[CRANK]** is activated.

When the generator is isolated from DC circuit, the **GENE** light comes on.

2 AC ELECTRICAL POWER

2.1 GENERAL

The AC power system is an optional installation required when the aircraft is equipped with an autopilot, gyroscopic instruments or specific equipment items. AC power is provided by a static inverter supplied with DC power.

- PERFORMANCE CHARACTERISTICS

Two different AC power systems are available depending on power required: 250 VA AC and 10 VA AC.

250 VA AC power system:

Power output	: 115 VAC	150 VA
	26 VAC	150 VA
	(with a max. total consumption of 250 VA)	
Frequency	: 400 Hz	

10 VA AC power system:

Power output	: 26 VAC	10 VA
Frequency	: 400 Hz	

2.2 DESCRIPTION AND DISTRIBUTION

Supplied by PP9 DC bus bar through a fuse or breaker located on the 31/32 α fuse or breaker panel (16 A fuse or breaker for 250 VA system, 2.5 A fuse or breaker for 10 VA system).

AC power distribution is protected by fuses or breakers located on the 31/32 α panel.

2.3 OPERATION

The inverter is switched on by pressing in the [INVERT] or [INV] pushbutton on the SCU.

2.4 CONTROL AND MONITORING

A **INV** caution light on the CWP indicates that the inverter is not operating, or a failure of the AC power generating system.

SECTION 7.12

AIR DATA SYSTEM

1 GENERAL

The air data system comprises the pitot tube, two static pressure ports, a bleed valve, an altimeter, a vertical speed indicator, an airspeed indicator and an OAT probe connected to the VEMD and FADEC.

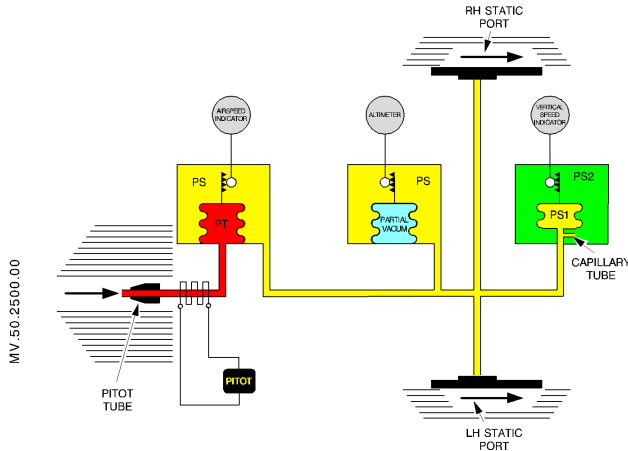


Figure 1: Pitot / static system

2 OPERATION

The pitot tube picks up the total pressure (Pt) which is transmitted to the airspeed indicator. The tube incorporates a heating element. An amber **PITOT** caution light on the caution and warning panel indicates that the heating system is not operating, [PITOT] in OFF position, or has failed.

The two static pressure ports under the cabin pick up the static pressure (Ps) which is transmitted to the conventional flying instruments (airspeed indicator, vertical speed indicator, altimeter) and to the VEMD for performance computation.

A bleed valve is used to drain any condensation water which may accumulate within the system.

SECTION 7.13

VENTILATION, HEATING AND DEMISTING

1 CABIN VENTILATION

Two separate circuits provide ventilation to the cabin:

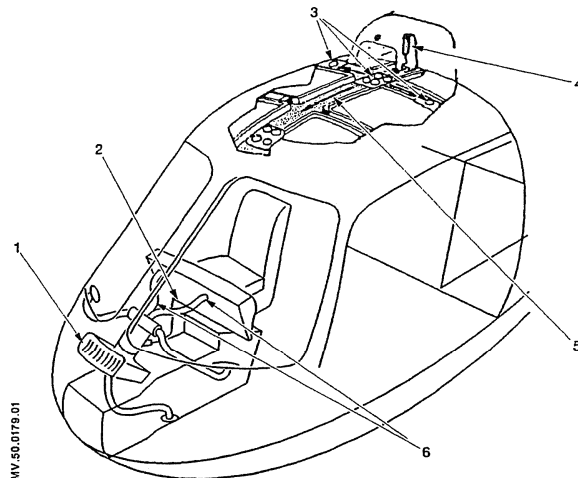
- Front ventilation

The air taken from the front cabin area flows through two ducts and is distributed to the crew. A pull-knob on the instrument panel controls the opening and adjustment of the ventilation circuit.

- Overhead ventilation

The air taken from the upper cabin area through a ram air scoop is then circulated to the air outlets via the structure posts.

Air is diffused by opening and the orientation of each air outlet.



- | | |
|--------------------------|-----------------------|
| 1 – Front aerator | 4 – Ram air scoop |
| 2 – Control pull-knob | 5 – Ventilation duct |
| 3 – Overhead air outlets | 6 – Front air outlets |

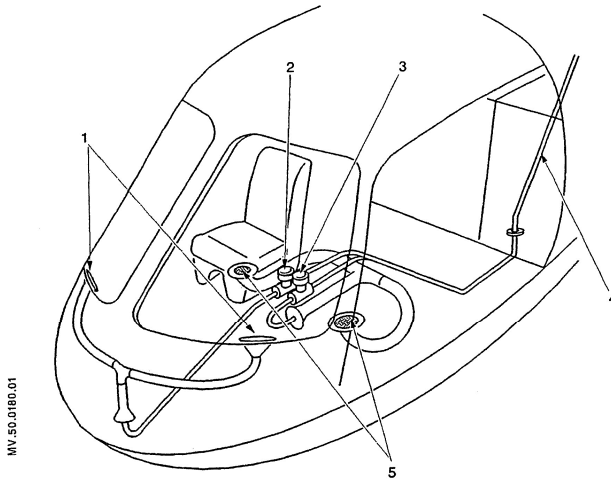
Figure 1: Cabin ventilation

2 HEATING AND DEMISTING SYSTEM

Heating and demisting are achieved by mixing hot P2 air bled from the engine with outside air drawn from under the cabin floor.

The air mixture is channelled through two separate ducts to the heating outlets provided under the front seats and to the demisting manifolds at the bottom of the front windscreen.

Two manually operated valves mounted on the P2 lines and located on the cabin floor are used to open and control warm air distribution.



- | | |
|-----------------------------|-----------------------|
| 1 - Demisting air diffusers | 4 - P2 air ducts |
| 2 - Demisting control valve | 5 - Heating diffusers |
| 3 - Heating control valve | |

Figure 2: Ventilation control

SECTION 7.14

APPAREO VISION 1000*

1 GENERAL

The APPAREO Vision 1000 installation is a cockpit imaging device.

The system captures:

- Aircraft position by GPS data,
- Flight attitudes,
- Cockpit imagery (instrument panel, console (partly), flight controls and partial exterior view),
- Ambient audio.

The flight data and imagery are stored on a crash-hardened memory module as well as on a removable SD card. The flight data can easily be transferred from the SD card to the Appareo's management and visualization software packages.

2 DESCRIPTION

The installation is composed of:

- A camera (4) located on the cabin ceiling equipped with:
 - A 28VDC Power port (1) connected to the 31/32α breaker panel,
 - An Ethernet port (2),
 - A GPS connection (3) linked to the GPS antenna (5),
 - A SD memory card (6),
 - A status indicator LED (7),
- A GPS antenna (5) located on the left side on the top of the glareshield.

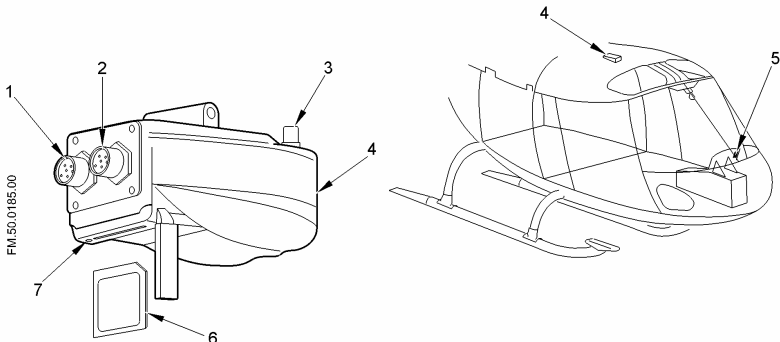


Figure 1: Appareo vision 1000 system

(*) If installed

3 OPERATION

The APPAREO vision 1000 installation does not require any flight crew attention during aircraft operation.
 For correct operation, a SD memory card (6) must be inserted in the camera (4).

NOTE

During low light operations, quality of images may be reduced.

- Status indicator (LED)

LED status	Description
Red	Fault detected.
Blue	Boot sequence.
Green	Normal operation.
Yellow	SD card not inserted, or SD card not formatted, or GPS position not received.
Not illuminated	Equipment not functioning.

SECTION 8

SERVICING

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SECTION 8.1

HANDLING

1 EQUIPMENT REQUIRED

- **For moving the aircraft by hand:**
 - Single or twin handling wheels,
 - Jacking lever.
- **For towing the aircraft with a tractor:**

The above-mentioned equipment, plus:

 - A towing bar installation.

2 HANDLING

2.1 MOVING THE HELICOPTER BY HAND

On prepared ground

- Position the ground handling wheels on the mounting studs according to aircraft balance,
- Install ground handling wheels (wheels outside skids, see detail B),
- Check that wheels are correctly locked (see detail A),
- Lift the aircraft onto its wheels using a jacking lever,
- Lock in this position with retaining pins.

On unprepared ground

- Use twin handling wheels,
- Proceed as before.

2.2 TOWING THE HELICOPTER WITH A TRACTOR

Prepare the aircraft as above and attach the towing bar.

Elastic cords are wrapped around the undercarriage front crosstube.

NOTE

Handles at tail boom must always be used to guide the aircraft when towed.

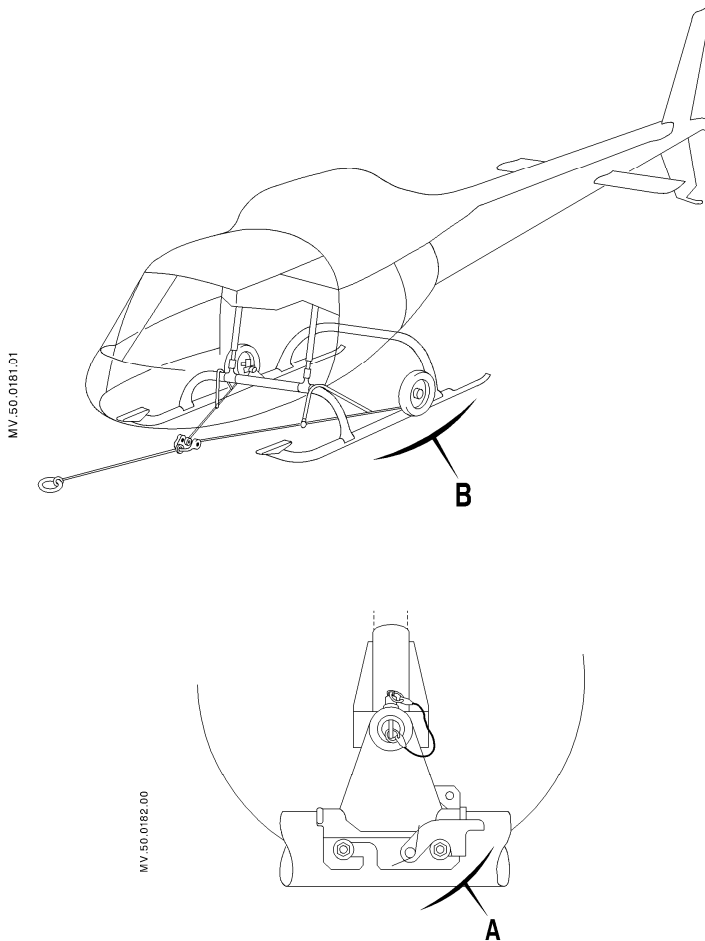


Figure 1: Towing and manual positioning device

SECTION 8.2

SERVICING INSTRUCTIONS

1 FUELS

The authorized fuels are given in SECTION 2 of the basic Flight Manual.

- Capacity (specific gravity: 0.79):

	Liters	US gal	UK gal	kg	lb
TOTAL FUEL TANK CAPACITY	540	143	119	427	940
NON-CONSUMABLE WITH STANDARD FUEL TANK	1.25	0.33	0.28	1	2.20
NON-CONSUMABLE WITH CRASH RESISTANT FUEL TANK	2	0.53	0.44	1.58	3.48
CONSUMABLE FUEL REMAINING WHEN FUEL COMES ON	60	15.8	13.1	47.4	104

2 FUEL ADDITIVES

The authorized anti-ice, fungicide and thermal stability fuel additives are given in SECTION 2 5 § 1 "APPROVED FUELS" of the basic Flight Manual.

If there is any doubt as to the concentration of additive in the contents of a fuel tank, the fuel is to be drained from the tank and replaced by fuel containing a known proportion of additive within the defined limits given in SECTION 2 of the Flight Manual unless it is possible to measure the concentration using a differential refractometer.

3 LUBRICANTS

- Engine oil system

Lubricants and commercial descriptions:

- Authorized lubricants Refer to SECTION 2 of the basic Flight Manual,
- Commercial descriptions..... Refer to the TURBOMECA publications.

Capacity:

Engine oil system capacity 6.2 liters (1.64 US gal, 1.36 UK gal).

- Transmission Components

Lubricants:

The authorized lubricants are given in SECTION 2 of the basic Flight Manual.

Capacity:

- Main gearbox (system included) 6.5 liters (1.7 US gal, 1.4 UK gal),
- Tail gearbox (system included) 0.33 liter (0.08 US gal, 0.07 UK gal).

4 HYDRAULIC FLUIDS

- Hydraulic Fluids

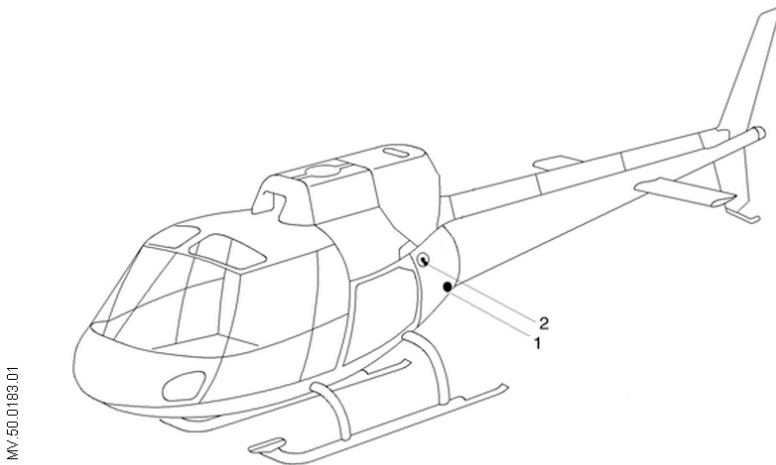
The authorized hydraulic fluids are given in SECTION 2 of the basic Flight Manual.

System:

- Total capacity of the system 3 liters (0.79 US gal, 0.66 UK gal).
- Operating pressure 40 bars (580 psi).

5 REFUELING

5.1 NORMAL REFUELING



MV 50.0183.01

Figure 1: Filler plug and electrostatic connector location

- Place the helicopter on a level surface,
- Connect the bowser earth cable to the electrostatic balance connector (1) on the helicopter,
- Check the quantity of fuel remaining in the tank on the VEMD fuel indicator,
- Observe the following safety precautions:
 - Ensure that the aircraft electrical power supply is switched off,
 - Place a fire extinguisher near the work area,
 - Strictly prohibit smoking in the safety area,
 - Prohibit the use of any means of lighting not conforming to the safety rules,
 - Ensure, during refueling (or defueling), that the bowser (or the defueling unit) is connected to the aircraft by the electrostatic balance connector (1),
 - Strictly prohibit draining of fuel tanks, whether partial or total, inside a hangar or shop.
- Fill the tank, monitoring the quantity of fuel delivered on the bowser flowmeter,
- Position and lock the filler plug (2),
- Disconnect the bowser earth connector from the aircraft electrostatic balance connector (1),
- Check that the difference in the VEMD fuel indicator readings corresponds to the quantity of fuel delivered and determines the corresponding weight.

5.2 REFUELING WITH ROTOR SPINNING

WARNING

REFUELING WITH ROTORS SPINNING IS PERMITTED ONLY AFTER PRIOR APPROVAL IS GIVEN BY THE AUTHORITY CONCERNED AND IN COMPLIANCE WITH OPERATIONAL REGULATIONS.

Observe the normal refueling safety precautions and strictly comply with the additional instructions defined below:

- Set the aircraft on a firm surface,
- Head aircraft into forward wind sector $\pm 45^\circ$ if wind above 15 kt (28 km/h),
- Lock the collective in full low pitch position,
- Pilot must remain at the flight controls during all refueling operations with rotors spinning,
- No radio transmission or operating of electrical switches,
- Limit refueling to 95% in order to prevent any fuel spillage,
- The pilot must always have someone in view who can signal to the mechanic to stop refueling,
- No one approaches the aircraft rotor disc unless acknowledged by the pilot,
- Make sure that the filler cap is closed and secured, report it to the pilot.

NOTE

For refueling with rotors spinning, it is preferable to set the twist grip to **FLIGHT** position. **IDLE** position may also be selected if required by local operational conditions.

Refueling operations with rotors set to **IDLE** should be conducted with wind velocity ≤ 15 kt (28 km/h) with a gust spread ≤ 5 kt (9 km/h) and should never be conducted on an elevated helipad or other structure where a vertical component of wind may be present.

Refueling with rotors spinning represents a potential hazard to the safety of the helicopter, its occupants and third-parties.

6 BATTERY CONNECTION

Prior to connect the main battery ground connection (large section wire), connect the small test ground wire and check visually that no pushbutton are set to **ON** in the cockpit System Control Unit.

SECTION 8.3

TEST SHEETS

1 GENERAL

The test sheets are intended to sum up the checks to be carried out in flight or on the ground, with rotors turning either after replacement of main components, or after an extensive operation, or further to periodic inspections.

The test sheets are in the form of reproducible sheets which can be filled in directly by the crew.

CAUTION

Since these checks do not form part of normal helicopter operation, they shall be carried out only by qualified personnel under the operator's responsibility.

2 LIST OF TEST SHEETS

- No. 0 FLIGHT REPORT
- No. 1 VEMD
- No. 2 GROUND RUN
- No. 3 HOVER FLIGHT
- No. 4 MAXIMUM CONTINUOUS POWER CLIMB
- No. 5 MAXIMUM TAKEOFF POWER CHECK
- No. 6 MAXIMUM CONTINUOUS POWER LEVEL FLIGHT
- No. 7 AUTOROTATION - 65 kt (120 km/h)

TEST SHEETS TO BE COMPLETED ACCORDING TO THE MAINTENANCE ACTION OR COMPONENT REPLACED:

The test items in the following table are a general proposal. Their extent may be adapted by the operator depending on the maintenance action performed.

TEST SHEETS No. →	0	1		2					3		4	5	6		7
MAINTENANCE ACTION OR COMPONENTS REPLACED ↓		A	B	A	B	C	D	E	A	B			A	B	
ENGINE, FADEC OR MODULE	●	/	◆	●	/	/	●	●	/	◆	◆	◆	/	●	/
EBCAU	●	/	/	●	/	/	●	/	/	/	/	/	/	/	/
MGB OR MODULE	●	/	/	●	●	●	◆	/	●	/	/	/	●	/	◆
MAIN ROTOR HUB OR BLADE CHANGE	●	/	/	/	●	●	/	/	●	/	/	/	●	/	◆
TAIL ROTOR – TAIL ROTOR DRIVE SHAFT	●	/	/	/	/	●	/	/	/	/	/	/	/	/	◆
VEMD	●	●	●	●	/	/	◆	/	/	/	/	/	/	◆	/

(◆) Refer to test sheet.

SHEET No 0		HELICOPTER AS 350 B3 Arriel 2B1		FLIGHT REPORT			
AIRCRAFT S/N :				VEMD		WEATHER	
AIRFIELD				VEMD Flight Number : []		Wind direction : []	
DATE				VEMD Flight Duration : []		Wind velocity : []	
CREW						OAT	
TAKE-OFF TIME				Cycles Ng []	Total []	QNH	
LANDING TIME				Cycles Nf []	Total []	QFE	
FLIGHT TIME						Hp	
ENGINE START						VEMD	
PILOT				Overlimit detected : YES [] NO []	Failures detected : YES [] NO []		
CREW MEMBER				Overlimit origin, value, duration : []	Failures codes and time : []		
WEIGHT				MAIN WORK COMPLETED BEFORE FLIGHT		REMARKS MADE BY CREW AFTER FLIGHT	
EQUIPPED EMPTY WEIGHT:							
CREW							
BALLAST							
ZERO FUEL WEIGHT							
FUEL							
TAKE-OFF WEIGHT							
CG							
UNITS USED (cross out as applicable)							
WEIGHT : [] kg [] lb							
FUEL : [] kg [] lb							
[] UK gal [] l [] US gal							
ALTITUDE : [] m [] ft							
AIRSPEED : [] kt [] km/h [] MPH				SPECIFIC OPTIONAL		CREW VISAS	

SHEET No 1A		HELICOPTER AS 350 B3 Arriel 2B1	VEMD CONFIGURATION AND MEMORY	
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED/LIMITATIONS	RESULTS OBTAINED
Before starting : Warning lights test: [W/LT tst]PRESS.			Test pattern on both screens and VEMD software serial number Identical variation on both screens Identical variation on both screens	Reference number : YES ☐ NO ☐ YES ☐ NO ☐
Brightness variation using lighting selector Brightness variation using +/- pushbutton on VEMD				
VEMD CONFIGURATION MODE (Refer to FILM SECTION 7)			Confirm options and units used on customer's aircraft. CONFIGURATION MENU pages appears: STARTER GENERATOR : FLOWMETER : SLING : SYSTEMS UNITS : ALTITUDE UNIT :	GOOD ☐ BAD ☐ 150 A ☐ 200 A ☐ I ☐ N/I ☐ I ☐ N/I ☐ IS ☐ IMPERIAL ☐ FEET ☐ METERS ☐ Kg/Lb ☐ L ☐ US G ☐ UK G ☐
• Select CONFIGURATION MODE ☐ OFF1 and ☐ OFF2 then hold ☐ SELECT and simultaneously ☐ ENTER				
VEMD MAINTENANCE MODE (Refer to FILM SECTION 7)			MAINTENANCE MENU pages appears: "FLIGHT REPORT" page ⇒ SELECT → ENTER Record last "VEMD flight" data : FLIGHT NUMBER DURATION CYCLES NG CYCLES NF OVERLIMIT DETECTED Record origin, value and duration FAILURE DETECTED Record code, type and time	GOOD ☐ BAD ☐ Number : Duration : Partial NG : Partial NF : Total NG : Total NF : YES ☐ NO ☐ YES ☐ NO ☐
• Select MAINTENANCE MODE ☐ OFF1 then hold ☐ OFF2 and ☐ SCROLL and simultaneously ☐ RESET				

SHEET No 1B		HELICOPTER AS 350 B3 Arriel 2B1	VEMD CONFIGURATION AND MEMORY	
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
VEMD CYCLE COUNTING (To be performed after a change of VEMD and after change of Engine or FADEC)			At VEMD removal : . Record engine total cycles After VEMD installation : . VEMD total cycles after 1st Start-up . VEMD total cycles after 2nd Start-up NOTE After the first start-up, only the total engine cycles of the new VEMD will be correct. The partial cycles will be correct after the second start-up.	NG : NF : NG : NF : NG : NF :

SHEET No 2A	HELICOPTER AS 350 B3 Arriel 2B1	GROUND RUN	
TEST PHASES AND REQUIREMENTS		RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
BEFORE ENGINE STARTING [EXT PWR/BATT] or [BAT/EPU] ON, twist grip set on IDLE VEMD pages displayed : - Upper screen : 3-info page - Bottom screen : vehicle page [HORN] switch : ON (activated) Twist grip set to FLIGHT Twist grip set back to IDLE		During FADEC self-test, red and amber GOV flash Fuel flow (FF) after stabilization = 60 kg/h +/- 5 (76 l/h +/- 6) After FADEC self-test : FADEC amber GOV and red GOV lights go off Free return to "FLIGHT" when out of "IDLE" position: TWGT GRIP light illuminates: "GONG" sounds: Record parameters before start : Before start check Battery Voltage ≥ 22 V - Starting selector ON : T4 regulated between 600 and 700°C - On warning panel : FUEL P and GENE lights go off FLI page displayed on VEMD ENG P light goes off MGB P light goes off NR ≤ 110 rpm (cold oil) NR ≤ 200 rpm (warm oil) SERVO* and HYDR lights go off (*) if installed Low NR warning sounds for $250 \leq \text{NR} \leq 360$ rpm	FF = GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ U min : V T4 max : °C GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ Alarm ☐ rpm Alarm OFF ☐ rpm FLI : FUEL QTY : % FF : °C END : % U/BUS : V I/OEN : A Eng oil b T : rpm NR : rpm
ENGINE STARTING Comply with Normal Procedures See FLM SECTION 4 - Ng ≈ 50 % - Ng = 60 % - Ng ≤ 68 % - NR ≤ 200 rpm - Engine in stabilized IDLE Record data when twist grip in FLIGHT position : NF / NR check (Pedals centered)		• Check: $377 \leq \text{Nf} \leq \text{NR} \leq 382$ rpm.	

SHEET No 2B		HELICOPTER AS 350 B3 Arriel 2B1	GROUND RUN	
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
Aircraft on ground, full low pitch. MAIN ROTOR TRACK Record track with stroboscope . TRACK CHECK : The setting must be carried out with the yellow (■) blade as reference.			Full low pitch NR stabilized NR = 379 +/- 3 rpm • Blades track difference is to be < ½ target.	

SHEET No 2C		HELICOPTER AS 350 B3 Arriel 2B1	GROUND RUN						
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED					
MAIN ROTOR BALANCING			NOTE Correct blade tracking is to be done prior rotor balancing. Rotor sleeves pre-loading condition: Yellow sleeve : 6 weights Red sleeve : 4 weights Blue sleeve : 0 weight		Unbalance		Sleeves loads adjustments		
					Amplitude	Phase	Red	Yellow	Blue
TAIL ROTOR DRIVE SHAFT BALANCING			To be performed only after maintenance action on tail rotor drive shaft. Long shaft: Accelerometer on 5th bearing Shaft rotation speed ≈ 6000 rpm Unbalance < 0.8 IPS		Unbalance		Loads adjustment		
					Amplitude	Phase	1	2	3
TAIL ROTOR BALANCING			To be performed only after maintenance action on tail rotor. Unbalance < 0.35 IPS		Unbalance		Loads adjustment		
					Amplitude	Phase	Point Nb	Load	

SHEET No 2D		HELICOPTER AS 350 B3 Arriel 2B1	GROUND RUN	
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
EBCAU CHECK (After Engine, fuel control or EBCAU maintenance action only). Plastic guard up, [EBCAU TEST] PRESS. [EBCAU TEST] PRESS again to end the test			“GONG” sounds Red GOV comes on Amber GOV comes on NR increases to 390 +/-3 rpm Red and amber GOV lights go off NR decreases to 379 +/-3 rpm	GOOD ☐ GOOD ☐ GOOD ☐ NR : rpm GOOD ☐ GOOD ☐ BAD ☐ BAD ☐ BAD ☐ BAD ☐
ENGINE SHUT - DOWN Comply with Normal Procedures. Refer to FLM SECTION 4 After 30 sec. ; - Starter switch on OFF position ; - Engine run-down time ; - Rotor brake efficiency ; - Flight report page ;			Check Ng at idle : 67 % ≤ Ng ≤ 69 %. - VEMD : 3-data mode for Ng ≤ 50 % Check GONG sounds for each red lights ON As specified in engine manufacturer maintenance manual Apply brake at 170 rpm : t ≤ 25 sec. Automatic “Flight Report” page display NR < 70 rpm. Record flight report data : FLIGHT NUMBER DURATION CYCLES NG CYCLES NF OVERLIMIT DETECTED Record origin, value and duration FAILURE DETECTED Record code, type and time	Idle Ng : % GOOD ☐ GOOD ☐ BAD ☐ BAD ☐ Run down time : sec. Braking time : sec. GOOD ☐ BAD ☐ Number : Duration : Partial NG: Partial NF: Total NG: Total NF: YES ☐ NO ☐ YES ☐ NO ☐

SHEET N° 2E		HELICOPTER AS 350 B3 Arriel 2B1	GROUND RUN : Check of Arriel 2B1 FADEC CHANNEL B Application of the procedures of worksheet 73.20.00 of AMM	
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
BEFORE ENGINE STARTING [BAT/EPU] and [DCT/BAT] or [EXT PWR BATT] set to ON. [BAT/EPU] and [DCT/BAT] and [EXT PWR BATT] set to OFF. Make DECU metering valve channel A unplugged. Refer to AMM 73.20.00 [BAT/EPU] and [DCT/BAT] or [EXT PWR BATT] set to ON. Read failure codes on VEMD 3-info page (in the FADEC data window). Remain 15 min. in this configuration.			- Red [GOV] and amber [GOV] lights go off. - Amber [GOV] warning light flashes. - FAIL 1 : 0008 - FAIL 2 : 0002	GOOD BAD GOOD BAD GOOD BAD GOOD BAD

SHEET No 2E		HELICOPTER AS 350 B3 Arriel 2B1	GROUND RUN : Check of Arriel 2B1 FADEC CHANNEL B Application of the procedures of worksheet 73.20.00 of AMM	
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
ENGINE STARTING Comply with normal procedures. Refer to FLM SECTION 4. Set twist grip to FLIGHT position. Remain at nominal NR for 3 min. Set twist grip back to IDLE position. Read failures codes on VEMD FLI page (in the FADEC data window). ENGINE SHUT-DOWN Comply with normal procedures. Refer to FLM SECTION 4. [BAT/EPU] and [DCT/BAT] or [EXT PWR BATT] set to OFF. Make DECU metering valve channel A connector re-plugged. [BAT/EPU] and [DCT/BAT] or [EXT PWR BATT] set to ON. [BAT/EPU] and [DCT/BAT] or [EXT PWR BATT] set to OFF.			- Red GOV and amber GOV warning lights remain off. - Amber GOV warning light flashes. - FAIL 1 : 0008 - FAIL 2 : 0002 - Amber GOV warning light flashes. - Red GOV and amber GOV warning lights go off.	GOOD BAD GOOD BAD GOOD BAD GOOD BAD GOOD BAD

SHEET No 3A	HELICOPTER AS 350 B3 Arriel 2B1	HOVER FLIGHT	
TEST PHASES AND REQUIREMENTS		RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
Aircraft on the ground GROUND RESONANCE CHECK On concrete ground : Check with cyclic movements ($\pm 5\%$ of the full course) at full low pitch and with TQ = 35%.		No diverging oscillations : convergence between 3 to 4 cycles Full low pitch : ☐ TQ = 35% : ☐ NOTE Some non diverging oscillations are permissible if the phenomenon stops when cyclic is set back to neutral position and collective at full low pitch.	GOOD ☐ BAD ☐ GOOD ☐ BAD ☐
TRACK AND BALANCE CHECKS Check main rotor balance. Check main rotor blades track.		Record data : - Unbalance to be < 0.2 IPS Record data using a stroboscope : - Blades track difference is to be $< \frac{1}{2}$ target.	Peak amplitude IPS Phase Track :

SHEET No 3B		HELICOPTER AS 350 B3 Arriel 2B1		HOVER FLIGHT	
TEST PHASES AND REQUIREMENTS		RESULTS TO BE OBTAINED / LIMITATIONS		RESULTS OBTAINED	
This test must be performed after major engine maintenance or after maintenance on the fuel control system. HOVER FLIGHT CHECK Increase collective to hover. at 5 ft (1.5 m) in 2 sec. HOVER CHECK		The rotor speed decay must be weak and will increase at steady state. No low NR audio warning.		GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ FLI FUEL QTY : NG : % FF: T4 : °C END: TQ : % U/BUS : V I/GEN : A Eng oil T : °C Eng oil P : b NR : rpm	

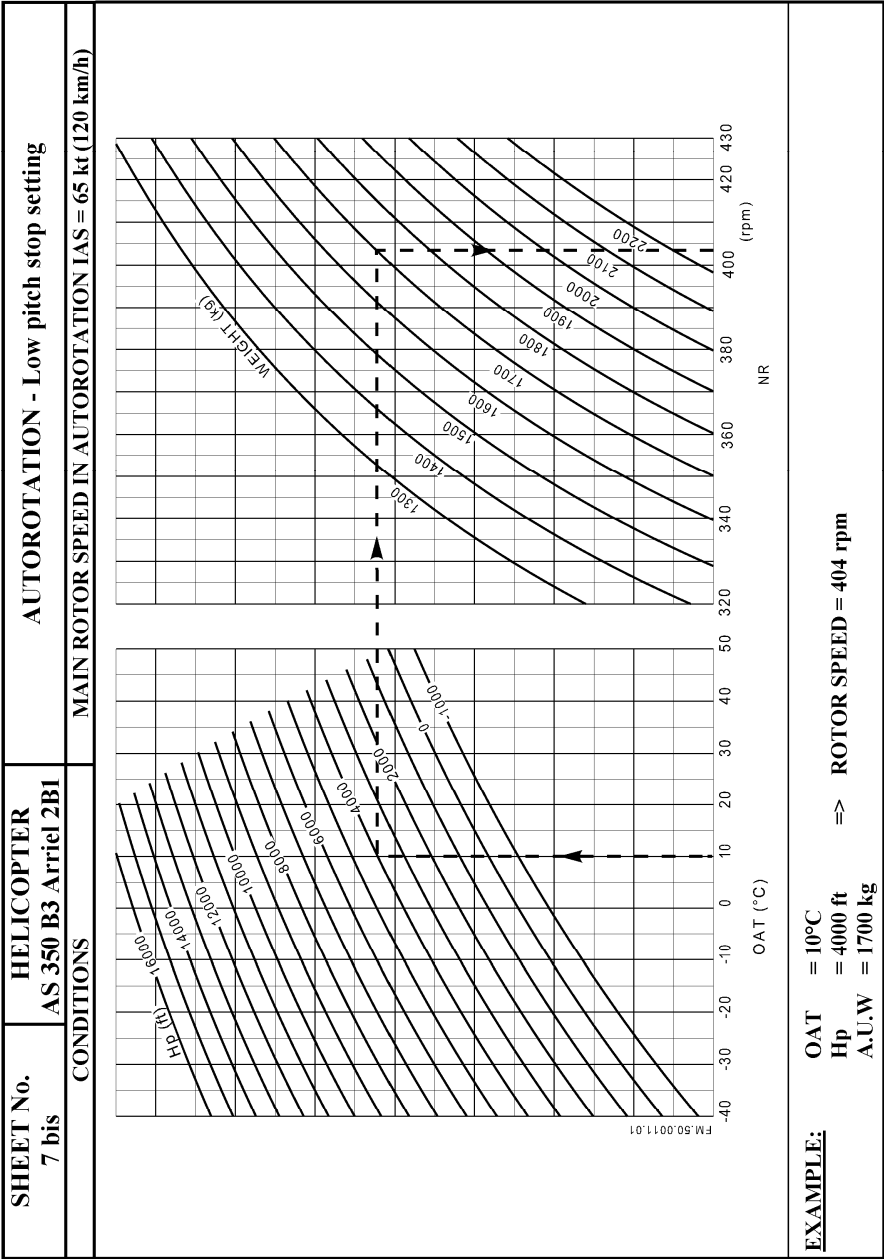
SHEET No. 4		HELICOPTER AS 350 B3 Arriel 2B1	MAXIMUM CONTINUOUS POWER CLIMB						
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED					
This test must only be performed if there is a doubt concerning the helicopter performance. Set power to obtain MCP limit CLIMB PERFORMANCE CHECK Set IAS kt = Vy = 65 kt at 0 Hp – (1 kt per 1000 ft) Set IAS km/h = Vy = 120 km/h at 0 Hp – (2 km/h per 300 m) No sideslip, no turbulence, no thermal activity.			Record data during continuous climb : FUEL QTY: AUW :						
Time :	Hp	OAT	Ng	t4	TQ	Rate of climb	ΔHp		
0									
0'30"									
1'									
1'30"									
2'									
2'30"									
3'									
3'30"									
4'									
The aircraft climb performance shall be at or above the performance given in SECTION 5 of the Flight Manual.			GOOD		BAD				

SHEET No 5		HELICOPTER AS 350 B3 Arriel 2B1	MAXIMUM TAKEOFF POWER	
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
This test must be performed after major engine maintenance or after maintenance on the fuel control system. <u>MAXIMUM TAKEOFF POWER CHECK</u> Check all P2 bleed air consumers off With IAS < 40 kt (74 km/h) increase collective to obtain TOP limit. NOTE It is recommended to perform this check with Hp ≥ 5000 ft (1524 m) to reach engine limit rather than torque limit. Max power audio warning check			See limitations FLM SECTION 2 Record data : CAUTION 10 < Max. FLI < 10.4 with t < 5 sec. Max power audio warning sounds for FLI > 10 for more than 1.5 sec.	Hp OAT FLI : Ng : % T4: °C TQ: % GOOD ☐ BAD ☐

SHEET No 6A		HELICOPTER AS 350 B3 Arriel 2B1	MAXIMUM CONTINUOUS POWER LEVEL FLIGHT	
TEST PHASES AND REQUIREMENTS		RESULTS TO BE OBTAINED / LIMITATIONS		RESULTS OBTAINED
MCP LEVEL FLIGHT Set power to obtain MCP, no turbulence, stabilize well before reading IAS.				IAS :
• <u>NR CHECK IN MCP LEVEL FLIGHT</u>		Adjust NR by NR trim to: Hp ≤ 1000 ft (300 m) : NR ≈ 390 rpm Hp = 5000 ft (1524 m) : NR ≈ 394 rpm Hp = 10000 ft (3000 m) : NR ≈ 396 rpm		Hp :
• <u>VIBRATION AND ROTOR TRACK AND BALANCE CHECK</u>				NR : rpm
Check 3 Ω Z vibration level (pilot seat)		Record data		3 Ω Z : IPS
Check rotor unbalance		Unbalance < 0.2 IPS		1 Ω Y : IPS Phase :
Check 1 Ω Z vibration level		Record 1 Ω nose section accelerometer data : - Using nose accelerometer diagram, determine red and blue blades tabs settings. • Max amplitude < 0.2 IPS Confirm settings using the stroboscope :		1 Ω Z : IPS Phase :
Check main rotor blades track		• Max rotor track < 1 target		Tab adjustments : ° Red : ° Blue : °
		NOTE The best setting is rarely obtained with a perfect track.		Track :
VNE WITH POWER ON From MCP level flight, increase speed to VNE		VNE kt = 155 kt at 0 Hp – (3 kt/1000 ft) VNE km/h = 287 km/h at 0 Hp – (5 km/h per 300 m)		Peak amplitude IPS Phase
Check main rotor unbalance.		Record data : • Unbalance < 0.3 IPS		1 Ω Y :
Check main rotor blades track.		Record data using a stroboscope : • Blades track difference is to be < 1 target.		Track :

SHEET No 6B		HELICOPTER AS 350 B3 Arriel 2B1	MAXIMUM CONTINUOUS POWER LEVEL FLIGHT																
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED															
POWER TRANSIENT CHECKS This test is not mandatory after VEMD maintenance. At Vy, no engine flame-out check : - Pitch decrease from MCP to full low pitch in 2 sec. - Pitch increase from TQ = 1 % to MCP in 3 sec. then in 1 sec.			No engine flame-out Minimum NR > 370 rpm No surge	GOOD ☐ BAD ☐ MIn NR <table><tr><td>rpm</td></tr><tr><td>rpm</td></tr></table> Max t4 <table><tr><td>°C</td></tr><tr><td>°C</td></tr></table> 3 sec. <table><tr><td>rpm</td></tr></table> 1 sec. <table><tr><td>rpm</td></tr></table>	rpm	rpm	°C	°C	rpm	rpm									
rpm																			
rpm																			
°C																			
°C																			
rpm																			
rpm																			
P2 AIR BLEEDS CHECKS Heating, demisting and sand filter (if installed) ON one by one Check t4 increase at constant power			Heating ON, t4 increases. Demisting ON, t4 increases. Sand filter ON, t4 increases, ☐ indication present on VEMD.	Δ t4 : <table><tr><td>°C</td></tr></table> Δ t4 : <table><tr><td>°C</td></tr></table> Δ t4 : <table><tr><td>°C</td></tr></table>	°C	°C	°C												
°C																			
°C																			
°C																			
BLEED VALVE CHECK Record P0 and P3 on FLI FADEC message window Record opening and closing Ng slow pitch changes			Opening and closing Ng values to be checked upon engine manufacturer maintenance manual Check Ng values in tolerance and presence of bleed valve open flag on VEMD	Hp : <table><tr><td></td></tr></table> P3 : <table><tr><td></td></tr></table> Opening Ng <table><tr><td>%</td></tr></table> Closing Ng <table><tr><td>%</td></tr></table> GOOD <table><tr><td>☐</td></tr></table> BAD <table><tr><td>☐</td></tr></table> GOOD <table><tr><td>☐</td></tr></table> BAD <table><tr><td>☐</td></tr></table> OAT : <table><tr><td></td></tr></table> PO : <table><tr><td></td></tr></table>			%	%	☐	☐	☐	☐							
%																			
%																			
☐																			
☐																			
☐																			
☐																			
ENGINE POWER CHECK Check all P2 bleed air consumers off. Set power to obtain MCP Stabilize at least 2 min. before switching to ENGINE POWER CHECK page (Use any of the SCROLL pushbuttons)			Note I/GEN before beginning power check. Record data after stabilization Check results by using the Flight Manual engine power check diagrams SECTION 5.	I/GEN : <table><tr><td>A</td></tr></table> IAS <table><tr><td></td></tr></table> NR <table><tr><td>rpm</td></tr></table> NG <table><tr><td>%</td></tr></table> NF <table><tr><td>rpm</td></tr></table> T4 <table><tr><td>°C</td></tr></table> Hp <table><tr><td>ft</td></tr></table> TQ <table><tr><td>%</td></tr></table> OAT <table><tr><td>°C</td></tr></table> T4 MARGIN <table><tr><td>°C</td></tr></table> TQ MARGIN <table><tr><td>%</td></tr></table> GOOD <table><tr><td>☐</td></tr></table> BAD <table><tr><td>☐</td></tr></table> GOOD <table><tr><td>☐</td></tr></table> BAD <table><tr><td>☐</td></tr></table>	A		rpm	%	rpm	°C	ft	%	°C	°C	%	☐	☐	☐	☐
A																			
rpm																			
%																			
rpm																			
°C																			
ft																			
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SHEET No 7		HELICOPTER AS 350 B3 Arriel 2B1	AUTOROTATION - 65 kt (120 km/h)	
TEST PHASES AND REQUIREMENTS			RESULTS TO BE OBTAINED / LIMITATIONS	RESULTS OBTAINED
LOW PITCH STOP SETTING This test must be performed after any maintenance action that may affect the rigging of the main rotor control system. Autorotation at IAS = 65 kt (120 km/h - 75 mph) If possible: · minimum weight < 1900 kg (4189 lb). · altitude: Hp ≤ 5000 ft (1524 m). Perform a steady state autorotation at IAS = 65 kt (120 km/h) and collective on the low pitch stop. Check that the engine is desynchronized (NR > NF). Best practice is to perform the test at conditions that lead to a NR between 400 and 415 rpm.			Max. NR: 430 rpm. NOTE If it is impossible to set the pitch on the stop without exceeding the max rpm, decrease the altitude or the weight if possible. Record data : If the free turbine is not desynchronized, increase Hp. NR must be in compliance with the value determined by means of following chart – 0 /+ 10 rpm. High NR audio warning sounds for NR ≥410 rpm VNE power off = 125 kt at 0 Hp - (3 kt per 1000 ft) VNE power off = 232 km/h at 0 Hp - (5 km/h per 300m) Constant heading maintained	FUEL QTY: Hp : OAT : AUW: NR : rpm Target NR rpm GOOD ☐ BAD ☐ GOOD ☐ BAD ☐ Hp : IAS : GOOD ☐ BAD ☐
Check high NR audio warning VNE WITH POWER OFF This test must be performed after maintenance action on the tail rotor control system. From level flight at IAS = VNE power off, decrease collective pitch to obtain TQ = 0 and NR ≈ 410 rpm. Check of left pedal margin				



SECTION 8.4

DAILY CHECKS

1 P INSPECTION

A pilot with the correct training and accreditation in compliance with the local maintenance regulations in force, has to refer to the Aircraft Maintenance Manual chapter 05-40 to perform Daily checks and the P inspection.

A pilot-owner of the helicopter has to refer to the EASA regulation and part M.A.803 Appendix VIII (List of inspections that can be done by a pilot-owner) or to equivalent local regulations, to identify if he can perform the Daily checks and the P inspection described in the Aircraft Maintenance Manual chapter 05-40.

SECTION 9

OPERATIONAL INFORMATION

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1 GENERAL.....	1
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3 OPERATION.....	2
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9.13 Not used

9.14 AIR AMBULANCE INSTALLATION

1 GENERAL.....	1
2 DESCRIPTION	1
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9.15 SKI INSTALLATION

1 GENERAL.....	1
2 DESCRIPTION	1
3 OPERATING PROCEDURE.....	1

9.16 Not used

9.17 AIR CONDITIONING SYSTEM

1 GENERAL.....	1
2 DESCRIPTION AND OPERATION.....	1
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4 PERFORMANCE DATA	2

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1 GENERAL.....	1
2 INSTALLATION AND OPERATION DESCRIPTION	1

SECTION 9.1

RECOMMENDATIONS FOR CARGO SLING OPERATIONS

1 PERSONNEL TRAINING

Cargo sling operations may only be conducted by pilots who already have considerable experience with this type of aircraft.

No pilot should perform solo external load flights without first having accomplished such operations with a qualified instructor.

Mechanics on ground duty must be fully informed by the pilot before each new operation, in particular as regards:

- Their position on the ground considering the proposed flight path,
- The direction in which to move away,
- The hook-up operation,
- Hand signals to be used or radio instructions,
- Protective equipment: Helmets, gloves, goggles (if applicable),
- The number of round trips between refueling operations,
- The manner of retrieving slings and nets.

2 MANDATORY PRE-OPERATIONAL CHECKS

- Helicopter condition:

In addition to the normal pre-flight checks of the helicopter, the release unit must be carefully inspected and the mechanism checked for correct release operation,

- Condition of sling equipment:

The nets, straps and slings must be examined thoroughly. Any worn or frayed components are to be discarded.

The cables, strops and shackles must be capable of carrying three times the maximum anticipated load.

- Preparation of loads:

Make sure that all participants are well aware of the weight of the loads. Ensure that the method of suspension is understood.

- Condition of loading and unloading areas:

Remove or tie down all that might be displaced by the rotor downwash.

- Total weight of helicopter with load:

Define maximum acceptable load compatible with terrain configuration and atmospheric conditions. The maximum all up weight of the aircraft is that at which OGE hover can be held over the higher of the take-off or landing platforms.

3 AIRBORNE LOADS

Heavy loads, such as bags of cement or drums of kerosene, which are carried in a net, present no particular problem.

Special precautions must be taken in the case of bulky loads, which have a tendency to oscillate and even to "float" during transport on the sling.

Permeability to air can have a stabilizing effect on a bulky load: for example, a cable-car should be carried with its doors open.

Never carry an airfoil alone: There is a great risk of the airfoil flying up into the tail rotor.

If several cables are used to sling the load, they must be long enough to form an angle of less than 45° between cables at the point of suspension under the helicopter; experience shows that oscillation of the load is thus less likely to occur.

On the other hand, if the load is slung on a single sling cable, it is preferable that a fairly short cable be used as there is then less risk of the load swinging, and it is easier to judge the height of the load during approach.

For the retrieval of crashed helicopters, it is generally possible to use a lifting ring on the rotor mast.

Airplanes are carried using straps passing under the fuselage or under the wings. The cables must be attached in such a way that the airplane is in a slight nose-down attitude when the helicopter is in hover.

4 IN-FLIGHT PRECAUTIONS

After hooking on the load, the ground mechanic is to check the position of the sling cables then move away. The pilot must then make sure that the mechanic has moved clear and then be advised by signs that he may lift off the load.

Power must be applied slowly enough to allow the helicopter to center itself above the load.

A vertical take-off must be made, avoiding dragging the load along the ground or striking any obstacle.

If the load starts to swing, slowly reduce speed or make a gentle left turn.

Approach must be made head into the wind with gradual reduction in airspeed, and transition into hover high enough above the ground to eliminate the risk of dragging the load.

Set the load down, then reduce collective pitch sufficiently to slacken the cables before opening the release unit hook; this also allows the pilot to ensure that the load is deposited. If the cables are long enough, move sideways a little before opening the hook, to prevent the ring and tackle from falling onto the freight.

Even after the mechanic has signaled that the load is released, move away as if it were not; this is an advisable precaution against possible misinterpretation of signals.

Never fly away with an empty net or an unballasted sling.

5 CARGO RING SIZE RULE

- Care must be taken when hooking a load on the cargo hook.
- It is the responsibility of the operator to ensure that the cargo hook will operate properly with each ring.
- A placard located on both sides of the A/C illustrates the following:
 - Cables, straps, ropes, etc. must not be used directly on the cargo hook,
 - Only one primary ring linked to a unique secondary ring shall be attached to the cargo hook.
- Cargo hook ring size shall obey the following rules:

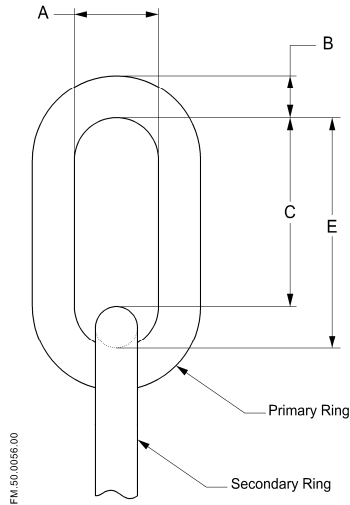


Figure 1: Cargo ring

Cargo Hook manufacturer	Cargo hook P/N	A	B	C	E
Breeze Eastern	17149-1	40mm-43mm	9.6mm-12.7mm	> 27.3mm	= A
Indraero SIREN	S 1609	50mm-80mm	12mm-28mm	> 50mm	< 80mm
Indraero SIREN	AS 21-5-7	40mm-110mm	16mm-20mm	> 40mm	< 110mm
Onboard Systems	528-023-51	> 50mm	< 26mm	> 55mm	Not specified

SECTION 9.2

EMERGENCY LOCATOR TRANSMITTER (KANNAD 406 AF-H OR 121 AF-H)

1 GENERAL

The KANNAD 406 AF-H or 121 AF-H radio beacon is an emergency transmitter which is used to locate the helicopter in case of an emergency.

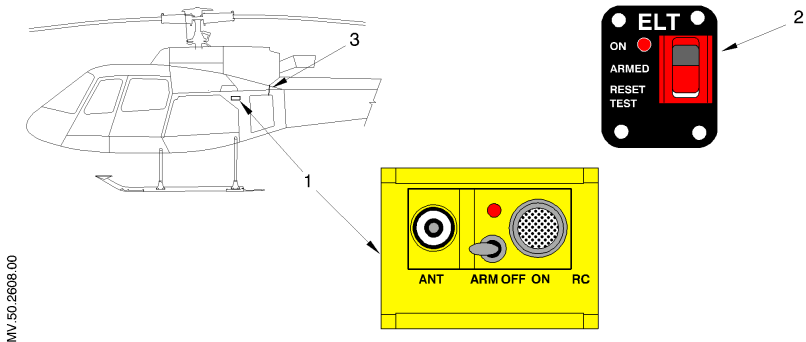
The KANNAD 121 AF-H radio beacon transmits simultaneously on the international frequencies of 121.5 MHz, 243 MHz.

The KANNAD 406 AF-H radio beacon transmits simultaneously on the international frequencies of 121.5 MHz, 243 MHz and 406.025 MHz.

The transmitter starts operating automatically in case of impact. It may be switched on manually via the switch located on the lower face of the transmitter or via the remote control switch located on the instrument panel.

2 DESCRIPTION AND OPERATION

- Components and location:
 - A transmitter (1), with an [ARM-OFF-ON] switch, attached to the structure inside the rear cargo hold,
 - An [ON-ARMED-RESET TEST] control switch located on the instrument panel (2),
 - An external antenna on the left side (3),



**Figure 1: Remote control switch and emergency locator
KANNAD 406 AF-H or 121 AF-H**

3 OPERATING PROCEDURE

- Pre-flight check:

Check the following on the transmitter:

- The switch is set to "ARM".

Check the following on the instrument panel

- The remote control switch is set to "ARMED".

- Functional test:

NOTE

Test procedure is to be conducted only once per month.

The self test mode is a temporary mode.

This mode is selected either:

- By switching the switch on the ELT from "OFF" to "ARM",
- By selecting "RESET TEST" on the remote control panel (provided that the switch on the ELT is in the "ARM" position).

The buzzer operates during the self test procedure.

CAUTION

As the ELT transmits on emergency frequencies, the self-test should be carried out only after authorisation by the control tower. If it is not possible to contact a control tower, the self-test must be carried out in the first five minutes of each hour.

NOTE

It is strictly prohibited to test the ELT by transmitting.

- Post-flight check:

After landing, set the VHF receiver to 121.5 MHz to ensure that the emergency locator transmitter has not accidentally been switched on.

NOTE

If the aircraft is to be grounded for a long time, set the switch on the ELT to the "OFF" position.

- Automatic operation:

The transmitter is activated automatically in the event of an impact, providing the switches are set to "ARMED".

NOTE

The "RESET TEST" position stops locator transmitting and resets the impact detector.

- Manual operation:

The unit may be activated manually by setting the control switch to "ON".

- Autonomous operation:

The transmitter may be used for self-contained operation on the ground as following:

- Remove the transmitter from its mounting bracket,
- Disconnect the coax from the aircraft antenna,
- Choose an unobstructed area,
- Extend the built-in tape antenna,
- Place the unit upright with the antenna upward,
- Switch on the transmitter by setting the **[ARM-OFF-ON]** switch to "ON".

SECTION 9.3

EMERGENCY LOCATOR TRANSMITTER (KANNAD INTEGRA AP-H)

1 GENERAL

The KANNAD INTEGRA AP-H radio beacon is an emergency transmitter with built-in GPS which is used to locate the helicopter in case of an emergency.

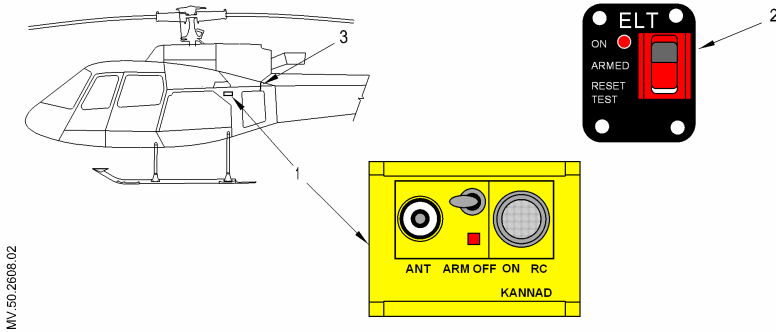
It transmits simultaneously on the international frequencies of 121.5 MHz and 406.025 MHz.

To avoid power consumption, the GPS receiver is not power supplied in "ARM" mode. After a crash (automatic activation) or manual activation, the GPS will try to acquire a position in continuous mode during one hour and by different sequences up to 24 hours of transmission. If the GPS receiver acquires a valid position, then the message will contain the true position in the next 406.025 MHz burst. If the GPS receiver does not acquire a valid position, then the message will contain the default value (GPS position not valid).

The transmitter starts operating automatically in case of impact. It may be switched on manually via the switch located on the lower face of the transmitter or via the remote control switch located on the instrument panel.

2 DESCRIPTION AND OPERATION

- Components and location:
 - A transmitter (1), with an **[ARM-OFF-ON]** switch, attached to the structure inside the rear cargo hold,
 - An auxiliary antenna attached to the transmitter,
 - An **[ON-ARMED-RESET TEST]** control switch located on the instrument panel (2),
 - An external antenna on the left side (3),



**Figure 1: Remote control switch and emergency locator
KANNAD INTEGRA AP-H**

3 OPERATING PROCEDURE

- Pre-flight check:
 - On the transmitter, check that the switch is set to "ARM",
 - On the instrument panel, check that the remote control switch is set to "ARMED".
- Functional test:

NOTE

Test procedure is to be conducted only once per month.

The self test mode is a temporary mode.

This mode is selected either:

- By switching the switch on the ELT from "OFF" to "ARM",
- By selecting "RESET TEST" on the remote control panel (provided that the switch on the ELT is in the "ARM" position).

The buzzer operates during the self test procedure.

NOTE

It is strictly prohibited to test the ELT by transmitting.

- Post-flight check:

After landing, set the VHF receiver to 121.5 MHz to ensure that the emergency locator transmitter has not accidentally been switched on.

NOTE

If the aircraft is to be grounded for a long time, set the switch on the ELT to the "OFF" position.

- Automatic operation:

The transmitter is activated automatically in the event of an impact, providing the switches are set to "ARMED".

NOTE

The "RESET TEST" position stops locator transmitting and resets the impact detector.

- Manual operation:

The unit may be activated manually by setting the control switch to "ON".

- Autonomous operation:

The transmitter may be used for self-contained operation on the ground as following:

- Remove the transmitter from its mounting bracket,
- Disconnect the coax from the aircraft antenna,
- Choose an unobstructed area,
- Connect the auxiliary antenna,
- Place the unit upright with the antenna upward,
- Switch on the transmitter by setting the **[ARM-OFF-ON]** switch to "ON".

SECTION 9.10

LOCATOR SEARCHLIGHT

1 GENERAL

The LOCATOR searchlight installation is intended to illuminate the ground using a swivelling light beam in order to facilitate certain missions (search and rescue, surveillance, etc.).

2 DESCRIPTION

This installation consists mainly of:

- A 450 W power light (Detail A) installed on the bottom, forward LH side of the lower structure, comprising:
 - A glass dome (3),
 - A swivelling parabolic reflector (4),
 - A fixed-arc lamp (5),
 - A housing (2),
 - A mount (1).
- A control handgrip (Detail B) which, when not used, is hooked onto a support located between the two seats,
- An amber light near the control handgrip support, which comes on to indicate that the searchlight is on.

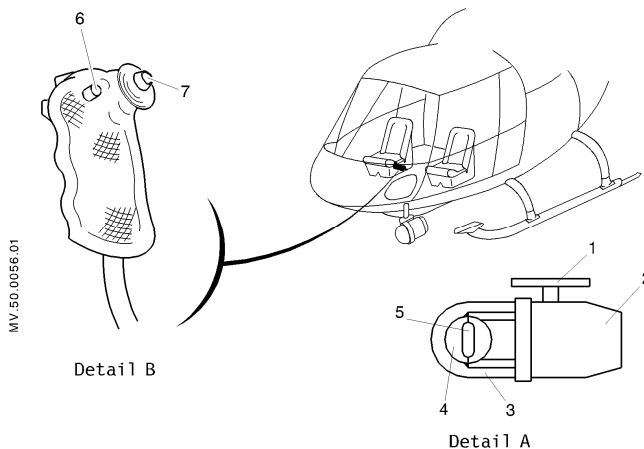


Figure 1: Additional LOCATOR searchlight

3 OPERATION

An ON/OFF pushbutton (6) located on the control grip is used to switch the LOCATOR searchlight installation on and off.

Full brightness is obtained 15 sec. after the searchlight has been switched on. This is confirmed by the amber indicator light which comes on.

A four-way button (7) is used to operate the reflector for orientating the light beam in the desired direction.

NOTE

To prevent any premature damage to the lamp it is advisable:

- **After the searchlight has been switched on, to wait 15 sec. before switching it off,**
- **After the searchlight has been switched off, to wait 30 to 60 sec. before switching it on again.**

SECTION 9.11

HAILERS

1 GENERAL

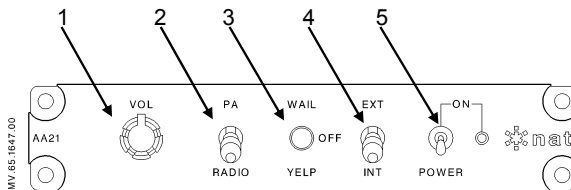
The hailers are intended to transmit either messages of a high sound level, or a continuous or intermittent signal (siren).

The effect of this optional equipment on the additional performance data is negligible.

2 DESCRIPTION

The system mainly consists of:

- An amplifier located in the LH side baggage hold,
- A dual hailer mounted on the landing gear rear cross beam,
- A control unit (AA21) located on the instrument panel,
- Fuses "HP" and "LS CTL" on the RH overhead panel 7a2.



- | | |
|--|--|
| 1- Volume level | 4- INT position is not connected |
| 2- Public address or radio mode | 5- System ON switch with green indicator |
| 3- 3-position switch : Continuous or intermittent mode | |

Figure 1: Public address control unit

System operation:

- ICS ON, set to "COM.3",
- control unit (AA21) ON.

NOTE

Switches 2 and 4 must be in position "PA" and "EXT" for the hailers to operate.

SECTION 9.12

ADDITIONAL LANDING LIGHT, CONTROLLABLE IN ELEVATION AND AZIMUTH

1 GENERAL

The additional landing light (3), controllable in elevation and azimuth, is an optional equipment designed for improving safety during approach and taxiing maneuvers.

This optional equipment is mounted on the underside of the lower structure, on the left of the fairing in front of the forward cross-tube of the landing gear.

Its power is 450 W.

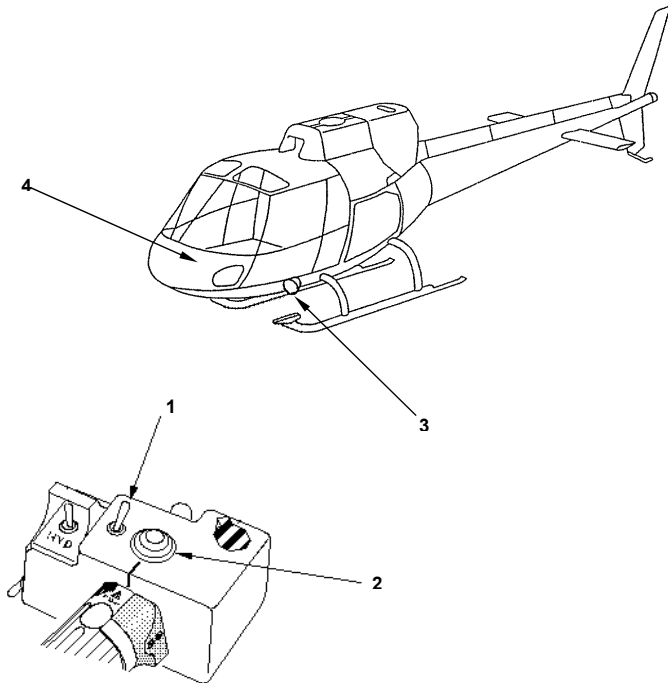


Figure 1: Additional landing light, controllable in elevation and azimuth

2 DESCRIPTION

The installation comprises:

- A retractable and swiveling light (3),
- A three-way switch (1) on each (pilot's and copilot's) collective pitch lever. This switch is used to switch the light on and retract it automatically,
- A four-way switch (2) on each (pilot's and copilot's) collective pitch lever. This switch is used to control the light in elevation and azimuth,
- A  "LAND LT" or "LDG LT" or "LIGHT" indicator light (4) on the instrument panel, which shows that one landing light is switched on.

3 OPERATION

The landing light is switched ON or OFF by means of the three-way switch **[ON-OFF-RETRACT]** on the collective pitch lever:

- When the switch is in the "ON" position, the landing light is switched on and the  "LAND LT" or "LDG LT" or "LIGHT" indicator light is on.
- When the switch is in the center "OFF" position, the landing light is switched off and the "LAND LT" or "LDG LT" or "LIGHT" indicator is off.
- The spring-loaded "RETRACT" position of the switch is used to switch off and retract the landing light automatically.

In this way, the landing light will always be switched off in the retracted position.

The extension and orientation of the landing light are controlled by means of the four-way switch.

4 PROTECTION OF THE CIRCUITS

The circuits are protected by means of:

- A 6.3 A fuse or breaker for the control circuits (located on the 31/32 α circuit breakers panel in the cockpit),
- A 20 A fuse or breaker for the lighting circuit (located in the 31/32 α panel).

SECTION 9.14

AIR AMBULANCE INSTALLATION

1 GENERAL

The air ambulance duty version is intended to carry one or two stretcher patients accompanied by one or two medical assistants seated on the RH rear bench seat.

2 DESCRIPTION

To be fitted out for the air ambulance role, it is necessary to remove the copilot's seat, the dual controls and if necessary the rear left hand bench seat.

The lower stretcher (6) rests on the cabin floor. It is secured with straps (3 and 5) to tie-down rings and brackets.

The top stretcher (1) is held by supports (2) on the rear bulkhead, a frame (4) at the front, and is secured by straps (7) to the floor-mounted tie-down rings.

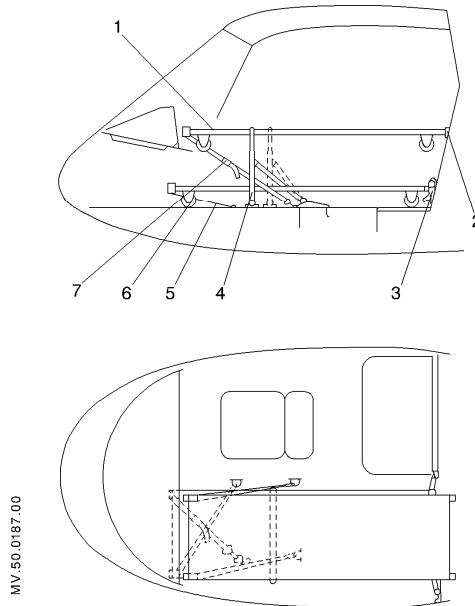


Figure 1: Air ambulance installation

3 OPERATING PROCEDURE

Three configurations are possible:

- 1 stretcher (upper or lower),
- 2 stretchers.

NOTE

If only one stretcher is being used it will save time to use the lower stretcher.

When not in service the stretchers are folded and stowed with their straps in the baggage hold. The upper stretcher support frame folds down onto the cabin floor.

The stretchers are installed in the following order:

1. Lower stretcher (6),
2. Upper stretcher (1).

3.1 PREPARATION OF THE CABIN

Installation of the air ambulance duty version requires a number of preliminary cabin alterations.

- Lower stretcher:
 - Remove: Dual controls, copilot's seat, seat cushions from LH rear passenger seat,
 - Fold up LH rear passenger seat against rear bulkhead.
- Upper stretcher:
 - Remove: Dual controls, (tail rotor control pedals need not be removed), copilot's seat, seat cushions from LH rear passenger seat, LH carpeting,
 - LH rear passenger seat remains open,
 - Raise the support to vertical position and secure.
- Upper and lower stretchers:
 - Remove: dual controls, copilot's seat, LH rear passenger seat cushions and seat, LH carpeting.

NOTE

For the "plush" version, both armrests of the LH rear passenger seat must be removed.

3.2 INSTALLING THE STRETCHERS

- Open the port side doors,
- Introduce the stretchers into the cabin forwards:
 - Place the lower stretcher on the cabin floor,
 - Place the upper stretcher on the support post.
- Engage the rear handles of the stretchers in the brackets on the rear bulkhead,
- Secure the retaining straps and hooks at the front and "PIP" pins at the rear.

CAUTION

The patients must be strapped to the stretchers and must be embarked feet forwards, head towards the tail.

SECTION 9.15

SKI INSTALLATION

1 GENERAL

The ski installation is intended for takeoff or landing on normal or snow-covered ground.

2 DESCRIPTION

The skis are secured to the skids via clamps.

SEFA skis have a glass-fibre/resin laminate structure and SURFAIR skis have a metal structure.

The rear spatula of the ski is reinforced with one or two struts.

SURFAIR skis enable 4 pairs of snow skis to be carried.

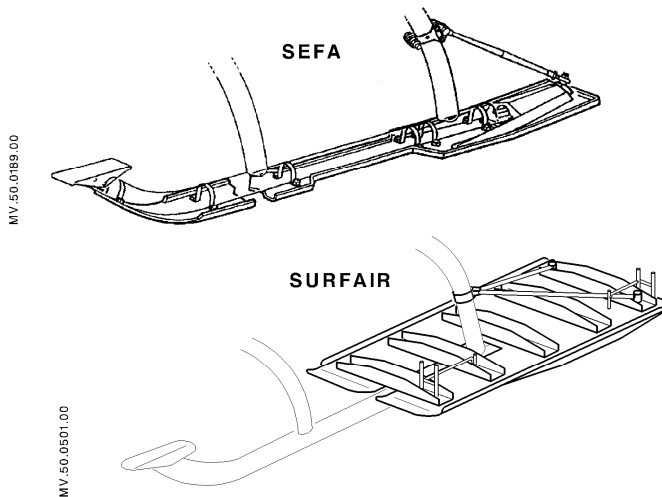


Figure 1: Ski installation

3 OPERATING PROCEDURE

Special attention is required regarding tail rotor ground clearance when landing in deep snow.

SECTION 9.17

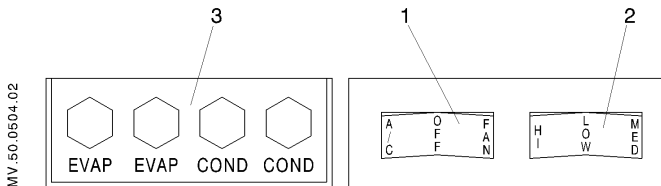
AIR CONDITIONING SYSTEM

1 GENERAL

The air conditioning system is intended to lower the ambient temperature within the cabin.

2 DESCRIPTION AND OPERATION

- Available to the pilot



1. Rocker switch:
 - OFF neutral (stops system operation),
 - FAN engaged (switches on ventilation),
 - A/C engaged (switches on ventilation and air conditioning).
2. Rocker switch:
 - LOW neutral (slow ventilation),
 - HI engaged (fast ventilation),
 - MED engaged (medium ventilation).
3. Protection fuses for condenser and fan blowers.

Figure 1: Air conditioning controls

- Available to the passengers

Six swivelling and adjustable outlets used to obtain the desired ventilation rate.

3 OPERATING PROCEDURE

System operation

- Set rocker switch 1:
 - To FAN to obtain cabin ventilation,
 - To A/C to obtain air conditioning.
- Select ventilation rate using rocker switch 2.

NOTE

It is recommended to close the external ventilation flap when using the system in the air conditioning mode, to get a better efficiency.

System shutdown

- Set rocker switch 1 to OFF (neutral position),
- Should the system fail, set rocker switch 1 to OFF.

4 PERFORMANCE DATA

The impact of the air conditioning system on the performance data given in the basic flight manual is negligible.

SECTION 9.18

NOISE REDUCTION

1 GENERAL

- Choose a flight path as far as possible from sensitive areas; otherwise, fly alongside the noisiest land routes (highways, railways),
- Fly at least at 1000 ft AGL,
- Fly if possible on the downwind side of sensitive areas,
- Maintain as much as possible steady flight, avoiding large pedal movements or over-control,
- Leave sensitive areas by turning to the right.

2 OPERATING IN SENSITIVE AREAS

- Fly-over:
 - Select IAS = 110 kt (204 km/h) for OAT = + 25°C,
 - Increase IAS slightly if OAT is higher, and inversely,
 - If possible, increase height to reduce the noise.
- Take-off and climbing from a helipad in a sensitive area:
 - After the shortest possible acceleration phase, once V_y is reached, set to MCP while maintaining V_y .
- Approach and landing on a helipad in a sensitive area:
 - Select IAS = 60 kt (111 km/h), with a rate of descent close to 1000 ft/min.
- Take-off from and landing on a helipad in a non sensitive area but adjacent to neighboring sensitive areas (seaside areas for example):
 - If possible choose a take-off path opposite to the sensitive area. Accelerate until V_y is reached, then start climbing at this speed with MCP,
 - If possible for landing, choose a path facing the sensitive area. Select V_y with a rate of descent close to 500 ft/min.
- Maneuvers near the ground (around hovering) in a sensitive area:
 - Avoid unnecessary hovering,
 - Avoid quick and repetitive pedal movements,
 - Prefer right spot turns.

SECTION 9.20

RADIO ALTIMETER (AHV 16)

1 GENERAL

The radio altimeter:

- Provides an accurate measurement of the aircraft height relative to the ground, regardless of the atmospheric conditions,
- Informs the crew when the aircraft descends below a decision height or below 100 ft*.

2 DESCRIPTION

2.1 COMPOSITION

The radio altimeter installation consists of:

- A transceiver,
- An indicator located on the instrument panel,
- An warning box,
- A transmission antenna,
- A reception antenna,
- 2.5 A circuit-breaker located on the 30A circuit breakers panel (on SCU LH side).

2.2 POWER SUPPLY

The system is powered by the aircraft 28 Vdc power system.

The location of the circuit breaker that protects this system is given in section 7.

(*) If installed

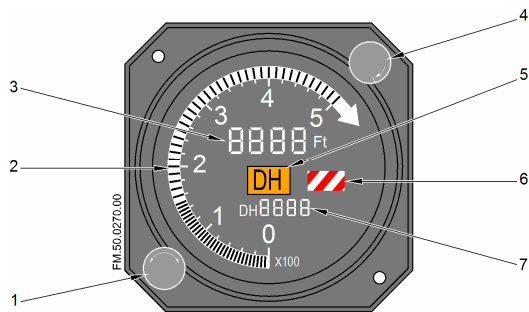
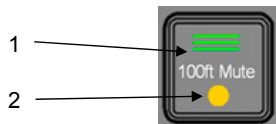


Figure 1: Radio altimeter indicator

Item	Description
1	"DH" selector: Decision height setting.
2	Scale 0 - 500 ft.
3	Height digital display (0 to 8000 ft).
4	"ON/OFF/TEST" selector.
5	"DH" indicator: The indicator flashes when the decision height is reached.
6	Flag: The flag appears if: - The system is faulty, - In test mode, - Above 8000 ft.
7	Decision height display.

2.3 CONTROL (ON SCU)



Push-button..... DISENGAGED:

- Indicator light (1) : remains off,
- Indicator light (2) : remains off, ("100 ft" audio alert is audible).

Push-button..... ENGAGED:

- Indicator light (1) : comes on,
- Indicator light (2) : comes on, ("100 ft" audio alert is muted).

3 OPERATION

3.1 GENERAL

The radio altimeter is switched on using the ON/OFF/TEST selector (4) on the indicator.

The DH knob (1) is used to set the decision height.

A specific audio alert (800 Hz modulation chopped at 5 Hz) is produced when the helicopter descends through the decision height.

NOTE

The DH audio alert is (re-)armed after the helicopter climbs at least 20 ft above DH.

When hovering low or when landing with the radio-altimeter antenna(s) above grass or plant cover, the radio-height may become invalid (flag appears). Such invalidities are more likely to occur in the presence of wind.

3.2 "100FT" AUDIO ALERT (IF INSTALLED)

A specific audio alert (800 Hz modulation chopped at 15 Hz) is produced when the helicopter descends through 100 ft.

For operations around 100 ft where the recurrence of the "100 ft" alert may hamper aircrew performance, the "100 ft mute" function may be used.

- "100 ft Mute" activation

[100ft Mute]ENGAGED, CHECK
the lights come on.



- "100 ft Mute" de-activation

[100ft Mute]DISENGAGED, CHECK
the lights go off.



NOTE

The "100 ft" audio alert is (re-)armed after the helicopter climbs above 120 ft.

4 TEST PROCEDURE

- Run-up checks:

Check the radio altimeter installation:

- 1. "ON/OFF/TEST" selector (4)..... ON,
- 2. "DH" selector (1) CHECK, $0 < DH < 100$ ft,
- 3. [100ft Mute] DISENGAGED,
- 4. "ON/OFF/TEST" selector (4)..... TEST, and CHECK:
 - Display of all front face segments,
(Test height, Flag, DH value),
 - "100 ft" audio alert,
 - "DH" audio alert.

SECTION 9.21

INTERCOM SYSTEM (GMA 340 H)

1 GENERAL

The Garmin GMA 340 H intercom system audio panel ensures the following main functions:

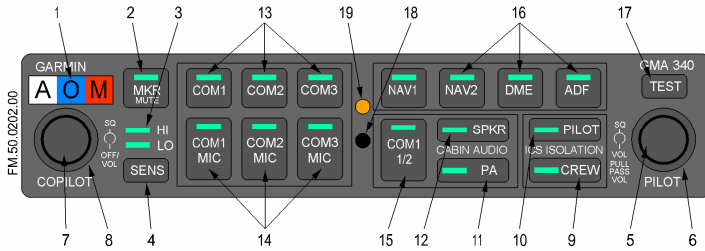
- Intercommunications between pilot, copilot and passengers,
- Radio-communication microphone and audio selection,
- Radio-navigation audio selection,
- Links with other audio inputs (music, public address, etc.).

The GMA 340 H features a failsafe mode in case the power supply is lost or the unit is inadvertently turned off.

2 INSTALLATION AND OPERATION DESCRIPTION

2.1 COMPONENTS AND LOCATION

- A panel located on the upper part of the center console with all necessary switches for available modes and audio sources selection,
- A 2.5 A circuit-breaker located on the 30A circuit breaker panel (on SCU RH side).



MKR

1. MKR beacon lights
2. MKR audio/ mute selector
3. MKR sensitivity indicator
4. MKR sensitivity selector

ICS

5. Pilot/Passengers volume (pull out to adjust passenger volume)
6. Pilot ICS squelch level (VOX)
7. Copilot volume
8. Pilot ICS squelch level (VOX)
9. Crew isolation mode
10. Pilot isolation mode
11. "Public Address" mode
12. "Speaker" mode

COM/NAV

13. COM receiver selectors
14. COM transmitter selectors
15. Split COM selector
16. NAV systems audio selectors

Other functions

17. Annunciator lights test pushbutton
18. Locking screw access
19. Photocell (automatic dimming)

Figure 1: GMA 340 H audio panel

2.2 OPERATING PROCEDURES

- Preflight check:

Check the following:

- GMA 340 unitON (copilot volume control turned clockwise past the click), pushbutton lights ON,
- ICS volumeSET for pilot, copilot and passengers,
- Radio channel.....SET upon request,
- Radio volume.....SET by selected COM volume control.

NOTE

The RH inner knob (5) is used for two functions:

- IN position: Pilot ICS volume setting,
- OUT position: Passenger ICS volume setting.

- GMA 340 failure or ICS power supply failure:
 - Turn off GMA 340 H (turn copilot ICS volume control to full counter-clockwise detent),
 - Intercom function is lost,
 - A failsafe circuit connects the pilot's headset and microphone directly to COM1.

NOTE 1**In case of ICS failure:**

- **COM1 is the only radio-communication system available,**
- **The RH pilot only has access to COM1.**

NOTE 2

Audio warnings are still available through the pilot headset only.

- Normal operating procedures:

The detailed operating procedures are described in the GMA 340 H Pilot's guide.

